BONUS QUESTION BANK

General Pediatrics Clinical Manual

550 Additional ABP-Style Board Review Questions with Full Explanations

Tobias R. Lindqvist, MD
Warren J. Oakhurst, DO

2027 Edition

How to Use This Bank

> **Open this file in your web browser** (Chrome, Safari, Edge) or in Acrobat. The links work there. They may not work in Google Drive previews or inside some apps.

This bank holds 550 single-best-answer questions written in the style of the board examination. Each has five options and a full explanation. The questions are new and do not repeat the review questions printed in the book.

To move around: every page has a bar at the bottom with links to the Contents, the start of the Questions, the start of the Answers, and this page. Under every question, tap **Answer** to jump to its explanation, and tap **Back to question** to return. Tap any subject in the Contents to open its questions or its answers.

To study: work one subject at a time, commit to an answer before you open the explanation, then read the explanation for the principle and for the trap in the closest wrong option. Finish with mixed sets from several subjects.

Each question carries a tag in brackets: Physiology and Pathophysiology, Epidemiology and Risk Assessment, Diagnosis, or Management and Treatment. The bank is an educational aid. It is not affiliated with, endorsed by, or approved by the American Board of Pediatrics. Check every dose and threshold against current labeling and local protocol.

Contents

Well-Child, Newborn, and Adolescent			
1. Preventative Pediatrics/Well-Child Care BQ-001 to BQ-066 (66 questions)	Questions	Answers	p. 6 / 162
2. Fetal and Neonatal Care BQ-067 to BQ-088 (22 questions)	Questions	Answers	p. 23 / 179
3. Adolescent Care BQ-089 to BQ-116 (28 questions)	Questions	Answers	p. 29 / 185
Infection, Mind, and Safety			
4. Infectious Diseases BQ-117 to BQ-155 (39 questions)	Questions	Answers	p. 37 / 192
5. Mental and Behavioral Health BQ-173 to BQ-205 (33 questions)	Questions	Answers	p. 48 / 203
6. Psychosocial Issues BQ-156 to BQ-172 (17 questions)	Questions	Answers	p. 58 / 212
8. Child Abuse and Neglect BQ-228 to BQ-244 (17 questions)	Questions	Answers	p. 69 / 223
24. Patient Safety, Quality Improvement, and Research Methods BQ-540 to BQ-550 (11 questions)	Questions	Answers	p. 158 / 309
23. Ethics BQ-529 to BQ-539 (11 questions)	Questions	Answers	p. 155 / 306
Organ Systems			
9. Orthopedics and Sports Medicine BQ-245 to BQ-266 (22 questions)	Questions	Answers	p. 74 / 228
10. Eye, Ear, Nose, and Throat BQ-267 to BQ-299 (33 questions)	Questions	Answers	p. 80 / 234
11. Cardiology BQ-300 to BQ-327 (28 questions)	Questions	Answers	p. 89 / 243
12. Pulmonology BQ-328 to BQ-355 (28 questions)	Questions	Answers	p. 97 / 250
13. Gastroenterology BQ-356 to BQ-382 (27 questions)	Questions	Answers	p. 105 / 258
14. Neurology BQ-383 to BQ-404 (22 questions)	Questions	Answers	p. 112 / 265
15. Skin/Dermatology BQ-405 to BQ-426 (22 questions)	Questions	Answers	p. 118 / 271
Acute, Complex, and Specialty Care			
7. Emergency and Critical Care BQ-206 to BQ-227 (22 questions)	Questions	Answers	p. 63 / 217
16. Hematology-Oncology BQ-427 to BQ-442 (16 questions)	Questions	Answers	p. 124 / 277

17. Allergy and Immunology BQ-443 to BQ-458 (16 questions)	Questions	Answers	p. 129 / 282
18. Endocrinology BQ-459 to BQ-474 (16 questions)	Questions	Answers	p. 134 / 287
19. Nephrology, Fluids, and Electrolytes BQ-475 to BQ-490 (16 questions)	Questions	Answers	p. 139 / 291
20. Genitourinary System BQ-491 to BQ-506 (16 questions)	Questions	Answers	p. 144 / 295
21. Genetics, Dysmorphology, and Metabolic Disorders BQ-507 to BQ-517 (11 questions)	Questions	Answers	p. 149 / 300
22. Rheumatology BQ-518 to BQ-528 (11 questions)	Questions	Answers	p. 152 / 303

Questions

Choose the single best answer. Tap Answer under any question to see its explanation.

Back to Contents

Domain 1. Preventative Pediatrics/Well-Child Care

Contents Answers for this subject

BQ-001. A 30-month-old girl is brought for a well-child visit. Her father has no concerns, and the pediatrician observes her combining two words, running, and pretending to feed a doll. Her last standardized developmental questionnaire was completed at 18 months and was normal. No questionnaire has been given since then. Which is the best next step? [Management and Treatment]

A. Give a standardized general developmental screening tool at this visit

B. Wait for the 36-month visit because surveillance found nothing abnormal today

C. Give only the autism-specific screening tool because that is the one due now

D. Refer her for a full developmental evaluation without any screening tool first

E. Ask the father to complete the CDC milestone checklist in place of a tool

Answer Contents

BQ-002. A 24-month-old boy is seen for a routine visit. At 18 months, a parent-completed autism screening questionnaire was negative. His mother now reports about 30 words, parallel play beside other children, and a normal response to his name. She has no concerns, and he has lost no skills. Which is the best next step? [Management and Treatment]

A. Accept the 18-month result as final and omit further autism screening

B. Repeat the autism-specific screening tool at this visit

C. Repeat the autism-specific screening tool only at the 30-month visit

D. Order a hearing test and delay any screening until the result returns

E. Rely on the mother's report because she voices no concerns at all today

Answer Contents

BQ-003. A pediatrician reviews the 2022 CDC milestone checklist with the mother of a healthy 15-month-old girl who is not yet taking steps alone. The mother says her older son also walked late and that half of children walk later than this. The checklist lists taking a few steps alone at 15 months. Which is the most accurate interpretation of this milestone? [Epidemiology and Risk Assessment]

A. About 50 percent of healthy children are expected to show the skill by this age

B. About 90 percent of healthy children are expected to show the skill by this age

C. Only children over the 97th percentile are expected to show the skill by this age

D. At least 75 percent of children are expected to show the skill by this age

E. This age marks the earliest point at which any child could show the skill

Answer Contents

BQ-004. A 6-month-old boy born 10 weeks early is brought for a routine visit. He sits with support, reaches for toys, and laughs but cannot roll over reliably. His mother worries that he lags behind other 6-month-olds in her playgroup, who sit alone. Which statement best explains why his skills should be judged against a corrected age of about 3.5 months? [Physiology and Pathophysiology]

A. Early birth permanently lowers the milestone standard for every skill and age

B. Neurologic maturation follows time since conception more closely than time since birth

C. Preterm infants acquire skills at a faster rate after discharge from the hospital nursery

D. Chronologic age is a less reliable measure than weight in preterm infants

E. Corrected age applies only to growth charts and never to developmental skills

Answer Contents

BQ-005. A 4-year-old girl is at the 3rd percentile for height with a normal growth velocity. Her father is 70 inches tall and her mother is 63 inches tall. The pediatrician wants to compare her height with her genetic potential. Using the standard formula for girls, which is her mid-parental target height? [Epidemiology and Risk Assessment]

A. 61.5 inches

B. 66.5 inches

C. 69 inches

D. 64 inches

E. 63 inches

Answer Contents

BQ-006. A 7-year-old boy is at the 2nd percentile for height, and his growth velocity has been normal across three years of plotted visits. His mother is 59 inches tall and his father is 66 inches tall. His bone age matches his chronologic age, and he has no signs of puberty yet. He is otherwise well. Which is the most likely diagnosis? [Diagnosis]

A. Constitutional delay of growth and puberty

B. Familial short stature

C. Growth hormone deficiency from a pituitary disorder

D. Acquired primary hypothyroidism with slowed growth

E. Celiac disease with chronic malabsorption

Answer Contents

BQ-007. A 6-year-old girl has fallen from the 30th percentile to under the 3rd percentile for height over 18 months. For the last month she has had morning headaches, and her mother notes that she holds books close to her face. The neurologic examination is otherwise normal. Which is the most likely cause of her poor growth? [Diagnosis]

A. Familial short stature with a low parental target height

B. Constitutional delay of growth and puberty with later catch-up

C. A mass in the hypothalamic or pituitary region

D. Primary hypothyroidism with slowed linear growth over a year

E. Celiac disease with chronic malabsorption of nutrients and minerals

Answer Contents

BQ-008. A 10-year-old girl has a BMI at the 96th percentile for age and sex. Her blood pressure and screening laboratory results are normal. Her mother says, "She is just growing out of it, so let's wait a year and watch." The girl enjoys sports and is willing to make changes. Which is the best next step? [Management and Treatment]

A. Recheck her BMI in one year before beginning any form of treatment

B. Prescribe weight-loss medication because her BMI is at the 96th percentile

C. Arrange metabolic and bariatric surgery evaluation to prevent future weight gain

D. Start intensive health behavior and lifestyle treatment for the whole family now

E. Place her on a strict calorie-restricted diet with weekly weighing at home

Answer Contents

BQ-009. A 10-year-old boy has a BMI of 31 kg/m2. The 95th percentile BMI for boys his age is 24.5 kg/m2. He feels well, and his mother asks how his weight is classified. His blood pressure is normal. Which is the most likely classification? [Diagnosis]

A. Severe obesity

B. Overweight

C. Obesity that stays under the severe range

D. Normal weight with high muscle mass

E. Severe obesity that calls for surgical treatment

Answer Contents

BQ-010. A 9-month-old girl has a weight that is 65 percent of the expected weight for her length. She is listless, has bruises of different ages on her trunk, and her mother offers little history. Outpatient feeding advice a month ago produced no weight gain. Which is the most likely diagnosis? [Diagnosis]

A. Cystic fibrosis with pancreatic insufficiency causing fat malabsorption

B. Congenital heart disease with poor feeding and heart failure

C. Growth hormone deficiency with a slowed growth rate

D. Severe malnutrition with suspected neglect or abusive injury

E. Celiac disease with a chronic malabsorption of gluten

Answer Contents

BQ-011. A 4-month-old boy has crossed two major percentiles downward in weight over 2 months, while his length and head circumference are preserved. He is alert and his examination is normal. His mother, who recently lost her job, says she adds extra water to each bottle so the formula will last longer. Which is the most likely cause of his poor weight gain? [Diagnosis]

A. Cystic fibrosis with fat malabsorption

B. Congenital hypothyroidism with slowed metabolism

C. Celiac disease with intestinal injury

D. Congenital heart disease with heart failure

E. Inadequate calories from diluted formula

Answer Contents

BQ-012. A 5-month-old exclusively breastfed girl was plotted on the older CDC 2000 weight-for-age chart. She appears to have drifted from the 60th to the 25th percentile since 2 months of age. She is vigorous, feeds well, and has normal stools and wet diapers. Which best explains the apparent drift? [Physiology and Pathophysiology]

A. Her milk supply has fallen since the second month, so she now takes in fewer calories each day

B. The older chart used a reference population that grew faster than healthy breastfed infants

C. Early fat malabsorption has slowed her weight gain after the first 2 months

D. Her weights were measured with clothes on, which lowers each plotted point

E. Subclinical hypothyroidism has slowed her growth away from her original channel

Answer Contents

BQ-013. A 7-year-old girl has had breast budding for 4 months. Her mother notes a recent growth spurt, and her height has risen from the 50th to the 80th percentile. Breast development has progressed from Tanner stage 2 to stage 3 over the last 3 months. She has no headaches or visual change. Which is the most likely diagnosis? [Diagnosis]

A. Central precocious puberty

B. Premature thelarche

C. Premature adrenarche

D. Constitutional growth acceleration

E. Exposure to an estrogen-containing cream

Answer Contents

BQ-014. A 6-year-old boy has had pubic hair and body odor for two months. His testes measure 2 mL each, his penis is prepubertal, and his growth velocity is normal. He has no headaches or visual change. Which mechanism most likely explains his findings? [Physiology and Pathophysiology]

A. Early activation of the hypothalamic-pituitary-gonadal axis with testicular growth

B. Early adrenal androgen secretion without activation of the gonadal axis

C. Autonomous testosterone production by a testicular tumor on one side

D. Excess growth hormone secretion from a pituitary adenoma in the brain

E. Absorption of androgen from a topical product used by a family member

Answer Contents

BQ-015. A 13-year-old girl, 6 months past her birthday, has no breast development. Her height tracks the 40th percentile, and she has no pubic hair. Her mother reached menarche at 15 years. Her examination is otherwise normal, and she asks whether something is wrong. Which description best fits her pubertal status? [Diagnosis]

A. Delayed puberty

B. Normal timing of puberty

C. Precocious puberty

D. Isolated premature adrenarche

E. Pubertal arrest after a normal start

Answer Contents

BQ-016. A 12-year-old boy has no pubic hair, and his mother is concerned that puberty has not started. On examination, each testis measures 4 mL with reddened, thinned scrotal skin, and the penis is still prepubertal. His height is on the 50th percentile. Which description best fits his pubertal status? [Diagnosis]

A. Genital Tanner stage 1, still prepubertal

B. Genital Tanner stage 2, the first sign of puberty

C. Genital Tanner stage 3, with penile lengthening and darker pubic hair

D. Delayed puberty needing endocrine testing now

E. Isolated adrenarche with no pubertal change

Answer Contents

BQ-017. A 12-year-old girl has breast tissue and areola that have enlarged together with no separation of their contours. Her pubic hair is darker, coarser, and curled over the mons. She has not had menarche. Which Tanner stage best describes her breast development? [Diagnosis]

A. Stage 2

B. Stage 4

C. Stage 3

D. Stage 5

E. Stage 1

Answer Contents

BQ-018. A 26-month-old boy walked at 13 months and ran by 20 months. Over three weeks he has become unsteady, drags his right leg, and no longer climbs onto the couch. He has no fever or recent trauma. His parents ask whether he is just going through a phase. Which best describes this change in his development? [Diagnosis]

A. Normal variation during a toddler growth spurt in height

B. Regression, a loss of motor skills the child had already gained

C. Isolated motor delay that needs only physical therapy to improve

D. Benign toddler gait that resolves with further maturation

E. Behavioral regression caused by stress in the home or at daycare

Answer Contents

BQ-019. A 20-month-old girl has 4 words and does not combine them. She was born at term weighing 2,300 g. Her parents speak two languages at home, and she is the third of three children. Her hearing has not been checked. Which feature most increases her risk of speech delay? [Epidemiology and Risk Assessment]

A. Exposure to two languages at home

B. Birth order as the third of three children

C. Birth weight of 2,500 g or less

D. Female sex

E. Birth at term gestation

Answer Contents

BQ-020. A 5-month-old boy's head circumference has climbed from the 50th percentile to over the 97th percentile in two months. He has vomited for 3 days, his anterior fontanelle is bulging and tense, and his eyes tend to deviate downward. Which mechanism most likely explains these findings? [Physiology and Pathophysiology]

A. Benign familial macrocephaly with normal intracranial pressure and a stable head growth channel

B. Premature closure of a skull suture that restricts growth of the brain

C. Viral gastroenteritis with dehydration and secondary sunken fontanelle changes

D. Normal catch-up growth of the head after a period of poor nutrition

E. Impaired circulation or absorption of cerebrospinal fluid, raising intracranial pressure

Answer Contents

BQ-021. A 12-month-old girl is seen for a well-child visit. Her mother has completed the CDC 12-month milestone checklist and marked every item as met, and the pediatrician's observation agrees. A general developmental screening tool was normal at 9 months. Which is the best next step? [Management and Treatment]

A. Continue surveillance today and give the next general screen at 18 months

B. Give a general developmental screening tool today to confirm the checklist

C. Give an autism-specific screening tool today as part of routine care

D. Request a full developmental evaluation to make sure nothing is missed

E. Give both a general and an autism screen at every visit from now on

Answer Contents

BQ-022. A 33-month-old boy had an abnormal general developmental screen at the 30-month visit. His hearing is normal, and a medical evaluation found no cause. His parents ask who will provide services now that he is approaching his third birthday. Which is the best next step? [Management and Treatment]

A. Wait until his third birthday and then apply only to a private therapy center

B. Defer any referral until a specialist has confirmed a specific diagnosis

C. Repeat the screening tool at 36 months and send him to the school district only if the result remains abnormal

D. Advise the family to enroll him in an ordinary preschool and review at age 5

E. Arrange an early intervention evaluation now, with planning for preschool special education at age 3

Answer Contents

BQ-023. A 26-month-old boy has a routine capillary blood lead level of 52 µg/dL. He is afebrile and playing, and his examination is normal. The family is renovating a house built in 1950, and he plays near sanded woodwork. Which is the best next step? [Management and Treatment]

A. Obtain a venous blood lead level within 48 hours and consult a toxicology expert

B. Obtain a venous blood lead level within 2 weeks and counsel the family about hand washing

C. Obtain a venous blood lead level within 1 month and inspect the home for paint

D. Obtain a venous blood lead level within 3 months and repeat the test then

E. Repeat the capillary level at the next routine visit and review the result

Answer Contents

BQ-024. The mother of a 2-year-old boy is relieved that his venous lead level is 3.0 µg/dL, which is under the current CDC reference value of 3.5 µg/dL. She asks whether this means his level is safe. The family lives in a 1960s apartment with peeling paint. Which statement about the reference value is most accurate? [Epidemiology and Risk Assessment]

A. It is the level under which lead causes no measurable harm in any child

B. It is the blood level at which chelation therapy should begin for any young child with known exposure at home

C. It is the average blood lead level among all children in the country

D. It is the level at which a child must be moved out of the home

E. It marks the 97.5th percentile of children's levels and is a trigger for action, not proof of safety

Answer Contents

BQ-025. A 4-year-old girl enrolled in Medicaid is new to the practice. Her records show no blood lead testing at 12 or 24 months. She is asymptomatic and lives in an older apartment. Her mother asks whether testing is still worthwhile at this age. Which is the best next step? [Epidemiology and Risk Assessment]

A. Obtain a blood lead level now because she was never tested

B. Skip testing because the routine testing ages have already passed

C. Test only if she develops symptoms such as irritability or abdominal pain

D. Wait until school entry and obtain the level at the kindergarten physical

E. Obtain only a hemoglobin, since a normal value excludes lead exposure

Answer Contents

BQ-026. A 13-month-old girl had a hemoglobin of 10.1 g/dL at her 12-month visit. Ferritin was low with a normal C-reactive protein. She started daily oral iron and cut back on milk. Today, one month later, her hemoglobin is 11.4 g/dL. Which best describes this result? [Diagnosis]

A. Spontaneous improvement unrelated to the iron she was given

B. Thalassemia trait with a stable hemoglobin level

C. A response that confirms iron deficiency anemia

D. Anemia of chronic inflammation that has begun to resolve

E. Lead toxicity improving because her milk intake was reduced

Answer Contents

BQ-027. A 10-year-old boy has a non-fasting non-HDL cholesterol of 152 mg/dL at his universal lipid screen. He is of normal weight and has no family history of early heart disease or high cholesterol. He is asymptomatic. Which is the best next step? [Management and Treatment]

A. Repeat a fasting lipid panel and average the results

B. Start a statin now because the level is over the cutoff

C. Discard the result because a non-fasting sample is not valid

D. Wait until the 17-to-21-year screen and repeat the test then

E. Order genetic testing for familial hypercholesterolemia right away

Answer Contents

BQ-028. A 15-year-old girl has a routine visit. Her BMI is at the 50th percentile, and she has no family history of early cardiovascular disease or high cholesterol, no diabetes, no hypertension, and no tobacco use. She had a normal universal lipid screen at age 10. Which is the best approach to lipid testing? [Epidemiology and Risk Assessment]

A. Order a fasting lipid panel today because she is now an adolescent

B. Perform the next universal lipid screen between 17 and 21 years

C. Repeat the lipid screen every year until she reaches age 21

D. Order a lipid panel today to confirm the result from age 10

E. Order apolipoprotein and lipoprotein testing as a baseline now

Answer Contents

BQ-029. A healthy 2-year-old boy, born at term, is seen for a well-child visit. He has no history of kidney disease, heart disease, or neonatal intensive care. His mother asks whether his blood pressure should be measured at every visit from now on. Which is the best answer? [Epidemiology and Risk Assessment]

A. Begin routine yearly measurement at age 3 unless a risk condition develops

B. Measure at every visit from now on to build a baseline

C. Begin routine yearly measurement at age 6 when he enters elementary school

D. Measure only if he becomes overweight or has symptoms

E. Measure only before sports participation in adolescence

Answer Contents

BQ-030. A 15-year-old boy has an average of three auscultatory blood pressure readings of 134/82 mmHg, taken with the correct cuff after 5 minutes of seated rest. He is asymptomatic. His BMI is at the 92nd percentile. Which best describes this blood pressure? [Diagnosis]

A. In the elevated blood pressure range

B. In the stage 1 hypertension range

C. In the stage 2 hypertension range

D. In the normal range for his age

E. In the hypertensive emergency range

Answer Contents

BQ-031. A 16-year-old girl is brought with a severe headache and blurred vision. Her auscultatory blood pressure is 190/125 mmHg and repeats at 188/124 mmHg. She is alert and has no focal weakness. Her mother asks whether to schedule outpatient ambulatory monitoring. Which is the best next step? [Management and Treatment]

A. Schedule ambulatory blood pressure monitoring at the clinic for the following week

B. Start lifestyle changes and recheck her pressure in three months

C. Ask her to measure blood pressure at home for two weeks

D. Send her to the emergency department for urgent blood pressure treatment

E. Order a renal ultrasound study and review it in the office tomorrow

Answer Contents

BQ-032. A 4-year-old girl has confirmed stage 2 hypertension on repeated auscultatory readings at two visits. The history is unremarkable, and arm and leg pressures are similar. Urinalysis, creatinine, electrolytes, and a lipid panel have been ordered. Which additional study is recommended at her age? [Management and Treatment]

A. Abdominal computed tomography with contrast

B. Overnight polysomnography

C. Skeletal survey for fractures

D. Bone age radiography

E. Renal ultrasonography

Answer Contents

BQ-033. A 3-week-old girl failed her newborn hearing screen in both ears, and a repeat screen at 2 weeks also failed. She feeds and grows well. Her grandmother says many babies pass later and the family should wait. Which is the best next step? [Management and Treatment]

A. Repeat the screen at the 6-month visit and act only if the result fails again then

B. Arrange diagnostic audiology now so that diagnosis is complete by 3 months of age

C. Wait for speech delay to appear before arranging any hearing testing

D. Fit hearing aids immediately without any diagnostic hearing testing

E. Delay evaluation until 12 months, when behavioral testing is easier

Answer Contents

BQ-034. A 13-year-old boy plays basketball. At his annual visit he reports that he fainted once while running in a game last month and had chest tightness first. His mother's cousin died suddenly at age 28 during exercise. His examination is normal. Which is the best next step? [Epidemiology and Risk Assessment]

A. Clear him for sports and advise more fluids before games

B. Reassure the family that fainting is vasovagal and recheck next year

C. Hold him from sports and arrange a cardiology evaluation

D. Obtain a lipid panel and clear him if the results are normal

E. Clear him for non-contact sports while waiting for a routine appointment

Answer Contents

BQ-035. A 4-month-old girl was born at term after an uncomplicated pregnancy. At her visit, the red reflex is bright in the right eye, but the left pupil shows a white reflex. She is otherwise well and follows faces. Her mother says it may be the camera light in photographs. Which best describes the left eye finding? [Diagnosis]

A. A normal variant that is common at this age and resolves later

B. A photographic artifact with no clinical meaning for her vision

C. A sign of a minor conjunctival infection that settles with drops

D. An abnormal red reflex needing prompt ophthalmology evaluation

E. An expected change as the lens matures by the age of 6 months

Answer Contents

BQ-036. A 3-year-old boy has an intermittent inward turn of the left eye noted since age 2. The red reflexes are normal and equal. His mother asks why the pediatrician stresses early vision screening. Which mechanism best explains why untreated strabismus can cause permanent vision loss in the deviated eye? [Physiology and Pathophysiology]

A. The retinal photoreceptors of the deviated eye degenerate from a chronic lack of light exposure there

B. The lens of the deviated eye becomes cloudy from persistent misalignment

C. The optic nerve of the deviated eye shrinks because of mechanical traction

D. The cornea of the deviated eye scars from chronic exposure to air

E. The developing visual cortex suppresses input from the misaligned eye, leaving it with poor acuity

Answer Contents

BQ-037. A 20-month-old girl has chalky white bands along the gumline of her upper front teeth. She carries a sippy cup of juice through the day and falls asleep with a bottle of milk. She has never seen a dentist. Which mechanism best explains these lesions? [Physiology and Pathophysiology]

A. Bacteria on the tooth surface ferment sugar to acid that demineralizes enamel

B. Excess systemic fluoride during enamel formation that mottles the surface of new teeth

C. Calcium deficiency from rickets that weakens the structure of enamel

D. Direct injury to the enamel from the nipple of the bottle

E. Viral infection of the gums that spreads onto the tooth surface

Answer Contents

BQ-038. A 3-year-old boy was adopted from another country two weeks ago. His records list vaccines, but the dates and products are unclear. He seems well, and his adoptive parents ask which tests he needs. Which is the best next step? [Management and Treatment]

A. Accept the vaccine record as written and test only if he develops a fever

B. Screen for hepatitis B, hepatitis C, HIV, tuberculosis, and intestinal parasites, and verify immunizations

C. Test only for tuberculosis, since the other infections would already have been identified at the time of adoption

D. Obtain only a blood lead level and a hemoglobin and defer other testing

E. Wait for 6 months before any testing so the family can adjust to him

Answer Contents

BQ-039. A 9-year-old boy is 53 inches tall and rides in the back seat. He has outgrown a forward-facing harness seat and now uses only the vehicle lap and shoulder belt. The shoulder belt crosses his neck and the lap belt rides over his abdomen. Which is the best advice? [Management and Treatment]

A. Continue the vehicle belt alone because 9 years is old enough for it

B. Move him to the front seat so that the shoulder belt crosses his shoulder and chest properly

C. Return him to a forward-facing harness seat until he reaches age 12

D. Tuck the shoulder belt under his arm so that it stops rubbing his neck

E. Use a belt-positioning booster seat until the belt fits, usually near 4 feet 9 inches

Answer Contents

BQ-040. A 5-month-old girl sleeps swaddled on her back in a flat crib in her parents' room. This week her mother noticed that she rolls from her back to her side while swaddled. Her mother asks whether swaddling lowers the risk of sudden infant death. Which is the best advice? [Management and Treatment]

A. Stop swaddling now because she is starting to roll, since swaddling does not lower the risk

B. Continue swaddling until 9 months of age, because it lowers the risk of sudden infant death in the crib

C. Wrap her more tightly so that she can no longer roll in the crib

D. Move her into the parents' bed so that rolling can be watched

E. Add a sleep positioner so that she stays on her back all night

Answer Contents

BQ-041. A 9-year-old girl with spastic cerebral palsy has eaten less for two weeks and cries during meals. Her parents say, "It is just her cerebral palsy getting worse." Her weight is stable, her abdomen is soft, and she grimaces when the clinician touches her left cheek. Which is the best next step? [Diagnosis]

A. Attribute the change to progression of cerebral palsy and give reassurance to the family

B. Plan gastrostomy tube placement to secure her nutrition

C. Increase the dose of the muscle relaxant to ease feeding

D. Defer further assessment for six months to find out whether the problem resolves

E. Evaluate for a new treatable problem, such as dental or gastrointestinal disease

Answer Contents

BQ-042. The parents of a healthy 3-year-old boy are frustrated because he still has daytime accidents several times a week. Training began at 30 months. He is otherwise well, with normal development and no constipation or urinary symptoms. They wonder whether he is delayed. Which is the best interpretation? [Epidemiology and Risk Assessment]

A. Delayed, because most boys are dry by day at 24 months

B. Abnormal, since daytime accidents at age 3 suggest a urologic disorder

C. A sign of failed technique, meaning the training should be started over from the beginning

D. Typical, since median daytime dryness is reached at about 35 months in boys

E. A behavioral disorder that calls for a referral to child psychiatry

Answer Contents

BQ-043. A 3-year-old girl has urinated in the toilet for 4 months, but she screams and hides when she needs a bowel movement and asks for a diaper. Her stools are firm and large, and she has no fever or vomiting. Her examination is otherwise normal. Which is the most likely cause of her stool refusal? [Diagnosis]

A. A urinary tract infection causing pain with stooling

B. Hirschsprung disease with a short aganglionic segment

C. Hypothyroidism with slowed intestinal transit

D. Celiac disease causing bowel urgency

E. Withholding stool because of painful, firm stools

Answer Contents

BQ-044. A 2-month-old boy is brought for his well-child visit. He is feeding and gaining weight well. His mother looks tired, avoids eye contact, and says she cries often and cannot sleep even when the baby does. She has no thoughts of harming herself or the baby. Which is the best next step? [Management and Treatment]

A. Reassure her that these feelings are the normal baby blues and will fade within a few weeks

B. Screen the mother with a validated depression tool at this visit and arrange follow-up

C. Defer any discussion of her mood until the 6-month visit

D. Focus the visit on the infant, since the mother is not the patient

E. Prescribe an antidepressant for her based on the history alone

Answer Contents

BQ-045. A girl weighing 1,800 g is born at 35 weeks to a mother who tests positive for hepatitis B surface antigen. The infant is stable in the nursery at 3 hours of age. The mother asks whether the baby should wait until she reaches 2,000 g before receiving any hepatitis B protection. Which is the best next step? [Management and Treatment]

A. Give hepatitis B vaccine together with hepatitis B immune globulin within 12 hours of birth

B. Give hepatitis B vaccine alone at the 1-month visit once she is feeding well

C. Wait until she reaches 2,000 g and then give the vaccine without hepatitis B immune globulin

D. Give hepatitis B immune globulin alone and begin the vaccine at 6 months

E. Test the infant for hepatitis B surface antigen and vaccinate only if negative

Answer Contents

BQ-046. An infant received hepatitis B vaccine at birth, at 5 weeks, and at 18 weeks of age. The third dose came more than 8 weeks after the second dose and more than 16 weeks after the first dose. Her mother asks whether the series is complete. Which is the best interpretation of the third dose? [Diagnosis]

A. Valid, because both minimum intervals between doses were met

B. Valid, because the birth dose may count as the third dose

C. Invalid, because it was given before 24 weeks of age

D. Invalid, because the series must restart from the first dose

E. Valid, but only if an antibody level is checked after the dose

Answer Contents

BQ-047. A 10-week-old boy was diagnosed with severe combined immunodeficiency after an abnormal newborn screen. He is afebrile and well. He is due for his 2-month vaccines, and his mother asks which should go ahead. Which is the most appropriate vaccination plan? [Management and Treatment]

A. Withhold the rotavirus vaccine because it is contraindicated in this condition

B. Give all of the vaccines that are due, including the rotavirus vaccine, at this visit

C. Withhold every vaccine until the immunodeficiency has been treated

D. Give only the rotavirus vaccine because it is taken by mouth

E. Give the rotavirus vaccine at half strength over two visits

Answer Contents

BQ-048. A healthy 16-month-old girl has never received Hib vaccine because her family recently immigrated. She is afebrile and has a normal examination. Her other vaccines are being reviewed separately. Which is the best plan for Hib vaccine? [Management and Treatment]

A. Give one dose today and a second dose in 8 weeks

B. Give one dose today and no further doses

C. Give a three-dose series at 4-week intervals

D. Give no dose, since she is older than 15 months

E. Restart a four-dose series on the standard infant schedule

Answer Contents

BQ-049. A 14-month-old boy has acute lymphoblastic leukemia and is receiving intensive chemotherapy that has profoundly weakened his immune system. His parents ask whether he can now receive the measles, mumps, rubella, and varicella vaccines that are due at his age. Which mechanism best explains why these live vaccines are contraindicated for him? [Physiology and Pathophysiology]

A. Chemotherapy drugs inactivate the vaccine viruses before an immune response can develop

B. Leukemic cells destroy the vaccine antigens inside the lymph nodes of the child

C. The weakened vaccine viruses can replicate unchecked when immunity is badly impaired

D. The vaccines stimulate abnormal white cells and trigger relapse of the leukemia

E. The vaccine viruses mutate into more virulent forms inside the bloodstream of the child

Answer Contents

BQ-050. An 18-month-old boy is brought for scheduled vaccines. He is lethargic with a temperature of 39.4 C (102.9 F), fast breathing, and poor oral intake, and he is being evaluated for pneumonia. His mother asks why the vaccines should wait. Which statement best explains the reason for deferring them at this visit? [Physiology and Pathophysiology]

A. It is a precaution, since illness can blur a reaction's diagnosis and blunt the response

B. Antibiotics given for pneumonia inactivate the antigens in every vaccine given that day

C. Fever over 38.0 C is a contraindication that applies to all vaccines in infants and toddlers

D. Inactivated vaccines can spread from the lungs to the blood during a bacterial infection

E. A child with pneumonia cannot make any antibody response to a vaccine

Answer Contents

BQ-051. A 6-month-old girl develops hives over her trunk, wheezing, and pallor 15 minutes after receiving vaccines in the office. She is alert but breathing with effort. Her mother is anxious and asks what is happening. Which is the most likely diagnosis? [Diagnosis]

A. Vasovagal syncope after the injections

B. Anaphylaxis to a vaccine

C. An expected local reaction with a mild fever

D. Serum sickness caused by vaccine components

E. A breath-holding spell triggered by pain

Answer Contents

BQ-052. A researcher finds 14 reports of febrile seizures after measles, mumps, and rubella vaccine in a national passive reporting system over one year. She wants to estimate the incidence per dose given and whether the vaccine caused the seizures. Which statement about this system is most accurate? [Epidemiology and Risk Assessment]

A. It gives true incidence per dose given because every vaccinated child is counted in it

B. It proves causation once several reports describe the same event

C. It accepts reports only from manufacturers, so the data are verified

D. It replaces the need for controlled studies of vaccine safety

E. It can signal events for further study but cannot give rates or prove causation

Answer Contents

BQ-053. A father declines all vaccines for his healthy 4-month-old daughter at two consecutive visits, citing a story about a neighbor's child. He has listened to counseling at each visit and stays calm. The practice values the relationship. Which is the best next step? [Management and Treatment]

A. Dismiss the family from the practice at this visit to protect the other patients in the waiting room

B. Accept the refusal and stop raising the subject at later visits

C. Document the refusal and the counseling, keep the relationship, and ask again at the next visit

D. Report the family to child protective services as medical neglect

E. Tell him the infant cannot be seen unless she receives the vaccines today

Answer Contents

BQ-054. A healthy 13-month-old boy is due for his first measles, mumps, rubella and varicella vaccines. His older brother has epilepsy that began with febrile seizures. The boy has had no seizures. His father asks which products to use for this visit. Which is the best plan? [Management and Treatment]

A. Give the combined measles, mumps, rubella, and varicella vaccine in one injection

B. Defer both vaccines until he is 4 years old to lower the seizure risk

C. Give the measles, mumps, and rubella vaccine and omit varicella vaccine

D. Give measles, mumps, and rubella and varicella vaccines as separate injections

E. Give the combined vaccine and add an antipyretic drug for several days

Answer Contents

BQ-055. A 5-day-old breastfed boy is jaundiced, vomiting, and feeding poorly, and his liver is enlarged. A newborn screen is positive for classic galactosemia, and enzyme testing confirms it. His mother is distraught and asks whether she can continue to breastfeed. Which is the best next step? [Management and Treatment]

A. Continue exclusive breastfeeding and add a vitamin supplement for the liver

B. Stop human milk, start a galactose-free formula, and involve a metabolic specialist today

C. Continue breastfeeding and give a standard cow's milk formula after each feed to add calories

D. Feed pumped human milk by bottle to protect the infant from reflux

E. Switch to a standard infant formula and recheck the screen in a month

Answer Contents

BQ-056. A 6-week-old formula-fed girl drinks about 20 ounces (600 mL) of iron-fortified, vitamin D-fortified formula daily and is growing well. Her father asks whether she needs a vitamin D supplement because the formula is fortified. Which is the best advice? [Management and Treatment]

A. Give no supplement, since all infant formula meets the daily vitamin D needs of an infant completely

B. Give a vitamin D supplement only after she is 6 months old

C. Give a daily vitamin D supplement, since she drinks less than about 27 ounces (800 mL) of formula

D. Give a vitamin D supplement only if she is also given human milk

E. Give a large weekly dose of vitamin D instead of a daily supplement

Answer Contents

BQ-057. A 10-month-old boy with dark skin has been exclusively breastfed and never given vitamin supplements. He has bowed legs, wide wrists, and a prominent beading along the ribs. He is not yet pulling to stand. Which mechanism best explains these findings? [Physiology and Pathophysiology]

A. Vitamin C deficiency that impairs collagen cross-linking in the bone matrix, capillaries, and gums

B. Vitamin D deficiency that impairs calcium and phosphate supply for mineralization of growing bone

C. Excess fluoride exposure that disturbs the structure of the growth plate

D. Hypothyroidism that slows the maturation of bone and cartilage

E. Vitamin A excess that accelerates bone resorption in the long bones

Answer Contents

BQ-058. A 7-month-old boy is exclusively breastfed and has poor weight gain, low muscle tone, and pallor. His mother follows a strict diet that excludes all animal products and takes no supplements. His hemoglobin is low and his mean cell volume is high. Which is the most likely diagnosis? [Diagnosis]

A. Vitamin B12 deficiency

B. Iron deficiency anemia

C. Vitamin D deficiency rickets

D. Thalassemia trait

E. Vitamin C deficiency

Answer Contents

BQ-059. A 4-year-old boy with autism eats only plain pasta and dry cereal and refuses every fruit and vegetable. He now has a limp, pain in both knees and ankles, and swollen, bleeding gums. He has had no fever or injury. Which is the most likely diagnosis? [Diagnosis]

A. Septic arthritis of the knee

B. Vitamin D deficiency rickets

C. Vitamin C deficiency

D. Juvenile idiopathic arthritis

E. Acute leukemia

Answer Contents

BQ-060. A 9-month-old girl is pale and irritable. Her parents started whole cow's milk at 7 months because formula was costly, and she now drinks about 28 ounces a day with few solids. Her hemoglobin is low with small red cells, and a stool test is positive for occult blood. Which mechanism best explains her anemia? [Physiology and Pathophysiology]

A. Hemolysis of red cells from an inherited enzyme deficiency

B. Impaired red cell production from vitamin B12 deficiency

C. Bone marrow failure from a congenital disorder of blood production that began in infancy

D. Intestinal blood loss from cow's milk combined with displacement of iron-rich foods

E. Blocked heme synthesis from chronic lead exposure in the home

Answer Contents

BQ-061. A 5-month-old boy has had constipation for 4 days and now has a weak cry, poor suck, drooping eyelids, and reduced tone. His grandmother has been dipping his pacifier in honey. He has no fever. Which mechanism best explains his illness? [Physiology and Pathophysiology]

A. Preformed toxin in the honey is absorbed and blocks acetylcholine release

B. Ingested spores germinate in the infant gut and release a toxin that blocks acetylcholine release

C. Autoantibodies attack the acetylcholine receptors at the neuromuscular junction of skeletal muscle

D. A virus invades the anterior horn cells of the spinal cord

E. Low potassium from vomiting weakens the skeletal muscles

Answer Contents

BQ-062. A 6-month-old girl has mild to moderate eczema controlled with moisturizers and an occasional low-potency topical steroid. She has no egg allergy or other food reaction and has started solids. Her parents ask when and how to introduce peanut. Which is the best advice? [Management and Treatment]

A. Obtain peanut-specific IgE testing or skin-prick testing before the first taste, whatever the eczema severity

B. Delay peanut until 3 years of age to lower the chance of reaction

C. Offer small whole peanuts as a finger food once she can grasp them

D. Perform a supervised oral peanut challenge in an allergy clinic first

E. Offer age-appropriate peanut, such as thinned smooth peanut butter, at home around 6 months without testing

Answer Contents

BQ-063. In the LEAP trial, among infants with a negative skin-prick test to peanut, peanut allergy at 60 months occurred in 13.7 percent of those who avoided peanut and 1.9 percent of those who ate it. A resident wants to know how many such infants must eat peanut early to prevent one case of allergy. Which is the closest number? [Epidemiology and Risk Assessment]

A. 3 infants

B. 9 infants

C. 7 infants

D. 19 infants

E. 53 infants

Answer Contents

BQ-064. An 18-month-old girl has eaten poorly since a diarrheal illness. Her weight-for-height z-score is -2.4, she has no edema, and her mid-upper arm circumference is 112 mm. She is alert and eating a little. Which is the most likely classification? [Diagnosis]

A. Moderate acute malnutrition by weight-for-height

B. Stunting with normal weight for height

C. Severe acute malnutrition by arm circumference

D. Normal nutritional status for age

E. Severe acute malnutrition by edema

Answer Contents

BQ-065. A 3-year-old boy with severe malnutrition from neglect is admitted and started on full-volume, high-calorie feeds. On day 3 he is lethargic, edematous, and weak, and laboratory tests show low phosphate, potassium, and magnesium. Which mechanism best explains these findings? [Physiology and Pathophysiology]

A. Excess urinary loss of electrolytes from untreated diabetes insipidus

B. Insulin-driven movement of electrolytes into cells during rapid refeeding

C. Poor absorption of electrolytes from intestinal damage by cow's milk protein

D. Adrenal insufficiency with salt wasting from lack of cortisol

E. Dilution of electrolytes from the intravenous fluids alone

Answer Contents

BQ-066. A healthy 8-month-old boy with two erupted teeth drinks well water from a private well. His mother asks for fluoride drops. His teeth are clean with no visible decay, and the family brushes twice daily with a rice-grain smear of fluoride toothpaste. Which is the best next step? [Management and Treatment]

A. Start daily fluoride drops now because private well water is usually deficient in fluoride content

B. Stop all fluoride products until he is 3 years of age

C. Measure the fluoride content of the well water and assess his caries risk before any supplement

D. Increase the toothpaste to an adult-sized amount twice daily

E. Replace toothpaste with a fluoride mouth rinse after meals

Answer Contents

Domain 2. Fetal and Neonatal Care

Contents Answers for this subject

BQ-067. A boy is born at 39 weeks after an uncomplicated labor. He cries at once, has good tone, and has a heart rate of 150 beats per minute and a pink color. His mother is stable. The obstetric team asks whether the cord can be clamped right away so the baby can be weighed. Which physiologic effect best explains the benefit of waiting at least 60 seconds? [Physiology and Pathophysiology]

A. Placental transfusion of blood to the infant while the lungs aerate and pulmonary blood flow rises

B. Transfer of maternal surfactant to the infant lungs through the umbilical vein during the first minute

C. Stimulation of the first breath by a rise in umbilical venous oxygen content after birth

D. Warming of the infant by blood returned from the placenta, which prevents later cold stress

E. Faster closure of the ductus arteriosus from prostaglandins carried back from the placenta

Answer Contents

BQ-068. A girl born at 40 weeks needed positive pressure ventilation for 3 minutes and then recovered. Her Apgar scores are 3 at 1 minute, 5 at 5 minutes, and 7 at 10 minutes. She is now pink, crying, and has a heart rate of 150 beats per minute. Her parents ask what the scores mean for her future. Which is the most appropriate next step? [Management and Treatment]

A. Begin therapeutic cooling at once, because a 5-minute score of 5 or lower confirms hypoxic injury

B. Tell the parents that the scores predict her neurologic outcome, which is likely to be good

C. Record the scores and take no further action, because she has recovered and looks well

D. Send an umbilical arterial blood gas and a placental examination, and explain the scores

E. Repeat the Apgar score every 5 minutes for the next hour to track her recovery from distress

Answer Contents

BQ-069. A girl born at 36 weeks and 2 days is 60 hours old and exclusively breastfed. Her weight is down 6 percent from birth, and her skin is yellow to the knees. The bilirubin is climbing toward the phototherapy threshold. Her mother and she share a compatible blood group, and the direct antiglobulin test is negative. Which is the most likely mechanism of her jaundice? [Physiology and Pathophysiology]

A. Accelerated red cell destruction from antibody coating of the infant red cells in the mother

B. Mechanical obstruction of bile flow from a congenital anomaly of the biliary tree

C. Slow hepatic conjugation combined with low milk intake that reduces bilirubin excretion

D. Inherited deficiency of an enzyme that protects red cells from oxidative injury

E. Overproduction of bilirubin from a large cephalohematoma that is resorbing slowly

Answer Contents

BQ-070. A boy born at 39 weeks to a mother with blood group O positive is 14 hours old. He is a first-born and feeds well, but his skin is yellow to the abdomen. A transcutaneous bilirubin is over the screening range, and his own blood group is A positive. Which additional test is the most appropriate to send with a serum bilirubin? [Diagnosis]

A. Urine culture on a catheter-collected specimen to look for urinary infection

B. Direct antiglobulin test on a sample of the infant's blood

C. Thyroid function tests on a sample of the infant's blood to look for hypothyroidism

D. Serum ammonia level on a sample drawn from the infant's heel to look for a metabolic cause

E. Chest radiograph to look for pulmonary disease contributing to the rising bilirubin

Answer Contents

BQ-071. A boy born at 39 weeks is 30 hours old and exclusively breastfed. His serum bilirubin is 11 mg/dL, and the nursery must choose which hour-specific phototherapy curve applies. The record lists several details of his course. Which finding places him on the curve with the lower treatment threshold? [Epidemiology and Risk Assessment]

A. A birth weight of 4,100 g, at the 90th percentile

B. A weight loss of 5 percent from birth weight

C. A serum albumin of 2.8 g/dL

D. Delivery by cesarean without labor

E. A first-born position in a family with no relevant history

Answer Contents

BQ-072. A boy born at 37 weeks received phototherapy starting at 36 hours of age for jaundice from ABO incompatibility. Treatment stopped this morning, after the bilirubin fell 2.4 mg/dL under the threshold that applied when it began. He is feeding well and his mother hopes to go home today. Which is the best next step? [Management and Treatment]

A. Repeat the bilirubin 6 to 12 hours after stopping and again the next day

B. Discharge him now with a routine visit in 1 week, because the level is under the threshold

C. Restart phototherapy at once to prevent a rebound rise in the bilirubin level

D. Repeat the bilirubin in 3 days at the first outpatient visit after discharge

E. Switch to formula feeding for 48 hours to hasten the clearance of bilirubin

Answer Contents

BQ-073. A boy born at 29 weeks has grunting, nasal flaring, and chest retractions beginning 30 minutes after birth. His oxygen requirement is rising. A chest radiograph shows low lung volumes with a diffuse granular pattern. Which is the most likely mechanism of his respiratory failure? [Physiology and Pathophysiology]

A. Retained fetal lung fluid caused by delayed absorption across the alveolar epithelium

B. Aspiration of stained amniotic fluid that blocks small airways and inactivates surfactant

C. Surfactant deficiency that raises alveolar surface tension and causes widespread atelectasis

D. Pulmonary vasoconstriction with right-to-left shunting of blood across the ductus arteriosus

E. Herniation of abdominal contents into the chest that compresses the developing lung

Answer Contents

BQ-074. A boy born at 41 weeks and 4 days has meconium-stained amniotic fluid and was limp at birth. At 3 hours he has a respiratory rate of 82 breaths per minute, grunting, and a barrel-shaped chest. A chest radiograph shows hyperinflated lungs with patchy opacities. Which is the most likely diagnosis? [Diagnosis]

A. Transient tachypnea of the newborn

B. Meconium aspiration syndrome

C. Respiratory distress syndrome from surfactant deficiency

D. Congenital diaphragmatic hernia on the left side

E. Early-onset pneumonia from maternal infection

Answer Contents

BQ-075. A term girl, born after an uncomplicated pregnancy, is 10 hours old and blue. Her right-hand saturation is 98 percent and her foot saturation is 86 percent. She breathes fast, has a loud second heart sound, and a chest radiograph shows clear lungs and a normal heart size. Oxygen raises her saturation only slightly. Which is the most likely mechanism of her cyanosis? [Physiology and Pathophysiology]

A. Sustained high pulmonary resistance with right-to-left shunting across the ductus

B. Alveolar fluid retention that produces ventilation-perfusion mismatch across both lungs

C. Surfactant deficiency that causes diffuse collapse of the terminal air spaces in both lungs

D. Air leak into the pleural space that collapses one lung and shifts the mediastinum

E. Bony obstruction of both posterior nasal passages that blocks airflow during quiet breathing

Answer Contents

BQ-076. A well-appearing boy born at 38 weeks was started on ampicillin and gentamicin at 2 hours of age after maternal fever in labor. A blood culture drawn before the first dose is sterile at 48 hours. He is feeding well, and his vital signs and examination are normal. Which is the most appropriate management? [Management and Treatment]

A. Continue the antibiotics for a full 7 days, because a sterile culture may miss early infection

B. Continue the antibiotics until a repeat C-reactive protein is normal on two consecutive days

C. Obtain a lumbar puncture before stopping the antibiotics to rule out meningitis

D. Stop the antibiotics and continue routine newborn care

E. Change to an oral antibiotic and discharge him on a 7-day course of treatment

Answer Contents

BQ-077. A girl born at 35 weeks and 5 days weighs 2.1 kg, under the 10th percentile for gestational age. At 3 hours she is alert and vigorous, with normal vital signs and a point-of-care glucose of 32 mg/dL. She has fed once. Which is the best next step? [Management and Treatment]

A. Give intravenous dextrose by protocol before any further feeding

B. Feed her and repeat the glucose after the feed

C. Collect a critical sample for insulin and cortisol and withhold feeds until it returns

D. Observe her without testing, because the value is a transitional change that will correct

E. Transfer her to intensive care for central venous access and a continuous infusion

Answer Contents

BQ-078. A boy born at term to a mother without diabetes is 4 days old and weighs 4.1 kg. He has needed intravenous dextrose to keep his glucose over 50 mg/dL since birth, and a glucose after a full feed this morning was 38 mg/dL. He is otherwise well. Which is the best next step in his evaluation? [Diagnosis]

A. Collect a critical sample for insulin and cortisol while the glucose is under 50 mg/dL

B. Begin a blood culture and broad-spectrum antibiotics to treat occult sepsis in the newborn

C. Reassure the family that glucose values are still adapting to life outside the womb at 4 days

D. Increase the feed volume and recheck the glucose at a clinic visit planned for next week

E. Order a karyotype to look for a chromosomal cause of low glucose

Answer Contents

BQ-079. A boy born at 35 weeks is 3 hours old in a cool delivery room. His axillary temperature is 35.4 C (95.7 F), and his glucose is 28 mg/dL. He weighs 2.4 kg, has little subcutaneous fat, and is jittery. Which is the most likely mechanism of his hypoglycemia? [Physiology and Pathophysiology]

A. Excess insulin secretion from hyperplasia of the pancreatic beta cells

B. Impaired glucose production from an inherited defect in a liver enzyme

C. Reduced glucose absorption caused by delayed maturation of the intestinal tract

D. Increased glucose consumption for heat production against small glycogen stores

E. Deficient cortisol secretion caused by hemorrhage into the adrenal glands

Answer Contents

BQ-080. A 6-day-old girl, born at term and exclusively breastfed, comes for a scheduled visit. She looks well, feeds eagerly, and has made 6 wet diapers daily. A rectal temperature is 38.2 C (100.8 F) and is the same on repeat. The examination is normal. Which is the best next step? [Management and Treatment]

A. Reassure the mother and advise lighter clothing, with a temperature recheck in 24 hours at home

B. Give an antipyretic and recheck her temperature at home in 4 hours, then return if still raised

C. Admit her for a full sepsis evaluation and start empiric antibiotics after cultures are drawn

D. Obtain a urine culture alone, and discharge her if the urinalysis is normal and she looks well

E. Discharge her with close telephone follow-up, because she is feeding well and has a normal examination

Answer Contents

BQ-081. A boy born at 39 weeks drools constantly and coughs and turns blue with his first feed. A nurse tries to pass a feeding tube, and it stops at 10 cm from the lips. The abdomen is soft and the anus is open. Which is the best next step in diagnosis? [Diagnosis]

A. Offer a second small feed with the baby held upright and watch for further choking spells

B. Pass a larger tube with firm pressure to clear the obstruction and reach the stomach

C. Obtain a barium swallow in the radiology suite using thin liquid barium in the baby's mouth

D. Start a proton pump inhibitor for reflux and reassess his feeding after one week of treatment

E. Obtain a chest and abdominal radiograph with the tube in place and keep him unfed

Answer Contents

BQ-082. A newborn girl born at 37 weeks has loops of thickened, matted bowel protruding through a defect to the right of the umbilical cord. No membrane covers the bowel. She is crying, and her vital signs are stable. Which is the best next step? [Management and Treatment]

A. Push the bowel back into the abdomen at the bedside and apply a tight abdominal binder to hold it in place

B. Apply dry gauze over the bowel, place her supine under a radiant warmer, and defer the surgical call until morning

C. Cover the bowel with warm, moist sterile dressings and plastic wrap, place a gastric tube, and call the surgeon

D. Give oral feeds to assess bowel function and obtain surgical advice only if she vomits after the feed

E. Leave the bowel exposed under a warmer to dry so that a protective crust forms before surgery

Answer Contents

BQ-083. A term boy turns blue with retractions when he is quiet or feeding, and he turns pink when he cries. A nurse cannot pass a suction catheter through either nostril. His lungs are clear, and his heart sounds are normal. Which is the most likely diagnosis? [Diagnosis]

A. Transient tachypnea of the newborn after delivery

B. Congenital diaphragmatic hernia on the left side

C. Laryngomalacia with collapse of the supraglottic tissue

D. Bilateral choanal atresia

E. Tetralogy of Fallot with right ventricular outflow obstruction

Answer Contents

BQ-084. A newborn is 1 day old with an enlarged phallus-like structure of 1.5 cm, fused labioscrotal folds, and no palpable gonads. The parents ask whether they have a son or a daughter, and they want the birth announcement sent today. Which is the best next step? [Management and Treatment]

A. Assign male sex because of the phallic structure and tell the family the baby is a boy

B. Assign female sex because the gonads are not palpable and tell the family the baby is a girl

C. Explain that a few tests are needed, withhold a sex assignment, and call endocrinology and genetics

D. Tell the family that the cause is unclear and discharge the baby with an outpatient visit next month

E. Reassure the family that the findings are normal for a newborn and need no testing

Answer Contents

BQ-085. Five newborns each have a normal hip examination on the first day of life. The pediatrician plans to arrange closer follow-up for hip dysplasia in one of them. Which infant is at highest risk and most deserves a careful hip plan? [Epidemiology and Risk Assessment]

A. A boy born by vertex delivery at 40 weeks with no family history of hip problems

B. A girl born by vertex delivery at 39 weeks with no family history of hip problems

C. A boy born by vertex delivery at 38 weeks to a mother with gestational diabetes

D. A boy born by vertex delivery at 41 weeks with a birth weight in the 50th percentile and no risk factors

E. A girl born breech at 39 weeks to a mother with childhood hip dysplasia

Answer Contents

BQ-086. A girl born at term fails the automated hearing screen in the left ear on day 2, and again on a rescreen before discharge. She is otherwise well. Her parents believe fluid will clear and favor waiting until she is older, and the birth hospital has no audiologist. Which is the best next step? [Management and Treatment]

A. Arrange diagnostic audiology to confirm or exclude hearing loss by 3 months of age

B. Repeat the screening test at 6 months of age and send her for audiology only if she fails it again

C. Delay hearing testing until she is 12 months old and has started to babble

D. Obtain tympanometry alone and treat any fluid before further hearing assessment

E. Refer her for ear tubes now, because fluid is the most likely cause of the failed screen

Answer Contents

BQ-087. A term boy was discharged at 18 hours of age. The dried blood spot for the state newborn screen was collected at 14 hours, soon after his first feeds, and the result is reported as normal. At the 3-day visit the pediatrician reviews the record. Which is the best next step? [Management and Treatment]

A. Accept the normal result as final, because a normal newborn screen needs no repetition

B. Repeat only the pulse oximetry and hearing screens, because the blood screen is complete

C. Arrange a repeat dried blood spot, because the first was taken before 24 hours of feeding

D. Wait for the family to return at 2 months and collect a repeat specimen at that visit

E. Order a full panel of confirmatory tests for every disorder in the state screening program

Answer Contents

BQ-088. A boy born at term passed pulse oximetry screening at 26 hours, with 97 percent in the right hand and 96 percent in a foot, and was discharged at 2 days. At 10 days he has poor feeding, fast breathing, and gray skin. His femoral pulses are faint, and the right arm blood pressure is higher than the leg pressure. Which is the most likely diagnosis? [Diagnosis]

A. Neonatal sepsis with septic shock

B. Transposition of the great arteries

C. Tricuspid atresia with a small right ventricle

D. Coarctation of the aorta

E. Total anomalous pulmonary venous return

Answer Contents

Domain 3. Adolescent Care

Contents Answers for this subject

BQ-089. A 14-year-old boy comes for his annual visit with his father. The boy is quiet. His father agrees that the visit should include time alone, and he steps out. You are about to ask about sexual activity, substance use, and mood. Which is the best next step before starting those questions? [Management and Treatment]

A. State what stays private and the limits, such as serious risk of harm and abuse
B. Promise that nothing he says will ever be shared with anyone, including his father
C. Ask the questions only after his father returns and gives his agreement to each topic in turn
D. Begin with the sexuality questions first, because they carry the greatest risk to his health
E. Postpone the confidential interview until he is 16 years old and more mature

Answer Contents

BQ-090. A 14-year-old girl asks for contraception without her parents' knowledge. She has one partner and is sexually active. Your clinic is in a state whose statutes you have not recently reviewed. Which statement is the most accurate about the legal framework for her request? [Epidemiology and Risk Assessment]

A. Every state requires a signature from a parent before any contraceptive method is prescribed
B. Federal law allows every minor to consent to any contraceptive method in every state
C. Minors can consent to contraception only when they are 16 years or older in every state
D. Minors can consent to contraception only after they have had a documented pregnancy
E. Authority of a minor to consent to contraception varies by state, so check state law

Answer Contents

BQ-091. A 16-year-old girl is tested for chlamydia and gonorrhea at a visit alone with you. Her mother has proxy access to the family patient portal, and laboratory results release there automatically. The girl asks you not to tell her mother. Which is the best next step? [Management and Treatment]

A. Let the results release as usual, because portal rules are set by the hospital and cannot be changed
B. Ask which phone number or address is safe for her results and turn off portal release for them
C. Tell the girl that results of infection tests always go to the parent who holds the insurance policy
D. Cancel the test and ask her mother to consent first, then repeat the test at the next visit
E. Ask her mother to avoid opening the portal, and trust that she will respect the request

Answer Contents

BQ-092. A 14.5-year-old boy is the shortest in his class and has no testicular enlargement, with testicular volume of 2 mL and no pubic hair. His growth velocity is steady, and his bone age is 12 years. His father started shaving at 17 years. He has no chronic symptoms, and the examination is otherwise normal. Which is the most likely diagnosis? [Diagnosis]

A. Hypogonadotropic hypogonadism from Kallmann syndrome with an absent sense of smell

B. Klinefelter syndrome with primary testicular failure from an extra X chromosome

C. Growth hormone deficiency with a decelerating growth velocity

D. Constitutional delay of growth and puberty

E. Hypothyroidism from autoimmune thyroiditis with a delayed bone age

Answer Contents

BQ-093. A 13.5-year-old girl has no breast development. Her height is at the 40th percentile and her weight is proportionate. She has Tanner stage 2 pubic hair, no chronic symptoms, and a normal examination. Her mother wants to wait until she is 15 years old. Which is the best next step in evaluation? [Diagnosis]

A. Reassure the family and recheck her in 12 months, because puberty can start any time before 16 years

B. Start estrogen therapy at once to induce breast development, then measure the hormones

C. Obtain brain imaging before any laboratory testing, because a central lesion is the most likely cause

D. Obtain pelvic ultrasonography alone and defer hormone measurements until the results return

E. Measure LH and FSH and obtain a bone age, and sort the cause as central or peripheral

Answer Contents

BQ-094. A 7-year-old girl has had breast budding for 3 months and a recent growth spurt. She has Tanner stage 3 breasts and early pubic hair, and her bone age is advanced by 2 years. A stimulation test shows a pubertal LH response. She is otherwise well. Which is the most likely mechanism of her early puberty? [Physiology and Pathophysiology]

A. Autonomous secretion of estrogen by an ovarian cyst, which acts independent of gonadotropins

B. Excess adrenal androgen production caused by an inherited block in the steroid synthesis pathway

C. Early activation of the hypothalamic pulse generator that drives gonadotropin release

D. Exposure to estrogen in a topical cream or similar product applied to her skin at home

E. Thyroid hormone deficiency that raises TSH and stimulates the ovary directly

Answer Contents

BQ-095. A 13-year-old boy has firm, tender, bilateral breast tissue 2 cm in diameter under each areola. His testes measure 8 mL, he has Tanner stage 3 pubic hair, and the examination is otherwise normal. His mother is worried. Which is the most likely mechanism of the breast tissue? [Physiology and Pathophysiology]

A. Estrogen secretion by a germ cell tumor in one testis

B. Fat deposition under the nipple from obesity in a boy with a high body mass index

C. Testicular failure caused by an extra X chromosome in a boy with Klinefelter syndrome

D. Prolactin secretion by a pituitary adenoma that stimulates breast tissue

E. A transient imbalance between estrogen and androgen effect during mid puberty

Answer Contents

BQ-096. A 15-year-old girl has had no periods for 4 months. Menarche was at 12 years, and her cycles had been about every 35 days. She denies sexual activity, her weight is stable, and her examination is normal. Which is the best first test? [Diagnosis]

A. Serum prolactin level measured in the morning after waking

B. Urine human chorionic gonadotropin test

C. Pelvic ultrasonography to look at the uterus and the ovaries

D. Thyroid-stimulating hormone level to look for hypothyroidism

E. Serum FSH and LH levels to look for ovarian insufficiency

Answer Contents

BQ-097. A 16-year-old ballet dancer has had no periods for 5 months. She trains 20 hours weekly, eats about 1,200 kcal daily, and her awake heart rate is 58 beats per minute. A pregnancy test is negative and her examination is otherwise normal. Which is the most likely mechanism of her amenorrhea? [Physiology and Pathophysiology]

A. Autonomous androgen secretion by a tumor of the ovary

B. Prolactin excess from a small pituitary adenoma

C. Outflow obstruction at the hymen that blocks menstrual flow

D. Suppressed GnRH pulsatility caused by low energy availability

E. Premature depletion of ovarian follicles with high gonadotropins

Answer Contents

BQ-098. A 14-year-old girl had menarche 2 years ago. Her cycles come every 35 to 45 days and last 5 days, with no heavy bleeding. She has mild acne, a normal BMI, no excess hair, and a normal examination. Her mother is worried about polycystic ovary syndrome. Which is the most likely explanation? [Diagnosis]

A. Physiologic anovulatory cycles in the early years after menarche

B. Polycystic ovary syndrome with sustained hyperandrogenism and cysts

C. Hypothyroidism from autoimmune thyroid disease

D. Premature ovarian insufficiency with a low estrogen level

E. Hyperprolactinemia from a pituitary microadenoma

Answer Contents

BQ-099. A 15-year-old girl has had vaginal bleeding for 11 days and soaks a pad every hour. She is pale and dizzy. Her heart rate is 128 beats per minute, her blood pressure is 88/50 mmHg, and her hemoglobin is 6.9 g/dL. A pregnancy test is negative. Which is the best next step? [Management and Treatment]

A. Prescribe oral iron alone and recheck her in 2 weeks, because the bleeding will settle

B. Admit her for monitored treatment of the bleeding and anemia, and send bleeding studies

C. Order pelvic ultrasonography and send her home to wait for the result before giving any treatment

D. Start an NSAID for the bleeding and observe her at home, with a return visit in a week

E. Perform dilation and curettage as first-line treatment to stop the bleeding quickly

Answer Contents

BQ-100. A 17-year-old girl smokes half a pack of cigarettes daily and asks to start a combined oral contraceptive. She has normal blood pressure, no migraine, no history of clots, and a BMI of 22 kg/m2. Which is the most accurate assessment of her eligibility under the CDC medical eligibility criteria? [Epidemiology and Risk Assessment]

A. Category 4, because the pill carries an unacceptable health risk in any smoker of her age

B. Category 1, because no restriction applies and cessation counseling is unnecessary

C. Category 2, because advantages generally outweigh risks, so the pill is acceptable

D. Category 3, because risks usually outweigh advantages and the pill should be avoided

E. Category 4 until she has stopped smoking for 12 consecutive months before starting

Answer Contents

BQ-101. A 16-year-old girl asks for a levonorgestrel intrauterine device. She is sexually active, has had two partners, and has never been tested for infection. She has no symptoms or pelvic pain, and her examination is normal. Which is the best next step? [Management and Treatment]

A. Collect chlamydia and gonorrhea tests at this visit and place the device without waiting

B. Delay placement until negative test results return, then schedule a separate placement visit

C. Treat her with empiric antibiotics for chlamydia and gonorrhea before placing the device

D. Offer only the oral pill, because intrauterine devices are unsuitable for sexually active teens

E. Perform cervical cytology before placement, because the device cannot be placed without it

Answer Contents

BQ-102. A 17-year-old girl had unprotected intercourse 3 days ago and asks about emergency contraception. After counseling, she chooses a copper intrauterine device, and she asks how it works. Her pregnancy test is negative. Which is the most likely mechanism by which the device prevents pregnancy? [Physiology and Pathophysiology]

A. Delay of ovulation by suppression of the LH surge from the pituitary gland

B. Thickening of cervical mucus by a progestin released from the device each day

C. Dissolution of an embryo that has already implanted in the lining of the uterus

D. Blockage of the fallopian tubes by scar tissue that forms over several weeks

E. A copper-induced inflammatory reaction that is toxic to sperm and eggs

Answer Contents

BQ-103. A 15-year-old girl has a positive urine pregnancy test at a visit alone with you. She is silent, then says, "I do not know what I am going to do." She has told no one. Which is the best next step? [Management and Treatment]

A. Advise her to continue the pregnancy and book prenatal care before she leaves today

B. Ask how she feels and review all her options privately, without steering her

C. Tell her that ending the pregnancy is her best choice because of her age

D. Call her mother immediately so that the adults can decide on a plan together

E. Ask her to return in 2 weeks after she has made her own decision

Answer Contents

BQ-104. A 17-year-old girl has gonorrhea detected on a vaginal swab at a routine screening visit, and the chlamydia result has not returned. She is well, weighs 62 kg, has no drug allergy, and is not pregnant. Which treatment is recommended under the 2021 CDC guidelines? [Management and Treatment]

A. A single intramuscular dose of ceftriaxone plus a single oral dose of azithromycin by mouth

B. A single oral dose of azithromycin alone, given while the chlamydia result is still pending

C. A single oral dose of cefixime alone, with no treatment for chlamydia

D. A single intramuscular ceftriaxone dose, plus doxycycline unless chlamydia is excluded

E. Doxycycline twice daily for 7 days alone, with no cephalosporin

Answer Contents

BQ-105. A 15-year-old sexually active girl has no symptoms and asks to be tested for infection. She has had two partners in the past year, uses condoms inconsistently, and is uncomfortable with a pelvic examination. Which is the most appropriate way to test her for chlamydia and gonorrhea? [Diagnosis]

A. A speculum examination with a cervical sample, because only a clinician can collect it

B. A urine culture sent to the laboratory, since both organisms grow readily in urine

C. A serum antibody test for both infections, repeated in 6 weeks to confirm the result

D. Cervical cytology with reflex testing, because the Pap test detects both infections

E. A self-collected vaginal swab for nucleic acid amplification testing

Answer Contents

BQ-106. A 16-year-old sexually active girl has had 3 days of lower abdominal pain and fever of 38.6 C (101.5 F). A pregnancy test is negative. She has lower abdominal tenderness and pain when the cervix is moved, with mucopurulent discharge at the cervix. Which is the most likely diagnosis? [Diagnosis]

A. Acute appendicitis with a localized abscess

B. Hemorrhage into a ruptured ovarian cyst

C. Acute cystitis from an ascending urinary infection

D. Pelvic inflammatory disease

E. Ectopic pregnancy with early leakage of blood

Answer Contents

BQ-107. A 17-year-old girl was treated for chlamydia 3 months ago, and her partner was treated at the same time. She has no symptoms. A nucleic acid test at the recommended retest interval is positive. Which is the most likely explanation? [Epidemiology and Risk Assessment]

A. Failure of the antibiotic regimen from resistant organisms

B. A false-positive result from residual nucleic acid in the genital tract

C. Reactivation of a latent infection that was never fully cleared

D. Contamination of the specimen during self-collection of the swab

E. Reinfection from an untreated or new partner

Answer Contents

BQ-108. A 13-year-old girl has chlamydia detected on screening. She says her boyfriend is 24 years old, is vague about when sex began, and looks fearful when her mother is mentioned. Which is the best next step? [Epidemiology and Risk Assessment]

A. Tell her mother the full details at once so that the family can respond together

B. Treat the infection and recheck her at the next annual visit without further questions

C. Advise her to stop seeing the partner and close the visit once the prescription is written

D. Interview her privately about coercion and safety, and follow the state reporting law

E. Ask her partner to come to the clinic to give his own account before any report is made

Answer Contents

BQ-109. The parents of an 11-year-old girl ask about HPV vaccination after reading that national schedules differ. She has had no doses. The pediatrician follows the AAP schedule, which recommends two doses starting at 9 to 12 years. Which is the best advice? [Management and Treatment]

A. Wait until she is 15 years old so that the vaccine protects her at the age of exposure

B. Delay vaccination until the schedule debate is settled by the national authorities

C. Give the vaccine only after cervical cytology is performed at 13 years of age

D. Give the vaccine only if she becomes sexually active and then start immediately

E. Give the first dose today and complete the series on the AAP schedule

Answer Contents

BQ-110. A 16-year-old girl tells you in a confidential interview that she drinks about four drinks at weekend parties a few times a month. She has no school or family problems. A validated screening tool places her use on the spectrum between experimentation and disorder. Which is the best next step? [Management and Treatment]

A. Refer her to an inpatient program for supervised treatment of alcohol dependence

B. Praise her answers and end the discussion, since her use is low and carries no real risk to her

C. Offer brief counseling with motivational interviewing, and ask about impaired drivers

D. Report her use to her school and her parents, since drinking at her age is illegal

E. Order liver enzymes and a toxicology panel before any further discussion of the use

Answer Contents

BQ-111. A 17-year-old boy says he sometimes takes pills from friends to relax. He has no history of opioid use and believes the pills are "just Xanax." He is not looking to quit. Which is the most important risk to discuss with him? [Epidemiology and Risk Assessment]

A. Rapid tolerance that leads to weekly increases in the dose he takes to relax

B. Cognitive dulling that lowers his performance in school and at any job he has

C. Legal consequences of possessing pills at school or on the street

D. Liver injury from repeated use of tablets with an unknown dose

E. Counterfeit pills may contain fentanyl, so naloxone should be on hand

Answer Contents

BQ-112. A 15-year-old boy on the wrestling team has lost 9 kg over 4 months and weighs 48 kg. His resting heart rate is 46 beats per minute, and his systolic pressure falls on standing. He avoids family dinners, exercises extra, and says he is "cutting." Laboratory results are normal. Which is the most likely diagnosis? [Diagnosis]

A. Restrictive eating disorder

B. Hyperthyroidism from Graves disease

C. Celiac disease with malabsorption of nutrients

D. Inflammatory bowel disease with chronic blood loss

E. Major depressive disorder with reduced appetite

Answer Contents

BQ-113. A 14-year-old girl with anorexia nervosa is admitted at a weight of 36 kg. On day 3 of nutritional rehabilitation she is weak and has edema, and her phosphate has fallen from 3.4 to 1.6 mg/dL. Which is the most likely mechanism? [Physiology and Pathophysiology]

A. Excess renal loss of phosphate from tubular injury after fluids

B. An insulin-driven shift of phosphate into cells

C. Impaired intestinal absorption of phosphate from villous atrophy

D. Dilution from rapid intravenous fluid given without any feeding

E. Excess vomiting of gastric contents with loss of phosphate and acid

Answer Contents

BQ-114. A 13-year-old girl has been diagnosed with anorexia nervosa. She is medically stable, with a normal heart rate, blood pressure, and temperature, and she weighs 82 percent of median BMI. Her parents ask which treatment has the best evidence. Which is the best treatment? [Management and Treatment]

A. Individual psychotherapy alone, with the parents kept out of the sessions

B. An antidepressant medicine as first-line therapy for weight restoration

C. Residential placement for 6 months before any outpatient treatment is tried

D. Family-based treatment in which the parents lead refeeding at home

E. Nutrition counseling for the patient alone, with no family involvement

Answer Contents

BQ-115. A 17-year-old male distance runner has a tibial stress fracture, frequent colds, and falling race times. His BMI is 18.5 kg/m2, he restricts calories to stay light, and a morning testosterone level is low. Which is the most likely diagnosis? [Diagnosis]

A. Overtraining syndrome caused by training load alone

B. Anabolic steroid withdrawal after a cycle of use

C. Primary testicular failure with an elevated LH

D. Celiac disease with malabsorption of calcium

E. Relative energy deficiency in sport

Answer Contents

BQ-116. An 18-year-old woman with type 1 diabetes attends her pediatric endocrinology clinic alone. Her mother, who attended every visit until now, telephones the clinic and asks for the latest laboratory results. The young woman has not given permission. Which is the best response? [Management and Treatment]

A. Release the results, because the mother has attended every visit since diagnosis

B. Release the results, because the mother pays for the care and the insurance premium each month

C. Explain that she is an adult, so results need her consent, and invite the mother to ask her

D. Release the results only if the mother promises to keep them private

E. Refuse all parental involvement from now on, even if the patient later agrees to it

Answer Contents

BQ-117. A 9-day-old boy, born at 39 weeks after an uncomplicated pregnancy, is brought in after his mother measured a rectal temperature of 38.2 C (100.8 F) at home. He is pink, vigorous, and feeding well. Urinalysis is negative, procalcitonin is 0.2 ng/mL, and C-reactive protein is 6 mg/L. His parents live nearby and ask to go home. Which is the best management? [Management and Treatment]

A. Discharge him with a recheck in 24 hours because every marker is under its cutoff and he feeds well

B. Give one dose of ceftriaxone and discharge him with a clinic visit in 24 hours

C. Repeat the urinalysis and procalcitonin in 12 hours before deciding about spinal fluid

D. Observe him in the department for six hours and discharge him if he stays afebrile

E. Obtain a blood culture and spinal fluid, then admit him and start parenteral antibiotics

Answer Contents

BQ-118. A 24-day-old girl, born at term, has a rectal temperature of 38.4 C (101.1 F) and looks well. Urinalysis is negative, but procalcitonin is 0.8 ng/mL. A lumbar puncture is attempted, and the sample is bloody and cannot be interpreted. Her parents are reliable and would like to take her home. Which is the best next step? [Management and Treatment]

A. Repeat the lumbar puncture at a clinic visit tomorrow and send her home meanwhile

B. Give one dose of ceftriaxone and discharge her with follow-up within 24 hours

C. Discharge her on oral antibiotics because the urinalysis is negative and she looks well

D. Hospitalize her, give parenteral antibiotics, and observe her until cultures are final

E. Discharge her without treatment and recheck her in 24 to 36 hours at the clinic

Answer Contents

BQ-119. A 52-day-old boy, born at term, has a rectal temperature of 38.3 C (100.9 F) and looks well. A catheterized urinalysis shows leukocyte esterase and many white cells. Procalcitonin is 0.2 ng/mL, absolute neutrophil count is 2,800 per mm3, and C-reactive protein is 8 mg/L. Blood and urine cultures have been sent. His parents are reliable. Which is the best management? [Management and Treatment]

A. Perform a lumbar puncture and admit him for parenteral antibiotics

B. Treat with an oral antibiotic and recheck him in 12 to 24 hours

C. Discharge him without antibiotics and recheck him in 24 to 36 hours

D. Give one dose of ceftriaxone and wait for the culture results at home

E. Admit him for 48 hours of observation without starting any antibiotic

Answer Contents

BQ-120. A 35-day-old girl, born at term, has a rectal temperature of 38.1 C (100.6 F) and looks well. Urinalysis is negative. The laboratory can run any inflammatory marker without delay. Her physician wants to follow the AAP guideline for well-appearing febrile infants and stratify her risk of invasive bacterial infection. Which combination of tests is preferred? [Diagnosis]

A. White blood cell count alone, interpreted with age-specific cutoffs for infants under two months old

B. Erythrocyte sedimentation rate together with the white blood cell count

C. Rectal temperature alone, using a threshold of 38.5 C (101.3 F) as the cutoff

D. Procalcitonin together with either absolute neutrophil count or C-reactive protein

E. Band count together with a urine dipstick, without any serum marker at all

Answer Contents

BQ-121. A 5-day-old boy, born at term, is seen in the newborn follow-up clinic with a rectal temperature of 38.2 C (100.8 F). He is alert, pink, and feeding well. His mother had no fever or infection around delivery. The 2021 febrile infant guideline does not cover infants of his age. His parents ask if a urine test is enough. Which is the best management? [Management and Treatment]

A. Perform a full sepsis evaluation with blood, urine, and spinal fluid, then admit him and treat

B. Obtain a urinalysis and procalcitonin, with discharge if both results are normal

C. Send a blood culture alone and arrange a recheck visit within 24 hours

D. Observe him at home with rectal temperature checks every four hours overnight and a call the next morning

E. Obtain a urine culture and give one intramuscular antibiotic dose before discharge

Answer Contents

BQ-122. A 12-day-old boy has a rectal temperature of 38.5 C (101.3 F) and three clustered vesicles on his scalp. His mother had a fever during labor. Cerebrospinal fluid shows 40 white cells per microliter with no organisms on Gram stain, and his alanine aminotransferase is elevated. Blood cultures are pending. Which is the best next step? [Management and Treatment]

A. Await the blood and cerebrospinal fluid cultures before adding any antiviral drug

B. Start oral valacyclovir and arrange outpatient review in 48 hours with his father

C. Start acyclovir now and send cerebrospinal fluid PCR, surface swabs, and blood PCR

D. Treat for bacterial meningitis alone because the Gram stain showed no organisms

E. Send maternal herpes serology and defer antiviral treatment until the result returns

Answer Contents

BQ-123. A 9-year-old girl returns from a camping trip in Connecticut with a 9 cm expanding red patch on her thigh that is clearer in the center. She has no fever, joint swelling, headache, or neurologic symptoms. Her parents ask for blood tests first to be sure of the diagnosis. Which is the best next step? [Diagnosis]

A. Send Lyme serology and treat only if the result is positive

B. Send serology with a confirmatory blot and begin treatment in two weeks

C. Give a single prophylactic dose of doxycycline and observe her for symptoms

D. Diagnose from the rash and start doxycycline without serologic testing

E. Obtain an electrocardiogram and a lumbar puncture before choosing any antibiotic

Answer Contents

BQ-124. A 10-year-old boy with fever and suspected pneumonia is in the emergency department. His temperature is 39.4 C (102.9 F), his oxygen saturation needs 2 L/min of oxygen, and he is drowsy but arousable. His blood pressure and capillary refill are normal for age. His Phoenix Sepsis Score is 2, and none of the points are cardiovascular. Which is the best classification? [Diagnosis]

A. Septic shock, because any score of 2 or higher defines shock

B. Infection without sepsis, because the Phoenix threshold for sepsis is a score of 3

C. Sepsis without septic shock

D. Septic shock, because respiratory points count the same as cardiovascular points

E. Sepsis only after the blood culture grows a bacterial pathogen from the sample

Answer Contents

BQ-125. A 3-year-old boy has had 2 days of fever and vomiting. He is confused and has a creatinine twice his baseline, but his heart rate is 130, capillary refill is 2 seconds, and his blood pressure is normal. The team judges that he has sepsis-associated organ dysfunction without shock. Cultures have been drawn. Which is the best timing for antimicrobials? [Management and Treatment]

A. Within 1 hour only if the blood culture becomes positive during that time

B. After the lumbar puncture and brain imaging are complete, within 12 hours

C. Within 24 hours, provided that he remains afebrile with intravenous fluids

D. As soon as possible after rapid evaluation, within 3 hours of recognition

E. Immediately before any assessment, because the 1 hour limit applies to every febrile child

Answer Contents

BQ-126. A 5-year-old girl, 20 kg, arrives at a rural hospital with no pediatric intensive care unit. She has fever, heart rate 170, blood pressure 70/30 mmHg, capillary refill of 5 seconds, and mottled skin. Antibiotics have been given, and transfer will take two hours. Which is the best fluid plan until the transport team arrives? [Management and Treatment]

A. Maintenance fluid alone until the transport physician arrives to decide about boluses

B. Boluses of 10 to 20 mL/kg over the first hour, to a total of up to 40 mL/kg, reassessing after each

C. A single 60 mL/kg bolus given as fast as possible, followed by a repeat bolus if perfusion stays poor

D. Boluses of 10 to 20 mL/kg to a total of 60 mL/kg before any other treatment

E. Withholding all fluid and starting dopamine through a peripheral line right away

Answer Contents

BQ-127. A 4-year-old girl, 16 kg, with septic shock from pneumonia receives three 20 mL/kg boluses in the first hour. Her capillary refill improves, but she then develops new crackles, an enlarged liver edge, and a rising oxygen requirement. Her blood pressure is now normal. Which is the most likely mechanism of her new findings? [Physiology and Pathophysiology]

A. Bacterial seeding of the liver and both lungs through the bloodstream, caused by the fluid infused

B. A transfusion reaction triggered by the crystalloid solution infused into her

C. Anaphylaxis to the antibiotic that was started before the first bolus

D. Volume given beyond her cardiac and pulmonary reserve, raising venous and capillary pressures

E. Primary adrenal insufficiency unmasked by the rapid expansion of her volume

Answer Contents

BQ-128. A 3-year-old boy, 14 kg, has fever, heart rate 185, cool mottled extremities, capillary refill of 5 seconds, and an irritable cry. His blood pressure is 88/50 mmHg, which is normal for age. A trainee suggests waiting for hypotension before treating him for shock. Which is the most likely explanation for his normal blood pressure? [Physiology and Pathophysiology]

A. Septic shock cannot be present while blood pressure remains normal for age

B. Tachycardia and vasoconstriction preserve pressure until compensation fails late in the course

C. Excess cardiac output from a raised stroke volume maintains his tissue perfusion

D. Fever causes vasodilation that offsets the circulatory failure and maintains pressure

E. Dehydration alone explains the findings and spares the peripheral vasculature and the myocardium entirely

Answer Contents

BQ-129. A 9-month-old uncircumcised boy has had fever to 39.2 C (102.6 F) for two days with no source on examination. A urine bag specimen from the clinic grew 100,000 colony-forming units per mL of Escherichia coli, and the bag urinalysis showed leukocyte esterase. He is drinking well. Which is the best next step? [Diagnosis]

A. Start antibiotics now and order voiding cystourethrography within the week

B. Repeat the bag collection and diagnose infection if the culture grows the same organism

C. Obtain a catheterized specimen for urinalysis and culture before making the diagnosis

D. Accept the bag result as confirmed and begin ten days of oral antibiotics

E. Order renal and bladder ultrasonography before any further urine testing is done

Answer Contents

BQ-130. A 6-year-old boy with sickle cell disease has fatigue and pallor five days after his brother had a slapped-cheek rash. His hemoglobin is 4.1 g/dL, down from a baseline of 8.5 g/dL, and his reticulocyte count is 0.1 percent. His spleen is not enlarged. Which is the most likely mechanism of his anemia? [Physiology and Pathophysiology]

A. Parvovirus B19 infection of erythroid precursors, halting red cell production

B. Splenic pooling of red cells during an acute sequestration episode

C. Autoantibody-mediated destruction of circulating red cells in the spleen

D. Increased hemolysis triggered by a viral vaso-occlusive event in the bone marrow and spleen

E. Occult gastrointestinal bleeding with gradual loss of iron and red cells

Answer Contents

BQ-131. A 14-month-old girl has her first febrile urinary tract infection, confirmed by a catheterized culture and treated with oral antibiotics. She is afebrile by day three. Renal and bladder ultrasonography shows moderate right hydronephrosis. Her parents ask whether any other study is needed now. Which is the best next study? [Diagnosis]

A. No further imaging, because ultrasonography has already been completed and the infection is responding

B. Repeat ultrasonography in one year, with no testing before that follow-up

C. Computed tomography of the abdomen with contrast to define the kidney anatomy

D. Renal scan to be obtained only if a second febrile infection happens

E. Voiding cystourethrography to look for reflux or obstruction of the right urinary tract

Answer Contents

BQ-132. A 5-year-old fully immunized girl has had two days of fever and cough, with crackles at the right base and an oxygen saturation of 97 percent. She is drinking well. A year ago she had a mild delayed rash with amoxicillin and no hives, swelling, or breathing difficulty, and the reaction was judged nonserious. Which is the best antibiotic? [Management and Treatment]

A. Oral azithromycin for five days, with a chest radiograph before she leaves

B. Oral cefpodoxime for the full course, with no routine blood tests or radiograph

C. Oral amoxicillin-clavulanate for the full course, with a blood count first

D. Oral trimethoprim-sulfamethoxazole for ten days, with a urinary antigen test

E. Oral penicillin V for the full course, with a blood culture before she leaves the department

Answer Contents

BQ-133. A 3-year-old fully immunized boy, 14 kg, is admitted with right lower lobe pneumonia, fever, and an oxygen requirement of 2 L/min. The chest radiograph shows lobar consolidation without effusion, and local pneumococcal resistance to penicillin is low. He is vomiting and cannot take oral medication. Which is the best initial antibiotic? [Management and Treatment]

A. Intravenous azithromycin alone, because it also covers atypical organisms

B. Intravenous ceftriaxone with vancomycin, because he needs oxygen therapy

C. Oral amoxicillin-clavulanate, because it adds activity against pneumococcus

D. Intravenous ampicillin, with the dose chosen by weight and local protocol

E. Intravenous clindamycin with gentamicin, because the lobe is fully consolidated

Answer Contents

BQ-134. A 4-year-old boy has had two days of runny nose, cough, and a temperature of 38.1 C (100.6 F). He is drinking and playing, and his ears, throat, and chest are normal. His mother asks for an antibiotic so he can return to preschool tomorrow. He is up to date on vaccines. Which is the best management? [Management and Treatment]

A. Prescribe amoxicillin for seven days to prevent a bacterial complication of what is probably a viral cold

B. Prescribe azithromycin for three days because the shorter course is more convenient

C. Explain that an antibiotic will not help and give a plan for symptoms and return precautions

D. Order a chest radiograph and blood count to decide whether treatment is needed

E. Give a single intramuscular dose of ceftriaxone to shorten the illness for the family

Answer Contents

BQ-135. A 2-year-old unimmunized girl has had three days of fever to 39.8 C (103.6 F), cough, runny nose, and red eyes. Today small white spots appear on the buccal mucosa and a red rash is spreading from her face down to her trunk. Her mother and the receptionist are standing in the waiting room. Which is the best next step? [Diagnosis]

A. Mask her, move her to an airborne isolation room, send PCR and IgM, and call the health department

B. Place her in a standard room with droplet precautions and wait for serology before calling the health department

C. Treat for scarlet fever with amoxicillin and send her home with return precautions

D. Send a throat culture and keep her in the waiting room until the result returns

E. Give MMR vaccine today and send her home to rest until the rash fades

Answer Contents

BQ-136. A 4-month-old boy, 6.5 kg, was in a clinic waiting room 4 days ago with a child later confirmed to have measles. He is well and afebrile, and his mother is breastfeeding. The health department lists him as a close contact and asks what protection he can still receive. Which is the most appropriate intervention? [Epidemiology and Risk Assessment]

A. MMR vaccine today, because it protects within 72 hours at any age

B. No intervention, because breastfed infants are protected by antibody in milk

C. Immune globulin, which works up to 6 days after the exposure

D. Vitamin A on two consecutive days to prevent disease after exposure

E. Two MMR doses given 28 days apart beginning this week

Answer Contents

BQ-137. An 8-month-old girl is scheduled to travel with her family to a country with ongoing measles transmission in three weeks. She has received all vaccines routine for her age. Her mother asks whether anything more is needed before departure. Which is the best advice? [Epidemiology and Risk Assessment]

A. Wait until 12 months, because MMR before that age is never recommended for any infant, even for travel

B. Give immune globulin before departure instead of the vaccine for her age

C. Give MMR now and count it as the first routine dose, with no later doses

D. Give vitamin A before travel to prevent measles during the trip abroad

E. Give a dose of MMR now; she still needs the routine doses at 12 to 15 months and 4 to 6 years

Answer Contents

BQ-138. A 12-year-old boy has had six months of declining school performance and personality change, and now has brief jerking movements of the trunk. He had measles at 14 months in his country of origin, before any vaccination. Electroencephalography shows periodic complexes. Which is the most likely mechanism of his illness? [Physiology and Pathophysiology]

A. Acute postinfectious demyelination triggered by a recent respiratory infection

B. Reactivation of latent herpes simplex virus in the temporal lobes of the brain

C. Persistent defective measles virus infection of neurons years after the original illness

D. Autoantibodies directed against neuronal NMDA receptors that act on the limbic system and the cortex

E. Prion protein misfolding that spreads slowly through neurons of the cortex

Answer Contents

BQ-139. A 7-year-old girl who has received two doses of varicella vaccine has 25 itchy lesions on her trunk, including papules, vesicles, and crusts at the same time. She has a temperature of 37.8 C (100.0 F) and feels otherwise well. A classmate had chickenpox two weeks ago. Which is the most likely diagnosis? [Diagnosis]

A. Hand, foot, and mouth disease from enterovirus infection of the skin

B. Bullous impetigo from Staphylococcus aureus colonization of the skin

C. Papular urticaria from multiple insect bites over several days' time

D. Breakthrough varicella

E. Allergic contact dermatitis from a new laundry detergent exposure

Answer Contents

BQ-140. A 9-year-old boy with one MMR dose has had two days of low-grade fever and swelling in front of the right ear and along the jaw. Swelling peaked yesterday. A buccal swab RT-PCR collected today returns negative. No other cause is evident, and the school asks whether he can return tomorrow. Which is the best next step? [Diagnosis]

A. Exclude mumps and return him to school tomorrow without restrictions

B. Treat the illness as mumps, isolate him until 5 days after swelling began, and report

C. Repeat the buccal swab in two weeks before taking any action at all

D. Start amoxicillin for bacterial parotitis and end isolation after the first dose

E. Give a third MMR dose today as outbreak control and send him back to school within the week

Answer Contents

BQ-141. A 29-year-old woman, one day after delivering a term infant, has a rubella IgG result showing no immunity. She is breastfeeding and has no fever. She has no prior vaccination record and plans to leave tomorrow. She asks whether anything is needed before she goes home. Which is the best next step? [Epidemiology and Risk Assessment]

A. Give MMR vaccine before discharge

B. Give immune globulin now and recheck her rubella titer at 6 weeks

C. Delay any vaccine until her infant is 12 months old and has been immunized

D. Offer no vaccine, because immunity develops naturally in the weeks after delivery

E. Give MMR only if she plans another pregnancy within the coming year

Answer Contents

BQ-142. A 3-year-old girl with asthma has had 3 days of fever to 39.5 C (103.1 F), cough, and body aches during influenza season. She is breathing comfortably and drinking well. The rapid influenza test result will not be available for several hours, and her mother asks whether treatment is still worthwhile at day 3. Which is the best management? [Management and Treatment]

A. Wait for the test and treat only if she has been ill for less than 48 hours

B. Start oseltamivir now without waiting for the test result

C. Give oseltamivir only if she is admitted to the hospital this week

D. Give amoxicillin for presumed influenza complicated by early pneumonia

E. Defer treatment because she is past 48 hours of illness and has improved

Answer Contents

BQ-143. A 3-day-old girl was born at 38 weeks in November. Her mother received the RSVpreF vaccine 6 days before delivery. The infant is well and feeding. Her mother asks why a long-acting antibody is recommended for her daughter despite her own vaccination. Which is the most likely reason the infant is not yet protected? [Physiology and Pathophysiology]

A. The vaccine antibody is destroyed by the infant's stomach acid after feeding

B. Maternal vaccination protects only infants who are born after the end of the winter RSV season each year

C. The vaccine produces immunity in the mother but never reaches the fetus

D. Placental antibody transfer needs time, and delivery within 14 days of vaccination is too soon

E. Only a live attenuated vaccine can cross the placenta to protect a newborn

Answer Contents

BQ-144. A 3-month-old boy, born in July at term, comes for a visit on October 6. His mother did not receive an RSV vaccine in pregnancy, and he has not received an RSV antibody. He is healthy and has no chronic conditions. His mother asks about RSV protection for the coming winter. Which is the best management? [Epidemiology and Risk Assessment]

A. Give no RSV product, because he was not born during the RSV season

B. Give nirsevimab or clesrovimab at this visit

C. Wait until his 8-month visit and give an RSV antibody then

D. Give him the maternal RSVpreF vaccine to protect him this winter

E. Give the antibody only if an RSV test is positive in a household member

Answer Contents

BQ-145. A 9-year-old girl has had 4 days of fever to 39.6 C (103.3 F), abdominal pain with vomiting, a rash, and lightheadedness. Her mother had COVID-19 five weeks ago. Her blood pressure is 82/40 mmHg, C-reactive protein is 14 mg/dL, and troponin is elevated. She has no focal tenderness, and the abdominal ultrasound is normal. Which is the most likely diagnosis? [Diagnosis]

A. Kawasaki disease with typical coronary involvement

B. Acute appendicitis with a developing perforation

C. Viral gastroenteritis with moderate dehydration

D. Acute rheumatic fever following streptococcal pharyngitis

E. Multisystem inflammatory syndrome in children

Answer Contents

BQ-146. An 11-day-old girl has microcephaly, petechiae, hepatosplenomegaly, and chorioretinitis. Saliva CMV PCR collected on day 3 of life was positive, and urine PCR confirmed it. She is feeding by tube and has no other illness. The family asks about treatment options. Which is the most appropriate therapy? [Management and Treatment]

A. Oral valganciclovir, usually continued for 6 months, with audiology every 6 months

B. Oral acyclovir for 21 days, with audiology only if she fails a screen

C. Intravenous ganciclovir for 10 days, followed by observation alone afterward

D. Supportive care alone, because treatment is reserved for infants older than one month

E. Intravenous immune globulin for 5 days, with audiology at 12 months of age

Answer Contents

BQ-147. A 4-year-old boy, 16 kg, received intravenous immune globulin and aspirin for complete Kawasaki disease. Two days after the infusion ended he is still febrile at 39.2 C (102.6 F), with persistent red lips and irritability. Echocardiography shows normal coronary arteries. Cardiology is available. Which is the best next step? [Management and Treatment]

A. Stop aspirin and observe him for another week before changing treatment

B. Give a second immune globulin infusion or pulse corticosteroids, with cardiology involvement

C. Switch to intravenous antibiotics for presumed bacterial lymphadenitis

D. Continue current care alone, because fever always resolves within five days of the first infusion

E. Repeat the echocardiogram in two months and make no change in therapy

Answer Contents

BQ-148. A 10-day-old girl, born at 38 weeks, has hydrocephalus, bilateral chorioretinitis, and scattered intracranial calcifications on head CT. Her mother kept cats at home, took no medications, and had a mild flu-like illness in the second trimester. The newborn hearing screen was passed. Which is the most likely diagnosis? [Diagnosis]

A. Congenital cytomegalovirus infection

B. Congenital rubella syndrome

C. Congenital toxoplasmosis

D. Neonatal herpes simplex encephalitis

E. Congenital Zika virus syndrome

Answer Contents

BQ-149. A 2-day-old boy, 3.0 kg, is born at term to a mother who received no prenatal care and had a reactive treponemal test at delivery. His examination is normal. His RPR titer is 1:32 and the maternal titer is 1:4. The team asks which evaluation is indicated. Which is the best next step? [Diagnosis]

A. Repeat the RPR at 3 months and treat only if the titer is still reactive

B. Spinal fluid analysis, blood count, and long-bone films, with 10 days of intravenous penicillin G

C. A single intramuscular dose of penicillin, with no spinal fluid analysis or radiographs before discharge

D. Ceftriaxone for 10 days and no lumbar puncture or radiographs afterward

E. Cerebrospinal fluid analysis only if he later develops a rash or snuffles

Answer Contents

BQ-150. A 3-week-old boy born to a mother with HIV has a reactive HIV antibody test at his first clinic visit. He is well and gaining weight, and his mother's viral load was suppressed through pregnancy. Her physician asks why the antibody test cannot settle the question of infection. Which is the most likely mechanism of the positive result? [Physiology and Pathophysiology]

A. Confirmed perinatal infection, since antibody tests are specific in newborns

B. Early seroconversion from breast milk exposure during the first week of life

C. Cross-reactivity with the maternal antiretroviral medications that remain in his blood after birth

D. Maternal IgG that crossed the placenta, which does not establish infection in the infant

E. Laboratory contamination, which is the usual cause of a reactive antibody test

Answer Contents

BQ-151. A 26-year-old woman with HIV has had a suppressed viral load on antiretroviral therapy throughout pregnancy and delivered a term infant today. She asks about breastfeeding. Her infant is well. She understands that milk can carry the virus and wants to know what the team recommends. Which is the best counseling? [Management and Treatment]

A. Advise against breastfeeding in all circumstances, since transmission through milk is certain

B. Advise breastfeeding with no infant prophylaxis, since suppression eliminates all risk of transmission

C. Advise breastfeeding only after the infant has a negative test at 18 months

D. Advise stopping her treatment during lactation to protect the infant from the drugs

E. Offer formula, donor milk, or breastfeeding by shared decision-making, with infant prophylaxis

Answer Contents

BQ-152. A 10-year-old boy, 34 kg, was bitten by an Ixodes tick that was attached for about 48 hours in a Lyme-endemic county. The tick was removed 24 hours ago. He has no rash or symptoms. His mother asks whether anything should be done now. Which is the best management? [Epidemiology and Risk Assessment]

A. Send Lyme serology today and treat only if the result comes back positive

B. Give a 14-day course of doxycycline to prevent infection

C. Give no treatment and no follow-up because he has no rash

D. Give a single dose of doxycycline now, within 72 hours of tick removal

E. Give amoxicillin for 28 days to prevent late Lyme disease

Answer Contents

BQ-153. A 14-year-old boy from Massachusetts has had two weeks of fatigue and today fainted while standing. He recalls a rash on his back last month that faded without treatment. His heart rate is 48 beats/min, and the electrocardiogram shows a PR interval of 320 milliseconds. Which is the most likely diagnosis? [Diagnosis]

A. Vasovagal syncope with athletic bradycardia

B. Acute rheumatic fever with early carditis

C. Lyme carditis

D. Viral myocarditis from enterovirus infection

E. Congenital long QT syndrome with arrhythmia

Answer Contents

BQ-154. A 2-year-old girl in Arkansas has had 4 days of fever and irritability, with a rash that began on her ankles and wrists and now includes petechiae. Platelets are 85,000 per microliter and sodium is 129 mmol/L. A tick was removed from her scalp last week. Which is the most likely mechanism of her rash, low platelets, and low sodium? [Physiology and Pathophysiology]

A. Rickettsial infection of vascular endothelium causing widespread vasculitis and leakage

B. Immune complex deposition following a recent streptococcal infection

C. Direct bone marrow invasion by a tick-borne virus suppressing platelets

D. Toxin release from Staphylococcus aureus colonizing the tick bite site

E. Autoimmune destruction of platelets with sodium wasting by the kidneys, unrelated to any infection

Answer Contents

BQ-155. A 7-month-old boy has had fever for 6 days. He has a rash and irritability but no conjunctivitis, oral changes, hand swelling, or lymphadenopathy. C-reactive protein is 5.2 mg/dL, ESR is 62 mm/hr, and echocardiography shows coronary artery dilation. Respiratory viral testing is negative. Which is the most likely diagnosis? [Diagnosis]

A. Roseola infantum with prolonged fever

B. Adenovirus infection with a viral exanthem

C. Incomplete Kawasaki disease

D. Measles in an unimmunized infant

E. Scarlet fever from group A Streptococcus

Answer Contents

Domain 5. Mental and Behavioral Health

Contents Answers for this subject

BQ-173. An 8-year-old girl is brought to clinic because of stomachaches nearly every school morning for 3 months. She cries and clings at drop-off, refuses sleepovers, and sleeps in her parents' room because she fears something will happen to them. Growth, stool pattern, and abdominal examination are normal, and the pain vanishes on weekends at home. Which is the most likely diagnosis? [Diagnosis]

A. Separation anxiety disorder

B. Generalized anxiety disorder

C. Irritable bowel syndrome with constipation

D. Functional abdominal pain disorder

E. Peptic ulcer disease with gastritis

Answer Contents

BQ-174. A 13-year-old boy has had 8 months of excessive worry about school, health, and family, with muscle tension, poor sleep, and missed classes. After 12 weekly sessions of exposure-based cognitive behavioral therapy, his symptoms have improved only partly and still impair daily function. He has no suicidal thoughts and no history of manic episodes. Which is the most appropriate management? [Management and Treatment]

A. Stop the psychotherapy and prescribe a daily benzodiazepine for symptoms

B. Add a selective serotonin reuptake inhibitor to continued psychotherapy

C. Replace the psychotherapy with an atypical antipsychotic at a low dose

D. Continue psychotherapy alone for another year before changing treatment

E. Start a stimulant medication to improve his concentration at school

Answer Contents

BQ-175. A 9-year-old girl attends a well visit with no complaints, and her parents report no school problems. Her clinic is revising its screening policy to follow the 2022 recommendations of the US Preventive Services Task Force for children and adolescents. Her parents ask which behavioral health screen applies to her at this age. Which screening is recommended for her at this visit? [Epidemiology and Risk Assessment]

A. Anxiety screening with a validated questionnaire

B. Depression screening with a validated questionnaire

C. Suicide risk screening with a validated checklist

D. Bipolar disorder screening with a mood checklist

E. Attention-deficit screening with a validated rating scale

Answer Contents

BQ-176. A 10-year-old boy has had a year of inattention, fidgeting, and unfinished work. Parent and teacher rating scales are both elevated, and symptoms began before age 7. Screens for anxiety, depression, trauma, and learning problems were negative, and hearing and vision are normal. His parents want treatment that works. Which is the best initial treatment plan? [Management and Treatment]

A. Parent training alone for 6 months before any medication is offered

B. A stimulant medication plus parent training in behavior management

C. Atomoxetine alone because stimulants are discouraged before age 12

D. Neuropsychological testing before any decision on treatment is made

E. Classroom accommodations alone, with medication reserved for adolescence

Answer Contents

BQ-177. A 12-year-old girl with ADHD has taken extended-release guanfacine for 8 months with good effect. Her family ran out of the medicine, and she missed every dose for 3 days. She now has headache and irritability, and her blood pressure is 142/92 mmHg, higher than her usual readings. Her examination is otherwise normal. Which is the most likely mechanism? [Physiology and Pathophysiology]

A. Rebound sympathetic activity after abrupt withdrawal of an alpha-2 agonist

B. Direct vasoconstriction from the drug accumulating in her peripheral tissues

C. Catecholamine excess from an undiagnosed adrenal tumor

D. Serotonin excess from interaction with another medicine

E. Renal artery stenosis unmasked when the drug was stopped

Answer Contents

BQ-178. A 4-year-old boy is brought for evaluation because he rarely answers to his name and repeats phrases from videos. He does not play with peers, avoids eye contact, and does not point to share interest. He lines up toy cars for an hour, becomes distressed if the route to preschool changes, and covers his ears at the hand dryer. His hearing screen is normal. Which is the most likely diagnosis? [Diagnosis]

A. Isolated expressive language disorder

B. Social anxiety disorder of childhood

C. Autism spectrum disorder

D. Attention-deficit/hyperactivity disorder

E. Sensorineural hearing impairment

Answer Contents

BQ-179. A 27-month-old boy attends a well visit. His parents completed the M-CHAT-R/F, and the total score is 5. They report that he says many words and plays with toys, but he rarely points to show them things. He has no loss of skills, and his hearing was recently reported as normal. Which is the best next step? [Diagnosis]

A. Rescreen at the 30-month visit and take no other action

B. Administer the structured M-CHAT-R/F Follow-Up interview

C. Refer straight to genetic testing with microarray and fragile X

D. Reassure the parents because the score is in the low-risk range

E. Order magnetic resonance imaging of the brain under sedation

Answer Contents

BQ-180. A 14-year-old girl has had 10 weeks of depressed mood, loss of interest, and falling grades, with early morning awakening. Her depression questionnaire score is in the moderate range. She has no suicidal thoughts and no personal or family history of mania. She and her parents agree to start psychotherapy and ask for a medicine to be added. Which medication is most appropriate? [Management and Treatment]

A. Paroxetine

B. Bupropion

C. Fluoxetine

D. Amitriptyline

E. Venlafaxine

Answer Contents

BQ-181. An 11-year-old boy has had repeated eye blinking, head jerks, and shoulder shrugs for 2 years. For the past 18 months he has also cleared his throat many times a day. The movements wax and wane, he can suppress them briefly, and examination, including gait and eye movements, is normal. School performance is good. Which is the most likely diagnosis? [Diagnosis]

A. Provisional tic disorder

B. Persistent motor tic disorder

C. Sydenham chorea after streptococcal infection

D. Absence seizures with eyelid fluttering

E. Tourette syndrome

Answer Contents

BQ-182. The mother of a 15-year-old boy with moderate depression hesitates to let him start an antidepressant because she read about the boxed warning. She asks what the underlying data actually showed. His examination is normal, and he denies current suicidal thoughts. Which statement best summarizes the data behind the warning? [Epidemiology and Risk Assessment]

A. Treated youth died by suicide at about twice the rate seen in those given placebo

B. The excess risk occurred only in children under 12 years and never in adolescents

C. Suicidal thinking occurred in under 1 percent of treated youth and equaled the placebo rate

D. About 4 percent of treated youth had suicidal thinking or behavior, twice placebo, and no suicides

E. The warning came from adult trials only and was applied to youth with no pediatric data or placebo group

Answer Contents

BQ-183. A 15-year-old girl has had 6 weeks of low mood, sleeping 12 hours a day, and falling grades. Her father was hospitalized twice for mania. She has never had a period of reduced need for sleep or unusual energy and denies suicidal thoughts. Her mother asks you to start an antidepressant today. Which is the best next step? [Management and Treatment]

A. Start fluoxetine now and review her in six weeks

B. Arrange psychiatry consultation before any antidepressant is started

C. Begin 6 to 8 weeks of watchful waiting with no other evaluation

D. Prescribe lithium to prevent a switch into mania before treatment

E. Reassure the family and repeat the questionnaire at the next annual visit

Answer Contents

BQ-184. A 12-year-old boy with type 1 diabetes is brought from a soccer match, shouting and refusing to be examined. He is sweating and trembling and cannot answer simple questions. He skipped breakfast but took his usual insulin dose. His heart rate is 128, and his breathing is quiet and unlabored. Which is the most likely cause of his agitation? [Diagnosis]

A. Primary psychosis

B. Diabetic ketoacidosis

C. Panic attack

D. Hypoglycemia

E. Sympathomimetic ingestion

Answer Contents

BQ-185. A 3-year-old girl with 3 days of fever and cough becomes restless and combative in the evening and pulls at her blanket and the monitor leads. Her respiratory rate is 56 with retractions and grunting, and pulse oximetry reads 82 percent on room air. Her glucose is normal. Which is the most likely mechanism of her agitation? [Physiology and Pathophysiology]

A. A primary anxiety disorder triggered by the illness

B. A fever-driven psychotic reaction to the infection

C. Cerebral hypoxia from impaired lung gas exchange

D. Cerebral hypoglycemia from poor oral intake

E. Withdrawal from a sedating medicine given at home

Answer Contents

BQ-186. A 12-year-old girl is in clinic for a sprained ankle. On a standard suicide risk screening tool she answers yes to a question about having thoughts of killing herself in the past week. When asked whether she is having these thoughts right now, she says yes. Her mother is in the waiting room, and no mental health clinician is on site. Which is the best next step? [Management and Treatment]

A. Give her a crisis line number and schedule a visit in one week

B. Complete a brief assessment in the room, write a safety plan, and send her home with her mother

C. Ask her to sign a contract promising not to hurt herself and then discharge her

D. Treat the answer as a negative screen because she has no plan or method

E. Keep her in sight, remove dangerous objects, and arrange an urgent mental health evaluation

Answer Contents

BQ-187. A 15-year-old boy is screened with a four-question suicide risk tool during a visit for acne. He answers no to the first, second, and fourth questions. When asked about thoughts of killing himself in the past week, he says, "I don't want to answer that." He is polite and calm, and his father is outside. Which is the best interpretation of the result? [Diagnosis]

A. A negative screen because no question was answered yes

B. An acute positive screen that requires emergency transfer to a psychiatric unit

C. A positive screen that calls for the acuity question and further assessment

D. An indeterminate screen to repeat at the next annual visit

E. A negative screen unless a parent also reports a concern

Answer Contents

BQ-188. A 14-year-old boy with autism is brought to the emergency department after he struck a classmate at school. He is pacing, clenching his fists, and shouting. His parents say he has hit staff at two prior placements and once broke a window. His vital signs are normal, and a medical cause has been excluded. Which feature of his history most strongly predicts an unsafe course? [Epidemiology and Risk Assessment]

A. His diagnosis of autism spectrum disorder

B. His male sex and his age in adolescence

C. Loud shouting at the time of triage

D. Referral from a school instead of from home

E. A history of aggression toward other people

Answer Contents

BQ-189. A 16-year-old girl screened non-acute positive for suicide risk. A brief safety assessment shows passive thoughts without a plan or intent, no past attempts, and supportive parents. Her father is present. The family keeps a loaded handgun in an unlocked bedroom drawer and her mother's sleeping pills in the bathroom. Which is the most important element of her safety plan? [Management and Treatment]

A. A signed agreement that she will not act on her thoughts

B. Daily review of her phone, messages, and social media accounts by both of her parents

C. A referral for therapy alone, with no change to the home setting

D. Locking up or removing the firearm and medicines at home, with her father's help

E. Instruction to keep the thoughts private until her next appointment

Answer Contents

BQ-190. A 5-year-old boy talks freely and loudly at home, but he has not spoken a word in kindergarten for 4 months. His teacher says he nods, points, and stays at the edge of the group, and he freezes when unfamiliar adults address him. Hearing and language testing are age appropriate, and he has no loss of skills. Which is the most likely diagnosis? [Diagnosis]

A. Autism spectrum disorder

B. Receptive language disorder

C. Selective mutism

D. Oppositional defiant disorder with refusal

E. Sensorineural hearing impairment

Answer Contents

BQ-191. The parents of a 7-year-old boy with monosymptomatic enuresis ask how a bed-wetting alarm could help, since he sleeps deeply and does not wake to a full bladder. He has no daytime symptoms, a normal examination, and a normal urine dipstick. They plan to use the alarm every night. Which is the most likely mechanism by which the alarm works? [Physiology and Pathophysiology]

A. Lowering nighttime urine production by raising antidiuretic hormone levels

B. Enlarging the bladder through repeated stretching during the night

C. Relaxing the external sphincter so that the bladder empties fully

D. Reducing his sleep depth through a rise in cortisol at night

E. Learned arousal or bladder inhibition in response to a full bladder

Answer Contents

BQ-192. A 10-year-old boy wets the bed four nights a week and has no daytime symptoms. A bladder diary shows normal daytime voided volumes and large urine volumes overnight. His urinalysis and examination are normal, and his fluid intake is typical. Which is the most likely mechanism of his bed-wetting? [Physiology and Pathophysiology]

A. Detrusor overactivity with a small functional bladder capacity

B. Obstruction of the bladder outlet from posterior urethral valves

C. Neurogenic bladder from an occult spinal dysraphism

D. Nocturnal polyuria with normal bladder capacity

E. Urinary tract infection with urgency at night

Answer Contents

BQ-193. A 9-year-old girl wets the bed three nights a week and has no daytime wetting or urgency. Her parents say she passes a hard stool about every third day and sometimes soils her underwear. Her examination is normal and a urine dipstick is negative. The family asks you to start treatment for the bed-wetting. Which is the best next step? [Management and Treatment]

A. Treat the constipation first, then reassess the bed-wetting

B. Start desmopressin tonight and address the stools later

C. Order renal and bladder ultrasonography before any treatment

D. Send her to urology for urodynamic testing of bladder function

E. Prescribe a bedtime tricyclic antidepressant for the bed-wetting

Answer Contents

BQ-194. A 16-year-old boy has had a 4-month decline in grades, with trouble concentrating, forgetfulness, and loss of motivation. His parents suspect ADHD. He had no attention problems before this year. At the visit his eyes are red, he asks for food, and he is slow to answer questions. His vital signs are normal. Which is the most likely explanation for his symptoms? [Diagnosis]

A. Newly emerging attention-deficit/hyperactivity disorder

B. Major depressive disorder with psychomotor slowing

C. Hypothyroidism with cognitive slowing

D. Absence seizures with brief lapses of awareness

E. Cannabis use affecting attention and memory

Answer Contents

BQ-195. A 9-year-old boy with ADHD started extended-release clonidine 2 weeks ago. At follow-up he is sleepy, his heart rate is 52, and his blood pressure is 86/48 mmHg, lower than his baseline. He is easily roused, with no chest pain or fainting. Which is the most likely mechanism of these findings? [Physiology and Pathophysiology]

A. Reduced central sympathetic outflow from alpha-2 receptor stimulation

B. Direct blockade of beta-adrenergic receptors in the cardiac conduction tissue

C. Blockade of norepinephrine reuptake in the brain's attention networks

D. A surge of dopamine release in the striatum and prefrontal cortex

E. Excess vagal tone from a cholinergic effect of the medicine itself

Answer Contents

BQ-196. A 13-year-old girl was diagnosed with Tourette syndrome at age 10. Her motor and vocal tics are mild, cause no distress, and do not interfere with school. Her mother asks what to expect in the coming years and whether medicine will be needed. Which statement about the outlook is most accurate? [Epidemiology and Risk Assessment]

A. Most patients have tics that worsen steadily through adolescence into adulthood

B. Tics almost always disappear completely by the age of 14

C. Tics improve only in children treated with antipsychotics

D. Vocal tics predict lifelong disability in nearly every patient

E. About 60 percent have minimal or mild tics within 6 years of evaluation

Answer Contents

BQ-197. A 17-year-old girl started sertraline 3 weeks ago for depression. Over 4 days she has slept 3 hours a night without fatigue, talks rapidly, jumps between topics, and spent her savings online, saying she feels unstoppable. She denies suicidal thoughts and drug use. Her examination and urine toxicology are normal. A maternal uncle has bipolar disorder. Which is the most likely diagnosis? [Diagnosis]

A. Serotonin syndrome from the antidepressant dose

B. Akathisia from sertraline alone

C. Generalized anxiety disorder with insomnia

D. A manic episode unmasked by the antidepressant

E. Stimulant intoxication from an unreported substance

Answer Contents

BQ-198. The mother of a 9-year-old boy reports that he is inattentive, forgets instructions, and loses homework. Her rating scale is elevated. At the visit he is calm and attentive, and hearing and vision are normal. Report cards show good grades and no teacher comments, and the teacher has not been contacted. Which is the best next step to establish a diagnosis of ADHD? [Diagnosis]

A. Obtain a teacher rating scale and school observations

B. Order neuropsychological testing before making a decision

C. Start a stimulant trial to learn whether he responds

D. Obtain an electroencephalogram to look for seizure activity

E. Order magnetic resonance imaging of the brain

Answer Contents

BQ-199. An 11-year-old girl has taken an extended-release stimulant for 3 weeks with good school results. Her mother reports that she skips lunch, eats a large dinner, and cannot fall asleep until 11 PM. Her weight is unchanged, and her blood pressure and pulse are normal for age. She denies sadness or worry. Which is the most likely explanation? [Physiology and Pathophysiology]

A. Early major depressive disorder

B. An emerging eating disorder

C. Hyperthyroidism unmasked by the stimulant medicine itself

D. Medicine-induced low blood glucose

E. Expected sympathomimetic effects of the stimulant

Answer Contents

BQ-200. A 17-year-old boy has taken fluoxetine at the highest tolerated dose for 8 weeks and attends weekly therapy for moderate depression, with no improvement. His mother confirms that he takes every dose. He denies suicidal thoughts and has had no periods of elevated mood. Which is the best next step? [Management and Treatment]

A. Double the dose immediately regardless of tolerability

B. Reassess the diagnosis, adherence, comorbid conditions, and bipolar symptoms

C. Stop the fluoxetine and start a stimulant for his low energy and poor concentration

D. Continue the same plan unchanged for six more months

E. Add an antipsychotic to boost the antidepressant response

Answer Contents

BQ-201. A 13-year-old boy is brought by his mother, who says he has been irritable and withdrawn for 2 months. He quit the basketball team, sleeps 11 hours but feels tired, eats little, and his grades fell from As to Cs. He says that nothing is fun. He denies drug use, and his examination and thyroid-stimulating hormone level are normal. Which is the most likely diagnosis? [Diagnosis]

A. Normal adolescent moodiness

B. Social anxiety disorder

C. Major depressive disorder

D. Attention-deficit/hyperactivity disorder

E. Bipolar disorder in a depressed phase

Answer Contents

BQ-202. A 15-year-old girl is brought to the emergency department by friends after a party. She is agitated and pacing, with wide pupils, a heart rate of 148, a blood pressure of 150/96 mmHg, and a temperature of 38.1 C (100.6 F). Her skin is moist and she has no focal neurologic findings. Her glucose is normal. Which is the most likely cause of her agitation? [Diagnosis]

A. Anticholinergic toxicity from an antihistamine

B. Opioid withdrawal after chronic use

C. Sedative-hypnotic overdose

D. Sympathomimetic intoxication

E. A first episode of primary psychosis

Answer Contents

BQ-203. A 12-year-old girl has an intense fear of vomiting in public. She has stopped eating in the cafeteria, riding the bus, and attending sleepovers. Each avoidance brings immediate relief, but her fear keeps growing and her therapist plans graded exposure. Which is the most likely mechanism by which her avoidance maintains the anxiety? [Physiology and Pathophysiology]

A. It strengthens the cortisol response of the adrenal gland

B. It prevents learning that the feared outcome does not occur

C. It permanently damages the brain regions that process fear and threat

D. It lowers the level of serotonin in the gut

E. It reduces the effect of any therapy she receives

Answer Contents

BQ-204. A 6-year-old boy has frequent tantrums, defies his parents daily, hits his sister, and ignores house rules. His teacher reports no behavior problems at school. Screens for ADHD, anxiety, trauma, language, and learning problems are negative, and the home is safe. His exhausted parents ask for medicine. Which is the best initial management? [Management and Treatment]

A. Start an antipsychotic to reduce his aggression

B. Prescribe a stimulant trial for impulsive behavior

C. Begin an SSRI for irritability, anger, and oppositional behavior

D. Refer him for inpatient psychiatric evaluation

E. Refer the parents for parent training in behavior management

Answer Contents

BQ-205. A 16-year-old girl with ADHD agrees to start a stimulant. She does well in school, vapes occasionally, and mentions that classmates ask to buy pills from friends who take medicine. Her heart examination and family history are unremarkable. Her mother asks which risk deserves the most attention in counseling. Which risk is most specific to prescribing a stimulant to her? [Epidemiology and Risk Assessment]

A. Permanent loss of several inches of adult height

B. Sudden cardiac death in an otherwise healthy teen

C. Physical dependence causing seizures after missed doses

D. Misuse and diversion of the prescription

E. Liver failure requiring transplant evaluation

Answer Contents

Domain 6. Psychosocial Issues

Contents Answers for this subject

BQ-156. A 7-year-old boy with persistent asthma is seen for his third emergency visit this year. His mother says the family lost their lease, is sharing a relative's apartment, and has moved three times since January. She describes visible mold in the apartment. He takes his controller medication reliably, and she has an eviction notice from a landlord. Which is the best next step? [Management and Treatment]

A. Increase his controller medication and repeat the same plan at every visit

B. File a neglect report because the home has visible mold and crowding

C. Advise the family to remain with the relative and revisit housing at the next annual visit

D. Refer the family to social work and a medical-legal partnership for housing and legal help

E. Document the housing history and address it at the next health supervision visit

Answer Contents

BQ-157. A pediatric practice uses the two-item Hunger Vital Sign at health supervision visits. A father of a 3-year-old answers never true to both statements. The nurse notes that his clothing is worn, though he has not mentioned hardship. The screen has a reported sensitivity of 97 percent and specificity of 83 percent. Which is the most accurate interpretation of his result? [Epidemiology and Risk Assessment]

A. Food insecurity is excluded, and no later screening is needed for this family

B. The screen is falsely negative whenever a parent's clothing appears worn or old

C. The result is positive because the sensitivity exceeds the specificity of the screen

D. The screen is unreliable because it applies only to families who receive public benefits now

E. Food insecurity is unlikely, although a small share of food-insecure families are missed

Answer Contents

BQ-158. A 30-year-old mother brings her 4-month-old daughter for a well visit. The infant is growing well. The mother screens positive for depression on a validated tool, reports feeling hopeless, and sleeps poorly. She denies thoughts of harming herself or the baby. She asks whether this is normal after having a baby. Which is the best next step? [Management and Treatment]

A. Tell her this is common after birth and recheck her at the 6-month visit

B. Refer the mother for treatment and arrange close pediatric follow-up for the infant

C. Start an antidepressant for the mother during the infant's visit without referral

D. Report her to child protective services for neglect because of her maternal depression

E. Defer the discussion because the infant is healthy and the visit is for the infant

Answer Contents

BQ-159. A 7-year-old girl has been clingier, has trouble sleeping alone, and has slipping grades for six weeks since her parents separated. Her parents argue in front of her at exchanges and use her to carry messages between homes. She eats well, has friends, and shows no self-harm or school refusal. Which is the most likely cause of her behavior? [Diagnosis]

A. Major depressive disorder that requires an antidepressant now

B. Separation anxiety disorder caused by a genetic predisposition

C. Attention-deficit/hyperactivity disorder that was unmasked by the move

D. Exposure to ongoing parental conflict during an expected period of adjustment

E. Early-onset conduct disorder resulting from poor parental limit-setting at home

Answer Contents

BQ-160. A 13-year-old girl's mother died eight months ago. Since then she has stopped seeing friends, her grades have fallen sharply, and she rarely eats a full meal. Last week she told her aunt that she wished to be with her mother. She denies a plan when asked directly. Which is the most likely diagnosis? [Diagnosis]

A. Normal grief at eight months, expected to resolve without intervention

B. Complicated grief with risk of self-harm

C. Adjustment disorder with mild anxiety that is expected to resolve soon

D. Bullying-related school avoidance leading to loss of appetite

E. Normal adolescent withdrawal related to puberty and peer change

Answer Contents

BQ-161. A 12-year-old boy tells you that classmates spread rumors about him, exclude him at lunch, and have done so since September. The leader is the captain of the football team. His mother has noticed stomachaches on school mornings. She asks whether this is normal teasing. Which feature best identifies this as bullying rather than ordinary peer conflict? [Diagnosis]

A. A single physical fight between two peers of similar strength that was resolved the same day

B. Occasional teasing between close friends who laugh together afterward

C. A disagreement that either child can end and resume freely

D. Aggression that occurs only online and never at school

E. Repeated aggression, including rumors and exclusion, that involves a power imbalance

Answer Contents

BQ-162. A 9-year-old boy was placed in foster care 3 weeks ago and had an initial health visit on day 2. His foster mother brings him now for an earache. The caseworker asks when the full evaluation is due and whether it can wait for his annual visit. Which is the best answer? [Management and Treatment]

A. Within 30 days of placement, covering medical, dental, developmental, and mental health needs

B. At the next annual well visit, about eleven months from now

C. Within 72 hours of placement, since the initial visit was already completed and a form was filed

D. Only after his birth parents have signed consent for each part

E. After six months in a stable placement, when records have arrived

Answer Contents

BQ-163. A pediatrician is preparing to see a 6-year-old girl entering foster care for the first time. She asks which problem is most likely to be present and should therefore be screened for at entry. The caseworker says the placement followed a short family crisis. Which statement best describes the girl's risk? [Epidemiology and Risk Assessment]

A. Most children in foster care have normal development, so screening can wait

B. A chronic condition affects fewer than 1 in 20 children in foster care

C. A significant mental health need is present in up to 80 percent of children in foster care

D. Documented abuse or neglect is present in fewer than 1 in 3, so the history is rarely relevant

E. Dental disease occurs in under 5 percent, so a dental referral can wait

Answer Contents

BQ-164. A 3-year-old girl living in a family shelter is seen for an ear infection. Her immunization record shows she is behind after three recent moves. Her mother has no working phone and no permanent mailing address, and no follow-up appointment is scheduled. She asks only for medicine for the ear. Which is the best management? [Management and Treatment]

A. Treat the infection and schedule a vaccine-only visit in three months' time at the main clinic

B. Defer vaccines until the family has stable housing and a working phone

C. Report the family to child protective services for medical neglect

D. Treat the infection, give catch-up vaccines now, and help with Medicaid and a contact route

E. Give a written summary only and ask the mother to call when she has a phone

Answer Contents

BQ-165. A mother who speaks Portuguese brings her 18-month-old son for a well visit. Her 15-year-old niece, who speaks English, offers to interpret, and the clinic has a video interpreter available. You need to discuss development, the family's housing, and the mother's own stress. Which is the best approach? [Management and Treatment]

A. Use the trained video interpreter for the whole visit

B. Let the niece interpret and confirm key points with a written handout afterward

C. Use the niece for routine questions and the interpreter only for sensitive topics

D. Speak slowly in English and ask the mother to nod if she understands

E. Ask the mother to bring an adult relative to the next visit and defer discussion

Answer Contents

BQ-166. A 10-year-old girl has had abdominal pain on school mornings for two months, with normal growth, a normal examination, and normal screening laboratory tests. Her grades have fallen, she avoids the school bus, and she has lost a jacket and lunch money twice. She says nothing is wrong at school. Which is the most likely explanation? [Diagnosis]

A. Celiac disease presenting with isolated chronic abdominal pain and school absence

B. Functional constipation with school-day exacerbation

C. Peer victimization (bullying) presenting with somatic symptoms

D. Peptic ulcer disease with morning symptoms

E. Generalized anxiety disorder unrelated to school events

Answer Contents

BQ-167. A 4-year-old boy's grandfather died two weeks ago, and his parents told him that Grandpa went to sleep. He now resists bedtime and asks every morning if Grandpa has woken up. He is otherwise playful and eating well. His parents ask what is driving his behavior. Which is the most likely explanation? [Physiology and Pathophysiology]

A. Complicated grief that requires antidepressant treatment

B. Oppositional behavior that is unrelated to the death

C. A language delay that limits his understanding of questions

D. He interprets the phrase literally and does not yet grasp that death is permanent

E. Imitation of a classmate's bedtime resistance and refusal to sleep alone at preschool

Answer Contents

BQ-168. A 27-year-old woman brings her 8-month-old son for a well visit. She is quiet, and her partner waits in the car. She has not mentioned any problems. The practice wants to address partner violence with every parent, in a way that helps even if no one discloses. Which is the best approach? [Management and Treatment]

A. Offer every parent private education and resource cards so that disclosure is not required

B. Ask each mother detailed questions about violence while the child and a relative are in the room

C. Ask only mothers who have visible bruises or who appear anxious

D. Invite the partner into the room to discuss the family's safety together

E. Report every family to child protective services as a precaution

Answer Contents

BQ-169. A mother tells you privately that her partner hits her. The clinic's patient portal gives both parents access to the child's chart, and her partner often reads it. She does not want to leave the home today and wants help for her 3-year-old, who sleeps poorly. Which is the best documentation and follow-up plan? [Management and Treatment]

A. Record her full account in the open chart where her partner can read it

B. Send safety instructions through the shared portal so both parents see them

C. Record nothing in the chart in any form, because any entry whatsoever would put the family at risk

D. Call the partner to verify what she told you before offering any resources

E. Write brief factual entries in protected notes and confirm a safe way to reach her

Answer Contents

BQ-170. A 5-year-old girl placed in foster care last week needs two overdue vaccines. The foster mother has an agency placement letter. The birth mother cannot be reached for two weeks, and the foster mother asks whether the vaccines have to wait. Who can usually give consent for routine care in this situation? [Management and Treatment]

A. Only the birth parents, so the vaccines must wait until they sign

B. A separate court order obtained for every single vaccine and for every later visit

C. The child welfare agency, usually through the foster parent or caseworker

D. The pediatrician, acting alone on the basis of medical need

E. The child herself, once she is able to say yes to the vaccine

Answer Contents

BQ-171. A 2-year-old boy is brought for a first visit in eight months. His father says they lost their car and could not get to appointments. The boy is growing well, has dental caries, and has no injuries or signs of inadequate supervision. The father works two jobs and asks for help getting care. Which best describes this situation? [Diagnosis]

A. Medical neglect that requires an immediate report to child protective services

B. Parental noncompliance that justifies discharging the family from the practice

C. A parenting problem that is best handled by advising the father to work fewer hours

D. A normal pattern of care for the age that needs no screening or resources

E. Unmet social needs that call for support and resources rather than a neglect report

Answer Contents

BQ-172. A mother and her 5-year-old son have recently arrived from Central America and present for a first visit. The mother appears worried and asks, through a trained interpreter, whether the clinic will report their immigration status. The boy is quiet and has not been in school. Which is the best response? [Management and Treatment]

A. Ask for documents to determine his insurance eligibility before proceeding

B. Ask the mother directly about her immigration status so that the correct paperwork can be started

C. Defer all questions about stress and trauma until a later visit

D. Explain that care does not depend on immigration status, and ask about food, housing, and trauma

E. Offer the visit only if a relative of the family can interpret

Answer Contents

Domain 7. Emergency and Critical Care

Contents Answers for this subject

BQ-206. A 14-month-old girl has had a barky cough and noisy breathing for 2 days. From the doorway she sits on her father's lap, looks around, and reaches for a toy. She has stridor when agitated, intercostal retractions, and nasal flaring. Her skin color is pink. Using the pediatric assessment triangle, which is the most likely impression? [Diagnosis]

A. Compensated shock

B. Decompensated shock

C. Respiratory failure

D. Primary brain or metabolic problem

E. Respiratory distress

Answer Contents

BQ-207. A 2-year-old boy is brought by his parents after he was found limp and hard to wake this morning following a 3-day illness with poor intake. From the doorway he does not interact or track, and his tone is poor. His breathing is quiet and unlabored, and his skin is pink with warm hands. Which is the best next step? [Diagnosis]

A. Begin bag-mask ventilation with oxygen

B. Check a bedside glucose

C. Give a 20 mL/kg isotonic crystalloid bolus

D. Perform rapid sequence intubation

E. Give naloxone intramuscularly

Answer Contents

BQ-208. A 4-year-old girl with 2 days of fever and vomiting is lethargic. Her heart rate is 168, her hands are cool, capillary refill is 5 seconds, and distal pulses are weak. Her blood pressure is 96/60 mmHg. A trainee says she cannot be in shock because her pressure is normal. Which process best explains her normal blood pressure? [Physiology and Pathophysiology]

A. Compensatory vasoconstriction and tachycardia that preserve pressure until late

B. Increased circulating volume from the infection itself

C. Complete loss of vascular tone with a high cardiac output and flushed, warm, well-perfused skin

D. Excess release of antidiuretic hormone raising her plasma volume

E. Fluid overload from prior intravenous fluid therapy

Answer Contents

BQ-209. An 8-year-old boy weighing 26 kg is found pulseless on the ward after a choking episode. Compressions and bag-mask ventilation are under way, and an intravenous line is in place. The monitor shows organized electrical activity at a rate of 40, and there is no palpable pulse. Which is the best next step? [Management and Treatment]

A. Defibrillate at 2 J/kg and resume compressions immediately after the shock

B. Give epinephrine 0.01 mg/kg intravenously as early as possible

C. Give amiodarone 5 mg/kg and shock after two minutes

D. Perform synchronized cardioversion at the lowest energy

E. Stop compressions for a 30 second pulse check first

Answer Contents

BQ-210. A 5-month-old girl with severe bronchiolitis becomes pale and unresponsive. Her heart rate falls from 150 to 48 per minute, with weak pulses and mottled skin. She is gasping, and oxygen saturation is 70 percent. Her airway is open but secretions are present. Which is the best next step? [Management and Treatment]

A. Give oxygen and begin bag-mask ventilation

B. Give epinephrine first and ventilate if there is no response

C. Begin chest compressions without ventilation

D. Perform synchronized cardioversion at once

E. Obtain a twelve-lead ECG before any intervention

Answer Contents

BQ-211. A 10-day-old boy was discharged at 2 days after an uncomplicated birth and fed well until 2 days ago. He is now gray and lethargic, with fast breathing, weak pulses in all limbs, cool extremities, and a liver edge 3 cm beyond the costal margin. He is afebrile, and his glucose, sodium, and potassium are normal. Which is the most likely diagnosis? [Diagnosis]

A. Late-onset bacterial sepsis with distributive shock

B. Inborn error of metabolism with hyperammonemia

C. Salt-wasting congenital adrenal hyperplasia

D. Viral myocarditis from enterovirus infection

E. Closing ductus in a ductal-dependent heart lesion

Answer Contents

BQ-212. A 16-year-old boy has a stab wound to the left chest. He is anxious, with a heart rate of 140 and a blood pressure of 78/50 mmHg. His neck veins are distended, left breath sounds are absent, and the trachea is shifted to the right. Which is the most likely mechanism of his shock? [Physiology and Pathophysiology]

A. Blood loss from a lacerated great vessel in the upper chest

B. Loss of vascular tone from spinal cord injury

C. Obstructed venous return from rising pressure in the chest

D. Poor myocardial contractility from cardiac contusion

E. Vasodilation from sepsis after the wound

Answer Contents

BQ-213. A 6-year-old boy develops hives, lip swelling, vomiting, and wheeze 10 minutes after eating a cookie containing peanuts at a party. He is anxious and speaks in short phrases. His heart rate is 140, and his skin is flushed. The school nurse has an epinephrine auto-injector in the bag. Which is the best next step? [Management and Treatment]

A. Give intramuscular epinephrine immediately

B. Give oral diphenhydramine and observe for 30 minutes

C. Give nebulized albuterol and recheck him

D. Give intravenous hydrocortisone before other treatment

E. Start an intravenous fluid bolus as the first treatment

Answer Contents

BQ-214. A 9-year-old girl had a viral illness 10 days ago. She now has fatigue and abdominal pain. Her heart rate is 150 and her blood pressure is 84/58 mmHg. Her extremities are cool, she has bibasilar crackles and a gallop, and her liver edge is 4 cm beyond the costal margin. Which type of shock is most likely? [Diagnosis]

A. Hypovolemic shock

B. Distributive shock from sepsis

C. Cardiogenic shock

D. Obstructive shock from tamponade

E. Distributive shock from anaphylaxis

Answer Contents

BQ-215. A 5-year-old girl has had fever for 1 day and now has a petechial rash, mottled skin, and a capillary refill of 5 seconds. Her heart rate is 170 and her blood pressure is 70/40 mmHg. Intraosseous access is in place and the first fluid bolus is running. A lumbar puncture and blood cultures have not been obtained. Which is the best next step? [Management and Treatment]

A. Wait for culture results before choosing antimicrobials

B. Give oral antibiotics once she can tolerate fluids

C. Give antimicrobials within the next three hours

D. Give broad-spectrum parenteral antimicrobials within one hour of recognition

E. Give dexamethasone first and delay antimicrobials until she has been stabilized

Answer Contents

BQ-216. A 4-month-old girl with a cold has nasal congestion, retractions, and noisy breathing. Her mother asks why a stuffy nose that bothers adults causes such distress in her baby. Mucosal swelling has narrowed her airway radius by half. Which principle best explains her distress? [Physiology and Pathophysiology]

A. Airway resistance rises in direct proportion to the loss of radius

B. Infants have thicker airway smooth muscle than adults do

C. Infant ribs are rigid and prevent chest expansion

D. Infants breathe too slowly to clear secretions

E. Airway resistance rises with the fourth power of the radius

Answer Contents

BQ-217. A 7-month-old boy suddenly cannot cry or cough while eating a piece of food in his high chair. He is awake, with a silent open mouth, dusky color, and weak effort. A parent calls out for help. You are the only trained person present, and he has not yet lost consciousness. Which is the best next step? [Management and Treatment]

A. Alternate five back blows with five chest thrusts

B. Give abdominal thrusts from behind him

C. Sweep his mouth blindly with a finger to find the object

D. Hold him upright and give sips of water

E. Begin chest compressions straight away

Answer Contents

BQ-218. A 20-month-old girl is found drowsy next to her grandmother's open pill organizer. Her respiratory rate is 8 per minute and shallow, and her pupils are 1 mm bilaterally. Heart rate is 90, saturation is 88 percent, and glucose is 94 mg/dL. Her skin is cool and dry. Which toxidrome best explains her findings? [Diagnosis]

A. Anticholinergic

B. Sympathomimetic

C. Cholinergic

D. Sedative-hypnotic

E. Opioid

Answer Contents

BQ-219. A 13-year-old girl is brought in after taking an unknown number of tablets from an over-the-counter sleep aid. Her temperature is 38.6 C (101.5 F), heart rate 138, and pupils are 8 mm. Her skin is flushed and dry, her bowel sounds are absent, and she is picking at the air and speaking nonsense. Which toxidrome best explains her findings? [Diagnosis]

A. Sympathomimetic

B. Cholinergic

C. Anticholinergic

D. Opioid

E. Sedative-hypnotic

Answer Contents

BQ-220. An 18-month-old boy has had drooling, refusal to eat, and a wet cough for 6 hours. His parents found a remote control with an open battery compartment. He is alert and his vital signs are stable. A chest radiograph shows a round radiopaque object in the upper esophagus with a double contour on the frontal view. Which is the most likely diagnosis? [Diagnosis]

A. Esophageal coin ingestion

B. Magnet ingestion with bowel pinching

C. Tracheal foreign body

D. Food bolus from eosinophilic esophagitis

E. Esophageal button battery ingestion

Answer Contents

BQ-221. A 2-year-old boy swallowed one tablet from his grandfather's pill box, which holds four different medicines. The identity of the tablet is unknown. He is smiling, and his examination and vital signs are normal. His mother asks whether any of the grandfather's medicines would make admission necessary despite his well appearance. Which ingested medicine carries the greatest risk of delayed deterioration? [Epidemiology and Risk Assessment]

A. A multivitamin without iron

B. A sustained-release calcium channel blocker

C. A calcium carbonate antacid in chewable form

D. A second-generation antihistamine

E. A stool softener

Answer Contents

BQ-222. A 2-year-old girl pulled a cup of hot tea onto her chest. She cried at once, and the skin of her anterior chest is red and painful, with intact and ruptured blisters that weep clear fluid. The skin blanches with pressure, and the burn covers an area about equal to three of her palms. Which is the most likely depth of the burn? [Diagnosis]

A. Partial-thickness burn

B. Superficial burn

C. Deep full-thickness burn

D. Full-thickness burn

E. Subdermal burn with exposed fat

Answer Contents

BQ-223. A burn consultation line receives calls about five children on the same afternoon, each with a scald from hot water at home. The on-call physician must decide which child needs immediate consultation with a burn center under American Burn Association criteria. Each child is otherwise well and has stable vital signs. Which child meets those criteria? [Epidemiology and Risk Assessment]

A. A 5-year-old with a 4 percent superficial burn of the arm

B. An 8-year-old with a 6 percent partial-thickness burn of the thigh

C. A 3-year-old with a 3 percent partial-thickness burn of the trunk

D. A 7-year-old with a full-thickness burn of the palm

E. A 12-year-old with a 5 percent partial-thickness burn of the calf

Answer Contents

BQ-224. A 6-year-old girl fell about 1 m (3 ft) from playground equipment and vomited once. She was not unconscious. She is alert with a Glasgow Coma Scale score of 15, has a mild headache, a normal neurologic examination, and no signs of a basilar skull fracture. Which is the best next step? [Management and Treatment]

A. Obtain an immediate head computed tomography scan

B. Discharge her at once with no further observation and no return instructions

C. Observe her in the emergency department and reserve imaging for worsening

D. Obtain magnetic resonance imaging under sedation

E. Obtain skull radiographs and discharge if normal

Answer Contents

BQ-225. A 9-year-old boy was thrown from his bicycle and struck his head. In the emergency department the physician applies the PECARN rule for children 2 years and older. The boy has vomited once and has a headache. Which finding would place him in the high-risk tier, where about 4.3 percent have a clinically important brain injury? [Epidemiology and Risk Assessment]

A. A single episode of vomiting

B. A severe headache

C. A brief loss of consciousness

D. A Glasgow Coma Scale score of 14

E. A fall from more than 1.5 m (5 ft)

Answer Contents

BQ-226. A 10-year-old boy is a restrained passenger in a high-speed motor vehicle crash. He groans and does not open his eyes to voice, and his Glasgow Coma Scale score is 7. He is breathing quickly and has a heart rate of 130. A cervical collar is in place. The neck examination is unreliable. Which is the best next step? [Management and Treatment]

A. Remove the collar after a normal clinical neck examination

B. Observe him in the resuscitation bay for four hours before deciding on imaging

C. Obtain cervical imaging only if neck pain develops

D. Defer cervical imaging until the head scan is read

E. Keep spinal immobilization, secure the airway, and image the cervical spine

Answer Contents

BQ-227. A 5-year-old boy is receiving moderate sedation for a laceration repair, with oxygen by nasal cannula. The nurse watching him reports a saturation of 98 percent, but he is snoring, his chest is barely moving, and the capnography waveform has flattened. Which is the most likely explanation for the normal saturation? [Physiology and Pathophysiology]

A. The oximeter is overreading because of motion artifact

B. Ketamine protects the airway against any decline in blood oxygenation

C. Hyperventilation is raising the blood oxygen content

D. Supplemental oxygen delays desaturation and hides hypoventilation

E. Snoring indicates that the airway is fully open

Answer Contents

Domain 8. Child Abuse and Neglect

Contents Answers for this subject

BQ-228. A 5-month-old boy is brought to the emergency department with a swollen, tender right thigh. His father says the baby rolled off the couch. Radiographs show a mid-shaft femur fracture. When the physician mentions a child abuse evaluation, the father becomes angry and says he will take the baby to a hospital across town. Which is the best next step? [Management and Treatment]

A. Make the report to child protective services now and inform the receiving hospital of the concern

B. Defer the report until the second hospital completes its own evaluation and reaches a firm conclusion

C. Accept the father's explanation and document that the family declined further testing

D. Ask the father to sign a release before any agency is contacted about the injury

E. Contact the family's regular clinician and wait for a reply before deciding about a report

Answer Contents

BQ-229. A 7-month-old girl is admitted with irritability. A full skeletal survey shows two healing posterior rib fractures. The explanation offered by her parents does not account for the injuries, a report has been filed, and she remains safely in the hospital. Brain imaging and laboratory studies are normal. Which is the most appropriate plan for further skeletal imaging? [Management and Treatment]

A. Repeat the full skeletal survey within 24 hours of the first

B. Obtain a single chest radiograph in 3 months

C. Perform a bone scan now to replace any repeat radiographs

D. Obtain further imaging only if new swelling or pain appears

E. Repeat the full skeletal survey in 2 to 3 weeks

Answer Contents

BQ-230. A 10-month-old boy is hospitalized with a skull fracture and several rib fractures from inflicted injury. His twin sister is in the clinic with the foster parents. She is playful, feeds well, and has a normal examination with no bruises or swelling. Which is the best management for the twin sister? [Management and Treatment]

A. Reassess her at the next well visit and image only if bruises appear

B. Obtain a bone scan only if she develops pain or limping

C. Obtain a full skeletal survey

D. Request head computed tomography alone, since her limbs look normal

E. Reassure the foster parents because she has no symptoms of injury

Answer Contents

BQ-231. A 4-month-old boy is brought in after a day of vomiting and poor feeding. He is lethargic with a bulging fontanelle. Computed tomography shows bilateral subdural hemorrhages of differing density. A dilated eye examination finds extensive multilayer retinal hemorrhages. The parents describe no trauma. Which is the most likely mechanism of the subdural hemorrhage? [Physiology and Pathophysiology]

A. Rupture of a congenital arteriovenous malformation within the brain cortex

B. Hemorrhagic transformation of a hypoxic-ischemic injury at birth

C. Slow venous leak from a dural sinus thrombosis caused by dehydration

D. Tearing of bridging veins by repeated acceleration and deceleration forces

E. Spontaneous bleeding from an acquired deficiency of clotting factors

Answer Contents

BQ-232. A 2-year-old boy is evaluated after a second femur fracture from a fall at low height. His sclerae appear blue-gray, his teeth are translucent and discolored, and his father has had hearing loss since his twenties. A skeletal survey shows generalized thin bones. Which is the most likely diagnosis? [Diagnosis]

A. Nutritional rickets from vitamin D deficiency

B. Osteogenesis imperfecta

C. Inflicted skeletal trauma as the only explanation

D. Hypophosphatasia from low alkaline phosphatase

E. Juvenile osteoporosis from prolonged corticosteroid use

Answer Contents

BQ-233. A 3-month-old boy has bruises over the left ear and the abdomen. His mother's brother has hemophilia, and she asks whether a bleeding disorder explains the marks. The infant is alert and feeding well. He cannot roll over. Which is the most appropriate management? [Management and Treatment]

A. Complete coagulation testing first and consider abuse only if the results are abnormal

B. Attribute the bruises to hemophilia and arrange hematology follow-up

C. Reassure the mother and recheck the bruises at a visit in one week

D. Evaluate for abuse and test for a bleeding disorder at the same time

E. Postpone any report until hematology finishes its consultation

Answer Contents

BQ-234. A 4-month-old girl is brought by a new foster parent who is worried about large blue-gray patches over the lower back and buttocks. The patches were recorded in the birth record, have not changed color, and are not tender. The infant is thriving and the rest of the examination is normal. Which is the most likely diagnosis? [Diagnosis]

A. Inflicted bruising from forceful gripping

B. Ecchymoses from a coagulation factor deficiency

C. Congenital dermal melanocytosis

D. Cutaneous venous malformation of the buttocks

E. Petechiae from immune thrombocytopenia

Answer Contents

BQ-235. A 12-year-old girl tells her pediatrician that a family friend touched her genitals 5 days ago and that it has happened before. She is calm, has no pain, bleeding, or discharge, and her general examination is normal. Her mother is supportive. Which is the best next step? [Management and Treatment]

A. Collect forensic evidence in the office today, because any disclosure within the past week requires an immediate emergency response

B. Start HIV prophylaxis and emergency contraception today as a precaution, since the contact was repeated and recent

C. Repeat a detailed interview with her mother present to confirm each point of her account before involving any agency

D. Reassure the family, delay a report until an examination is abnormal, and reassess in 6 months for behavior change

E. Report to child protective services and arrange a scheduled specialist evaluation with trauma-focused mental health referral

Answer Contents

BQ-236. A 5-year-old girl has had 2 weeks of vaginal discharge. Her mother knows of no abuse, and the child, asked open questions, denies it. Examination shows a frothy yellow-green discharge with mild erythema. A wet mount shows motile trichomonads. Which is the most likely source of the infection? [Epidemiology and Risk Assessment]

A. Perinatal transmission from her mother at birth

B. Sexual contact with an infected person

C. Poor perineal hygiene with ordinary bathing

D. Exposure to bubble bath in a shared tub

E. Overgrowth of normal flora after a recent antibiotic

Answer Contents

BQ-237. A 2-year-old boy is seen for the first time in 14 months. He has missed all immunization visits, has several untreated cavities, and weighs less than the 3rd percentile. His mother works nights, and he is often left alone with his 6-year-old sister. No injuries are found. Which form of maltreatment does this most likely represent, and is it also the form most often confirmed among victims nationally? [Epidemiology and Risk Assessment]

A. Physical abuse

B. Sexual abuse

C. Fabricated illness by a caregiver

D. Neglect

E. Emotional abuse

Answer Contents

BQ-238. A 4-month-old boy is found at a routine visit to have a healing humerus fracture and a bruise on the cheek, and the history offered does not explain either. He is alert, feeds well, has a flat fontanelle, and has a normal neurologic examination. A skeletal survey is being arranged. His physician asks whether brain imaging is needed. Which is the best next step? [Epidemiology and Risk Assessment]

A. Obtain neuroimaging despite the normal neurologic examination

B. Defer imaging unless vomiting or irritability develops later

C. Obtain cranial ultrasonography through the open fontanelle only

D. Substitute a dilated eye examination for brain imaging

E. Repeat the neurologic examination in 24 hours and then decide

Answer Contents

BQ-239. A 3-year-old girl is brought for a well visit. Her mother mentions one bruise, a 3 cm purple mark behind the left ear, and attributes it to a fall from a swing yesterday. The child walks and talks normally, and the rest of the skin examination is normal. According to a validated bruising rule for children under 4 years, which feature of this bruise most raises concern for abuse? [Epidemiology and Risk Assessment]

A. Its purple color

B. Its size of 3 cm

C. Its location behind the ear

D. Its position on the left side

E. The mother's report of a playground fall

Answer Contents

BQ-240. A 9-month-old girl is admitted for faltering growth. Her weight fell from the 40th percentile to under the 3rd percentile over 4 months, and tests for medical causes are unrevealing. On the ward she takes every feed eagerly and gains 250 g in 5 days. At home she gained poorly on the same formula. Which is the most likely diagnosis? [Diagnosis]

A. Celiac disease with malabsorption

B. Cystic fibrosis with malabsorption

C. Acid reflux disease with feeding aversion

D. Cow's milk protein allergy to the formula

E. Inadequate nutrition provided at home

Answer Contents

BQ-241. A 5-month-old girl has a subdural hemorrhage on computed tomography after an unexplained episode of lethargy. The parents describe no trauma, and a skeletal survey is pending. A pediatric ophthalmologist performs a dilated retinal examination of both eyes. Which finding would most strongly support inflicted head trauma? [Diagnosis]

A. A single small flame hemorrhage beside the optic disc

B. A few preretinal hemorrhages confined to the posterior pole

C. Numerous hemorrhages in several retinal layers reaching the periphery

D. Retinal hemorrhages seen through an undilated pupil on a screening look

E. A small hemorrhage that resolves in a few days

Answer Contents

BQ-242. A pediatrician has found a healing rib fracture in a 3-month-old boy, and a report is required. The infant's father, calm but defensive, asks whether the pediatrician thinks he hurt his son. Which is the best response? [Management and Treatment]

A. "I am required by law to report injuries like this for every baby, and a team will help work out what happened."

B. "I have not decided who caused this, so I will wait until I am completely certain before telling any agency anything."

C. "You need to explain the injury to my satisfaction before I decide whether a report is necessary."

D. "Your account is acceptable, so I will report only if the later tests confirm abuse."

E. "I believe someone in your family did this, and you should expect the baby will be taken today."

Answer Contents

BQ-243. A 31-year-old mother brings her 5-year-old son for a well visit. Her partner stays in the room and answers most questions. She avoids eye contact and flinches when he raises his voice, and she has disclosed nothing. The clinic follows a universal education approach. Which is the best next step? [Management and Treatment]

A. Ask her directly, in front of her partner and her son, whether he is violent toward her

B. Offer a resource card to every parent at the visit without requiring disclosure

C. Ask the 5-year-old what happens at home while his mother is present

D. Confront the partner about the signs of intimate partner violence

E. Report the mother to child protective services for exposing her son to violence

Answer Contents

BQ-244. A 9-year-old girl tells her pediatrician, who asked "Is anything bothering you?", that her stepfather "comes in my room at night and touches me." She is calm. The pediatrician has called child protective services and is now writing the record. Which is the best chart entry? [Management and Treatment]

A. A summary stating that she was sexually abused by her stepfather, a conclusion confirmed by the history

B. A note naming the stepfather as the offender and recording the pediatrician's belief about guilt

C. A paraphrase of the disclosure that omits the question asked, to keep the entry brief

D. Her exact words in quotation marks with the question that prompted them, plus factual observations

E. A brief note that she made a statement, with details left for the investigators

Answer Contents

BQ-245. A 6-year-old boy has had a painless limp for 5 weeks, with occasional knee pain after running. He is afebrile and active. Abduction and internal rotation of the right hip are limited. A radiograph of the pelvis shows a small, dense, flattened right femoral head. Which is the most likely mechanism of this disease? [Physiology and Pathophysiology]

A. Posterior and inferior slippage of the femoral head through the growth plate under body weight

B. Bacterial seeding of the hip joint from the bloodstream

C. Temporary loss of blood supply to the femoral head, with bone softening followed by healing

D. Self-limited synovial inflammation that follows a viral infection

E. Failure of the acetabulum to deepen around a loose femoral head

Answer Contents

BQ-246. A 5-year-old girl has had a limp for 2 days after a cold. Her temperature is 36.9 C, and she bears weight with a limp. Hip motion is mildly limited by pain. The erythrocyte sedimentation rate is 14 mm/hour and the white cell count is 8,200 per mm3. Hip ultrasonography shows a small effusion. Which is the most likely diagnosis? [Diagnosis]

A. Septic arthritis of the hip

B. Transient synovitis of the hip

C. Legg-Calve-Perthes disease

D. Slipped capital femoral epiphysis

E. Osteomyelitis of the femoral neck

Answer Contents

BQ-247. A 2-year-old boy has refused to walk since this morning. His temperature is 38.9 C (102.0 F), and hip movement is painful. The erythrocyte sedimentation rate is 28 mm/hour and the white cell count is 10,400 per mm3. Radiographs of the pelvis show no fracture. Which is the best next step? [Management and Treatment]

A. Prescribe ibuprofen and recheck him in 3 days

B. Start empiric oral antibiotics and discharge him with close follow-up in clinic

C. Obtain hip ultrasonography and involve orthopedics if an effusion is found

D. Arrange a bone scan in one week to look for a stress injury

E. Obtain magnetic resonance imaging of the lumbar spine first

Answer Contents

BQ-248. A 4-year-old girl will not bear weight on the right leg. Her temperature is 39.0 C (102.2 F), the erythrocyte sedimentation rate is 52 mm/hour, and the white cell count is 11,000 per mm3. Her parents ask how likely it is that the hip joint is infected. Using the Kocher predictors, which is the best estimate? [Epidemiology and Risk Assessment]

A. About 93 percent

B. About 3 percent

C. About 40 percent

D. Under 1 percent

E. About 99.6 percent

Answer Contents

BQ-249. A fully immunized 14-month-old girl has refused to bear weight on her left leg for 3 days. Her temperature is 37.9 C, and the left ankle is swollen and warm. The sedimentation rate is 32 mm/hour and the white cell count is 11,000 per mm3. Fluid from the ankle, inoculated into a blood culture bottle, later grows a gram-negative coccobacillus. Which is the most likely organism? [Diagnosis]

A. Staphylococcus aureus

B. Kingella kingae

C. Salmonella species

D. Streptococcus pneumoniae

E. Hemophilus influenzae type b

Answer Contents

BQ-250. A 9-year-old boy has had fever of 38.8 C (101.8 F) and left shin pain for 4 days and now limps. The distal tibia is tender and warm. The C-reactive protein is elevated, and radiographs of the tibia are normal. Blood cultures have been drawn. Which is the most appropriate next imaging study? [Diagnosis]

A. Repeat plain radiographs of the tibia tomorrow

B. Ultrasonography of the left knee joint

C. Computed tomography of the tibia with contrast

D. Magnetic resonance imaging of the left leg

E. Plain radiographs of the opposite tibia for comparison

Answer Contents

BQ-251. A 6-year-old girl was admitted 3 days ago with tibial osteomyelitis. Her fever resolved at 48 hours, her pain has eased, and her C-reactive protein is falling. The blood culture grew methicillin-susceptible Staphylococcus aureus, and magnetic resonance imaging showed no abscess. She is eating well. Which is the best management? [Management and Treatment]

A. Continue intravenous antibiotics through a central venous line for a full 6 weeks as an inpatient

B. Stop all antibiotics now, since her fever has resolved

C. Repeat magnetic resonance imaging every day until the marrow edema clears

D. Proceed to surgical debridement of the tibia without delay

E. Switch to an oral antibiotic chosen by susceptibilities and arrange close follow-up

Answer Contents

BQ-252. A 5-month-old girl born breech has asymmetric thigh skin folds noted by her parents. Examination shows limited abduction of the left hip, and the left thigh looks shorter when the knees are flexed. Which is the best imaging study at this visit? [Diagnosis]

A. Ultrasonography of both hips

B. Magnetic resonance imaging of the pelvis under sedation

C. Technetium bone scan of the pelvis

D. Anteroposterior radiograph of the pelvis

E. Computed tomography of both hips

Answer Contents

BQ-253. A 12-year-old girl who weighs 84 kg has had 3 weeks of vague left groin pain and a limp. Flexing the left hip makes the thigh swing outward. The anteroposterior view of the pelvis looks equivocal, and the frog-leg lateral view shows posterior displacement of the left femoral head on the neck. Which is the most appropriate management? [Management and Treatment]

A. Stop weight bearing and arrange urgent orthopedic surgical fixation

B. Allow weight bearing as tolerated with outpatient physical therapy and a recheck

C. Prescribe ibuprofen and recheck in 2 weeks

D. Fit a hip abduction brace and review in 4 weeks

E. Aspirate the hip under ultrasound guidance to exclude infection

Answer Contents

BQ-254. A 10-year-old boy falls from a bicycle. His distal radius is tender and angled. A radiograph shows a fracture line through the growth plate that exits through a triangular fragment of the metaphysis, and the epiphysis is intact. Which is the Salter-Harris classification? [Diagnosis]

A. Type IV

B. Type I

C. Type V

D. Type III

E. Type II

Answer Contents

BQ-255. An 11-year-old girl rolls her ankle during a soccer match. She has tenderness and mild swelling directly over the distal fibular growth plate and bears weight with pain. The ligament tests are stable, and ankle radiographs are normal. Which is the most likely diagnosis? [Diagnosis]

A. Anterior talofibular ligament sprain

B. Salter-Harris type I fracture of the distal fibula

C. Osteochondritis dissecans of the talus

D. Avulsion fracture at the base of the fifth metatarsal

E. Tarsal coalition with a peroneal spasm

Answer Contents

BQ-256. A 9-year-old boy has severe pain 6 hours after a forearm fracture was casted. Analgesia does not help. His fingers are swollen, and the pain increases sharply when the examiner passively extends them. Pulses are present. Which is the most likely mechanism of his pain? [Physiology and Pathophysiology]

A. Occlusion of the brachial artery by a displaced fragment of the fractured forearm bones

B. Direct pressure on the skin from a rough cast edge

C. Transection of a nerve at the fracture site

D. Rising pressure within a closed fascial space that compromises tissue perfusion

E. Expected inflammation after reduction of the fracture

Answer Contents

BQ-257. An 18-month-old boy who started walking 4 months ago refuses to bear weight after his foot caught in a toy and his leg twisted. He has no fever. There is tenderness over the distal tibia and little swelling. A radiograph shows a nondisplaced spiral fracture of the distal tibial shaft. The history fits the injury. Which is the most likely diagnosis? [Diagnosis]

A. Toddler's fracture

B. Osteomyelitis of the tibia

C. Septic arthritis of the ankle

D. Inflicted tibial fracture

E. Transient synovitis of the hip

Answer Contents

BQ-258. A 13-year-old boy has a left thoracic curve with a Cobb angle of 22 degrees found at a sports physical. He has had back pain for 2 months that wakes him at night. The neurologic examination is normal. Which is the best next step? [Management and Treatment]

A. Fit a thoracolumbar brace now and review him with a repeat radiograph in 6 months

B. Observe and repeat the standing radiograph in 6 months

C. Obtain magnetic resonance imaging of the spine to look for another cause

D. Start a core strengthening program as the only treatment

E. Refer for spinal fusion at the next orthopedic visit

Answer Contents

BQ-259. A 5-week-old boy has held his head tilted to the right with the chin turned to the left since birth. A firm, mobile 2 cm lump is felt in the right sternocleidomastoid muscle. The eyes and neurologic examination are normal, and there is mild flattening of the right side of the head. Which is the best management? [Management and Treatment]

A. Teach a daily stretching and positioning program and examine the hips

B. Fit a rigid cervical collar to hold the head straight for the next 3 months

C. Order magnetic resonance imaging of the brain and neck

D. Schedule surgical release of the muscle

E. Reassure and recheck at 12 months without treatment

Answer Contents

BQ-260. A newborn boy has a right foot that points downward and inward. The heel is small and narrow, and the foot cannot be brought to a neutral position by gentle pressure. The remainder of the examination is normal. Which is the best management? [Management and Treatment]

A. Observe with home stretching and recheck at 6 months

B. Prescribe corrective shoes to wear in the first year

C. Perform surgical release of the tight tendons and ligaments in the newborn period

D. Reassure the parents that the foot will correct with growth

E. Begin serial casting with the Ponseti method in the first weeks of life

Answer Contents

BQ-261. A 13-year-old boy who plays soccer has had 6 weeks of pain at the front of the knee that worsens with running and kneeling. He has a tender bony prominence at the tibial tubercle. Knee motion is full, there is no effusion, and the ligaments are stable. A radiograph shows fragmentation of the tubercle. Which is the most likely diagnosis? [Diagnosis]

A. Sinding-Larsen-Johansson disease

B. Patellar tendon rupture

C. Osgood-Schlatter disease

D. Osteochondritis dissecans of the femoral condyle

E. Septic arthritis of the knee

Answer Contents

BQ-262. A 16-year-old distance runner has had 3 months of shin pain and 5 months without menses. She restricts food, and her body mass index is 17 kg/m2. Hopping reproduces the pain, and magnetic resonance imaging shows a tibial stress fracture. Which is the most likely mechanism underlying both her amenorrhea and her fracture? [Physiology and Pathophysiology]

A. Primary ovarian failure with early loss of estrogen

B. Energy intake too low for training, lowering estrogen and thinning bone

C. A prolactin-secreting pituitary adenoma

D. Hyperthyroidism with accelerated bone turnover

E. Vitamin D intoxication with excess calcium absorption and bone mineralization

Answer Contents

BQ-263. A 15-year-old girl felt a pop when she planted her foot and pivoted during a basketball game. Within 2 hours the knee was swollen and tense, and she could not continue. The Lachman test shows increased anterior translation with a soft end point. Which is the most likely diagnosis? [Diagnosis]

A. Patellofemoral pain syndrome with maltracking of the kneecap

B. Lateral collateral ligament sprain

C. Prepatellar bursitis

D. Septic arthritis of the knee

E. Anterior cruciate ligament tear with hemarthrosis

Answer Contents

BQ-264. A 16-year-old boy comes for a preparticipation evaluation before football. He has no symptoms and no family history of heart disease. On examination, he has a grade 2/6 systolic murmur at the left lower sternal border that grows louder when he moves from squatting to standing. Which is the best next step? [Epidemiology and Risk Assessment]

A. Clear him and repeat the evaluation in two years

B. Clear him after a resting electrocardiogram alone

C. Obtain echocardiography before clearing him for sports

D. Clear him but restrict him to noncontact sports

E. Order an exercise stress test and clear him if it is normal

Answer Contents

BQ-265. A school asks whether every athlete needs an electrocardiogram. A 14-year-old girl, with no symptoms and no family history of early cardiac death or heart disease, wants to play basketball. Her history, blood pressure, and cardiac examination are normal. Which is the best decision? [Epidemiology and Risk Assessment]

A. Obtain a routine resting electrocardiogram first

B. Obtain an echocardiogram before clearance

C. Request an exercise stress test before the season

D. Clear her for play without an electrocardiogram

E. Refer her to cardiology before she plays

Answer Contents

BQ-266. A 14-year-old football player was struck in the head and removed from play with headache, dizziness, and a gap in memory for the event. Three days later he still has a mild headache. His parents ask to keep him in a dark room until he feels well. Which is the best advice? [Management and Treatment]

A. Allow light activity as tolerated and permit staged return to sport only after symptoms have cleared

B. Keep him in a darkened room with no screens, books, or exercise until every symptom has completely gone

C. Allow return to practice once the headache is mild

D. Obtain a baseline cognitive test today to decide on clearance

E. Obtain head computed tomography today before any activity

Answer Contents

Domain 10. Eye, Ear, Nose, and Throat

Contents Answers for this subject

BQ-267. A 4-year-old girl has had right ear pain for 1 day and a temperature of 38.2 C (100.8 F) after a cold. She is playful and drinking well. Pneumatic otoscopy of the right ear shows a mildly bulging, immobile, opaque drum with fluid, and the left ear is normal. Her mother prefers to avoid antibiotics if it is safe and can reach the clinic at any time. Which is the best management? [Management and Treatment]

A. Prescribe a 10-day course of amoxicillin-clavulanate and arrange a recheck visit in 2 weeks if she improves

B. Refer for tympanostomy tubes because this is her first episode of ear pain

C. Give analgesia and watch for 48 to 72 hours, adding an antibiotic if she worsens or fails to improve

D. Prescribe an oral decongestant and antihistamine for 7 days

E. Prescribe a 5-day course of azithromycin and recheck only if symptoms persist

Answer Contents

BQ-268. A 14-month-old boy has a temperature of 39.4 C (102.9 F) and bilateral ear pain for 2 days. Pneumatic otoscopy shows bulging, immobile, red drums in both ears. He finished a course of amoxicillin for the same problem 3 weeks ago and has no drug allergy. Which is the best antibiotic management? [Management and Treatment]

A. Amoxicillin-clavulanate at high dose for 10 days

B. Amoxicillin alone at high dose for 10 days

C. Azithromycin once daily for 5 days

D. Watchful waiting for 72 hours with analgesia only

E. Topical antibiotic ear drops in both ears

Answer Contents

BQ-269. An 18-month-old boy is brought in because he screamed during a diaper change and his mother noticed his right ear was bright red. His temperature is 37.2 C. The right tympanic membrane is diffusely red, but it is translucent with clear landmarks and moves briskly with a pneumatic puff. Which is the most likely explanation for the color of the drum? [Physiology and Pathophysiology]

A. Early acute otitis media with pus in the middle ear

B. Otitis media with effusion behind the drum

C. Vasodilation from crying and vigorous handling

D. Bullous myringitis from a viral or mycoplasma infection

E. A cholesteatoma pressing on the drum

Answer Contents

BQ-270. A 3-year-old girl had acute otitis media treated 6 weeks ago. She is now well, afebrile, and without ear pain, but her parents worry that she hears poorly. Both drums are amber and retracted, with a visible air-fluid level and sluggish movement on a pneumatic puff. Which is the most likely diagnosis? [Diagnosis]

A. Acute otitis media

B. Tympanosclerosis

C. Cholesteatoma

D. Chronic suppurative otitis media

E. Otitis media with effusion

Answer Contents

BQ-271. A 4-year-old boy has had documented fluid behind both drums for 3 months after an ear infection. He has no ear pain or fever. His parents say he turns the television up loud. A trial of nasal steroid spray prescribed elsewhere has not helped. Which is the best next step? [Management and Treatment]

A. Prescribe a 10-day course of amoxicillin

B. Prescribe a short course of oral prednisolone

C. Continue the nasal steroid for 3 more months

D. Arrange age-appropriate hearing testing

E. Refer for adenoidectomy without further testing

Answer Contents

BQ-272. A 2-year-old girl has had four episodes of acute otitis media in 12 months, the latest 3 weeks ago. Today she is well. Both drums are normal and mobile, and tympanometry shows no fluid. Her parents request tympanostomy tubes. Which is the best management? [Management and Treatment]

A. Place bilateral tympanostomy tubes now because of the four recorded episodes

B. Defer tubes because the middle ears are free of fluid today

C. Start a daily low-dose antibiotic for 6 months

D. Perform an adenoidectomy alone

E. Order computed tomography of the temporal bones

Answer Contents

BQ-273. A 3-year-old boy with tympanostomy tubes placed 8 months ago develops purulent drainage from the left ear for 2 days. He is afebrile and playful, with no swelling or tenderness behind the ear. Which is the best treatment? [Management and Treatment]

A. Oral amoxicillin at high dose for 10 days with a recheck in 2 days

B. Oral prednisolone with an antihistamine

C. Urgent removal of the tube in the operating room

D. Topical antibiotic ear drops that are safe for use with tubes

E. Admission for intravenous antibiotics

Answer Contents

BQ-274. A 3-year-old girl has had fever and right ear pain for 5 days. The right pinna is now pushed outward and forward, with tender swelling and redness behind the ear. Her temperature is 39.3 C (102.7 F). Which is the most likely diagnosis? [Diagnosis]

A. Acute otitis externa

B. Acute mastoiditis

C. Furunculosis of the canal

D. Cervical lymphadenitis

E. Acute parotitis

Answer Contents

BQ-275. A 5-week-old girl failed automated hearing screening in both ears at birth and has had no follow-up. Her parents say she startles at loud noises and appears to hear. Her examination is normal. Which is the most appropriate next step? [Management and Treatment]

A. Repeat the automated screen at the routine 6-month well visit and review the result then

B. Reassure the parents because she startles

C. Wait until speech is delayed before acting

D. Arrange diagnostic audiology now so that diagnosis is completed by 3 months of age

E. Refer for cochlear implant evaluation

Answer Contents

BQ-276. A 30-month-old boy says about 10 words and does not combine them. His social interaction and motor skills are typical for age, and he has had two ear infections. His parents ask what to do first. Which is the best step? [Management and Treatment]

A. Refer for formal audiologic testing

B. Begin speech therapy and review in 6 months

C. Send to developmental pediatrics without hearing testing

D. Order magnetic resonance imaging of the brain

E. Wait until age 3 years to see whether speech catches up

Answer Contents

BQ-277. A 2-year-old boy has had a fever and a sore throat for 1 day. His 7-year-old brother had group A streptococcal pharyngitis confirmed by culture 3 days ago. The younger child has a red throat with exudate and no cough or runny nose. Which is the best next step? [Epidemiology and Risk Assessment]

A. Prescribe amoxicillin without any test

B. Reassure the family that streptococcal infection is rare at this age

C. Order an antistreptolysin O titer

D. Perform a rapid antigen test for group A streptococcus

E. Order a complete blood count with differential

Answer Contents

BQ-278. A 9-year-old girl has streptococcal pharyngitis confirmed by a rapid test. Three years ago she developed a mild, delayed, itchy rash with penicillin, with no hives, swelling, or trouble breathing. She is otherwise healthy and has no other drug allergies. Which is the best antibiotic choice? [Management and Treatment]

A. Cephalexin for 10 days

B. Azithromycin for 5 days

C. Clindamycin for 10 days

D. Doxycycline for 10 days

E. Trimethoprim-sulfamethoxazole for 10 days

Answer Contents

BQ-279. A 16-year-old boy has had 4 days of severe right throat pain, fever of 39.2 C (102.6 F), a muffled voice, and difficulty opening his mouth. The right tonsil is swollen, and the uvula is pushed to the left. He has taken amoxicillin for 2 days without relief. Which is the most likely diagnosis? [Diagnosis]

A. Streptococcal pharyngitis with treatment failure

B. Infectious mononucleosis

C. Epiglottitis

D. Peritonsillar abscess

E. Retropharyngeal abscess

Answer Contents

BQ-280. A 7-year-old boy has had four documented episodes of streptococcal pharyngitis in the past year, each with fever over 38.3 C (101 F), exudate, and a positive test. He has no history of periodic fever syndrome, peritonsillar abscess, or antibiotic allergy. His parents ask for tonsillectomy. Which is the best recommendation? [Management and Treatment]

A. Watchful waiting with documentation of each future episode

B. Tonsillectomy now, because four documented episodes in a year is enough to qualify

C. Adenoidectomy alone

D. Prophylactic penicillin every day for a year

E. Polysomnography before deciding on surgery

Answer Contents

BQ-281. A 5-year-old girl with sickle cell disease snores loudly every night, has witnessed pauses in breathing, and has enlarged tonsils. She is restless in sleep and breathes through her mouth. Her parents ask about tonsillectomy. Which is the best next step? [Epidemiology and Risk Assessment]

A. Schedule adenotonsillectomy without further testing

B. Obtain overnight home pulse oximetry only

C. Start an intranasal steroid for 3 months

D. Order polysomnography before surgery

E. Watch until she is 8 years old

Answer Contents

BQ-282. A 6-year-old boy is going home after tonsillectomy for obstructive sleep apnea. His parents ask about pain medicine, and a relative offers leftover codeine cough syrup. The operation was uneventful, and he takes no other medicines. Which is the best plan? [Management and Treatment]

A. Give the leftover codeine cough syrup as needed

B. Give aspirin with food for pain

C. Give no analgesia to avoid masking bleeding

D. Give tramadol as the first-line drug

E. Use scheduled ibuprofen and acetaminophen

Answer Contents

BQ-283. A 5-year-old boy had a tonsillectomy 8 days ago. At home he vomits bright red blood and has blood in his mouth. The bleeding has stopped, but he looks pale and his heart rate is 128 per minute. Which is the best next step? [Management and Treatment]

A. Reassure the family that this is normal scab separation and give extra fluids at home

B. Prescribe an oral antibiotic and recheck in 2 days

C. Send him to the emergency department for evaluation by the operating surgeon

D. Have the family give ibuprofen and observe at home

E. Schedule a clinic visit at the end of the week

Answer Contents

BQ-284. A 7-year-old girl had a cold with nasal discharge and cough that improved by day 6. On day 8 she develops a new fever of 39.1 C (102.4 F), headache, increased purulent nasal discharge, and worsening cough. Which is the most likely diagnosis? [Diagnosis]

A. Acute bacterial sinusitis

B. Uncomplicated viral upper respiratory infection

C. Allergic rhinitis

D. Pertussis

E. Adenoiditis with otitis media

Answer Contents

BQ-285. A 3-year-old girl in group child care has had a temperature of 39.5 C (103.1 F) and thick yellow nasal discharge for 4 days. She looks ill. She has no drug allergy, took no antibiotics in the past 30 days, and has no eye findings. Which is the best treatment? [Management and Treatment]

A. Amoxicillin alone for 5 days

B. Amoxicillin-clavulanate for 10 days

C. Azithromycin for 3 days

D. Observation for 3 more days

E. Trimethoprim-sulfamethoxazole for 10 days

Answer Contents

BQ-286. A 14-year-old boy has had 10 days of fever and purulent nasal discharge. Today he developed a severe frontal headache, photophobia, and a generalized seizure. His eyes look normal and move freely, with no eyelid swelling. He is drowsy but arousable. Which is the most likely complication? [Diagnosis]

A. Preseptal cellulitis of the upper eyelid and cheek

B. Allergic fungal sinusitis with thick eosinophilic mucin

C. Migraine with aura complicated by a provoked seizure

D. Intracranial extension of the sinus infection

E. Orbital cellulitis with spread behind the orbital septum

Answer Contents

BQ-287. A 9-year-old boy has clear watery rhinorrhea, itching of the nose and eyes, and runs of sneezing every spring. The turbinates are pale and boggy. He has no fever. Which is the most likely mechanism of his symptoms? [Physiology and Pathophysiology]

A. Viral replication in the respiratory epithelium

B. IgE-mediated histamine release from mast cells after allergen exposure

C. Bacterial invasion of the sinus mucosa

D. Autonomic hyperreactivity of nasal mucosa without any allergic sensitization

E. Delayed T-cell reaction to a topical nasal medicine

Answer Contents

BQ-288. A 2-year-old girl has had 5 days of foul-smelling discharge from the right nostril only. She is afebrile and eating well, with no cough or sneezing. No one has seen her put anything in her nose. Which is the most likely diagnosis? [Diagnosis]

A. Allergic rhinitis

B. Adenoid hypertrophy

C. Nasal foreign body

D. Choanal atresia

E. Bacterial sinusitis

Answer Contents

BQ-289. An 18-month-old boy has had four nosebleeds in a month, each lasting more than 10 minutes. His mother notes easy bruising on his shins, and her father bled for a long time after dental surgery. Which is the best next step? [Epidemiology and Risk Assessment]

A. Reassure the mother that nosebleeds are usually from nose picking

B. Cauterize the septum at the next visit

C. Obtain computed tomography of the sinuses

D. Recommend saline spray and humidification only

E. Evaluate for a bleeding disorder with laboratory testing

Answer Contents

BQ-290. A 15-year-old boy has had recurrent right-sided nosebleeds for 4 months and progressive right nasal obstruction. He has no history of nose picking or trauma. Anterior rhinoscopy shows normal mucosa and no bleeding site, and there is no family history of bleeding. Which is the most likely diagnosis? [Diagnosis]

A. Digital trauma of the anterior septum

B. Allergic rhinitis

C. Juvenile nasopharyngeal angiofibroma

D. Nasal foreign body

E. Septal hematoma

Answer Contents

BQ-291. A 5-year-old boy has a 2 cm nontender midline neck mass at the level of the hyoid bone. It rises when he swallows and when he sticks out his tongue. The overlying skin is normal. Which is the most likely diagnosis? [Diagnosis]

A. Infantile hemangioma

B. Lymphatic malformation

C. Reactive lymph node

D. Branchial cleft cyst

E. Thyroglossal duct cyst

Answer Contents

BQ-292. An 8-year-old girl has had a tender 3 cm left cervical node and low-grade fever for 2 weeks. A kitten was adopted 3 weeks ago, and a healing papule is present on her left forearm. Which organism is the most likely cause? [Diagnosis]

A. Bartonella henselae

B. Staphylococcus aureus

C. Mycobacterium avium complex

D. Epstein-Barr virus

E. Toxoplasma gondii

Answer Contents

BQ-293. A 2-year-old boy has a firm, nontender 3 cm submandibular mass that has grown over 6 weeks. The overlying skin is violaceous, and he is afebrile. A course of cephalexin produced no change. Which is the most likely diagnosis? [Diagnosis]

A. Staphylococcal adenitis with early abscess formation

B. Cat-scratch disease

C. Nontuberculous mycobacterial lymphadenitis

D. Kawasaki disease with cervical lymphadenopathy

E. Hodgkin lymphoma of the cervical lymph nodes

Answer Contents

BQ-294. A 4-year-old girl has a misaligned left eye. Acuity is 20/20 in the right eye and 20/80 in the left. Both eyes are structurally normal. Which is the most likely mechanism of the reduced vision in the left eye? [Physiology and Pathophysiology]

A. Degeneration of the left retina

B. Opacity of the left cornea

C. Thickening of the left lens

D. Atrophy of the left optic nerve from years of disuse and poor stimulation

E. Cortical suppression of the image from the misaligned eye

Answer Contents

BQ-295. A 5-month-old boy has had constant tearing of the right eye and mucoid crusting on waking since 3 weeks of age. The eye is white, the cornea is clear, and he is not bothered by light. Which is the best management? [Management and Treatment]

A. Prescribe topical antibiotic drops for 2 weeks and recheck after the course

B. Arrange urgent probing under anesthesia

C. Teach massage and plan probing if it persists at 12 to 18 months

D. Obtain computed tomography of the orbits

E. Refer for dacryocystorhinostomy

Answer Contents

BQ-296. A boy born at 27 weeks of gestation is now 4 weeks old and still in the neonatal unit. His parents ask what eye care he needs. His red reflexes are present. Which is the best plan? [Epidemiology and Risk Assessment]

A. Arrange ophthalmologist retinal examinations on the retinopathy of prematurity schedule

B. Check the red reflex at each routine visit and rely on it as the screening test in infancy

C. Plan instrument-based screening at 12 months of age

D. Begin letter acuity testing at 3 years of age

E. Defer examination until he is discharged from the nursery

Answer Contents

BQ-297. A 4-year-old boy swallowed three small round magnets from a toy set about 6 hours ago and has had mild abdominal pain since. He is afebrile and the abdomen is soft. A radiograph shows a cluster of metal objects in the abdomen. Which mechanism explains the serious danger? [Physiology and Pathophysiology]

A. Leakage of a heavy metal that causes systemic toxicity

B. Obstruction of the esophagus by an object of large size lodged at the narrowing

C. Chemical burn from an alkaline discharge on the mucosa

D. Attraction across separate bowel loops that traps the wall and causes necrosis

E. Intestinal stasis from spasm of the pyloric sphincter

Answer Contents

BQ-298. A 3-month-old boy has had inspiratory stridor since 2 weeks of age. It is louder when he cries, feeds, or lies on his back and quieter when he is prone. He is growing well and has no cyanosis. Which is the most likely mechanism of his stridor? [Physiology and Pathophysiology]

A. Fixed narrowing of the subglottis by a growing infantile hemangioma of the airway

B. Inward collapse of floppy supraglottic tissue during inspiration

C. Collapse of the lower trachea during expiration

D. Paralysis of both vocal cords

E. Swelling of the subglottic tissue from a viral infection

Answer Contents

BQ-299. A 4-year-old girl, unvaccinated by her parents' choice, has had a temperature of 40 C (104 F) for 8 hours with sore throat, drooling, and a muffled voice. She sits leaning forward, looks toxic, and has soft inspiratory stridor. Which is the best next step? [Management and Treatment]

A. Call anesthesiology and otolaryngology to secure the airway before any throat examination

B. Examine the throat with a tongue depressor and a bright light to confirm the diagnosis at once

C. Send her to radiology alone for a lateral neck radiograph

D. Start nebulized epinephrine and discharge her with oral antibiotics

E. Draw blood cultures and wait for the results before acting

Answer Contents

Domain 11. Cardiology

Contents Answers for this subject

BQ-300. A 5-year-old boy is seen for a school physical. He is healthy, grows well, and has no symptoms. While he sits upright, you hear a soft continuous hum beneath the right clavicle that grows louder when he turns his head to the left. The hum disappears when he lies flat and when you press gently on the neck veins. Pulses and blood pressure are normal. Which is the most likely diagnosis? [Diagnosis]

A. Patent ductus arteriosus with continuous flow

B. Venous hum, which is an innocent murmur

C. Small ventricular septal defect with turbulent flow

D. Aortic stenosis with a systolic ejection murmur

E. Pulmonary flow murmur from brisk right-sided flow

Answer Contents

BQ-301. A 6-year-old girl has had frequent colds and tires faster than her classmates during play. She is thin but not cyanotic. The second heart sound is widely split and does not change with breathing, and a soft systolic ejection murmur is heard at the upper left sternal border. Which is the most likely mechanism of the fixed splitting of the second heart sound? [Physiology and Pathophysiology]

A. Delayed aortic valve closure from left ventricular outflow obstruction in all phases

B. Early pulmonary valve closure from raised pulmonary vascular resistance that persists through every phase of respiration

C. Delayed right bundle conduction that lengthens right ventricular ejection only in inspiration

D. Reversed ductal flow that equalizes pressure in both great arteries during systole

E. Left-to-right shunting that keeps right ventricular volume high throughout the respiratory cycle

Answer Contents

BQ-302. A 6-week-old boy, born at term, now sweats and breathes fast during feeds and has gained little weight in 3 weeks. His newborn examination found no murmur. Today a harsh holosystolic murmur is heard at the left lower sternal border, the liver edge is 3 cm under the costal margin, and oxygen saturation is 98 percent. Which is the most likely explanation for the timing of his symptoms? [Physiology and Pathophysiology]

A. Closure of the ductus arteriosus that cuts off blood flow to the lower body and the abdominal organs

B. Rising pulmonary pressure that reverses the shunt from left to right to right to left

C. Progressive narrowing of the aortic valve that raises left ventricular afterload

D. Falling pulmonary vascular resistance that lets more blood cross the defect from left to right

E. Loss of maternal antibody that allows viral infection of the heart muscle

Answer Contents

BQ-303. A 10-year-old boy has a blood pressure of 148/92 mmHg in the right arm at each of three visits. His only complaint is leg cramps when he runs. Radial pulses are strong, and femoral pulses are weak and delayed. A soft systolic murmur is heard over the left back. He is not obese. Which is the most likely diagnosis? [Diagnosis]

A. Renal artery stenosis from fibromuscular dysplasia of the vessel

B. Coarctation of the aorta distal to the left subclavian artery

C. Primary hypertension related to family history and diet

D. Pheochromocytoma with episodic catecholamine release

E. Hypertrophic cardiomyopathy with a dynamic outflow gradient

Answer Contents

BQ-304. A term newborn, 14 hours old, is deeply blue with little respiratory distress. Saturation is 70 percent in the right hand and in a foot, and it rises only to 74 percent with 100 percent oxygen. A single loud second heart sound is heard, and there is no murmur. The chest film shows a narrow mediastinum and normal lung fields. Which is the most likely diagnosis? [Diagnosis]

A. Persistent pulmonary hypertension after meconium aspiration

B. Neonatal pneumonia with early respiratory failure

C. Total anomalous pulmonary venous return with obstruction

D. Transposition of the great arteries

E. Coarctation of the aorta with early ductal closure

Answer Contents

BQ-305. A well-appearing term newborn is screened at 24 hours of life on room air. Saturation is 97 percent in the right hand and 86 percent in a foot. Pulses are normal in all limbs, there is no murmur, and the infant feeds well. The nurse asks what to do. Which is the best next step? [Management and Treatment]

A. Repeat the measurement in 1 hour and discharge if the foot reaches 95 percent

B. Discharge with routine follow-up, since the infant looks well

C. Arrange an echocardiogram now, because the screen is positive

D. Start prostaglandin E1 and transfer without any imaging

E. Place the infant in a warm oxygen hood and repeat in 12 hours

Answer Contents

BQ-306. An 8-month-old girl with known tetralogy of Fallot cries hard during a blood draw and becomes deeply blue and limp. Her breathing is fast and deep, saturation is 62 percent, and the systolic murmur has become quiet. She is still responsive. Which is the best next step? [Management and Treatment]

A. Hold her in the knee-to-chest position, give oxygen, and calm her

B. Give a diuretic by intravenous push to relieve pulmonary congestion

C. Give a cardiac glycoside by intravenous push to improve contractility

D. Start prostaglandin E1 to reopen the ductus arteriosus

E. Begin chest compressions and prepare for defibrillation

Answer Contents

BQ-307. A boy born at 28 weeks of gestation is 9 days old and needs little oxygen. He now has bounding pulses, an active precordium, and a widened pulse pressure. A continuous murmur is heard at the left infraclavicular area. Which is the most likely mechanism of the wide pulse pressure? [Physiology and Pathophysiology]

A. Reduced stroke volume from outflow obstruction at the aortic valve

B. Raised systemic vascular resistance from a surge of catecholamines released during respiratory distress

C. Right-to-left shunting through the ductus at the end of systole

D. Loss of ventricular filling from a pericardial effusion under pressure

E. Diastolic runoff of blood from the aorta into the pulmonary circulation through the ductus

Answer Contents

BQ-308. A 2-month-old girl with trisomy 21 feeds in short bouts, sweats with feeds, and has gained 100 g in 3 weeks. Her respiratory rate is 64 per minute and her liver edge is 3 cm under the costal margin. She is not blue. A harsh systolic murmur is heard at the left sternal border with a single loud second sound. The ECG shows a superior QRS axis. Which is the most likely diagnosis? [Diagnosis]

A. Isolated patent ductus arteriosus with a small shunt

B. Tetralogy of Fallot with severe pulmonary stenosis

C. Atrioventricular septal defect

D. Coarctation of the aorta with early ductal closure

E. Hypoplastic left heart syndrome with ductal dependence

Answer Contents

BQ-309. A newborn is found to have trisomy 21 on genetic testing. At 2 days of age, the infant feeds well, is pink, and has normal pulses, no murmur, and a saturation of 97 percent. The parents ask whether his heart needs further assessment. Which is the best response? [Epidemiology and Risk Assessment]

A. Reassurance is enough, with testing only if symptoms appear later

B. Pulse oximetry at each visit is enough, with imaging only if the saturation falls under 95 percent

C. An ECG alone is enough, with imaging if the axis is abnormal

D. Testing can wait until age 1 year, when the heart is fully grown

E. An echocardiogram is advised soon, because many defects are silent in the first days

Answer Contents

BQ-310. A boy born at term passed pulse oximetry screening at 24 hours and went home. At 6 days of age he is gray and limp and breathing fast, with a heart rate of 190 per minute and a low blood pressure. Femoral pulses are absent, and the liver edge is 4 cm under the costal margin. Which is the most likely mechanism of his collapse? [Physiology and Pathophysiology]

A. Early closure of the foramen ovale that traps blood in the left atrium

B. Rising pulmonary vascular resistance that cuts blood flow to the left heart

C. Closure of the ductus arteriosus that had been supplying the lower body

D. Reopening of the ductus arteriosus that floods the lungs with blood

E. Toxin-mediated depression of the heart muscle from group B streptococcus

Answer Contents

BQ-311. An 11-year-old girl had a brief episode of racing heartbeat that began and ended suddenly during a sleepover. She is well now. A resting ECG shows a PR interval of 0.08 seconds, a slurred upstroke at the start of the QRS complex, and a normal corrected QT interval. Which is the most likely diagnosis? [Diagnosis]

A. First-degree atrioventricular block with a prolonged conduction time

B. Wolff-Parkinson-White pattern with an accessory pathway

C. Long QT syndrome with delayed ventricular repolarization

D. Brugada pattern with a coved ST elevation in the right chest

E. Right bundle branch block with a terminal delay in the QRS

Answer Contents

BQ-312. A stable 9-year-old boy has a narrow-complex tachycardia at 200 per minute that started suddenly during a game. Bearing down has not helped. The nurse asks how adenosine will end the rhythm. Which is the most likely mechanism of action? [Physiology and Pathophysiology]

A. Transient block of atrioventricular node conduction that breaks the re-entry circuit

B. Prolonged ventricular repolarization that suppresses the early premature beats that start the loop

C. Blockade of sodium channels in the accessory pathway tissue itself

D. Direct slowing of the sinus node so that it overrides the focus

E. Reduced sympathetic tone that lowers the rate of atrial firing

Answer Contents

BQ-313. A 5-month-old girl has fed poorly for 10 hours. She is gray and drowsy, with a capillary refill of 5 seconds and a blood pressure of 60/30 mmHg. The monitor shows a regular narrow-complex rhythm at 250 per minute that does not vary. There is no intravenous access. Which is the best next step? [Management and Treatment]

A. Repeated vagal maneuvers for several more minutes

B. Rapid adenosine once an intravenous line is secured

C. Synchronized cardioversion at 0.5 to 1 J/kg

D. Intravenous verapamil given slowly over 5 minutes

E. A rapid fluid bolus of 20 mL/kg before any shock

Answer Contents

BQ-314. A 14-year-old boy with a repaired congenital heart defect has palpitations. He is alert, with a blood pressure of 108/64 mmHg and good perfusion. The monitor shows a regular wide-complex tachycardia at 190 per minute, and the QRS duration is 0.14 seconds. Which is the best next step? [Management and Treatment]

A. Obtain expert cardiology help before giving any antiarrhythmic drug

B. Repeat vagal maneuvers for several more minutes before any further therapy is considered

C. Give intravenous verapamil slowly over 2 minutes

D. Give an unsynchronized shock as for ventricular fibrillation

E. Start an oral beta blocker and discharge him after an hour

Answer Contents

BQ-315. A 12-year-old girl lost consciousness while swimming laps and was pulled from the pool by a lifeguard. She woke within a minute and has no seizure history. She is well now. Her father says a paternal aunt drowned at age 25 for no known reason. A resting ECG shows a corrected QT interval of 490 ms. Which is the most likely diagnosis? [Diagnosis]

A. Vasovagal syncope provoked by cold water

B. Hypertrophic cardiomyopathy with outflow obstruction

C. Epilepsy with a generalized tonic-clonic seizure

D. Wolff-Parkinson-White pattern with atrial fibrillation

E. Long QT syndrome with torsades de pointes

Answer Contents

BQ-316. An 8-year-old boy has genetically confirmed long QT syndrome type 1 after a faint during a swimming lesson. He has been symptom-free since. His corrected QT interval is 500 ms. The family asks about the initial drug treatment. Which is the best choice? [Management and Treatment]

A. A selective beta-1 blocker such as metoprolol

B. A nonselective beta blocker such as nadolol

C. A class III antiarrhythmic such as amiodarone

D. A class IC antiarrhythmic such as flecainide

E. A cardiac glycoside such as digoxin

Answer Contents

BQ-317. A 7-year-old girl faints when a fire alarm rings at school and is found to have a corrected QT interval of 510 ms. Monitoring records a twisting ventricular tachycardia that stops by itself. Which is the most likely mechanism of the arrhythmia? [Physiology and Pathophysiology]

A. Re-entry through an accessory atrioventricular pathway

B. Enhanced automaticity of the atrioventricular node

C. Delayed repolarization that lets early afterdepolarizations trigger ventricular tachycardia

D. Slowed ventricular depolarization from sodium channel block that widens the QRS complex and slows conduction

E. Scar-related re-entry around a prior ventricular incision

Answer Contents

BQ-318. A 14-year-old girl feels warm and nauseated and sees tunnel vision while standing in a long line at a hot assembly. She slumps to the floor, looks pale and sweaty, and wakes within 30 seconds. She has no chest pain or palpitations and no family history of sudden death. The examination, orthostatic vital signs, and ECG are normal. Which is the most likely diagnosis? [Diagnosis]

A. Long QT syndrome with exertional arrhythmia

B. Vasovagal syncope with a typical prodrome

C. Hypertrophic cardiomyopathy with outflow obstruction

D. Generalized epilepsy with brief loss of awareness

E. Postural tachycardia syndrome with chronic fatigue

Answer Contents

BQ-319. A 12-year-old boy has had 3 days of sharp left-sided chest pain that is worse with deep breathing and with twisting. Pressing on the left costochondral junctions reproduces it. He has no fever, no exertional symptoms, and no family history of heart disease. Vital signs and heart sounds are normal. Which is the most likely cause of his pain? [Epidemiology and Risk Assessment]

A. Myocarditis after a recent viral illness

B. Pericarditis with a small effusion

C. Coronary artery anomaly with ischemia

D. Musculoskeletal chest wall pain

E. Panic disorder with hyperventilation

Answer Contents

BQ-320. A 14-year-old girl has had chest tightness during her last three track practices. It stops with rest. Her examination and resting ECG are normal. There is no family history of cardiac disease or sudden death. Which is the best next step? [Management and Treatment]

A. Reassure her and allow full activity, since the resting ECG is normal and she feels well afterward

B. Obtain an echocardiogram and hold her from vigorous exercise until it is reviewed

C. Prescribe an antacid for 2 weeks and allow full activity

D. Order a chest film and allow limited activity in practice

E. Review her in 6 months with a repeat resting ECG

Answer Contents

BQ-321. A 13-year-old boy has had 3 days of chest pain and tiredness after a week of fever and cough. He is pale, with a heart rate of 120 per minute. The ECG shows diffuse ST changes, the troponin is raised, and the chest film shows a mildly enlarged heart. Which is the best next test to assess his condition? [Diagnosis]

A. Cardiac catheterization with coronary angiography

B. Cardiac magnetic resonance imaging

C. Exercise stress testing

D. Echocardiography

E. Twenty-four-hour ambulatory rhythm monitoring

Answer Contents

BQ-322. A 15-year-old basketball player becomes light-headed at the end of a game. His father died suddenly at age 38. A harsh systolic murmur is heard at the left sternal border; it grows louder when he stands and softer when he squats. The ECG shows left ventricular hypertrophy with deep narrow Q waves. Which is the most likely diagnosis? [Diagnosis]

A. Hypertrophic cardiomyopathy with outflow obstruction

B. Valvular aortic stenosis with a fixed gradient

C. Ventricular septal defect with a left-to-right shunt

D. Dilated cardiomyopathy with reduced contractility

E. Innocent flow murmur of a trained athlete

Answer Contents

BQ-323. A 17-year-old soccer player with known hypertrophic cardiomyopathy has a systolic murmur at the left sternal border. He is asymptomatic, and his last echocardiogram showed asymmetric septal thickening. During the strain phase of a Valsalva maneuver, the murmur becomes louder. Which is the most likely mechanism? [Physiology and Pathophysiology]

A. Larger ventricular volume that stretches the outflow tract and increases obstruction

B. Higher afterload that pushes the septum away from the outflow tract

C. Smaller ventricular volume that narrows the outflow tract and increases obstruction

D. Increased venous return that boosts forward flow across the valve

E. Faster filling that lengthens diastole and opens the outflow tract

Answer Contents

BQ-324. A 12-year-old boy who is overweight has an office blood pressure of 146/92 mmHg measured with a standard child cuff. His upper arm is large, and the bladder of the cuff covers only a small part of its circumference. He has no symptoms. Which is the best next step? [Diagnosis]

A. Start an ACE inhibitor today and review in 3 months

B. Order ambulatory monitoring after this single elevated reading and start treatment if it is high

C. Send him to the emergency department for urgent control

D. Reassure the family and recheck at the next annual visit

E. Repeat the measurement with a larger cuff and confirm it by auscultation

Answer Contents

BQ-325. A 2-year-old girl, born at 29 weeks of gestation, comes for a routine visit. She is well and grows normally. Her examination is normal, and she has no history of kidney or heart disease. Her mother asks whether blood pressure needs to be checked at her age. Which is the best response? [Epidemiology and Risk Assessment]

A. Measure it at every visit, because birth before 32 weeks is a risk condition

B. Measure it first at age 3 years, as is advised for all healthy children at well visits

C. Measure it only if she develops headaches or poor feeding

D. Measure it once at age 5 years and then every year

E. Measure it only if she is later found to have kidney disease

Answer Contents

BQ-326. A 13-year-old girl is sent from school after a severe headache and vomiting. Her auscultated blood pressure is 186/124 mmHg and is similar on repeat. She is drowsy and reports blurred vision. Her pulses are normal. Which is the best next step? [Management and Treatment]

A. Prescribe an oral antihypertensive and recheck in 1 week

B. Arrange a nephrology visit within 1 week and ambulatory monitoring of her pressure at home

C. Counsel on diet and activity and recheck in 6 months

D. Reassure her that this is white coat hypertension

E. Send her to the emergency department for monitored blood pressure control

Answer Contents

BQ-327. A 15-year-old runner was hospitalized with myocarditis 4 months ago. He now feels well and asks to rejoin the team. Echocardiography at discharge showed recovered function, and he has had no palpitations or fainting since. His family has read that athletes must rest for 6 months. Which is the best approach? [Management and Treatment]

A. Clear him now, since 3 months have already passed

B. Wait until exactly 6 months have passed and then allow full return to competition without any further testing or review

C. Arrange cardiology reassessment with echocardiography, rhythm monitoring, and exercise testing before any return

D. Allow him to play if he stays under 85 percent of his maximum effort

E. Prohibit all competitive sport for life to avoid any risk

Answer Contents

BQ-328. A 3-year-old boy wheezes with every cold, 5 or 6 times a year, with cough and fast breathing lasting about a week. Between colds he is well, sleeps through the night, and plays normally. He has no eczema. He uses albuterol with a valved holding chamber during colds. Which addition to his plan is the most appropriate? [Management and Treatment]

A. A daily high-dose inhaled corticosteroid taken throughout the year

B. A short course of oral corticosteroid started at the first sign of a cold and repeated with each illness

C. A daily oral antibiotic through the winter months to prevent infection

D. A daily long-acting bronchodilator taken throughout the year

E. A short course of daily inhaled corticosteroid started at the first sign of a cold

Answer Contents

BQ-329. A 9-year-old girl uses albuterol once a week and does not wake at night. Over the past 12 months she needed oral steroid courses twice and made one emergency visit, each time during a cold. Her mother thinks her asthma is mild. Which is the best assessment of her asthma? [Epidemiology and Risk Assessment]

A. Her asthma is mild intermittent because daytime symptoms are rare

B. Her exacerbations place her at high risk, so a daily controller is warranted despite few symptoms

C. Her risk is low, because the attacks were triggered by viral colds

D. Her risk cannot be judged until spirometry is repeated in 6 months

E. Her pattern shows exercise-induced bronchoconstriction alone, which carries a low risk of attacks and needs only a reliever

Answer Contents

BQ-330. An 8-year-old boy with asthma has had cough and wheeze for 6 hours. He speaks in short phrases, has retractions, and has a saturation of 92 percent. He improved a little after one albuterol treatment by spacer. Which is the most appropriate next step? [Management and Treatment]

A. Wait for the third albuterol treatment before giving a steroid

B. Give an inhaled corticosteroid alone in place of a systemic dose

C. Give an oral corticosteroid now and repeat albuterol by spacer

D. Start intravenous magnesium and prepare for intubation

E. Add an oral antibiotic for presumed bacterial bronchitis

Answer Contents

BQ-331. A 10-year-old girl with a severe asthma attack has received three doses of albuterol with ipratropium and an oral steroid over the past hour. She still has retractions, speaks single words, and has a saturation of 89 percent on oxygen. Which is the best next step? [Management and Treatment]

A. Give intravenous magnesium sulfate and arrange intensive care review

B. Repeat the same treatments and reassess in 2 hours

C. Discharge her with an action plan, since maximal therapy has been given

D. Start an oral antibiotic for presumed pneumonia

E. Start a long-acting beta agonist and observe for 4 hours

Answer Contents

BQ-332. A 7-year-old boy with asthma has had wheeze for two days. Now he is drowsy, sitting upright, and unable to speak, and his chest is almost silent on auscultation. Saturation is 88 percent. Which is the most likely explanation for the silent chest? [Physiology and Pathophysiology]

A. Resolution of bronchospasm after the earlier bronchodilator treatment

B. Hyperventilation that has washed out carbon dioxide and eased his symptoms through better gas exchange

C. A small pneumothorax that has muffled breath sounds on one side only

D. Mucus in the upper airway that has blocked sound transmission

E. Airflow so reduced that little air moves, signaling impending respiratory failure

Answer Contents

BQ-333. A 6-year-old girl has coughed during and after gym class for a year. The cough is dry, worse in cold air, and wakes her twice a month. Examination between episodes is normal. Her father has hay fever. Which test is the most useful to support the diagnosis? [Diagnosis]

A. Spirometry with a bronchodilator response

B. Chest computed tomography with contrast of the lungs

C. Sweat chloride measurement

D. Bronchoscopy with lavage

E. Chest radiograph in both views

Answer Contents

BQ-334. An 8-month-old boy has had a low-pitched expiratory noise since early infancy. It is louder when he cries or feeds and during colds. He is thriving. The sound is monophonic, and three trials of albuterol have not helped. Which is the most likely diagnosis? [Diagnosis]

A. Early-onset asthma with recurrent wheeze

B. Cystic fibrosis with chronic airway infection

C. Tracheomalacia

D. Bronchiolitis obliterans after a viral illness

E. Foreign body in the lower airway

Answer Contents

BQ-335. A 3-month-old girl has had 3 days of cough and nasal congestion. Respirations are 62 per minute with mild retractions, and saturation is 95 percent on room air. She has taken less than half of her usual feeds and has had two wet diapers in 24 hours. Which is the best management? [Management and Treatment]

A. Give nebulized albuterol every 4 hours until she is eating well and her wheeze has cleared

B. Provide nasal suction and give fluids by nasogastric tube or admit for hydration

C. Give oral dexamethasone daily for 3 days at home

D. Start oral amoxicillin and obtain a chest radiograph

E. Give hypertonic saline in clinic and discharge her home

Answer Contents

BQ-336. A 6-week-old boy, born at term in October, is healthy and breastfed. His mother asks how to lower his risk of severe RSV illness this season. She did not receive an RSV vaccine during pregnancy. Which is the best recommendation? [Epidemiology and Risk Assessment]

A. Give daily oral ribavirin throughout the winter

B. Give monthly azithromycin during the winter months

C. Give routine nebulized hypertonic saline each morning

D. Give a long-acting RSV monoclonal antibody before or early in the season

E. Keep him out of all public places and away from other children until he is 2 years old

Answer Contents

BQ-337. A 20-month-old girl has a barking cough and a hoarse voice after 2 days of a cold. Stridor is audible at rest, and she has mild retractions. She is alert, drinks well, and is not drooling. Temperature is 38.1 C (100.6 F). Which is the best management? [Management and Treatment]

A. Give nebulized albuterol and an oral antibiotic

B. Give a single dose of dexamethasone and nebulized epinephrine

C. Obtain a neck radiograph before any treatment is given

D. Give a single dose of dexamethasone and discharge her at once

E. Admit her for early intubation to secure the airway

Answer Contents

BQ-338. A 4-year-old boy had a croup-like cough for 3 days. He now has a temperature of 39.6 C (103.3 F), looks toxic, and has stridor with thick purulent secretions. Dexamethasone and two epinephrine nebulizations gave little relief. Which is the most likely diagnosis? [Diagnosis]

A. Spasmodic croup with a repeat episode

B. Foreign body lodged in the larynx

C. Epiglottitis with rapid supraglottic swelling

D. Bacterial tracheitis

E. Viral croup that has not yet peaked

Answer Contents

BQ-339. A 3-year-old fully immunized girl has had 2 days of fever of 38.9 C (102.0 F) and cough. Her respiratory rate is 36 per minute, with no retractions, and saturation is 97 percent. There are focal crackles at the right base. She drinks well and is alert. Which is the best management? [Management and Treatment]

A. Obtain a chest radiograph and a blood count before any treatment is started at home

B. Prescribe oral azithromycin as first-line therapy

C. Admit her for intravenous ceftriaxone and oxygen

D. Give supportive care only, since most pneumonia is viral

E. Treat at home with oral amoxicillin and give return precautions

Answer Contents

BQ-340. A 10-year-old boy has had 8 days of low-grade fever, headache, sore throat, and a dry cough that is slowly worsening. He is active and eating. The chest has scattered crackles, and a radiograph shows patchy interstitial infiltrates. Which pathogen is the most likely cause? [Epidemiology and Risk Assessment]

A. Streptococcus pneumoniae

B. Mycoplasma pneumoniae

C. Respiratory syncytial virus

D. Staphylococcus aureus

E. Moraxella catarrhalis

Answer Contents

BQ-341. A 5-year-old girl is on day 3 of oral amoxicillin for lobar pneumonia, but her fever of 39.5 C (103.1 F) persists. She has new pleuritic pain and rising effort. Percussion is dull and breath sounds are decreased at the right base. Which is the best next step? [Management and Treatment]

A. Double the amoxicillin dose and review in 1 week

B. Switch to oral azithromycin alone and review in 3 days

C. Obtain chest imaging to look for a parapneumonic effusion and arrange admission

D. Add a nebulized bronchodilator and continue the current drug

E. Observe at home for another 48 hours before deciding whether to change her treatment

Answer Contents

BQ-342. A 6-week-old girl has poor weight gain despite a voracious appetite, bulky greasy stools, and skin that tastes salty. The family recently moved, and her newborn screening result cannot be found. Which test is the most appropriate to confirm the suspected diagnosis? [Diagnosis]

A. Celiac antibody panel

B. Stool culture with ova and parasite testing

C. Abdominal ultrasound for pyloric stenosis

D. Thyroid function tests

E. Sweat chloride measurement

Answer Contents

BQ-343. A 14-month-old boy with cystic fibrosis has a chronic wet cough, and cultures grow Staphylococcus aureus. His mother asks why his mucus is so thick and hard to clear. He is gaining weight on enzyme treatment. Which is the most likely mechanism? [Physiology and Pathophysiology]

A. Failure of ciliary beating caused by absent dynein arms

B. Overproduction of immunoglobulin that thickens the mucus gel

C. Defective chloride transport across epithelium that reduces water in the airway surface liquid

D. Loss of surfactant that causes alveoli to collapse at end-expiration and stiffens the lungs of the infant

E. Destruction of elastin by neutrophils in healthy lung tissue

Answer Contents

BQ-344. A 5-year-old boy has a chronic wet cough, digital clubbing, and bilateral nasal polyps. His weight is at the 5th percentile, and he has frequent bulky loose stools. Newborn screening was reported as normal. Which is the most likely diagnosis? [Diagnosis]

A. Primary ciliary dyskinesia

B. Allergic rhinitis with asthma

C. Protracted bacterial bronchitis

D. Cystic fibrosis

E. Common variable immune deficiency

Answer Contents

BQ-345. A 4-month-old girl had a positive newborn screen for cystic fibrosis. A sweat chloride was 78 mmol/L, and genetic testing found two known disease-causing variants. She is gaining weight and has mild loose stools. Which is the best next step? [Management and Treatment]

A. Repeat the sweat test in 6 months before making any decision about her care or follow-up

B. Send her to a cystic fibrosis center to begin multidisciplinary care

C. Observe at home until respiratory symptoms appear

D. Start long-term oral antibiotic prophylaxis alone

E. Order bronchoscopy to look for early lung disease

Answer Contents

BQ-346. A 7-month-old girl born at 25 weeks of gestation has bronchopulmonary dysplasia and needed home oxygen. She is feeding well and gaining weight, and her parents are anxious about the coming winter. Her parents ask how to reduce her risk of hospital admission with respiratory infection this winter. Which advice is the most effective? [Epidemiology and Risk Assessment]

A. Keep her on a home cardiorespiratory monitor throughout the winter months and during every nap

B. Give a daily oral antibiotic throughout the winter months

C. Give a daily oral steroid to prevent airway inflammation

D. Stop all contact with other children until age 2 years

E. Give the recommended immunizations and RSV prevention, and keep the home free of smoke

Answer Contents

BQ-347. A 2-year-old boy with cerebral palsy coughs and chokes with thin liquids. He has a weak cough and drools often. Over the past year he has had three right lower lobe pneumonias. Growth is normal and he has no clubbing. Which test is the most appropriate next? [Diagnosis]

A. Sweat chloride measurement

B. Spirometry with bronchodilator testing

C. Skin prick testing for aeroallergens

D. A videofluoroscopic swallow study

E. Chest computed tomography with contrast

Answer Contents

BQ-348. A 3-year-old girl has had a daily wet cough for 8 weeks. She has finished two successive 2-week courses of an appropriate antibiotic, and the cough has not changed. Growth is normal, there is no clubbing, and the chest film is normal. Which is the best next step? [Management and Treatment]

A. Repeat the same antibiotic for another 4 weeks

B. Start a daily inhaled corticosteroid for presumed asthma

C. Arrange further evaluation, including consideration of bronchoscopy with lavage

D. Prescribe a cough suppressant and review in 3 months

E. Reassure the parents that the cough will clear on its own within the next year or so

Answer Contents

BQ-349. An 8-month-old boy is eating crackers when he suddenly cannot cry or cough. He is conscious, with weak movements and a blue color, and makes no sound. He had been well that morning and has no history of reflux or airway problems. His father calls for help. Which is the best next step? [Management and Treatment]

A. Give abdominal thrusts from behind while holding him upright

B. Sweep the mouth with a finger to remove the object

C. Give rescue breaths and check for a pulse

D. Give five back blows alternating with five chest thrusts

E. Call emergency services and wait for their arrival

Answer Contents

BQ-350. An 18-month-old boy has a cough since choking on a peanut yesterday. He has been coughing intermittently and is otherwise comfortable, with a normal temperature. A chest film taken in expiration shows a hyperinflated left lung with the heart pushed to the right. Which is the most likely mechanism of the hyperinflation? [Physiology and Pathophysiology]

A. Complete obstruction with absorption of trapped air and collapse of the lung segment beyond the object

B. A pleural effusion that compresses the lung and pushes the heart

C. A tension pneumothorax from alveolar rupture during choking

D. Reactive bronchospasm that is uniform across both lungs

E. A ball-valve obstruction that lets air in on inspiration but traps it on expiration

Answer Contents

BQ-351. A 3-month-old girl has had noisy breathing since 2 weeks of age. It is a high-pitched, fluttering inspiratory sound that is louder when she is supine, crying, or feeding. She gains weight well, has no retractions, and has a normal cry. Which is the most likely diagnosis? [Diagnosis]

A. Laryngomalacia

B. Subglottic stenosis

C. Vocal cord paralysis

D. Vascular ring

E. Subglottic hemangioma

Answer Contents

BQ-352. A 4-month-old boy has biphasic stridor that has worsened over 4 weeks. He has had two episodes of barking cough labeled as croup. A raised red lesion is present on his chin and lower face. His cry is normal. Which is the most likely diagnosis? [Diagnosis]

A. Laryngomalacia with reflux

B. Viral croup with recurrent episodes

C. Vocal cord paralysis from birth injury

D. Tracheomalacia with a fixed expiratory sound

E. Subglottic hemangioma

Answer Contents

BQ-353. A 40-day-old boy, born at 36 weeks of gestation, is brought in after being found pale and limp for 30 seconds before recovering. This was the first event. The examination, growth, and vital signs are normal. Which statement best describes his risk category? [Epidemiology and Risk Assessment]

A. Higher risk, because he is younger than 60 days

B. Lower risk, because the event lasted less than 1 minute

C. Lower risk, because he was born after 32 weeks of gestation

D. Higher risk, because he was born before 37 weeks of gestation

E. Lower risk, because the examination is normal

Answer Contents

BQ-354. A 5-year-old girl has a headache, nausea, and drowsiness, and she looks flushed. Her family has used a gas heater in a closed room for two nights, and her brother has similar symptoms. Pulse oximetry reads 99 percent on room air. Which best explains the normal reading? [Physiology and Pathophysiology]

A. Hyperventilation has raised arterial oxygen content past normal values and flushed the skin pink and warm

B. Cold hands have caused a falsely high reading

C. Methemoglobin has absorbed light and raised the reading

D. Anemia has lowered the oxygen demand of the tissues

E. Carboxyhemoglobin absorbs light like oxyhemoglobin, so the oximeter overestimates saturation

Answer Contents

BQ-355. A 9-year-old boy with Duchenne muscular dystrophy has had morning headaches and growing drowsiness. His breath sounds are clear, but his breathing is slow and shallow. Saturation is 94 percent on oxygen, and a venous gas shows a carbon dioxide level of 72 mmHg. Which best describes his respiratory failure? [Physiology and Pathophysiology]

A. Hypoxemic failure from fluid filling the alveoli

B. Hypoxemic failure from ventilation-perfusion mismatch in the lung

C. Compensated metabolic acidosis with normal ventilation

D. Hypercapnic failure from weakness of the respiratory pump

E. Hypoxemic failure from a right-to-left cardiac shunt

Answer Contents

Domain 13. Gastroenterology

Contents Answers for this subject

BQ-356. A 9-day-old boy born at term has had two episodes of green vomit in the past 6 hours after feeding well until today. He is irritable, his abdomen is mildly distended and tender, and a small amount of dark stool is in the diaper. Heart rate is 190 per minute and capillary refill is 4 seconds. After placing an intravenous line and giving an isotonic bolus, which is the best next step? [Management and Treatment]

A. Start a proton pump inhibitor and observe for 24 hours at home

B. Obtain abdominal ultrasonography to measure the pyloric muscle thickness

C. Obtain an urgent upper gastrointestinal contrast study with surgical consultation

D. Change to a protein hydrolysate formula, add a thickener, and recheck weight in one week at the clinic

E. Send stool for culture and admit for supervised oral rehydration

Answer Contents

BQ-357. A 10-month-old boy has had three episodes of sudden screaming in the past 4 hours, with his knees drawn up. Between spells he is pale and limp. He vomited once, and his abdomen is soft with a vague mass in the right upper quadrant. He has had no fever. Which is the best next step? [Diagnosis]

A. Abdominal ultrasonography

B. Barium enema as the first study before any other imaging

C. Plain radiographs of the abdomen only, with discharge if normal

D. Oral antibiotics for presumed gastroenteritis with a recheck in two days

E. Upper gastrointestinal contrast study with small-bowel follow-through

Answer Contents

BQ-358. A term newborn girl has not passed meconium by 56 hours of life. Her abdomen is distended, and she vomited green fluid once. On rectal examination, the finger is gripped by a tight, empty rectum, and a gush of foul-smelling stool and gas follows its withdrawal. Her sacrum looks normal. Which is the most likely diagnosis? [Diagnosis]

A. Functional constipation

B. Meconium plug syndrome from maternal diabetes

C. Congenital hypothyroidism

D. Anorectal malformation with a rectoperineal fistula

E. Hirschsprung disease

Answer Contents

BQ-359. A 3-month-old girl is brought by her father because she spits up after most feeds. She is content, feeds eagerly, and has gained 900 g in the past 4 weeks. The spit-up is effortless, milky, and never green or bloody. Her examination, including the abdomen, is normal. Which is the most likely explanation for her symptoms? [Physiology and Pathophysiology]

A. Hypertrophy of the pyloric muscle narrowing the gastric outlet

B. Eosinophilic inflammation of the esophageal mucosa triggered by food allergy and ongoing exposure

C. Transient relaxation of an immature lower esophageal sphincter with large liquid feeds

D. Intermittent partial obstruction from intestinal malrotation

E. Delayed gastric emptying from an underactive thyroid gland

Answer Contents

BQ-360. A 7-week-old boy is fussy during and after feeds, arches his back, and has gained only 200 g in 3 weeks. Symptoms began 2 weeks after a cow's milk-based formula was started. Vomiting is nonbilious and not forceful, and stools have occasional streaks of mucus. Examination is otherwise normal. Which is the most likely diagnosis? [Diagnosis]

A. Cow's milk protein allergy

B. Pathologic acid reflux from sphincter dysfunction

C. Hypertrophic pyloric stenosis

D. Infantile colic

E. Congenital lactase deficiency

Answer Contents

BQ-361. The parents of a healthy 10-day-old firstborn boy are feeding him well and he has no vomiting. A relative's infant needed surgery for pyloric stenosis, and the parents ask what in the first two weeks of life would raise their son's own risk. He has a small chance of contact with a coughing adult. Which exposure is of greatest concern? [Epidemiology and Risk Assessment]

A. A single dose of intramuscular vitamin K at birth

B. Oral erythromycin given for pertussis prophylaxis

C. Phototherapy for physiologic jaundice

D. Exclusive breastfeeding on demand

E. Hepatitis B vaccine at birth

Answer Contents

BQ-362. A 3-year-old girl weighing 14 kg has had 36 hours of watery diarrhea and two episodes of vomiting. She is thirsty and irritable, with dry lips, fewer tears, and normal capillary refill. She drank a few sips without vomiting in clinic. Heart rate and blood pressure are normal. Which is the most appropriate management? [Management and Treatment]

A. Oral rehydration solution in small, frequent amounts, with replacement of ongoing losses

B. Intravenous isotonic bolus followed by admission for maintenance fluids

C. Apple juice diluted by half, given freely until the diarrhea stops and her thirst has resolved

D. Nasogastric feeding of standard formula at full strength overnight

E. Bowel rest with clear liquids only for 24 hours before any solid food

Answer Contents

BQ-363. A 2-year-old boy has had 3 days of vomiting and diarrhea. He is lethargic and cool, with mottled skin, deep rapid breathing, and a capillary refill of 5 seconds. Heart rate is 190 per minute and blood pressure is low for age. Bedside glucose is normal. Intravenous access is in place. Which is the best next step? [Management and Treatment]

A. Oral rehydration solution by nasogastric tube over four hours

B. Single dose of ondansetron followed by sips of oral rehydration solution

C. Rapid isotonic crystalloid bolus with reassessment after each bolus

D. Hypotonic maintenance fluid at twice the usual rate

E. Stool culture and empiric antibiotic therapy before any fluid

Answer Contents

BQ-364. A 4-year-old girl has had 2 days of bloody diarrhea with severe cramps and a low-grade fever of 38.0 C (100.4 F). She is alert, drinking, and urinating. A stool Shiga toxin assay is pending. A resident suggests starting an oral antibiotic. Which is the most appropriate management? [Management and Treatment]

A. Start oral azithromycin now to shorten the illness

B. Supportive care with fluids, withholding antibiotics and antimotility drugs

C. Give loperamide to reduce stool frequency and cramps

D. Give bismuth subsalicylate to coat the inflamed bowel

E. Start oral ciprofloxacin now and discontinue it when the cultures are negative

Answer Contents

BQ-365. A 3-year-old boy has become pale and irritable 6 days after an episode of bloody diarrhea. He has passed little urine for a day and has puffy eyelids. Hemoglobin is 7.1 g/dL, platelet count is 48,000 per mm3, and creatinine is 1.8 mg/dL. The blood smear shows fragmented red cells. Which is the most likely mechanism? [Physiology and Pathophysiology]

A. Autoantibody destruction of platelets and red cells

B. Consumption of clotting factors from sepsis

C. Immune complex deposition in the glomerulus after a streptococcal infection

D. Bone marrow suppression from a viral illness

E. Toxin-mediated endothelial injury causing microvascular thrombosis

Answer Contents

BQ-366. A 4-year-old girl has had hard stools for 5 months since starting preschool, where she avoids the bathroom. She now has liquid stool in her underwear several times a day and seems unaware of it. Her mother is worried about diarrhea. Examination shows a palpable stool mass in the lower abdomen and a full rectum. Which is the most likely mechanism of the soiling? [Physiology and Pathophysiology]

A. Liquid stool leaking around an impacted rectal mass

B. Rapid colonic transit from a viral infection

C. Malabsorption of fat from small bowel injury

D. Aganglionic segment with a failed relaxation reflex

E. Intentional withholding in defiance of toilet training

Answer Contents

BQ-367. A 2-year-old boy has passed a large amount of dark red stool twice today without pain. He is pale but alert, and his abdomen is soft and nontender. Heart rate is 140 per minute with normal blood pressure. Hemoglobin is 8.0 g/dL. He has had no fever, diarrhea, or vomiting. Which is the most likely diagnosis? [Diagnosis]

A. Juvenile polyp

B. Anal fissure

C. Intussusception

D. Meckel diverticulum

E. Infectious bacterial colitis

Answer Contents

BQ-368. A 4-year-old boy has had three episodes of small amounts of bright red blood on the stool over 2 weeks. Stools are soft and painless to pass, and blood is mixed on the surface with some mucus. He is thriving, with normal hemoglobin. The perianal skin is intact, and abdominal examination is normal. Which is the most likely diagnosis? [Diagnosis]

A. Meckel diverticulum

B. Juvenile polyp

C. Anal fissure

D. Crohn disease

E. Intussusception

Answer Contents

BQ-369. A 6-week-old boy, exclusively breastfed, has had streaks of blood and mucus in about half of his stools for 10 days. He feeds well, gains weight appropriately, and is happy. His abdomen is soft, his perianal skin is intact, and hemoglobin is normal. His mother drinks milk daily. Which is the most likely diagnosis? [Diagnosis]

A. Anal fissure

B. Bacterial colitis from an enteric pathogen

C. Intussusception

D. Meckel diverticulum

E. Food protein-induced proctocolitis

Answer Contents

BQ-370. An 18-month-old girl cries during stooling and has had bright red streaks on the surface of formed stool for 3 weeks. Her mother says she holds back stool. The abdomen is soft, and growth is normal. Examination shows a small tear at the midline posterior anal margin. Which is the best next step? [Management and Treatment]

A. Daily polyethylene glycol to soften stool, with a barrier ointment

B. Topical hydrocortisone cream applied to the anal margin for 3 months

C. Colonoscopy to exclude inflammatory bowel disease

D. Rectal suction biopsy to exclude Hirschsprung disease

E. Stool cultures and a course of oral antibiotics

Answer Contents

BQ-371. A 4-year-old girl has had poor weight gain, bloating, and loose stools for 6 months while eating wheat. Her total serum IgA level is undetectable, and tissue transglutaminase IgA is negative. Her older brother has celiac disease. Which is the best next step? [Diagnosis]

A. Accept the negative result and begin a gluten-free diet

B. Measure deamidated gliadin IgA as the confirmatory test and then decide on diet

C. Measure an IgG-based celiac antibody, such as tissue transglutaminase IgG

D. Check HLA typing alone to exclude celiac disease

E. Repeat tissue transglutaminase IgA in one year

Answer Contents

BQ-372. A 2-year-old boy is brought for his well visit. His parents are worried about his eating. He eats a limited variety, refuses some foods on first offer, and takes 5 to 6 foods consistently. He is active, his weight and length follow his growth curve, and his examination is normal. Which is the most likely diagnosis? [Diagnosis]

A. Avoidant restrictive food intake disorder

B. Normal selective eating of toddlerhood

C. Oral-motor feeding disorder

D. Celiac disease

E. Zinc deficiency

Answer Contents

BQ-373. An 8-year-old girl has had abdominal pain and iron deficiency anemia for a year. She eats wheat daily. Total IgA is normal and tissue transglutaminase IgA is 4 times the upper limit of normal. The family agrees to further testing. Which is the best next step? [Management and Treatment]

A. Starting a gluten-free diet now and repeating serology in six months

B. Colonoscopy with ileal intubation and random colonic biopsies

C. Upper endoscopy with multiple duodenal and bulb biopsies

D. Diagnosis of celiac disease without further testing

E. Capsule endoscopy before any biopsies

Answer Contents

BQ-374. A 13-year-old boy has had 5 months of intermittent abdominal pain, loose stools, and a 4 kg weight loss. He has grown only 2 cm in the past year. He has perianal skin tags and a draining fistula. Calprotectin is elevated and C-reactive protein is high. Which is the most likely diagnosis? [Diagnosis]

A. Ulcerative colitis

B. Crohn disease

C. Celiac disease

D. Functional abdominal pain

E. Infectious bacterial colitis

Answer Contents

BQ-375. A 10-year-old girl has had 3 weeks of bloody diarrhea with urgency and nocturnal stools. She has lost 2 kg. Stool cultures, Clostridioides difficile testing, and parasite studies are negative. Hemoglobin is 10 g/dL and fecal calprotectin is very high. Which is the most likely diagnosis? [Diagnosis]

A. Crohn disease with small-bowel involvement

B. Infectious colitis from Shiga toxin-producing Escherichia coli

C. Ulcerative colitis

D. Celiac disease

E. Eosinophilic gastroenteritis

Answer Contents

BQ-376. A newborn girl is delivered at term to a mother who tested positive for hepatitis B surface antigen during pregnancy. The infant weighs 3.2 kg and is well. Which is the best management in the first 12 hours of life? [Management and Treatment]

A. Hepatitis B vaccine and hepatitis B immune globulin

B. Hepatitis B vaccine alone, deferred until two months of age

C. Hepatitis B immune globulin alone, with vaccine at one month

D. Observation with a hepatitis B surface antigen check at six months

E. Oral antiviral therapy for the infant for six weeks

Answer Contents

BQ-377. A child care center reports that three children aged 2 to 4 years have developed vomiting and dark urine, and one was jaundiced. A 4-year-old boy in the room has mild fever and nausea. His mother asks how the illness spreads and what can protect the other children. Which is the most likely route of transmission? [Epidemiology and Risk Assessment]

A. Respiratory droplets from coughing and sneezing

B. Blood exposure from shared toys that have been contaminated by cuts on the hands

C. Vertical transmission during birth

D. Fecal-oral spread through contaminated hands, food, or objects

E. Bites from mosquitoes during outdoor play

Answer Contents

BQ-378. A 3-week-old boy, exclusively breastfed, has had yellow skin since the first week. He feeds eight times daily, has yellow seedy stools, and has regained his birth weight. Total bilirubin is 9.0 mg/dL with a direct bilirubin of 0.4 mg/dL. Examination is normal, and urine is pale. Which is the most likely diagnosis? [Diagnosis]

A. Biliary atresia

B. Neonatal hepatitis from a congenital infection

C. Choledochal cyst

D. Galactosemia

E. Breast milk jaundice

Answer Contents

BQ-379. A 9-week-old girl has biliary atresia confirmed on cholangiography. Her parents ask why surgery cannot wait until she is a few months older, when she would be stronger. She has pale stool, mild hepatomegaly, and a direct bilirubin of 5.5 mg/dL. Which is the best explanation for the urgency of the operation? [Physiology and Pathophysiology]

A. Progressive duct obliteration scars the liver, and success falls after about 60 days

B. Bile acids damage the intestine and cause perforation within weeks

C. The hepatic artery thromboses after the second month of life

D. Persistent conjugated bilirubin causes kernicterus unless the ducts are drained early

E. The gallbladder ruptures if the operation is delayed

Answer Contents

BQ-380. A 3-year-old boy has had 3 days of fever, vomiting, and refusal to walk. He is ill-appearing, with a temperature of 39.2 C (102.6 F), a distended abdomen, and rigidity with guarding. Heart rate is 160 per minute. White blood cell count is 22,000 per mm3. Ultrasonography shows a noncompressible tubular structure with free fluid. Which is the best next step? [Management and Treatment]

A. Discharge with oral analgesia and a recheck in the clinic in 48 hours if pain persists

B. Oral antibiotics alone with outpatient review

C. Barium enema to look for appendiceal filling

D. Laxatives for presumed constipation with abdominal distension

E. Intravenous fluids and antibiotics with urgent surgical consultation

Answer Contents

BQ-381. A 14-year-old girl has had sudden severe right lower abdominal pain for 6 hours with nausea and one episode of vomiting. Her last menstrual period was 7 weeks ago. She is sexually active and uses no contraception. She is afebrile, with right lower quadrant tenderness. Which is the best next step? [Diagnosis]

A. Abdominal CT with contrast before any laboratory test

B. Urine pregnancy test

C. Pelvic ultrasonography without laboratory studies

D. Observation at home with oral analgesia for 24 hours

E. Diagnostic laparoscopy immediately without any imaging

Answer Contents

BQ-382. A 9-year-old girl has had episodes of periumbilical abdominal pain for 5 months, lasting under an hour, usually on school mornings. She does not wake at night with pain and has no vomiting, fever, weight loss, or blood in the stool. Her growth is normal and examination is unremarkable. Which is the best next step? [Diagnosis]

A. Upper endoscopy with biopsies

B. Abdominal CT with contrast

C. Empiric trial of a proton pump inhibitor for 8 weeks

D. Reassurance with a pain diary and no further testing

E. Celiac, inflammatory, and infectious stool laboratory panel

Answer Contents

BQ-383. A 20-month-old boy had a 3-minute generalized seizure with a temperature of 39.8 C (103.6 F) due to roseola. He recovered fully within 30 minutes and has normal development. His father had febrile seizures as a child. His parents ask how likely it is that another febrile seizure will occur. Which is the most accurate estimate? [Epidemiology and Risk Assessment]

A. Nearly every child has another febrile seizure within a year

B. About one in twenty children has another febrile seizure

C. Recurrence only occurs if the fever is not treated with an antipyretic

D. Recurrence is certain if the family history is positive

E. About one in three children has another febrile seizure

Answer Contents

BQ-384. A 2-year-old girl had a seizure with a temperature of 39.5 C (103.1 F). Her mother saw the right arm and face twitching for about 25 minutes before it stopped. She was sleepy for 2 hours afterward. She has had no prior seizures. She now has a runny nose and her neck is supple. Which is the most likely diagnosis? [Diagnosis]

A. Complex febrile seizure

B. Simple febrile seizure

C. Childhood absence epilepsy

D. Breath-holding spell

E. Infantile epileptic spasms

Answer Contents

BQ-385. An 8-month-old boy had a generalized seizure lasting 3 minutes with a temperature of 39.4 C (102.9 F). Because of a family move, he has not received any Hib or pneumococcal vaccines. He is now drowsy but arousable, with no obvious source of fever. His fontanelle is flat and his neck is supple. Which is the best next step? [Management and Treatment]

A. Obtain an electroencephalogram

B. Obtain a brain MRI under sedation

C. Perform a lumbar puncture

D. Start daily phenobarbital for prevention

E. Send the family home with antipyretic advice

Answer Contents

BQ-386. A 6-year-old boy weighing 20 kg has had a generalized tonic-clonic seizure for 6 minutes in the emergency department. He is cyanotic and unresponsive. Two attempts at intravenous access have failed, and a glucose check is normal. Which is the best next step? [Management and Treatment]

A. Give a loading dose of phenytoin by the intraosseous route first

B. Wait for intravenous access before giving any medication

C. Give oral levetiracetam by mouth with sips of water

D. Give intramuscular haloperidol to settle the movements

E. Give intramuscular midazolam

Answer Contents

BQ-387. A 6-month-old girl was developing normally, but over 3 weeks she has had clusters of sudden flexion of the trunk and arms, mostly after waking, up to 10 times in a row. She has stopped rolling over and smiling less. Her mother thought it was colic. EEG shows a chaotic, high-amplitude background. Which is the most likely diagnosis? [Diagnosis]

A. Infantile epileptic spasms

B. Benign sleep myoclonus of infancy

C. Acid reflux with Sandifer posturing

D. Childhood absence epilepsy

E. Febrile seizures

Answer Contents

BQ-388. A 5-month-old boy was developing normally until 3 weeks ago, when he lost his social smile and began having clusters of brief extensor spasms. EEG is hypsarrhythmic. Examination shows four hypopigmented ash leaf macules under a Wood lamp. Echocardiography shows a cardiac rhabdomyoma. Which is the best initial medication? [Management and Treatment]

A. Carbamazepine

B. Vigabatrin

C. Ethosuximide

D. Phenytoin

E. Valproate

Answer Contents

BQ-389. An 8-year-old boy is seen after two episodes of waking at night with drooling, gurgling, and twitching of the right side of his face. He could not speak during the events and was aware. Development and examination are normal. EEG shows centrotemporal spikes during sleep. Which is the most likely diagnosis? [Diagnosis]

A. Juvenile myoclonic epilepsy

B. Panayiotopoulos syndrome with autonomic seizures

C. Night terrors

D. Self-limited epilepsy with centrotemporal spikes

E. Childhood absence epilepsy

Answer Contents

BQ-390. A 15-year-old girl has had jerks of her arms in the morning, spilling her breakfast, and one generalized tonic-clonic seizure after a late night. EEG shows 4 to 6 Hz generalized polyspike and wave. A resident proposes valproate because it is most effective. She is sexually active and does not use contraception. Which counseling point is most important? [Epidemiology and Risk Assessment]

A. Valproate has no effect on hormonal contraceptives or pregnancy outcomes in girls

B. Valproate carries a major risk of birth defects, so alternatives are preferred

C. Valproate is safe in pregnancy if taken with food

D. Valproate causes absence seizures to worsen in girls

E. Valproate must be stopped at puberty

Answer Contents

BQ-391. A 9-year-old girl with childhood absence epilepsy has more frequent staring spells after her primary care clinician started carbamazepine 6 weeks ago. EEG now shows continuous 3 Hz generalized spike and wave. Her examination is unchanged. Which is the most likely explanation for the worsening? [Physiology and Pathophysiology]

A. She has developed a second, focal epilepsy that the new drug has unmasked in the temporal lobe

B. The drug dose is too low for her weight

C. She has developed a drug-induced psychosis

D. The seizures are psychogenic

E. The drug aggravates generalized spike-wave discharges in this syndrome

Answer Contents

BQ-392. A 10-year-old boy had his first generalized tonic-clonic seizure while awake, lasting 2 minutes, with a postictal period of 30 minutes. Glucose and electrolytes are normal. He has no fever or head injury, and his examination and development are normal. His parents ask what testing is reasonable. Which is the best next step? [Diagnosis]

A. Electroencephalogram

B. Lumbar puncture

C. Daily antiseizure medication started at once

D. Brain MRI under general anesthesia only

E. Repeat observation with no further testing ever

Answer Contents

BQ-393. A 13-year-old girl has had episodes of throbbing unilateral headache with nausea and light sensitivity that last 4 to 6 hours, about twice a month. Examination, including fundi, is normal. Rest in a dark room helps a little. Her mother asks which medicine works best for an attack. Which is the best recommendation? [Management and Treatment]

A. Daily amitriptyline for prevention

B. Oral opioid analgesic for each attack

C. Routine brain MRI before any treatment

D. Ibuprofen taken at the start of the headache

E. Caffeine tablets taken when the headache peaks

Answer Contents

BQ-394. A 12-year-old girl has had 1 day of fever, severe headache, vomiting, and light sensitivity. Her temperature is 39.4 C (102.9 F), and she resists neck flexion. She is drowsy but arousable. Blood pressure is normal. Capillary refill is 2 seconds. Which is the best next step? [Management and Treatment]

A. Blood cultures and prompt intravenous antibiotics, with lumbar puncture when safe

B. Outpatient NSAIDs and a headache diary

C. Brain MRI first, with antibiotics started only after the scan result has been reviewed

D. Oral sumatriptan for presumed migraine

E. Reassurance and a recheck in 48 hours

Answer Contents

BQ-395. A 14-year-old girl has had a dull daily headache for 4 months. She takes ibuprofen or acetaminophen five days a week and has missed school. Her neurologic examination, including fundi, is normal, and the headache is not worse in the morning or when lying down. Which is the most likely diagnosis? [Diagnosis]

A. Idiopathic intracranial hypertension

B. Medication-overuse headache

C. Posterior fossa tumor

D. Chronic subdural hematoma

E. Sinusitis

Answer Contents

BQ-396. A 13-year-old boy collides heads with a teammate at soccer and has had headache, fogginess, and light sensitivity for 2 days. He did not lose consciousness. Head CT in the emergency department showed no abnormality. His parents ask why he feels unwell if the scan is normal. Which is the best explanation? [Physiology and Pathophysiology]

A. A small bleed that is too small to be seen on any scan

B. A permanent loss of neurons that the scan is too insensitive to show at this stage

C. A temporary disturbance of brain function without visible structural injury

D. A skull fracture that will appear on later films

E. A purely psychological reaction to the collision

Answer Contents

BQ-397. A 6-year-old boy has had 3 days of weakness and unsteadiness after a diarrheal illness. He now struggles to stand and has absent knee and ankle reflexes with symmetric leg weakness. Sensation is intact, and his breathing is comfortable. His pulse varies widely between measurements. Which is the best next step? [Management and Treatment]

A. Discharge with physical therapy follow-up

B. Admit for monitoring of respiratory and autonomic function and start immunotherapy

C. Start high-dose oral corticosteroids as an outpatient with weekly neurologic review

D. Obtain an outpatient EMG within a month

E. Observe at home until he cannot walk

Answer Contents

BQ-398. A 4-year-old boy has been slow to walk and now has difficulty climbing stairs. He pushes on his thighs to stand from the floor, and his calves look enlarged. Creatine kinase is 18,000 U/L, many times the upper limit of normal. Which is the most likely mechanism of the elevated creatine kinase? [Physiology and Pathophysiology]

A. Autoimmune destruction of the postsynaptic receptors at the neuromuscular junction

B. Degeneration of anterior horn cells

C. Demyelination of peripheral nerves

D. Leakage of enzyme through fragile muscle membranes that lack dystrophin

E. Failure of acetylcholine release from nerve terminals

Answer Contents

BQ-399. A 3-month-old boy is constipated for 4 days and now has a weak cry, poor suck, drooping eyelids, and decreased movement that began in the face and has moved to the limbs. He is afebrile and alert. His mother recently stopped formula and introduced a home-prepared cereal. Which is the most likely mechanism of the weakness? [Physiology and Pathophysiology]

A. Autoantibodies against acetylcholine receptors

B. Loss of motor neurons from an SMN1 deletion

C. A toxin that blocks acetylcholine release at the neuromuscular junction

D. Absence of dystrophin in muscle membranes

E. Immune-mediated demyelination of peripheral nerves after a recent viral infection

Answer Contents

BQ-400. A 4-month-old boy was born at 28 weeks and had a periventricular hemorrhagic injury. He is alert but his legs are stiff and cross when lifted. MRI shows periventricular white matter injury, and the General Movements Assessment finds absent fidgety movements. His parents are told to wait for a missed walking milestone. Which is the most appropriate next step? [Epidemiology and Risk Assessment]

A. Wait until 12 months before any referral

B. Refer now for early intervention and therapy while confirming the diagnosis

C. Reassure that preterm infants always catch up

D. Start botulinum toxin injections now

E. Order repeat MRI every month until she is walking and then reassess her tone

Answer Contents

BQ-401. A newborn has an open lumbosacral myelomeningocele with a sac covered by thin membrane. He is lying prone with a sterile moist dressing over the lesion. The nurse asks about precautions to take when handling him during the many procedures that lie ahead. Which is the most appropriate precaution? [Epidemiology and Risk Assessment]

A. Place him supine on a flat bed

B. Apply a tight compression dressing

C. Delay all surgical consultation for several weeks

D. Use latex-free gloves and equipment

E. Give routine folic acid to the infant

Answer Contents

BQ-402. An 18-month-old girl attends a well visit. Her parents report that she points rarely, does not respond to her name consistently, and says only a few words. She walks well and has normal hearing screening results. The clinician has no screening tool in the room. Which is the most appropriate step? [Epidemiology and Risk Assessment]

A. Wait until age 4 before deciding whether screening is needed

B. Reassure that late talkers are never autistic

C. Administer a validated autism screening tool today

D. Order a brain MRI

E. Start a gluten-free diet

Answer Contents

BQ-403. A 2-year-old boy says about five words and does not combine them. He follows simple commands, plays with other children, and points to things. His birth history is normal. He has had several ear infections. His mother asks what to do first. Which is the best next step? [Management and Treatment]

A. Wait for the child to outgrow the delay by age 4

B. Order a brain MRI

C. Order fragile X testing before hearing assessment

D. Start speech therapy only after a psychiatric evaluation

E. Formal audiologic evaluation

Answer Contents

BQ-404. A 4-year-old girl with a ventriculoperitoneal shunt placed for congenital hydrocephalus has had 2 days of headache, vomiting, and increasing sleepiness. Her temperature is 37.2 C (99.0 F). Her eyes show downward gaze limitation, and the shunt site is not red. Which is the most likely diagnosis? [Diagnosis]

A. Viral gastroenteritis

B. Shunt malfunction

C. Migraine

D. Shunt infection

E. Medulloblastoma

Answer Contents

Domain 15. Skin/Dermatology

Contents Answers for this subject

BQ-405. A 10-month-old boy has had itchy, dry, red patches on his trunk and limbs since 3 months of age. His mother applies moisturizer daily but avoids the prescribed steroid ointment because she fears thin skin and stunted growth. He sleeps poorly, and the skin is cracked and weeping in places. There is no fever. Which is the best counseling? [Management and Treatment]

A. Stop moisturizers during flares to avoid irritation

B. Use a high-potency steroid on all of his skin daily for several months at a time

C. Avoid all steroids and use only fragrance-free soaps

D. Replace baths with daily scrubbing of the affected skin

E. Apply a mid-potency steroid to red patches until smooth and moisturize daily

Answer Contents

BQ-406. A 5-year-old girl with long-standing eczema has a 24-hour history of fever of 39.0 C (102.2 F) and sudden worsening of the rash on her face and neck. Examination shows clusters of uniform punched-out vesicles and crusted erosions, with some coalescing. She looks unwell and is eating poorly. Which is the best next step? [Management and Treatment]

A. Start antiviral therapy for herpes simplex today and send a swab for PCR

B. Increase the topical steroid potency on the face and review the rash in one week

C. Prescribe oral cephalexin alone for presumed impetigo

D. Apply a topical antifungal and recheck in 3 days

E. Wait for the PCR result before starting any medicine

Answer Contents

BQ-407. A 3-day-old term girl has developed scattered red blotches with small yellow-white pustules in their centers on her trunk and limbs. She feeds well, is afebrile, and is active. The palms and soles are spared. The blotches appear in different places each day. Which is the most likely diagnosis? [Diagnosis]

A. Neonatal herpes simplex infection

B. Erythema toxicum neonatorum

C. Transient neonatal pustular melanosis

D. Staphylococcal scalded skin syndrome

E. Miliaria crystallina

Answer Contents

BQ-408. A 10-day-old term boy has a cluster of vesicles on a red base on his scalp. For 12 hours he has fed poorly and been lethargic. His temperature is 38.2 C (100.8 F). His mother had a cold sore 2 weeks before delivery. Which is the best next step? [Management and Treatment]

A. Apply topical antiviral cream to the scalp lesions and review in the clinic in three days

B. Prescribe oral antibiotics for impetigo and discharge

C. Reassure that this is erythema toxicum

D. Await culture results before starting any medicine

E. Admit for sepsis evaluation, send herpes PCR, and begin intravenous acyclovir

Answer Contents

BQ-409. A 4-week-old boy has thick, greasy, yellow scale on the scalp and salmon-colored patches behind the ears and in the skin folds. He is feeding well and is not scratching or fussy. His weight gain is normal. Which is the most likely diagnosis? [Diagnosis]

A. Atopic dermatitis of infancy

B. Tinea capitis

C. Seborrheic dermatitis

D. Psoriasis

E. Scabies

Answer Contents

BQ-410. A 3-year-old boy has several flaccid, thin-walled blisters on his abdomen and buttocks, some ruptured and leaving a shiny red base with a thin brown crust. He is afebrile and playing well. Another child at his nursery had similar lesions. Which is the most likely cause of the blisters? [Physiology and Pathophysiology]

A. Antibodies against desmosomes in the epidermis

B. Herpes simplex virus replication in keratinocytes

C. A drug reaction with cytotoxic T cell injury

D. An exfoliative toxin from Staphylococcus aureus

E. Mite burrowing in the stratum corneum

Answer Contents

BQ-411. A 6-year-old girl was treated with permethrin cream for scabies two weeks ago, along with her family, and her bedding was washed. Her itch is still present, though no new burrows have appeared and no household member has symptoms. She is otherwise well. Which is the best explanation for her ongoing itch? [Physiology and Pathophysiology]

A. Failure of permethrin because the mites are resistant to all topical therapy

B. Secondary infection with group A streptococcus

C. A delayed allergic reaction to mite antigens that lingers after treatment

D. Reinfestation from a pet cat

E. Contact dermatitis from the bedding detergent

Answer Contents

BQ-412. A 5-year-old boy has about 20 small, firm, pearly, dome-shaped papules with a central dimple on his trunk and axillae for 4 months. He is otherwise healthy and not bothered by them. His mother asks if he must stay out of school and whether treatment is required. Which is the best management? [Management and Treatment]

A. Oral antiviral therapy for 2 weeks and removal from school until lesions clear

B. Systemic antibiotics for 10 days

C. Surgical excision of all lesions

D. Observation, because the lesions resolve on their own, with hygiene advice

E. Hospital admission for isolation

Answer Contents

BQ-413. A 7-year-old boy has a boggy, tender, 4 cm scalp mass with pustules and broken hairs. There is hair loss over the area and enlarged posterior cervical nodes. He has been afebrile. A course of oral cephalexin for 1 week did not help. Which is the most likely diagnosis? [Diagnosis]

A. Bacterial scalp abscess with cellulitis

B. Kerion from tinea capitis

C. Alopecia areata

D. Seborrheic dermatitis

E. Cutaneous leishmaniasis

Answer Contents

BQ-414. A 4-year-old girl has had comedones and papules on her cheeks and forehead for 3 months. She also has axillary odor and pubic hair. Her height is over the 97th percentile and her growth velocity is increased. Her breast tissue is not developed. Which is the most appropriate next step? [Epidemiology and Risk Assessment]

A. Reassurance that this is a normal variant of early childhood skin and needs no testing

B. Oral isotretinoin

C. A trial of oral doxycycline for acne

D. Daily scrubbing with an abrasive cleanser

E. Evaluation for androgen excess, including bone age and hormone testing

Answer Contents

BQ-415. A 16-year-old boy has painful nodules and cysts on the face and back with scarring. He has used a topical retinoid with benzoyl peroxide for 4 months and oral doxycycline for 4 months, with little improvement. He feels low and avoids school photographs. Which is the best next step? [Management and Treatment]

A. Increase the topical benzoyl peroxide concentration alone and continue for 6 months

B. Add a second oral antibiotic for another 6 months

C. Switch to daily scrubbing with exfoliating cleansers

D. Oral isotretinoin through a registered risk-management program

E. Continue current therapy for another year

Answer Contents

BQ-416. A 2-month-old girl has a bright red, raised 2 cm lesion on her thigh that was a faint pale patch at birth. It has doubled in size over the past 3 weeks. She is otherwise well and the lesion is not ulcerated. Her parents ask what will happen to it. Which is the most accurate explanation of its natural course? [Physiology and Pathophysiology]

A. Rapid growth until about 5 months of age, then slow shrinking over years

B. Steady growth throughout early childhood with no shrinking until adolescence

C. Immediate disappearance within one month

D. Permanent fixed size and color for life

E. Rapid malignant change in adolescence

Answer Contents

BQ-417. A 3-month-old boy has a flat pink-purple birthmark covering the left forehead and upper eyelid, present since birth. It has not changed in color or size, apart from growing with him. His parents ask which complications to watch for. Which pair of problems should be screened for? [Epidemiology and Risk Assessment]

A. Airway obstruction and heart failure

B. Hypothyroidism and liver hemangiomas

C. Deafness and kidney stones

D. Scoliosis and cafe-au-lait macules

E. Glaucoma and seizures

Answer Contents

BQ-418. A 6-year-old girl has seven tan flat patches larger than 5 mm on her trunk and legs, and freckling in both axillae. Her height is on the 40th percentile and development is normal. Her father has similar spots and learning difficulties. Which is the most likely diagnosis? [Diagnosis]

A. McCune-Albright syndrome

B. Tuberous sclerosis complex

C. Sturge-Weber syndrome

D. Neurofibromatosis type 1

E. Normal pigmentation variant

Answer Contents

BQ-419. A 2-day-old Black girl has a flat, blue-gray, ill-defined 8 cm patch over her lower back and buttocks. It is not tender, and the parents fear it is a bruise from the delivery. Her examination is otherwise normal. Which is the best explanation for the patch? [Physiology and Pathophysiology]

A. Bleeding into the skin from birth trauma that will turn green and yellow in a week

B. A sign of a coagulation disorder

C. A pigmented tumor that will grow

D. A bacterial infection of the skin

E. Dermal melanocytes that stopped migrating on their way to the epidermis

Answer Contents

BQ-420. A 3-month-old girl has seven small red hemangiomas on her face, trunk, and arms. She is feeding and gaining weight well, and her heart rate and breathing are normal. The liver edge is palpable 3 cm past the costal margin. Which is the best next step? [Epidemiology and Risk Assessment]

A. Skin biopsy of the largest lesion to confirm the diagnosis

B. Abdominal ultrasonography to look for hepatic hemangiomas

C. Brain MRI with contrast

D. Observation alone with no further evaluation

E. Bone marrow aspiration

Answer Contents

BQ-421. A 3-month-old boy has a rapidly enlarging red, raised hemangioma on the left upper eyelid, partly covering the pupil when the eye is open. He closes the eye when light is shone in the room. He has no other lesions and is otherwise well. Which is the best next step? [Management and Treatment]

A. Reassurance and review at 12 months

B. Topical antibiotic ointment to the lesion

C. Urgent referral to ophthalmology and a hemangioma specialist

D. Surgical excision of the whole lesion at 5 years of age after growth stops

E. Compression bandage over the eye

Answer Contents

BQ-422. A 7-year-old boy has itchy, raised, red plaques on the trunk and arms that appeared 2 days after a cold. Individual lesions disappear within a few hours and new ones appear elsewhere. He has no lip swelling, wheeze, or vomiting. Which is the most likely mechanism of the individual lesions? [Physiology and Pathophysiology]

A. Cytotoxic T cell attack on keratinocytes

B. Bacterial toxin splitting the superficial epidermis

C. Immune complex deposition in small and medium-sized arteries of the skin and joints

D. Mast cell release of histamine with transient leakage from dermal vessels

E. Mite burrowing in the epidermis

Answer Contents

BQ-423. A 6-year-old girl develops itchy hives all over her body 10 minutes after eating a peanut cookie at a party. Then she has lip swelling, repeated vomiting, and wheezing. Her blood pressure is 80/45 mm Hg, which is low for her age. Which is the best next step? [Management and Treatment]

A. Intramuscular epinephrine

B. Oral cetirizine and observation at home

C. Oral prednisolone and a recheck in a week

D. Intravenous diphenhydramine alone

E. Inhaled albuterol alone

Answer Contents

BQ-424. A 15-year-old boy has had fever, sore throat, and tender neck nodes for 5 days. His clinician prescribed amoxicillin for presumed streptococcal pharyngitis. On day 3 he develops a diffuse pink rash of macules and papules. He has no hives, wheeze, or swelling. A monospot test is positive. Which is the most likely explanation for the rash? [Physiology and Pathophysiology]

A. True IgE-mediated penicillin allergy that will persist for the rest of his life

B. Stevens-Johnson syndrome

C. Scarlet fever from group A streptococcus

D. A non-allergic rash that often follows aminopenicillins in mononucleosis

E. Serum sickness from the first dose

Answer Contents

BQ-425. A 3-year-old girl is brought 10 days after starting cefaclor for otitis media. She has fever, swollen painful knees and ankles, and itchy, purple-centered plaques on her trunk. The mouth and eyes are normal. Urinalysis is normal. Which is the most likely diagnosis? [Diagnosis]

A. Stevens-Johnson syndrome

B. Kawasaki disease

C. Acute rheumatic fever

D. Henoch-Schonlein purpura with arthritis

E. Serum sickness-like reaction

Answer Contents

BQ-426. An 8-year-old girl has a round, smooth, hairless patch 3 cm across on her scalp that her mother noticed 2 weeks ago. There is no scale, redness, or itch, and short broken hairs taper toward the scalp at the edge of the patch. The skin surface is normal. Which is the most likely diagnosis? [Diagnosis]

A. Tinea capitis

B. Trichotillomania

C. Alopecia areata

D. Traction alopecia

E. Telogen effluvium

Answer Contents

BQ-427. A 7-year-old girl with sickle cell anemia has had 2 days of fever to 38.9 C (102.0 F), cough, and chest pain. She received an oral opioid for a pain episode yesterday. Her respiratory rate is 36 per minute, and her oxygen saturation is 88 percent on room air. A chest radiograph shows a new right lower lobe infiltrate. Which is the most appropriate management? [Management and Treatment]

A. Give oxygen, a cephalosporin plus a macrolide, and incentive spirometry, with early hematology input

B. Give oxygen and oral amoxicillin alone, then discharge her after four hours if she looks comfortable and alert

C. Give a large intravenous fluid bolus, withhold all opioids, and repeat the radiograph in 24 hours at home

D. Transfuse two units of red cells at once whatever her hemoglobin level, and defer all antibiotic therapy

E. Begin high-dose intravenous corticosteroids and hold antibiotics until sputum culture results are available

Answer Contents

BQ-428. A 14-month-old boy with sickle cell anemia is brought in after 2 hours of fussiness and sudden pallor. He is lethargic, with a heart rate of 190 per minute and a capillary refill of 4 seconds. A firm mass fills the left upper abdomen and was not present at his last visit. Hemoglobin is 3.9 g/dL (baseline 8.2), and the reticulocyte count is 18 percent. Which is the most likely diagnosis? [Diagnosis]

A. Aplastic crisis due to parvovirus B19 infection

B. Acute splenic sequestration crisis

C. Acute chest syndrome with hypoxemia

D. Hemolytic crisis from coexisting G6PD deficiency

E. Overwhelming pneumococcal sepsis with shock

Answer Contents

BQ-429. A 2-year-old girl with sickle cell anemia has had two episodes of painful hand swelling. Her hemoglobin is 8.0 g/dL and she is growing well. Her hematologist starts hydroxyurea, and her parents ask how it works. After 3 months of therapy, her laboratory results will show a higher mean corpuscular volume and a rise in one specific hemoglobin fraction. Which is the most likely mechanism by which the drug reduces vaso-occlusive events? [Physiology and Pathophysiology]

A. Replacement of hemoglobin S with hemoglobin A in mature red cells

B. Dissolution of sickled cells that are already lodged in small vessels

C. Restoration of splenic filtering of encapsulated bacteria

D. Increased fetal hemoglobin that inhibits hemoglobin S polymerization

E. Blockade of platelet aggregation in the small arterioles

Answer Contents

BQ-430. A 5-year-old boy with sickle cell anemia has no neurologic symptoms and a normal examination. At his annual transcranial Doppler screening, the time-averaged mean velocity in the middle cerebral artery is 215 cm per second. His hemoglobin is 7.9 g/dL, and he takes hydroxyurea and daily penicillin. His parents ask what should happen next. Which is the best next step? [Management and Treatment]

A. Obtain a screening head CT scan and start daily aspirin therapy for stroke prevention

B. Repeat the screening study in 12 months because he has no neurologic symptoms

C. Add a daily antiplatelet agent and continue annual screening at the usual interval

D. Give a single red cell transfusion today and discharge him without specialist input

E. Send him to hematology to consider chronic transfusion therapy for stroke prevention

Answer Contents

BQ-431. A 5-year-old boy develops dark urine and jaundice 2 days after starting nitrofurantoin for a urinary infection and eating fava beans. Hemoglobin is 6.4 g/dL, reticulocytes are 9 percent, and indirect bilirubin is elevated. The direct antiglobulin test is negative. His smear shows bite cells, and special staining shows Heinz bodies. A maternal uncle had similar episodes. Which is the most likely diagnosis? [Diagnosis]

A. Glucose-6-phosphate dehydrogenase deficiency

B. Hereditary spherocytosis with a hemolytic crisis

C. Warm autoimmune hemolytic anemia in a child

D. Paroxysmal nocturnal hemoglobinuria with hemolysis

E. Hemolytic uremic syndrome after an infection

Answer Contents

BQ-432. A 9-year-old boy with hereditary spherocytosis, whose hemoglobin is usually 10 g/dL, has pallor and fatigue after a sibling had a febrile illness with a facial rash. His hemoglobin today is 4.2 g/dL, and the reticulocyte count is 0.2 percent. His spleen is palpable at its usual size, and he has no jaundice beyond his baseline. Which is the most likely mechanism of this fall in hemoglobin? [Physiology and Pathophysiology]

A. Pooling of red cells in an acutely enlarged spleen

B. Autoantibody-mediated destruction of mature red cells

C. Oxidant injury to the red cell membrane from a new drug exposure

D. Viral infection of red cell precursors that halts production

E. Chronic blood loss from the gastrointestinal tract

Answer Contents

BQ-433. A healthy 2-year-old girl has a mean corpuscular volume of 62 fL, a normal ferritin, and an elevated hemoglobin A2 on electrophoresis. Her parents, who are both asymptomatic, have the same microcytosis and elevated hemoglobin A2, so each carries a beta globin gene mutation. They ask about a future pregnancy. What is the chance that a future child will have beta thalassemia major? [Epidemiology and Risk Assessment]

A. 0 percent

B. 25 percent

C. 50 percent

D. 75 percent

E. 100 percent

Answer Contents

BQ-434. An 11-month-old boy who recently began cruising has a warm, swollen right knee after a minor fall and refuses to bear weight. His parents report prolonged oozing after circumcision. The platelet count and prothrombin time are normal. The activated partial thromboplastin time is prolonged and corrects when his plasma is mixed with normal plasma. Which is the best next test? [Diagnosis]

A. Factor VIII and factor IX activity assays

B. Platelet function analysis with aggregation studies

C. Diagnostic arthrocentesis of the swollen right knee

D. Fibrinogen level and thrombin time measurements

E. Ristocetin cofactor activity as a single test

Answer Contents

BQ-435. A 13-year-old girl has had heavy periods since menarche at age 12, soaking a pad every hour for 8 days at a time. She has nosebleeds that last 15 minutes and bruises easily. Her mother has similar bleeding. Hemoglobin is 9.4 g/dL, and the platelet count, prothrombin time, and activated partial thromboplastin time are normal. Which is the best next step? [Diagnosis]

A. Reassure her and begin oral iron without further testing

B. Give a platelet transfusion before the next period to prevent heavy bleeding

C. Order a factor VIII assay and no other studies

D. Send von Willebrand factor antigen, activity, and a factor VIII level

E. Perform a bone marrow aspirate to evaluate platelet production

Answer Contents

BQ-436. A 2-year-old girl had a cold 10 days ago. For 2 hours she has had persistent bleeding from both nostrils despite firm pressure, and she has petechiae on her trunk. She is afebrile and alert, and the examination is otherwise normal. Platelets are 7,000 per mm3. The hemoglobin, white count, and smear are otherwise normal. Which is the most appropriate management? [Management and Treatment]

A. Observe her at home with bleeding precautions until the platelet count recovers on its own

B. Transfuse platelets at once and repeat the count in one hour

C. Apply local measures, give corticosteroids, and arrange hematology follow-up

D. Perform bone marrow aspiration before giving any drug treatment at all

E. Plan an elective splenectomy within the week to prevent further bleeding

Answer Contents

BQ-437. A 4-week-old exclusively breastfed boy born at home is brought in with vomiting, lethargy, pallor, and a bulging fontanelle. He received no injections after birth. The platelet count and fibrinogen are normal. The prothrombin time is markedly prolonged and the activated partial thromboplastin time is prolonged. Head ultrasonography shows an intracranial hemorrhage. Which is the most likely underlying mechanism? [Physiology and Pathophysiology]

A. Deficiency of vitamin K-dependent clotting factors

B. Consumption of clotting factors in disseminated intravascular coagulation

C. Congenital absence of factor VIII activity

D. Autoantibody destruction of circulating platelets

E. Deficient von Willebrand factor-mediated platelet adhesion

Answer Contents

BQ-438. A healthy 4-year-old boy of African ancestry has a preoperative blood count showing a white cell count of 4,300 per mm3 with 21 percent neutrophils and 2 percent bands. Hemoglobin and platelets are normal. He has had no unusual infections, mouth ulcers, or fevers, and his father has had similar counts. Which is the most likely explanation? [Epidemiology and Risk Assessment]

A. Cyclic neutropenia with a 21-day periodicity

B. Severe congenital neutropenia with marrow arrest

C. Benign ethnic neutropenia with normal marrow reserve

D. Autoimmune neutropenia with antineutrophil antibodies

E. Early acute leukemia with marrow replacement

Answer Contents

BQ-439. A 6-year-old boy with acute lymphoblastic leukemia is receiving a red cell transfusion. Twenty minutes into the unit he develops a temperature of 38.9 C (102.0 F), chills, back pain, and dark urine. His blood pressure falls from 100/60 to 80/45 mm Hg, and his heart rate rises to 140 per minute. Which is the most likely diagnosis? [Diagnosis]

A. Acute hemolytic transfusion reaction

B. Febrile nonhemolytic transfusion reaction

C. Allergic transfusion reaction with urticaria

D. Transfusion-associated circulatory overload

E. Transfusion-related acute lung injury

Answer Contents

BQ-440. A 2-year-old girl has had 3 weeks of irritability, low-grade fever, and bruising around both eyes. Examination shows a firm, nontender abdominal mass that crosses the midline, and she has rapid, chaotic eye movements with jerking of the limbs. Urine catecholamine metabolites are elevated. Which is the most likely diagnosis? [Diagnosis]

A. Wilms tumor

B. Acute lymphoblastic leukemia

C. Hepatoblastoma

D. Burkitt lymphoma

E. Neuroblastoma

Answer Contents

BQ-441. A 12-year-old boy has a rapidly enlarging abdominal mass, and biopsy shows Burkitt lymphoma. Before any chemotherapy, potassium is 6.3 mmol/L, phosphate is 8.2 mg/dL, calcium is 7.1 mg/dL, uric acid is 14 mg/dL, and creatinine is 1.6 mg/dL (baseline 0.6). His urine output is low. Which is the best next step? [Management and Treatment]

A. Start vigorous intravenous hydration without potassium or phosphate, check G6PD status, and give rasburicase

B. Restrict fluids to protect his kidneys and begin allopurinol alone, with repeat chemistry panels in the morning

C. Begin chemotherapy immediately and recheck the chemistry panel after the first cycle of treatment

D. Give intravenous calcium gluconate for the low calcium and hold all other treatment until morning

E. Arrange hemodialysis now before giving any fluid or drug therapy to the child

Answer Contents

BQ-442. An 18-month-old girl is thriving, but her parents notice that her right pupil looks white in flash photographs while the left looks red. Her right eye has begun to turn inward, and the red reflex is absent on that side. She was born at term. Which diagnosis must be excluded first? [Diagnosis]

A. Congenital cataract

B. Retinoblastoma

C. Persistent fetal vasculature

D. Ocular toxocariasis

E. Retinopathy of prematurity

Answer Contents

Domain 17. Allergy and Immunology

Contents Answers for this subject

BQ-443. A 12-year-old girl at school eats a cashew-containing snack bar. Within 10 minutes she has generalized hives and vomits twice. She is alert, her breathing is unlabored, and her blood pressure is normal. She has a known tree nut allergy, and the school nurse has an epinephrine autoinjector. Which is the best next step? [Management and Treatment]

A. Give an oral second-generation antihistamine and observe for one hour

B. Give intramuscular diphenhydramine and call her parents to come and take her home from school

C. Give intramuscular epinephrine into the anterolateral thigh and call emergency services

D. Give oral prednisone and have her rest in the nurse's office for four hours

E. Offer water and an antiemetic and recheck her skin after thirty minutes

Answer Contents

BQ-444. A 6-year-old girl weighing 22 kg is stung by a wasp at a picnic. Within minutes she has generalized hives, throat tightness, and vomiting. The clinic stocks epinephrine at 1 mg/mL, and her heart rate is 135 per minute. Which is the most appropriate intramuscular epinephrine dose? [Management and Treatment]

A. 0.022 mg

B. 0.22 mg

C. 0.5 mg

D. 2.2 mg

E. 0.15 mg

Answer Contents

BQ-445. The parents of a healthy 6-month-old boy ask when to give peanut. He has mild eczema that responds to moisturizer and a low-potency topical steroid, tolerates fruit, vegetables, and rice cereal, and has never reacted to a food. Neither parent has a food allergy. Which is the best next step? [Management and Treatment]

A. Delay peanut until after his first birthday

B. Give the first peanut dose in the clinic after a skin prick test

C. Order a food allergy panel before any peanut exposure at home

D. Wait until his eczema has completely cleared before starting any peanut at home or elsewhere

E. Introduce peanut-containing foods at home with other solids, without testing

Answer Contents

BQ-446. A 4-year-old girl eats scrambled egg several times a week without symptoms. A food panel ordered at a well visit shows an egg white-specific IgE of 3.2 kU/L. Her eczema is mild, and her examination is normal. Her parents ask whether she must stop eating egg. Which is the best interpretation of this result? [Physiology and Pathophysiology]

A. Silent egg allergy that requires an epinephrine autoinjector

B. Egg allergy that requires strict egg avoidance from now on

C. Sensitization without clinical allergy, so she can keep eating egg

D. Early egg allergy that should be monitored with repeat panels every year

E. Food protein-induced enterocolitis triggered by egg exposure

Answer Contents

BQ-447. A 10-year-old boy was treated in the emergency department for anaphylaxis after eating a pecan. He needed two doses of intramuscular epinephrine to clear wheeze and hypotension. He has now been symptom free for one hour, and his mother asks to go home. Which feature of this episode most increases his risk of a biphasic reaction? [Epidemiology and Risk Assessment]

A. Needing a second dose of epinephrine to control symptoms

B. Male sex in a child of school age with no other risk features

C. A family history of seasonal hay fever in a first-degree relative

D. A history of mild seasonal rhinitis during the past year

E. Age older than 10 years at the time of the reaction

Answer Contents

BQ-448. A 7-month-old girl is fed rice cereal for the first time. Two hours later she has repeated vomiting, then becomes pale and limp. She has no hives, wheeze, or lip swelling. In the emergency department she is lethargic and tachycardic, and she improves after an intravenous saline bolus. Which is the most likely diagnosis? [Diagnosis]

A. IgE-mediated food allergy with anaphylaxis

B. Food protein-induced enterocolitis syndrome

C. Eosinophilic esophagitis with food refusal

D. Hypertrophic pyloric stenosis with vomiting

E. Oral allergy syndrome after a raw fruit

Answer Contents

BQ-449. A 9-year-old boy has sneezing, an itchy nose, and congestion that are present year-round and worse in spring. He snores and sleeps poorly. The nasal mucosa is pale and boggy, with clear drainage and a transverse nasal crease. He takes diphenhydramine at night and is drowsy at school. Which is the best treatment? [Management and Treatment]

A. Increase the diphenhydramine to twice a day

B. Add montelukast as the first-line medication

C. Switch to a second-generation oral antihistamine alone

D. Send him for allergen immunotherapy before starting any medication for his symptoms at all

E. Start a daily intranasal corticosteroid and teach aim away from the septum

Answer Contents

BQ-450. A 3-year-old boy has had 5 days of foul-smelling, blood-tinged discharge from the right nostril only. He has no fever, sneezing, or nasal itching, and the left nostril is clear. He is otherwise well. Which is the most likely diagnosis? [Diagnosis]

A. Allergic rhinitis with seasonal symptoms

B. Viral upper respiratory tract infection

C. Retained nasal foreign body

D. Acute bacterial sinusitis

E. Adenoid hypertrophy

Answer Contents

BQ-451. An 11-year-old girl has had daily itchy hives for 8 weeks. Each welt fades within a day without bruising, and she has no trigger and no other symptoms. Her examination is normal. She takes standard-dose cetirizine daily with only partial relief. Which is the best next step? [Management and Treatment]

A. Increase the second-generation antihistamine up to four times the standard dose

B. Order a broad food-specific IgE panel to find the trigger

C. Start a prolonged daily course of oral prednisone for symptom control over several months

D. Add omalizumab at once, without changing the antihistamine

E. Replace cetirizine with a first-generation antihistamine at bedtime

Answer Contents

BQ-452. A 12-year-old boy has had three episodes of lip and hand swelling with severe cramping abdominal pain, each lasting 2 to 3 days. He has no hives or itching. One episode did not respond to diphenhydramine, a steroid, or epinephrine. His father has similar episodes. Which is the best next test? [Diagnosis]

A. Serum tryptase level

B. Food-specific IgE panel

C. Total serum IgE level

D. Stool studies for parasites

E. Serum C4 level

Answer Contents

BQ-453. A 15-year-old boy with proteinuric kidney disease started enalapril 3 weeks ago. He wakes with swollen lips and tongue. He has no hives, itching, or wheeze, and he ate no new foods. Which mediator is most responsible for this swelling? [Physiology and Pathophysiology]

A. Histamine released from mast cells

B. IgE cross-linking on basophils

C. Bradykinin accumulating from reduced breakdown

D. Leukotriene release from activated eosinophils in the tissues

E. Immune complex deposition in small vessels

Answer Contents

BQ-454. A 3-month-old boy has had oral thrush for 6 weeks, chronic diarrhea, and a fall in weight from the 40th to under the 3rd percentile. He had pneumonia last month. His newborn screen was reported as normal, and he has had no live vaccines yet. Which is the best first test? [Diagnosis]

A. Complete blood count with differential

B. Sweat chloride test for cystic fibrosis

C. Quantitative serum IgE level

D. Tetanus antibody titer level

E. Upper gastrointestinal series

Answer Contents

BQ-455. A 2-year-old boy has had three deep skin abscesses, a Staphylococcus aureus liver abscess, and an Aspergillus lung infection. His growth is normal. His lymphocyte count and immunoglobulin levels are normal. Which component of immunity is most likely defective? [Physiology and Pathophysiology]

A. Antibody production by B cells

B. Helper function of T cells

C. Activity of the complement pathway

D. Activation of mast cells by antigen exposure in tissue

E. Killing of ingested microbes by phagocytes

Answer Contents

BQ-456. A 4-year-old girl attends preschool and has had 5 ear infections and 2 pneumonias in the past 12 months. Her growth is normal. She has no thrush, deep abscesses, or family history of immune disease. Her examination today is normal. Which is the best next step? [Epidemiology and Risk Assessment]

A. Reassure the family that this is typical of preschool exposure

B. Start continuous prophylactic antibiotics through the winter

C. Arrange evaluation by an immunologist because two warning signs apply

D. Order a food allergen panel to look for a hidden trigger

E. Recommend tonsillectomy and adenoidectomy for the recurrent ear infections

Answer Contents

BQ-457. A 9-year-old boy takes trimethoprim-sulfamethoxazole for a skin infection. On day 9 he has fever, a painful blistering rash, erosions of the lips and mouth, and red eyes. His blood pressure is normal. Which is the best next step? [Management and Treatment]

A. Stop the drug permanently and arrange urgent hospital care

B. Switch to an oral antihistamine and continue the antibiotic course

C. Continue the antibiotic and add a topical corticosteroid to the rash

D. Hold the drug for two days, then rechallenge under observation

E. Treat with oral antibiotics at home and review him in one week

Answer Contents

BQ-458. An 11-year-old boy has biopsy-proven eosinophilic esophagitis, with more than 15 eosinophils per high-power field after a proton pump inhibitor trial. His family prefers dietary treatment. His food-specific IgE tests and skin prick tests are positive for wheat, soy, and egg. Which is the best approach to choosing foods to remove? [Management and Treatment]

A. Remove the foods that were positive on his IgE and skin prick tests, such as wheat and soy

B. Send a food-specific IgG panel and remove every positive food

C. Start empiric elimination of common triggers such as milk and repeat endoscopy

D. Keep an unrestricted diet and use a symptom diary to judge response

E. Remove only the single food that he reports as the worst trigger

Answer Contents

BQ-459. A 13-year-old girl weighing 40 kg is in diabetic ketoacidosis. She has received a saline bolus, and her perfusion has improved. One hour after fluids began, her venous pH is 7.20, glucose is 450 mg/dL, and potassium is 4.2 mmol/L. Her bicarbonate is 9 mmol/L. Which is the best way to begin insulin? [Management and Treatment]

A. Start an intravenous insulin infusion at 0.4 units per hour

B. Start an intravenous insulin infusion at 2 units per hour with no bolus

C. Start an intravenous insulin infusion at 8 units per hour with no bolus

D. Give a 4-unit intravenous bolus and then infuse 2 units per hour

E. Give 10 units of insulin subcutaneously and recheck the glucose in four hours

Answer Contents

BQ-460. A 12-year-old girl in diabetic ketoacidosis has received saline boluses for an hour. Her venous pH is 7.12, glucose is 410 mg/dL, and serum potassium is 3.0 mmol/L. Her electrocardiogram shows flattened T waves with U waves, and her urine output is adequate. The insulin infusion is ready. Which is the best next step? [Management and Treatment]

A. Start the insulin infusion and replace potassium afterward

B. Give an insulin bolus to correct the acidosis faster

C. Give sodium bicarbonate and withhold potassium until the pH is normal

D. Begin potassium replacement and hold insulin until the level rises

E. Switch to hypotonic fluid without potassium to avoid worsening overload

Answer Contents

BQ-461. A 7-year-old boy admitted with severe diabetic ketoacidosis has improved: after 6 hours of treatment his pH is 7.22 and glucose is 290 mg/dL. He suddenly reports a severe headache, vomits, and becomes irritable and then drowsy. His heart rate falls from 120 to 62 per minute, and his oxygen saturation drops. Which is the best next step? [Management and Treatment]

A. Obtain a head CT scan before giving any treatment for brain swelling

B. Increase the insulin infusion to clear the acidosis faster

C. Give a 20 mL/kg normal saline bolus to restore perfusion

D. Give intravenous lorazepam for presumed seizure activity

E. Give mannitol or 3% saline now and obtain imaging afterward

Answer Contents

BQ-462. A 9-month-old girl has a seizure after a 4-hour gap between feeds. Her plasma glucose is 28 mg/dL (1.6 mmol/L). A critical sample drawn at that moment shows a low beta-hydroxybutyrate, low free fatty acids, a detectable insulin level, and normal cortisol and growth hormone. Which mechanism best explains these findings? [Physiology and Pathophysiology]

A. Failure of cortisol-driven gluconeogenesis during the fast

B. A block in mitochondrial fatty acid oxidation during the fast

C. Depleted glycogen stores after prolonged fasting and vomiting

D. Insulin secretion that blocks glucose release, lipolysis, and ketogenesis

E. Growth hormone deficiency that impairs fat mobilization during the overnight fast

Answer Contents

BQ-463. A 15-year-old boy with obesity and acanthosis nigricans has had 2 months of thirst and frequent urination. Random glucose is 330 mg/dL and A1C is 10.4 percent. He has no ketones, and his venous pH and bicarbonate are normal. His mother has type 2 diabetes. Which is the most appropriate initial management? [Management and Treatment]

A. Metformin and a lifestyle program alone for three months

B. Basal insulin, metformin, and a lifestyle program together

C. A lifestyle program alone with a recheck of A1C in three months

D. A sulfonylurea as monotherapy with a lifestyle handout

E. Referral for bariatric surgery with no medication now

Answer Contents

BQ-464. A 9-year-old boy is diagnosed with type 1 diabetes after 3 weeks of polyuria, and his mild ketoacidosis has resolved. His mother has Hashimoto thyroiditis. He has no abdominal pain, diarrhea, or poor growth, and his examination is normal. The team plans to screen for coexisting autoimmune disease at diagnosis. Which pair of tests is most appropriate? [Epidemiology and Risk Assessment]

A. Antinuclear antibody and rheumatoid factor, with an inflammatory marker panel

B. Morning cortisol and adrenal antibodies, with a corticotropin stimulation test

C. Lipid profile and urine albumin-to-creatinine ratio, with a retinal examination

D. Tissue transglutaminase IgA with total IgA, and thyroid antibodies with TSH

E. Stool calprotectin and fecal elastase

Answer Contents

BQ-465. A 3-week-old girl born at term has prolonged jaundice, constipation, poor feeding, a hoarse cry, a large tongue, and an umbilical hernia. She was born at term after an uncomplicated pregnancy and passed meconium on the first day. She sleeps most of the day. Which is the most likely diagnosis? [Diagnosis]

A. Congenital hypothyroidism

B. Biliary atresia

C. Congenital adrenal hyperplasia

D. Hirschsprung disease

E. Galactosemia

Answer Contents

BQ-466. An 11-year-old girl has had 18 months of slowing growth, with a height velocity of 3 cm per year, while her weight rose by 6 kg. She is tired, cold, and constipated. The thyroid is diffusely enlarged, firm, and painless. TSH is 38 mIU/L and free T4 is low. Which is the most likely diagnosis? [Diagnosis]

A. Graves disease

B. Subacute thyroiditis

C. Hashimoto thyroiditis

D. Central hypothyroidism

E. Growth hormone deficiency

Answer Contents

BQ-467. A 15-year-old girl has had 4 months of weight loss despite a good appetite, heat intolerance, tremor, and a resting heart rate of 120 per minute. She has a diffuse goiter and mild proptosis. TSH is under 0.01 mIU/L, and free T4 is high. Which is the most likely mechanism of her hyperthyroidism? [Physiology and Pathophysiology]

A. Autoimmune destruction of thyroid follicles with leakage of stored hormone

B. A TSH-secreting adenoma of the anterior pituitary gland

C. An autonomous hyperfunctioning nodule in one thyroid lobe

D. Excess iodine intake that overrides normal hormone synthesis

E. Autoantibodies that stimulate the thyrotropin receptor

Answer Contents

BQ-468. A term newborn has an enlarged phallus-like structure measuring 2.5 cm, fused labioscrotal folds, a single perineal opening, and no palpable gonads. Sodium is 136 mmol/L and potassium is 5.4 mmol/L at 2 days of age. The parents ask whether they have a son or a daughter. Which is the best next step? [Diagnosis]

A. Assign male sex today based on the appearance of the genitalia

B. Plan early genital surgery and discharge the family home

C. Start hydrocortisone and fludrocortisone empirically before any laboratory tests are drawn or returned

D. Send a karyotype and 17-hydroxyprogesterone, monitor electrolytes, and involve a specialist team

E. Reassure the family and repeat the examination at the six-week visit

Answer Contents

BQ-469. A 6-year-old girl's height has tracked just under the 3rd percentile since age 2, with a growth rate of 6 cm per year. Her mother is 150 cm tall and her father is 163 cm, and both grew steadily. Her bone age matches her chronologic age, and TSH is normal. Which is the most likely diagnosis? [Diagnosis]

A. Growth hormone deficiency

B. Constitutional delay of growth

C. Familial short stature

D. Turner syndrome

E. Hypothyroidism

Answer Contents

BQ-470. A girl aged 7 years 8 months has had breast budding for 3 months. Her breast development is Tanner stage 2, with no pubic hair or axillary odor. She has no headache or visual change, and her height velocity is 6 cm per year. The remainder of her examination is normal. Which is the best management? [Management and Treatment]

A. Reexamine her every 4 to 6 months to watch for progression

B. Begin a GnRH agonist immediately to preserve her adult height

C. Obtain a brain MRI scan this week to exclude a lesion

D. Start estrogen blockade after a bone age study

E. Discharge her from care with no follow-up visits

Answer Contents

BQ-471. A 14-year-old boy is the shortest in his class. He is healthy, with prepubertal testes and no pubic hair. His growth rate is 4.5 cm per year, and his bone age is 12 years. His father reached his final height at age 19. Which is the most likely cause of his delayed puberty? [Epidemiology and Risk Assessment]

A. Klinefelter syndrome due to an extra X chromosome

B. Isolated GnRH deficiency with absent pubertal onset

C. Primary hypothyroidism with a high TSH level

D. Growth hormone deficiency with a low IGF-1 level

E. Constitutional delay of growth and puberty

Answer Contents

BQ-472. A 3-year-old boy has bowed legs and poor growth despite 6 months of vitamin D treatment. His 25-hydroxyvitamin D and calcium are normal, phosphate is low, and parathyroid hormone is normal. His father is short with bowed legs. Wrist radiographs show widened, frayed growth plates. Which is the most likely mechanism? [Physiology and Pathophysiology]

A. Dietary deficiency of vitamin D

B. Renal wasting of phosphate

C. Deficient parathyroid hormone secretion

D. Dietary deficiency of calcium

E. Malabsorption of fat-soluble vitamins

Answer Contents

BQ-473. A 5-week-old boy has a generalized seizure. He has a small jaw and a cardiac murmur. The facial nerve twitches when tapped in front of the ear, and his electrocardiogram shows a long QT interval. Ionized calcium and parathyroid hormone are low, phosphate is high, and magnesium is normal. Which is the most likely underlying cause? [Diagnosis]

A. Vitamin D deficiency in the newborn

B. X-linked hypophosphatemia from phosphate wasting

C. Hypoparathyroidism as part of DiGeorge syndrome

D. Dietary calcium deficiency in a breastfed infant

E. Pseudohypoparathyroidism with end-organ resistance

Answer Contents

BQ-474. A thin 3-year-old boy is found lethargic at 6 a.m. after 2 days of gastroenteritis with poor intake. His glucose is 38 mg/dL (2.1 mmol/L), and he improves after dextrose. The critical sample shows a high beta-hydroxybutyrate, low insulin, a high cortisol, and normal lactate. His development is normal. Which is the most likely diagnosis? [Diagnosis]

A. Ketotic hypoglycemia of early childhood

B. Congenital hyperinsulinism

C. Primary adrenal insufficiency

D. Fatty acid oxidation disorder with low ketones

E. Sulfonylurea ingestion by the child

Answer Contents

BQ-475. A 3-week-old exclusively breastfed boy has lost 14 percent of his birth weight. His skin feels doughy and he is irritable. Serum sodium is 164 mEq/L. After a small isotonic bolus restores perfusion, the team plans to lower the sodium over about 48 hours rather than within a few hours. Which is the most likely mechanism by which rapid correction would cause seizures? [Physiology and Pathophysiology]

A. Sudden loss of potassium from brain cells into the plasma space

B. Excess antidiuretic hormone release from the posterior pituitary gland

C. Rapid shrinkage of brain cells as water leaves them for plasma

D. Movement of water into brain cells that adapted to high osmolality

E. Falling plasma calcium from dilution of the extracellular space

Answer Contents

BQ-476. An 8-year-old boy with Burkitt lymphoma received his first chemotherapy 12 hours ago. He now has muscle weakness. Serum potassium is 7.6 mEq/L, and the ECG shows peaked T waves and a widened QRS complex. Urine output is adequate and the sample was not hemolyzed. Which is the most appropriate first intervention? [Management and Treatment]

A. Insulin with glucose to shift potassium into the cells

B. Sodium polystyrene sulfonate by mouth to bind potassium

C. Inhaled albuterol to drive potassium into the cells

D. A repeat sample to confirm the value before any treatment

E. Intravenous calcium to stabilize the cardiac membrane

Answer Contents

BQ-477. A 9-year-old girl who weighs 28 kg is 14 hours after tonsillectomy and is receiving half-isotonic saline with dextrose. She has vomited twice, reports headache, and now has a generalized seizure that stopped after a few minutes. Serum sodium is 118 mEq/L. Which is the most appropriate management? [Management and Treatment]

A. Fluid restriction alone with a repeat sodium in 6 hours

B. Intravenous lorazepam alone, followed by a head CT before other treatment

C. A small intravenous bolus of 3 percent saline

D. A large bolus of isotonic saline infused over 1 hour

E. Oral salt tablets with a change to oral fluids only

Answer Contents

BQ-478. A 9-month-old girl has poor growth and vomiting but no diarrhea. Blood gas shows pH 7.28 and bicarbonate 14 mEq/L. Sodium is 136 mEq/L, chloride 112 mEq/L, and potassium 3.1 mEq/L, giving an anion gap of 10. Urine pH is 6.8. Which is the most likely mechanism of the acidosis? [Physiology and Pathophysiology]

A. Excess lactic acid from tissue hypoperfusion

B. Ketone production from fasting in a vomiting infant

C. Bicarbonate loss through stool from unrecognized diarrhea

D. Failure of the distal tubule to excrete acid into the urine

E. Retention of uremic acids from falling glomerular filtration

Answer Contents

BQ-479. A 20-month-old boy has had watery stools and vomiting for 2 days. He weighs 11 kg, which is 6 percent less than his last well weight. He is irritable but drinks eagerly, with dry mucous membranes, few tears, and capillary refill of 2 to 3 seconds. His breathing is normal. Which is the best next step? [Management and Treatment]

A. Small frequent amounts of oral rehydration solution by spoon or syringe

B. An intravenous isotonic bolus followed by admission to the ward

C. Undiluted apple juice offered freely as tolerated by mouth

D. Loperamide to slow stool losses, with fluids held for 4 hours

E. A sports drink in unrestricted amounts, followed by a clear liquid diet

Answer Contents

BQ-480. An 8-year-old boy has had cola-colored urine and puffy eyelids for 2 days. He had a sore throat about 2 weeks ago. Blood pressure is 134/88 mm Hg, over the 95th percentile for his height. Urine shows red cell casts and 2+ protein. C3 is low and creatinine is 0.9 mg/dL. Which is the most likely diagnosis? [Diagnosis]

A. IgA nephropathy

B. Alport syndrome

C. Minimal change nephrotic syndrome

D. Post-infectious glomerulonephritis

E. Hemolytic uremic syndrome with microangiopathy

Answer Contents

BQ-481. A 14-year-old boy has had two episodes of visible tea-colored urine, each beginning the day after a cold. He feels well between episodes. Blood pressure is normal, C3 is normal, and urine protein-to-creatinine ratio is 0.4 mg/mg. Microscopy shows dysmorphic red cells. Which is the most likely diagnosis? [Diagnosis]

A. IgA nephropathy

B. Post-infectious glomerulonephritis

C. Alport syndrome

D. Hypercalciuria

E. Urolithiasis

Answer Contents

BQ-482. A 10-year-old boy has persistent microscopic hematuria with dysmorphic red cells. An audiogram shows high-frequency sensorineural hearing loss, and his maternal uncle began dialysis in his twenties. His blood pressure and creatinine are currently normal and he has no rash. Which is the most likely mechanism of his kidney disease? [Physiology and Pathophysiology]

A. Immune complex deposition after streptococcal infection

B. Deposition of IgA-containing immune complexes in the glomerular mesangium after infection

C. Excess urinary calcium crystallizing within the tubules

D. Autoantibody attack on the glomerular basement membrane

E. Abnormal type IV collagen in the glomerular and cochlear basement membranes

Answer Contents

BQ-483. A tall 13-year-old girl has 1+ protein on a random school dipstick. Her blood pressure, creatinine, and urine sediment are normal. A random urine protein-to-creatinine ratio is 0.6 mg/mg, and a first-morning sample the next day is 0.1 mg/mg. Which is the most likely diagnosis? [Diagnosis]

A. Minimal change nephrotic syndrome

B. Orthostatic proteinuria

C. Focal segmental glomerulosclerosis

D. Reflux nephropathy with scarring

E. Post-infectious glomerulonephritis

Answer Contents

BQ-484. A 6-year-old boy with relapsing nephrotic syndrome has fever of 39.2 C (102.6 F), abdominal pain, and tense ascites. Ascitic fluid contains many neutrophils, and culture grows Streptococcus pneumoniae. Which is the most likely reason he is susceptible to this infection? [Physiology and Pathophysiology]

A. Excess immunoglobulin production by activated B cells in the spleen

B. Impaired neutrophil production from the high-dose steroid course alone

C. Urinary loss of immunoglobulins and complement, with edema fluid as a growth medium

D. Direct bacterial invasion from raised intestinal pressure caused by chronic constipation and bowel distension

E. Infected mesenteric thrombi from the hypercoagulable state of nephrosis

Answer Contents

BQ-485. A 5-year-old girl who has had vomiting and diarrhea for 3 days took ibuprofen for fever. She is thirsty with dry mucous membranes and has passed little urine. Creatinine is 1.1 mg/dL, up from a baseline of 0.4 mg/dL. Urinalysis is bland. Which is the most likely mechanism of the rise in creatinine? [Physiology and Pathophysiology]

A. Obstruction of both ureters by crystals precipitated from concentrated urine in a dehydrated child

B. Direct tubular toxicity from a recent aminoglycoside course

C. Immune complex deposition in the glomerular capillaries

D. Microangiopathic injury to the glomerular capillaries from toxin

E. Reduced glomerular perfusion after NSAID blockade of prostaglandin-mediated dilation

Answer Contents

BQ-486. A 10-year-old boy who weighs 34 kg is 10 hours after spinal fusion. His urine output has averaged 0.3 mL/kg/hour for the past 8 hours. Creatinine is 0.5 mg/dL, unchanged from his baseline. Blood pressure and heart rate are normal. Which is the most appropriate interpretation? [Diagnosis]

A. No kidney injury, because the creatinine has stayed exactly at the baseline value

B. Normal postoperative antidiuresis that requires no evaluation

C. Chronic kidney disease, because of persistently low output

D. Acute kidney injury by the urine output criterion, with creatinine lagging

E. Nephrotic syndrome, because of postoperative fluid retention

Answer Contents

BQ-487. A 4-year-old boy has blood pressure readings that exceed the stage 2 threshold for his age, sex, and height at three visits, each confirmed by auscultation. He is lean, asymptomatic, and had a urinary infection in infancy. The examination is normal. Which is the best next step? [Diagnosis]

A. Start an ACE inhibitor and recheck in one year

B. Renal and bladder ultrasound with urinalysis and creatinine

C. Reassurance, because primary hypertension is usual at this age

D. Polysomnography for suspected sleep apnea

E. Repeat measurements at the next annual visit

Answer Contents

BQ-488. A 15-year-old girl has clinic blood pressure readings of 136/84, 134/82, and 138/86 mm Hg at three visits, each confirmed by auscultation. She is asymptomatic with a body mass index of 31 kg/m2. Her family history includes hypertension. Which is the most accurate classification of her blood pressure? [Epidemiology and Risk Assessment]

A. Normal blood pressure

B. Elevated blood pressure

C. Stage 1 hypertension

D. Stage 2 hypertension

E. White coat hypertension

Answer Contents

BQ-489. A 4-month-old exclusively breastfed boy with dark skin, who receives no vitamin D supplement, has a generalized seizure. Glucose and sodium are normal. Ionized calcium is low at 0.8 mmol/L. Wrist radiographs show cupping and fraying of the metaphyses. Which is the most likely mechanism of the seizure? [Physiology and Pathophysiology]

A. Hyperammonemia from impaired hepatic urea synthesis after a protein-rich feed

B. Hypoglycemia from fasting between feeds

C. Brain swelling from excess free water

D. Reduced intestinal calcium absorption from vitamin D deficiency

E. Hypoparathyroidism from absent parathyroid tissue

Answer Contents

BQ-490. A 5-week-old boy has projectile vomiting after every feed and a palpable olive-shaped mass. Sodium is 133 mEq/L, potassium 2.9 mEq/L, chloride 86 mEq/L, and bicarbonate 34 mEq/L. He is hungry after vomiting and mildly dehydrated. Which is the most likely mechanism of the hypokalemia? [Physiology and Pathophysiology]

A. Aldosterone-driven renal potassium loss in a volume-depleted infant

B. Large potassium losses directly in the vomited gastric fluid over several days

C. Shift of potassium out of cells by acidemia

D. Inadequate potassium intake during a brief illness

E. Excess potassium loss in stool from secondary diarrhea

Answer Contents

BQ-491. A term newborn boy has a weak urinary stream and a palpable bladder. Prenatal ultrasound showed bilateral hydronephrosis, and a postnatal ultrasound shows bilateral dilation with a thick-walled bladder. He has no fever and is otherwise well. Which is the best next step? [Management and Treatment]

A. Observe urine output for 48 hours before ordering further tests

B. Order a nuclear renal scan before any bladder drainage

C. Place a bladder catheter, then arrange a voiding cystourethrogram with urology

D. Start continuous antibiotic prophylaxis and plan a repeat ultrasound at 6 months of age

E. Reassure that transient dilation is likely and repeat imaging at 6 weeks

Answer Contents

BQ-492. A 3-year-old girl has had three febrile urinary infections in 5 months. A voiding cystourethrogram shows grade IV reflux on the right. Renal function is normal, blood pressure is normal, and her parents are worried about long-term kidney damage. Which is the most appropriate management? [Management and Treatment]

A. Reassurance and no treatment, because reflux resolves with growth

B. Continuous antibiotic prophylaxis with referral to urology for follow-up

C. Immediate bilateral ureteral reimplantation without a trial of prophylaxis

D. Repeat voiding studies every 3 months with no antibiotic

E. Cranberry supplements alone, with urine cultures only for fever

Answer Contents

BQ-493. A 9-month-old boy born at term has a left testis that cannot be palpated in the scrotum or inguinal canal at three consecutive visits, including examinations in a warm room. The right testis is normal. Which is the best next step? [Management and Treatment]

A. Scrotal and inguinal ultrasound before any referral

B. A course of human chorionic gonadotropin to induce descent

C. Reassurance and repeat examination at age 3 years

D. Observation until puberty for spontaneous descent of the testis

E. Referral to pediatric urology for orchiopexy

Answer Contents

BQ-494. A 4-year-old boy's right testis cannot be found in the scrotum while he is standing in the cold examination room. When he sits cross-legged in a warm room, the testis is palpable in the groin and is guided easily into the scrotum, where it stays without tension. Which is the most likely diagnosis? [Diagnosis]

A. Undescended testis in the inguinal canal

B. Retractile testis

C. Ectopic testis in the perineum

D. Testicular agenesis

E. Communicating hydrocele

Answer Contents

BQ-495. A 7-week-old boy born at 32 weeks has been irritable for 5 hours and has vomited twice. A firm, tender bulge is present in the right groin and extends into the scrotum, and the abdomen is mildly distended. Both testes are palpable and the cremasteric reflex is present. Which is the most likely diagnosis? [Diagnosis]

A. Torsion of the spermatic cord with ischemia

B. Acute inguinal lymphadenitis

C. Communicating hydrocele

D. Acute epididymitis

E. Incarcerated inguinal hernia

Answer Contents

BQ-496. A term newborn boy has a urethral opening on the underside of the glans, a hooded dorsal foreskin, and two descended testes. The pregnancy and family history are unremarkable. His parents ask to schedule routine circumcision before discharge, and the nurse notes the urine stream is directed downward. Which is the best advice? [Management and Treatment]

A. Proceed with routine circumcision now because the hooded foreskin is redundant tissue

B. Perform a bedside meatotomy to move the opening forward

C. Obtain a karyotype and endocrine panel before any other step

D. Delay circumcision and arrange pediatric urology repair in early childhood

E. Reassess at age 5 years and repair only if he has symptoms

Answer Contents

BQ-497. A 9-year-old boy has had left scrotal pain for 1 day that began gradually and is now moderate. He has no nausea. A small tender nodule at the upper pole of the testis shows a blue discoloration through the skin. The cremasteric reflex is present and urinalysis is normal. Which is the most likely diagnosis? [Diagnosis]

A. Torsion of the spermatic cord with testicular ischemia

B. Torsion of the appendix testis

C. Epididymitis

D. Incarcerated inguinal hernia

E. Hydrocele

Answer Contents

BQ-498. A 5-year-old boy has had 2 days of gradually increasing right scrotal swelling and tenderness with a temperature of 38.4 C (101.1 F). The epididymis is swollen and tender, the cremasteric reflex is present, and urinalysis shows pyuria. He had a febrile urinary infection at age 2. Which is the most appropriate management? [Management and Treatment]

A. Emergency surgical exploration of the scrotum

B. Observation without testing, because the course is benign

C. Scrotal ultrasound alone, then discharge without antibiotics

D. Treat as epididymitis and image the urinary tract for an anomaly

E. Antibiotics alone, with no imaging of the urinary tract unless he relapses

Answer Contents

BQ-499. A 3-month-old boy has painless scrotal swelling that transilluminates. His parents note it is larger by evening and smaller in the morning. The testis can be felt normally, and there is no bowel in the sac. Which is the most likely mechanism of the swelling? [Physiology and Pathophysiology]

A. Patent processus vaginalis allowing peritoneal fluid to enter the scrotum

B. Twisting of the spermatic cord on its vascular pedicle, cutting off blood flow

C. Bacterial infection of the epididymis from urinary reflux

D. Dilated veins of the pampiniform plexus around the cord

E. Failure of the testis to complete its descent through the inguinal canal into the scrotum

Answer Contents

BQ-500. A 14-month-old uncircumcised girl has had fever of 39.3 C (102.7 F) for 2 days without a clear source. She is playful and well hydrated, and the examination is normal. Her parents ask how urine should be collected. Which is the best next step? [Diagnosis]

A. Obtain a bagged specimen and send it for culture alone before treatment

B. Wait for a clean-catch sample once she is toilet trained

C. Start ceftriaxone empirically without any urine testing

D. Obtain urine by catheterization for urinalysis and culture

E. Use a dipstick on a bagged specimen to guide treatment

Answer Contents

BQ-501. A male newborn at 14 hours of age has mild unilateral hydronephrosis noted on prenatal ultrasound. A postnatal ultrasound performed now shows normal-appearing kidneys. He is feeding well, voiding normally, and has no family history of kidney disease. The parents ask whether anything more is needed before discharge. Which is the best next step? [Diagnosis]

A. Discharge him with no further imaging

B. Voiding cystourethrography now

C. Repeat the ultrasound at 3 to 7 days of age

D. Continuous antibiotic prophylaxis and no further imaging

E. A nuclear renal scan at 1 month

Answer Contents

BQ-502. A term newborn has a proximal hypospadias with a small phallus and a nonpalpable left testis. The right testis is palpable in the inguinal canal. There is no pigmentation of the genitalia and no dysmorphic features. The family asks what should be done before they leave the hospital. Which is the best next step? [Diagnosis]

A. Evaluate for a disorder of sex development with karyotype and hormone tests

B. Schedule circumcision before discharge for hygiene and to allow easier repair

C. Reassure the parents and review at 6 months of age

D. Start testosterone injections now to enlarge the phallus before surgical repair

E. Perform scrotal ultrasound alone at 1 year of age

Answer Contents

BQ-503. A 3-month-old boy has persistent moderate left hydronephrosis with a dilated renal pelvis and calyces. The ureter and bladder look normal, and a voiding cystourethrogram is normal. He is growing well and has had no infections. Which is the best next step? [Diagnosis]

A. Continuous antibiotic prophylaxis alone with no further imaging or urology review

B. A second voiding cystourethrogram in 3 months

C. Percutaneous renal biopsy with electron microscopy

D. Bilateral ureteral reimplantation

E. A nuclear renal scan to measure split function, with urology review

Answer Contents

BQ-504. A 6-year-old girl has had isolated microscopic hematuria on three urinalyses over 3 months. Blood pressure is normal, urine protein is negative, and the red cells are round without casts. Creatinine is normal and urinalyses in her parents and sibling are normal. Which is the best next step? [Diagnosis]

A. Percutaneous renal biopsy with electron microscopy

B. Cystoscopy under anesthesia

C. A trial of oral prednisone

D. A spot urine calcium-to-creatinine ratio

E. Monthly C3 measurements

Answer Contents

BQ-505. A 2-year-old girl has a second febrile urinary tract infection in 4 months, each caused by Escherichia coli. She improves quickly with treatment and is growing normally. Her parents ask whether any imaging beyond the ultrasound is needed. Renal and bladder ultrasound is normal. Which is the best next step? [Diagnosis]

A. No further imaging, because the ultrasound is normal

B. A nuclear renal scan during the current infection

C. Cystoscopy under anesthesia

D. A CT scan of the abdomen with contrast

E. Voiding cystourethrography

Answer Contents

BQ-506. Parents of a 5-month-old boy with a left undescended testis ask what long-term problems the condition can cause if surgery is delayed or never done, and the family wants to understand the reasons for repair. Which pair of risks should be discussed? [Epidemiology and Risk Assessment]

A. Renal scarring and hypertension

B. Infertility and testicular cancer

C. Delayed puberty and short stature

D. Hip dysplasia and leg length difference

E. Urinary incontinence and nocturnal enuresis

Answer Contents

BQ-507. A 3-day-old girl has swelling of the backs of both hands and feet and a webbed neck. Femoral pulses are weaker than brachial pulses, and the arm blood pressure is higher than the leg pressure. Which cardiac lesion is most likely? [Diagnosis]

A. Atrioventricular septal defect

B. Tetralogy of Fallot

C. Supravalvular aortic stenosis

D. Coarctation of the aorta

E. Truncus arteriosus

Answer Contents

BQ-508. A 3-day-old boy with a cleft palate and truncus arteriosus has a generalized seizure. Glucose is normal, and ionized calcium is low at 0.7 mmol/L. Magnesium is normal and the infant is otherwise well. Which is the most likely mechanism of the seizure? [Physiology and Pathophysiology]

A. Hypoglycemia from congenital hyperinsulinism

B. Hyperammonemia from a urea cycle defect

C. Vitamin D deficiency passed from a deficient mother during pregnancy

D. Hyponatremia from excess antidiuretic hormone

E. Hypoparathyroidism from small or absent parathyroid tissue

Answer Contents

BQ-509. A 3-year-old girl has rapid weight gain and constant food seeking. She was floppy and fed poorly as an infant. Genetic testing confirms an abnormality of chromosome 15q11-q13, and her parents ask how the disorder arises. Which is the most likely mechanism of her condition? [Physiology and Pathophysiology]

A. Loss of function of the maternally inherited genes at 15q11-q13

B. A CGG repeat expansion in the FMR1 gene

C. A microdeletion of chromosome 7q11.23

D. Loss of function of the paternally inherited genes at 15q11-q13

E. An extra copy of chromosome 21

Answer Contents

BQ-510. A 10-month-old boy has irritability and constipation. He has an elfin face and is unusually friendly with strangers. A systolic ejection murmur is heard at the upper right sternal border, and serum calcium is 12.6 mg/dL. Which is the most likely diagnosis? [Diagnosis]

A. Williams syndrome

B. Prader-Willi syndrome

C. Fragile X syndrome

D. Turner syndrome

E. Angelman syndrome

Answer Contents

BQ-511. A 13-year-old boy has intellectual disability, a long face, and prominent ears. He had delayed speech as a toddler and now shows hand-flapping and social anxiety. His maternal uncle has a similar learning disability, and his mother is concerned about the family history. Which is the most likely diagnosis? [Diagnosis]

A. Prader-Willi syndrome

B. Williams syndrome

C. Klinefelter syndrome

D. Down syndrome

E. Fragile X syndrome

Answer Contents

BQ-512. A 3-year-old boy has global developmental delay. He has no dysmorphic features, no clear syndromic pattern, a normal examination, and normal metabolic screening. His growth is normal, and his parents ask which genetic test should come first. Which genetic test is the best first-tier choice? [Diagnosis]

A. Karyotype

B. Chromosomal microarray

C. Exome sequencing

D. Testing of a single gene such as FMR1 alone

E. Targeted testing for an aneuploidy panel

Answer Contents

BQ-513. A 7-day-old breastfed girl is jaundiced, feeds poorly, and vomits. She is lethargic with an enlarged liver. Urine tests positive for reducing substances but negative on a glucose-specific dipstick. A blood culture has been sent. Which is the best next step? [Management and Treatment]

A. Continue breast milk and begin phototherapy alone, repeating the bilirubin in 12 hours

B. Switch to a protein-free formula and start oral antibiotics

C. Stop lactose feeds and start antibiotics covering Gram-negative bacteria

D. Delay antibiotics until the culture result returns

E. Change to a lactose-containing hydrolyzed formula and send cultures

Answer Contents

BQ-514. A 4-day-old term boy was well at discharge and now vomits, is lethargic, and breathes deeply. Blood gas shows pH 7.12 and bicarbonate 8 mEq/L, with an anion gap of 28 mEq/L. Urine ketones are large, glucose is 60 mg/dL, and ammonia is moderately elevated. Blood culture is negative so far. Which is the most likely diagnosis? [Diagnosis]

A. Urea cycle defect

B. Organic acidemia

C. Fatty acid oxidation defect

D. Congenital adrenal hyperplasia

E. Neonatal sepsis with lactic acidosis

Answer Contents

BQ-515. A 10-month-old boy is found lethargic after a day of vomiting and poor intake. Glucose is 28 mg/dL and urine ketones are negative. He has a mildly enlarged liver, and an older sibling died suddenly in infancy. Which is the most likely mechanism of the hypoglycemia? [Physiology and Pathophysiology]

A. Excess insulin secretion from overactive pancreatic beta cells in infancy

B. Failure of cortisol production by the adrenal glands

C. Impaired fatty acid oxidation limiting ketone production during fasting

D. Deficiency of growth hormone from pituitary dysfunction

E. Excess glycogen storage blocking release of glucose

Answer Contents

BQ-516. A 12-day-old boy is lethargic and vomiting with fast breathing. Blood gas shows pH 7.15, and ammonia is markedly elevated. His newborn screen, collected at 12 hours of life, was reported as normal. A colleague says metabolic disease is excluded. Which is the best interpretation of the screening result? [Epidemiology and Risk Assessment]

A. It excludes metabolic disease in this infant, so the evaluation should focus on infection alone

B. It may be falsely reassuring because of early collection, so diagnostic testing is needed

C. It requires a repeat screen alone before any further testing

D. It is reliable at any age, so look for a formula preparation error

E. It proves tandem mass spectrometry detects every treatable disorder

Answer Contents

BQ-517. A 6-day-old term girl is feeding well and has a normal examination. Her newborn screen, collected at 36 hours, is reported positive for phenylketonuria. The state program asks for follow-up, and the parents have called to ask whether to worry. Which is the best next step? [Diagnosis]

A. Repeat the filter paper screen at the 2-month visit

B. Reassure the family because the infant appears well

C. Start a low-protein diet at home on the parents' own judgment

D. Obtain a head MRI to assess the infant for early brain injury before any laboratory tests

E. Send confirmatory plasma amino acids and contact the metabolic specialist

Answer Contents

Domain 22. Rheumatology

Contents Answers for this subject

BQ-518. A 4-year-old boy has had daily fever spikes to 40 C (104 F) for 3 weeks, returning to normal between spikes. A salmon-colored rash appears with each spike and fades. His joints are mildly stiff. ESR and ferritin are raised, white cell count is 22,000 per mm3, and cultures are negative. Which is the most likely diagnosis? [Diagnosis]

A. Systemic juvenile idiopathic arthritis

B. Kawasaki disease

C. Acute lymphoblastic leukemia

D. PFAPA syndrome

E. Oligoarticular juvenile idiopathic arthritis

Answer Contents

BQ-519. A 7-year-old girl with systemic juvenile idiopathic arthritis has persistent fever and new bruising. Her platelet count has fallen from 450,000 to 120,000 per mm3, ESR has dropped from 80 to 20 mm/hour, and ferritin has risen to 8,000 ng/mL. Fibrinogen is falling. Which is the most likely mechanism of this change? [Physiology and Pathophysiology]

A. Autoantibody-mediated destruction of platelets and other circulating blood cells

B. Uncontrolled macrophage and T cell activation with massive cytokine release

C. Replacement of the bone marrow by leukemic blast cells

D. Marrow suppression from methotrexate toxicity

E. Bacterial sepsis with disseminated intravascular clotting

Answer Contents

BQ-520. A 6-month-old boy has had fever for 7 days. He has mild bilateral conjunctival injection and a faint rash, with no other typical features. CRP is 82 mg/L, hemoglobin is 9.4 g/dL, albumin is 2.6 g/dL, and urine shows sterile pyuria. Which is the best next step? [Management and Treatment]

A. Observe for 2 more days at home until the classic criteria are fully met

B. Start oral antibiotics for presumed bacterial infection

C. Give aspirin alone and discharge with close follow-up

D. Obtain an echocardiogram and treat with intravenous immunoglobulin

E. Perform a lumbar puncture for suspected meningitis

Answer Contents

BQ-521. A 3-year-old girl with Kawasaki disease has an echocardiogram at diagnosis. The left anterior descending coronary artery has a Z-score of 6.1 and an absolute diameter of 4.5 mm. The right coronary artery is normal. Which is the most accurate classification of this coronary finding? [Epidemiology and Risk Assessment]

A. Medium aneurysm

B. Small aneurysm

C. Dilation only

D. Giant aneurysm

E. Normal coronary artery

Answer Contents

BQ-522. A 7-year-old boy has palpable purpura on the buttocks and legs, colicky abdominal pain, and swollen painful ankles for 2 days. He is afebrile, the platelet count is 310,000 per mm3, and blood pressure is normal. Which is the best next step? [Management and Treatment]

A. Platelet transfusion to correct the bleeding tendency before any other step

B. Urinalysis and blood pressure now, then repeat urine checks for months

C. Bone marrow aspiration to exclude leukemia

D. Start oral prednisone to prevent nephritis

E. Skin biopsy before any other test is ordered

Answer Contents

BQ-523. A 3-year-old girl has had fever to 39.8 C (103.6 F) every 4 weeks for 6 months, each episode lasting 5 days with aphthous ulcers, pharyngitis, and tender cervical nodes. Throat cultures are negative, and blood counts are normal during attacks. She grows well and is healthy between episodes. A single dose of prednisone ended the last attack. Which is the most likely diagnosis? [Diagnosis]

A. Familial Mediterranean fever

B. Cyclic neutropenia

C. Recurrent streptococcal pharyngitis

D. PFAPA syndrome

E. Systemic juvenile idiopathic arthritis

Answer Contents

BQ-524. An 8-year-old girl of Armenian ancestry has had episodes of fever and severe abdominal pain lasting 1 to 3 days, recurring every few weeks and resolving without treatment. During attacks she lies still with a tender abdomen. She is well between episodes and grows normally, and her brother has similar attacks. Which is the most likely diagnosis? [Diagnosis]

A. PFAPA syndrome

B. Cyclic neutropenia

C. Familial Mediterranean fever

D. Systemic juvenile idiopathic arthritis

E. Recurrent acute appendicitis

Answer Contents

BQ-525. A 7-year-old girl has a purple discoloration of the upper eyelids and flat-topped scaly papules over the knuckles and elbows. She has difficulty climbing stairs and uses her hands to rise from the floor. Creatine kinase is raised. Which is the most likely diagnosis? [Diagnosis]

A. Systemic lupus erythematosus

B. Juvenile dermatomyositis

C. Atopic dermatitis

D. Guillain-Barre syndrome

E. Systemic sclerosis

Answer Contents

BQ-526. A 14-year-old girl has a malar rash, arthritis, thrombocytopenia, hypertension, and 3+ protein on urinalysis, and a skin examination shows photosensitivity. Anti-double-stranded DNA antibodies are strongly positive, and C3 and C4 levels are low. Which is the most likely mechanism of the low complement levels? [Physiology and Pathophysiology]

A. Reduced hepatic synthesis from cirrhosis

B. Urinary loss of complement proteins through the heavily protein-leaking glomerulus

C. Inherited deficiency of an early complement component

D. Consumption of complement by immune complexes activating the classical pathway

E. Autoantibody neutralization of complement by a drug reaction

Answer Contents

BQ-527. A 5-year-old girl has had a swollen, warm right knee for 10 weeks with morning stiffness and a mild limp. Two weeks of naproxen gave partial relief. Only that joint is involved, ANA is positive, and a slit-lamp examination has been arranged. Which is the best addition to treatment? [Management and Treatment]

A. Long-term daily oral prednisone

B. Anakinra

C. Oral cyclophosphamide

D. Splinting of the knee in extension for several months

E. Intra-articular triamcinolone hexacetonide

Answer Contents

BQ-528. A 5-year-old boy has bilateral calf pain on most evenings that eases with massage and is gone by morning. He runs and plays all day, has no limp, and has normal growth and examination. There is no fever, weight loss, or pallor. Which is the most likely diagnosis? [Diagnosis]

A. Acute lymphoblastic leukemia

B. Growing pains

C. Septic arthritis

D. Juvenile dermatomyositis

E. Osteoid osteoma

Answer Contents

Domain 23. Ethics

Contents Answers for this subject

BQ-529. A 3-year-old boy with leukemia needs a lumbar puncture today. His parents speak limited English. The 12-year-old sister who accompanies them offers to interpret the consent discussion because she is fluent and calm, and the parents appear to trust her. Which is the best next step? [Management and Treatment]

A. Accept the sister's help because she knows the family well

B. Proceed with the procedure using gestures and written pictures

C. Use a qualified medical interpreter for the discussion

D. Ask the father to sign an English form after a brief summary

E. Use a translation app on the clinician's phone

Answer Contents

BQ-530. A 10-year-old boy is brought by ambulance after a bicycle crash. He is unresponsive with blood pressure of 62/30 mm Hg and a distended, rigid abdomen. Imaging shows splenic rupture. His parents cannot be reached by telephone. He needs transfusion and emergency surgery. Which is the best next step? [Management and Treatment]

A. Treat under implied consent while continuing to reach the parents

B. Wait for a parent to arrive and sign written consent before any procedure

C. Obtain a court order authorizing surgery before starting

D. Ask the ethics committee to approve treatment first

E. Provide supportive fluids only until the parents are located

Answer Contents

BQ-531. A 9-year-old girl has diabetic ketoacidosis with a pH of 7.05. Her parents decline insulin and ask for herbal remedies instead. The team has explained that she may die without insulin. The parents remain firm. Which is the best next step? [Management and Treatment]

A. Honor the parents' refusal and discharge her with herbal medicine and close follow-up

B. Keep talking, involve the ethics service, and seek legal help given the imminent risk

C. Sedate her and give insulin without informing the parents

D. Provide fluids only until the parents agree to insulin

E. Ask the girl to make the decision herself

Answer Contents

BQ-532. A 16-year-old boy with newly diagnosed Crohn disease refuses infliximab. He can repeat the risks and benefits and gives consistent reasons for choosing no treatment, but says, "Serious complications happen to other people, not me." Which element of decision-making capacity is most in question? [Diagnosis]

A. Appreciating how the information applies to himself

B. Understanding the information

C. Reasoning about the options he has been given

D. Stating a consistent choice

E. Reaching the age of legal majority for medical decisions

Answer Contents

BQ-533. A 15-year-old girl tells you during time alone that she has a plan to overdose on pills tonight. She asks you not to tell her parents. Before the visit you explained the limits of confidentiality, and she appears sincere and distressed. Which is the best response? [Management and Treatment]

A. Keep the secret as she asked, and schedule a follow-up visit in about one to two weeks

B. Telephone her parents at once without first telling her what you plan to say to them

C. Promise secrecy in return for her agreement not to act on the plan tonight

D. Tell her you must involve her parents and arrange urgent mental health evaluation

E. Inform only her school counselor and take no further steps

Answer Contents

BQ-534. A 16-year-old girl is starting hormonal contraception under her state's minor consent law. Her mother has a proxy portal account that shows visit notes and results, and insurance statements are mailed to the family home. The girl wants the visit kept private. Which is the best next step? [Management and Treatment]

A. Review what the proxy portal shows and ask how she wants results and bills handled

B. Advise her to avoid using the patient portal and all insurance billing systems entirely

C. Document the visit fully in a note that the proxy can view

D. Inform her mother because the portal would reveal the visit

E. Cancel the prescription because privacy cannot be protected

Answer Contents

BQ-535. A 9-year-old girl has severe neurologic impairment after an anoxic injury, with recurrent aspiration pneumonia and frequent admissions. The team recommends a palliative care consultation. Her parents decline, saying, "Palliative means you are giving up on her." Which is the best response? [Management and Treatment]

A. Respect the refusal and revisit the topic only if she later enters hospice care at home or in a facility

B. Tell them it is limited to children who are expected to die soon

C. Insist on the consultation because the team requires it

D. Explain that palliative care adds symptom control and family support alongside treatment

E. Offer hospice enrollment instead of the consultation

Answer Contents

BQ-536. A 4-month-old infant is ventilated after a cardiac arrest with severe hypoxic-ischemic brain injury, and the prognosis is uncertain. His parents hesitate to continue ventilation after the first week because "once it starts it can never be stopped." Which statement is best? [Management and Treatment]

A. Once ventilation has been started it cannot ethically be stopped for any reason or prognosis

B. The parents must choose between full treatment and comfort care today

C. A time-limited trial is acceptable, as withdrawing is ethically equivalent to not starting

D. Only a court may allow withdrawal of ventilation

E. Comfort care ends when ventilation is withdrawn

Answer Contents

BQ-537. Parents of a 3-year-old with severe chronic lung disease want a tracheostomy and long-term ventilation. The team believes the burdens outweigh the benefits. After three family meetings, discussion has stalled and both sides are frustrated. Which is the best next step? [Management and Treatment]

A. Proceed with the team's preferred plan without further talks

B. Transfer care to another hospital without discussing it

C. Agree to the parents' request and end further discussion

D. Seek a court order before any other step

E. Request an ethics service consultation

Answer Contents

BQ-538. A 14-year-old boy with relapsed leukemia and his parents agree to a do-not-resuscitate order. His mother asks, "Does this mean you will stop treating him?" He is on his third chemotherapy regimen, is awake and alert, and his father is anxious about stopping care. Which is the best explanation? [Management and Treatment]

A. It covers resuscitation only; other care continues and the plan can be revisited

B. It ends all treatment aimed at the leukemia and moves him to comfort care only from tonight

C. It is permanent and cannot be changed

D. It moves his care to hospice automatically

E. It stops pain medicine so that he can be kept alert

Answer Contents

BQ-539. A healthy 12-year-old boy is invited to join a study of a new asthma drug that requires two extra blood draws and offers him no direct benefit. His parents consent. He says he does not want to take part. Which is the best next step? [Management and Treatment]

A. Enroll him because the parents have given permission

B. Offer a gift card to persuade him to agree

C. Enroll him after giving a mild sedative before the draws

D. Honor his refusal and do not enroll him

E. Ask the review board to override his refusal

Answer Contents

Domain 24. Patient Safety, Quality Improvement, and Research Methods

Contents Answers for this subject

BQ-540. A pharmacist reviewing a morning order for gentamicin in a 3-year-old notices a dose ten times the intended amount caused by a misplaced decimal point. She contacts the prescriber, and the order is corrected before the drug is dispensed. Which term best classifies this event? [Epidemiology and Risk Assessment]

A. Preventable adverse event

B. Near miss

C. Sentinel event

D. Adverse drug reaction

E. Unavoidable complication

Answer Contents

BQ-541. A 6-year-old boy had surgery on the wrong side, and the hospital classifies the event as sentinel. The family has been told. Leadership meets the same day and asks how to find out why it happened. Which is the best next step? [Management and Treatment]

A. Perform a root cause analysis of system and process failures

B. Dismiss the surgeon at once pending an investigation of the event

C. Review only the surgeon's training records

D. Send all staff a reminder email to be more careful

E. Wait for the family to file a complaint

Answer Contents

BQ-542. A pediatric hospital finds that preventable adverse events cluster around shift change, when information about sick patients is lost or misunderstood, and the safety committee has asked for a plan. Leadership wants an intervention with the best evidence for reducing medical errors. Which is the best choice? [Management and Treatment]

A. A reminder email telling clinicians to be more careful

B. Longer shifts to reduce the number of handoffs per day

C. Written reprimands for clinicians who omit information at handoff

D. Handoffs delegated to the least experienced clinician

E. A standardized handoff format combined with team training

Answer Contents

BQ-543. A hospital medicine team plans to test a new discharge checklist on its own unit to improve timely follow-up appointments, and hopes to present the results. The members are unsure whether institutional review board oversight is required. Which is the best next step? [Management and Treatment]

A. Start the project without telling anyone

B. Assume it is research because they hope to present it

C. Ask the institutional review board to determine whether review is required

D. Assume it is quality improvement and therefore that no approval of any kind is needed

E. Obtain written consent from every family before any change

Answer Contents

BQ-544. A team aims to shorten the time to first antibiotic for children with suspected sepsis in the emergency department. The process measure is minutes to the first dose, and the team has agreed on an aim and a baseline. Which is the most suitable balancing measure? [Management and Treatment]

A. The proportion given antibiotics who prove to have no bacterial infection

B. Median minutes from arrival to the first antibiotic dose

C. Thirty-day mortality among children with sepsis

D. The proportion of children with a completed sepsis screening checklist at triage

E. The proportion of nurses trained in the new pathway

Answer Contents

BQ-545. After a clinic changes how it hands out written asthma action plans, a run chart of the monthly percentage of children with a plan shows nine consecutive points over the baseline median, beginning the month after the change. Which is the most appropriate interpretation? [Diagnosis]

A. Random variation around the baseline median

B. A measurement error in data collection

C. A shift caused by seasonal change alone

D. A sustained shift that is unlikely to be random variation

E. A result that is meaningful only after a randomized trial

Answer Contents

BQ-546. Researchers want to study whether prenatal exposure to a medication is linked to a rare birth defect that occurs in about 1 in 20,000 births. Resources are limited, and a randomized trial would be unethical. Which study design is most efficient for this question? [Epidemiology and Risk Assessment]

A. Randomized controlled trial

B. Prospective cohort study

C. Case-control study

D. Cross-sectional survey

E. Case series

Answer Contents

BQ-547. In a case-control study of mothers of infants with limb defects and mothers of healthy infants, interviews show that case mothers report more medication use in pregnancy. Medical records show equal prescribing in both groups. Which bias most likely explains this finding? [Epidemiology and Risk Assessment]

A. Lead-time bias

B. Confounding by maternal age

C. Selection bias in choosing controls

D. Publication bias

E. Recall bias

Answer Contents

BQ-548. A new screening test is evaluated in 200 children. Of 50 children with the condition, 45 test positive. Of 150 children without the condition, 20 test positive. The investigators want to report how well it rules in disease. What is the specificity of the test? [Epidemiology and Risk Assessment]

A. 90 percent

B. 87 percent

C. 69 percent

D. 96 percent

E. 13 percent

Answer Contents

BQ-549. In a randomized trial, 12 percent of control children and 8 percent of children given a new preventive intervention were hospitalized during follow-up of one year, and the two groups were otherwise similar. How many children must be treated with the intervention to prevent one hospitalization? [Epidemiology and Risk Assessment]

A. 4

B. 12

C. 33

D. 50

E. 25

Answer Contents

BQ-550. A randomized trial of a new drug reports a relative risk of the primary outcome of 0.80, with a 95 percent confidence interval of 0.60 to 1.10. The primary outcome is clinically important and the sample size was modest. Which is the best interpretation of this result? [Epidemiology and Risk Assessment]

A. The drug significantly reduces the outcome by 20 percent

B. The data show no clear difference, because the interval includes 1

C. The drug increases the outcome by 10 percent

D. There is a 95 percent probability that the drug truly works in all children

E. The study made a type I error

Answer Contents

Answers and Explanations

Each explanation names the mechanism, why the correct option is right, the trap in the closest wrong option, and what to do at the bedside.

Back to Contents

Answers: Domain 1. Preventative Pediatrics/Well-Child Care

Contents Questions for this subject

BQ-001. Answer: A. Give a standardized general developmental screening tool at this visit
Surveillance is a continuous process that depends on the examiner and the length of the visit, so it can miss a quiet or typical-looking child. A validated general screen is due at 9, 18, and 30 months whatever the parent says, and this is the 30-month visit. The closest trap is the milestone checklist, which supports surveillance but is not a validated tool. Autism screening belongs at 18 and 24 months. At the bedside, give a parent-completed questionnaire today, score it, and start a referral if the result falls in the referral range.
Key principle: A validated general developmental screen is due at 30 months even when surveillance is reassuring.
Back to question Contents

BQ-002. Answer: B. Repeat the autism-specific screening tool at this visit
Autism signs can emerge after 18 months, and a single negative screen cannot cover a child whose differences appear later. For this reason the AAP schedules autism-specific screening at both 18 and 24 months. The trap is trusting the earlier result or the absence of parent concern, since parents often miss subtle social patterns in a short visit. At the bedside, give the Modified Checklist for Autism in Toddlers, Revised with Follow-Up today, and if it is positive, arrange diagnostic evaluation and early intervention the same week.
Key principle: Autism-specific screening is repeated at 24 months even after a negative 18-month result.
Back to question Contents

BQ-003. Answer: D. At least 75 percent of children are expected to show the skill by this age
The revised checklists place each milestone at the age when most children, at least 75 percent, can do it. Earlier lists used the median age, which meant half of healthy children had not yet reached the skill and invited waiting. The mother is describing that older standard, and it is the closest trap. A miss now means most children her age can do what this girl cannot. At the bedside, examine tone, strength, reflexes, and hips, give a screening tool, and arrange early intervention rather than waiting.
Key principle: A missed 2022 milestone means that most children of that age have the skill, so look harder rather than wait.
Back to question Contents

BQ-004. Answer: B. Neurologic maturation follows time since conception more closely than time since birth
Brain maturation proceeds with gestational age, so an infant born 10 weeks early has had about 2.5 months less time to mature than the calendar suggests. His chronologic age is 6 months, and subtracting the weeks of prematurity gives a corrected age near 3.5 months, at which sitting with support is expected. The trap is judging him against term peers, which labels a healthy infant as delayed. At the bedside, plot growth and milestones at corrected age and keep surveillance going, since a true delay still appears on that scale.
Key principle: Judge growth and development of a preterm infant at corrected age.
Back to question Contents

BQ-005. Answer: D. 64 inches

A child's growth channel follows the parents' heights, so the target gives a reference for judging a short child. For a girl, subtract 5 inches from the father's height and average the result with the mother's: 70 minus 5 is 65, and 65 plus 63 divided by 2 is 64 inches. The closest trap is 66.5 inches, which averages the parents without the sex adjustment. At the bedside, compare her percentile with this target and follow her velocity; a normal rate with a low target points to a familial pattern.

Key principle: For girls, average the mother's height with the father's height minus 5 inches.

Back to question Contents

BQ-006. Answer: B. Familial short stature

Children inherit a growth channel from their parents, so a child of short parents who tracks a low percentile at a normal rate fits familial short stature. The target height of 65 inches is low, and his bone age matches his age, so he will not have extra time to catch up. Constitutional delay is the closest trap, but it brings a delayed bone age and a late-maturing parent. At the bedside, calculate the target, plot the growth rate every 6 months, and test further only if his velocity falls.

Key principle: Familial short stature combines a low target height, a normal growth rate, and a bone age that matches age.

Back to question Contents

BQ-007. Answer: C. A mass in the hypothalamic or pituitary region

A mass near the hypothalamus and pituitary can disturb growth hormone output and press on the optic pathways, which explains slowed growth together with headaches and vision change. Familial and constitutional patterns keep a steady growth rate, and neither causes neurologic symptoms. Hypothyroidism and celiac disease slow growth, but they do not produce visual change. Treating or waiting on first-line tests is the trap, since the central cause matters most. At the bedside, document the fall across percentiles, arrange urgent specialist evaluation, and tell the family the date.

Key principle: Poor growth with headaches or vision change suggests a central cause and needs urgent specialist referral.

Back to question Contents

BQ-008. Answer: D. Start intensive health behavior and lifestyle treatment for the whole family now

A BMI at or over the 95th percentile defines obesity, which rarely resolves without treatment. The 2023 AAP guideline therefore advises treating now, with intensive health behavior and lifestyle treatment for the whole family, at least 26 hours of face-to-face contact over 3 to 12 months. Waiting a year is the closest trap, because nothing supports watchful waiting. Medication starts at 12 years and surgery evaluation at 13. At the bedside, ask permission to discuss weight, use non-blaming words, and enroll the family in a program.

Key principle: Obesity in childhood calls for prompt, intensive, family-based lifestyle treatment with no watchful waiting.

Back to question Contents

BQ-009. Answer: A. Severe obesity

In children 2 to 19 years, obesity begins at the 95th percentile, and severe obesity is a BMI of 120 percent of the 95th percentile or more, or 35 kg/m2 or more. For this boy, 120 percent of 24.5 is 29.4, and his BMI of 31 is over it. The trap is calling him simply obese because he is over the 95th percentile, which misses the second threshold. At the bedside, begin intensive family-based treatment, screen for comorbid conditions, and note that surgery evaluation applies from age 13.

Key principle: Severe obesity is a BMI of at least 120 percent of the 95th percentile or at least 35 kg/m2.

Back to question Contents

BQ-010. Answer: D. Severe malnutrition with suspected neglect or abusive injury

A weight under 70 percent of the expected weight for length defines severe malnutrition, and bruises of different ages with a guarded history suggest neglect or injury. Children with failure to thrive are more often abused, and poverty is the largest single risk factor. Organic disease is the trap, since laboratory panels find a cause in under 1 percent and nothing here points to one. At the bedside, admit her for observed feeding, examine her undressed, and report the concern to child protective services as the law requires.

Key principle: Severe malnutrition, failed outpatient care, or suspected neglect or injury calls for admission.

Back to question Contents

BQ-011. Answer: E. Inadequate calories from diluted formula

Too little food reaching the child is the usual cause of failure to thrive, and over 80 percent of affected children have no medical disease. Formula mixing errors and food insecurity are common routes. Here the dilution lowers energy intake, and the pattern of preserved length and head size fits recent undernutrition. The trap is chasing a rare organic disease, when the history already gives the cause. At the bedside, watch a bottle being made, correct the mixing, and connect the family with food assistance.

Key principle: In failure to thrive, the feeding history usually reveals the cause before any test does.

Back to question Contents

BQ-012. Answer: B. The older chart used a reference population that grew faster than healthy breastfed infants

The WHO 2006 standards describe how breastfed infants grow under good conditions, while the older CDC reference included many formula-fed infants who gain faster after the first months. A healthy breastfed infant can therefore seem to falter on the older chart while following her own normal channel. The trap is blaming her milk supply and adding formula. At the bedside, replot every measurement on the WHO standards for children under 2 years, reassure the mother, and weigh her again at the next visit.

Key principle: A breastfed infant who seems to falter on the older CDC curve is often following the WHO curve.

Back to question Contents

BQ-013. Answer: A. Central precocious puberty

Puberty before age 8 in a girl is precocious, and the combination of rapid progression and accelerated height points to activation of the hypothalamic-pituitary-gonadal axis. Isolated premature thelarche tends to stay static or regress slowly and does not accelerate growth. Adrenarche gives pubic hair and odor without breast tissue. At the bedside, stage the breast, plot the growth line, obtain a bone age, and arrange endocrine referral, and ask about headaches and vision since a central lesion can drive early puberty.

Key principle: Breast development before age 8 with rapid progression and growth acceleration suggests central precocious puberty.

Back to question Contents

BQ-014. Answer: B. Early adrenal androgen secretion without activation of the gonadal axis

Adrenarche is the rise in adrenal androgens, and it runs on its own schedule, separate from the gonadal axis. It produces pubic hair and body odor without testicular enlargement. His testes are well under the 4 mL that marks true puberty, and his growth is steady, which argues against a gonadal source. The trap is early activation of the gonadal axis, which would enlarge the testes. At the bedside, measure testicular volume, plot growth, and follow him, with bone age if growth quickens.

Key principle: Pubic hair alone does not mean puberty has begun, since adrenarche is separate from the gonadal axis.

Back to question Contents

BQ-015. Answer: A. Delayed puberty

Delayed puberty in a girl means no breast development by age 13, or no menarche by 16, or no progression over 3 years once puberty starts. She is past 13 with no thelarche, so she meets the definition. A mother who matured late suggests a constitutional pattern, but that is a cause and not a definition. Arrest would require a start that then stalls. At the bedside, check the growth line, bone age, and family history, reassure her honestly, and follow her closely.

Key principle: No breast development by age 13 in a girl defines delayed puberty.

Back to question Contents

BQ-016. Answer: B. Genital Tanner stage 2, the first sign of puberty

In boys, the first true sign of puberty is testicular enlargement to a volume of 4 mL or more, and genital stage 2 adds reddened scrotal skin. Pubic hair can lag behind because it reflects adrenal androgens and not the testes. At 12, he sits well within the normal 9 to 14 age window. The trap is waiting for pubic hair or penile growth as proof. At the bedside, record genital and pubic hair ratings separately, reassure his mother, and review the growth line at the next visit.

Key principle: Testicular volume of 4 mL or more is the first sign of puberty in boys.

Back to question Contents

BQ-017. Answer: C. Stage 3

Tanner staging rates breast and pubic hair separately from stage 1, prepubertal, to stage 5, adult. In stage 3 the breast and areola enlarge as one mass, and pubic hair becomes darker, coarser, and curled. Stage 4 brings a secondary mound formed by the areola, and stage 2 is only a breast bud. Her findings fit stage 3 in both ratings. The trap is stage 4, chosen because menarche usually follows soon after. At the bedside, inspect and palpate, record both ratings, and describe puberty to her before the next change arrives.

Key principle: Tanner stage 3 shows breast and areola enlarging as one mass with darker, curled pubic hair.

Back to question Contents

BQ-018. Answer: B. Regression, a loss of motor skills the child had already gained

He once ran and climbed, and now he cannot. A loss of skills, especially with weakness on one side, is regression and is never benign. It can reflect a mass, a stroke, a spinal cord problem, or neurologic disease, and the cause cannot be known at the visit. The closest trap is a delay that therapy alone can fix, since a delay is a failure to gain skills and not a loss. At the bedside, examine tone, strength, reflexes, and head size, arrange evaluation the same day, and give the family a date.

Key principle: Loss of skills is regression, and it changes the plan at that visit with no waiting.

Back to question Contents

BQ-019. Answer: C. Birth weight of 2,500 g or less

Risk factors for speech delay include male sex, sustained hearing problems, and a birth weight of 2,500 g or less. Bilingual homes and birth order do not raise the risk, and female sex is protective rather than harmful. Her weight of 2,300 g therefore stands out. The trap is blaming two languages, which delays referral and discourages the family. At the bedside, test hearing, refer for speech-language evaluation and early intervention together, and tell the family to keep speaking the language they know best.

Key principle: Low birth weight, male sex, and hearing loss raise the risk of speech delay, while bilingualism does not.

Back to question Contents

BQ-020. Answer: E. Impaired circulation or absorption of cerebrospinal fluid, raising intracranial pressure

A head that crosses percentile lines quickly, with a tense fontanelle, vomiting, and downward deviation of the eyes, points to hydrocephalus, in which obstructed flow or reduced absorption of cerebrospinal fluid raises pressure and enlarges the skull. Familial macrocephaly follows a stable channel, and dehydration sinks the fontanelle. The trap is catch-up growth, which does not cause vomiting or eye signs. At the bedside, arrange neurologic evaluation that day with imaging, and keep the infant under observation.

Key principle: Rapid head growth with a bulging fontanelle and vomiting needs a neurologic evaluation the same day.

Back to question Contents

BQ-021. Answer: A. Continue surveillance today and give the next general screen at 18 months

Surveillance runs at every visit, whereas validated screening is tied to fixed ages: 9, 18, and 30 months for the general tool and 18 and 24 months for autism. At 12 months no tool is due, and a completed checklist with agreeing observation supports a reassuring result. The trap is adding a tool at every visit, which costs time without the evidence behind the fixed schedule. At the bedside, document the surveillance, ask again about parent concerns, and book the 18-month visit with both tools planned.

Key principle: Between the fixed screening ages, surveillance alone carries developmental monitoring at each visit.

Back to question Contents

BQ-022. Answer: E. Arrange an early intervention evaluation now, with planning for preschool special education at age 3

Federal law ensures early intervention services from birth to 36 months and preschool special education after that, and neither requires a confirmed diagnosis. A failed screen begins an evaluation and does not end it. Waiting for a label or a repeat screen wastes the months that remain in the early window. At the bedside, send the referral today, ask the program to plan the transition to the school district, and give the family a date for the evaluation.

Key principle: Refer for early intervention without waiting for a diagnosis, and plan the move to preschool special education at 3.

Back to question Contents

BQ-023. Answer: A. Obtain a venous blood lead level within 48 hours and consult a toxicology expert

Lead interferes with heme synthesis and neural development, and the speed of confirmation rises with the level. A capillary result of 45 µg/dL or more calls for a venous sample within 48 hours, and values this high need a toxicologist for decisions about chelation. Waiting weeks treats the number as routine and leaves him exposed. At the bedside, send the venous sample today, stop his access to the renovation dust, and arrange the toxicology consultation before the result returns.

Key principle: A capillary lead level of 45 µg/dL or more needs venous confirmation within 48 hours.

Back to question Contents

BQ-024. Answer: E. It marks the 97.5th percentile of children's levels and is a trigger for action, not proof of safety

The reference value is a population measure: in 2021 the CDC set it at the 97.5th percentile of blood lead levels in children 1 to 5 years. It identifies children with higher exposure than most and prompts action, but no level of lead is known to be harmless. Treating it as a safe threshold is the trap, and it leaves a child in a leaded environment. Chelation is reserved for very high levels. At the bedside, identify and remove the paint hazard, clean dust, check nutrition, and repeat testing as advised.

Key principle: The blood lead reference value is a trigger for action, not a safe level.

Back to question Contents

BQ-025. Answer: A. Obtain a blood lead level now because she was never tested

Children enrolled in Medicaid are tested at 12 and 24 months, or between 24 and 72 months if they were never tested. This girl falls in that catch-up window. Lead exposure is usually silent, so waiting for symptoms means waiting for injury. A hemoglobin cannot exclude it, since lead anemia appears late and mild. The trap is believing the missed ages closed the opportunity. At the bedside, send a blood lead level today, ask about housing age and renovation, and act on the result by its level.

Key principle: A Medicaid-enrolled child who was never tested is screened for lead between 24 and 72 months.

Back to question Contents

BQ-026. Answer: C. A response that confirms iron deficiency anemia

Iron deficiency leaves red cell production short of its raw material, and replacing iron lets the marrow catch up. A rise of about 1 g/dL after a month of iron confirms the diagnosis, and her rise of 1.3 g/dL fits. The low ferritin with a normal C-reactive protein already pointed to deficiency. Inflammation raises ferritin, so it does not explain her low value. At the bedside, continue iron by weight until stores recover, keep milk to a modest amount, and recheck later.

Key principle: A hemoglobin rise of about 1 g/dL after a month of iron confirms iron deficiency anemia.

Back to question Contents

BQ-027. Answer: A. Repeat a fasting lipid panel and average the results

A non-fasting non-HDL cholesterol is an acceptable first test, and a value of 145 mg/dL or more, or an LDL cholesterol of 130 mg/dL or more, calls for repeat testing. Because a single result varies, a fasting panel is repeated and the results are averaged before the child is labeled. Drug treatment is generally held until age 10 unless the disorder is severe, and lifestyle changes come first. The trap is starting a statin on one number. At the bedside, order the fasting panel and review diet and activity.

Key principle: A non-HDL cholesterol of 145 mg/dL or more is repeated as a fasting panel and averaged.

Back to question Contents

BQ-028. Answer: B. Perform the next universal lipid screen between 17 and 21 years

Universal lipid screening happens once at 9 to 11 years and once at 17 to 21 years. Between 12 and 16, testing is selective and depends on risk factors such as family history, diabetes, hypertension, obesity, or smoking. She has none, so the interval needs no testing. The trap is treating age alone as the indication, which adds cost and false positives. At the bedside, ask again about family history and risk factors, and plan the second screen at a visit between 17 and 21 years.

Key principle: Between 12 and 16 years, lipids are tested only when risk factors are present.

Back to question Contents

BQ-029. Answer: A. Begin routine yearly measurement at age 3 unless a risk condition develops
The 2017 AAP guideline asks for blood pressure at every yearly visit from age 3. Younger children are measured only when they carry a risk condition, such as prematurity, congenital kidney disease, or a transplant. A healthy term toddler has none, so yearly screening starts one year from now. The trap is measuring at every visit, which has no benefit in a low-risk child. At the bedside, reassure the mother, note the age, and plan the first measurement at the 3-year visit.
Key principle: Blood pressure is measured yearly from age 3, and earlier only in children with risk conditions.
Back to question Contents

BQ-030. Answer: B. In the stage 1 hypertension range
From age 13, blood pressure is classified by fixed values: normal under 120/80, elevated 120/under 80 to 129/under 80, stage 1 at 130/80 to 139/89, and stage 2 at 140/90 or more. His average of 134/82 falls in the stage 1 range. The trap is calling it elevated, which applies only to readings under 130 systolic with a diastolic under 80. At the bedside, confirm on repeat visits, advise lifestyle changes, and start the evaluation for causes if the range persists.
Key principle: In adolescents 13 and older, 130/80 to 139/89 mmHg is the stage 1 range.
Back to question Contents

BQ-031. Answer: D. Send her to the emergency department for urgent blood pressure treatment
A pressure over 180/120 mmHg in an adolescent, especially with headache and visual symptoms, signals hypertensive emergency with a risk of end-organ injury. These are the findings that make the usual stepwise confirmation unsafe. Ambulatory monitoring and home readings are tools for diagnosis in stable patients and cannot be allowed to delay care. At the bedside, call for emergency transport, avoid sending her home, and give the receiving team the readings and symptoms.
Key principle: A blood pressure over 180/120 mmHg in an adolescent is an emergency.
Back to question Contents

BQ-032. Answer: E. Renal ultrasonography
In children, confirmed hypertension is often secondary to kidney or vascular disease, and the younger the child, the more likely this is. For this reason the evaluation includes a history, an examination with arm and leg pressures, urinalysis, creatinine, electrolytes, and a lipid panel, and renal ultrasonography under age 6. Computed tomography exposes her to radiation without answering the first question. At the bedside, order the ultrasound study, confirm with ambulatory monitoring, and involve nephrology.
Key principle: Children with confirmed hypertension under age 6 need renal ultrasonography.
Back to question Contents

BQ-033. Answer: B. Arrange diagnostic audiology now so that diagnosis is complete by 3 months of age
The 1-3-6 benchmark of the Joint Committee on Infant Hearing sets screening by 1 month, diagnosis by 3 months, and intervention by 6 months. Language develops fastest in infancy, and unaddressed loss costs those months. Waiting for speech delay is the trap, since that signal arrives after the window has closed. Hearing aids need confirmed thresholds first. At the bedside, schedule pediatric audiology, request early intervention services, and tell the family the dates.
Key principle: Newborn hearing follows 1-3-6: screen by 1 month, diagnose by 3, and intervene by 6.
Back to question Contents

BQ-034. Answer: C. Hold him from sports and arrange a cardiology evaluation

Exertional syncope or chest pain, and a relative who died suddenly at a young age, are the cardiac arrest risk questions asked at yearly visits from 11 to 21 years. A positive answer points to an inherited arrhythmia or cardiomyopathy that can cause sudden death during exertion. The trap is reassurance based on a normal examination, since many such conditions have a normal examination. At the bedside, stop sports participation, arrange cardiology evaluation promptly, and explain that clearance depends on that assessment.

Key principle: Exertional fainting or chest pain, or a relative's sudden death at a young age, needs cardiac evaluation before sports.

Back to question Contents

BQ-035. Answer: D. An abnormal red reflex needing prompt ophthalmology evaluation

A normal red reflex means light passes through a clear lens and vitreous to a healthy retina. A white, dull, or absent reflex can come from cataract, retinoblastoma, or other disease that blocks the path, and delay risks both vision and life. The trap is attributing the finding to the camera or to a minor eye infection. At the bedside, send her to a pediatric ophthalmologist promptly, tell the family why a white reflex is never a watch-and-wait finding, and give them a date.

Key principle: A white or abnormal red reflex always needs prompt pediatric ophthalmology evaluation.

Back to question Contents

BQ-036. Answer: E. The developing visual cortex suppresses input from the misaligned eye, leaving it with poor acuity

During early childhood the visual cortex is still learning, and when the two eyes send conflicting images it favors the straight eye and suppresses the other. The suppressed eye develops poor acuity, called amblyopia, even though the eye itself is structurally normal. Treatment works only while the visual system is still plastic, which is why screening starts early. The trap is a structural explanation in the eye. At the bedside, refer him to pediatric ophthalmology promptly and ask about family history of amblyopia.

Key principle: Amblyopia results from cortical suppression in a developing visual system, so screening aims to find it while treatment works.

Back to question Contents

BQ-037. Answer: A. Bacteria on the tooth surface ferment sugar to acid that demineralizes enamel

Dental decay begins when bacteria in plaque ferment sugars and produce acid, and frequent exposure keeps the pH low so that enamel loses mineral. Juice sipped through the day and milk at bedtime supply constant sugar, and chalky white bands are the first stage of that demineralization. The trap is fluorosis, which appears as mottling and not as decay at the gumline. At the bedside, stop the bedtime bottle and the constant sippy cup, brush with fluoride toothpaste, apply fluoride varnish, and book a dental visit.

Key principle: Frequent sugar exposure feeds plaque bacteria that make acid and demineralize new teeth.

Back to question Contents

BQ-038. Answer: B. Screen for hepatitis B, hepatitis C, HIV, tuberculosis, and intestinal parasites, and verify immunizations

Children adopted internationally come from settings with a higher burden of hepatitis B, hepatitis C, HIV, tuberculosis, and intestinal parasites, and an immunization record from abroad cannot be trusted until it is checked. Waiting for symptoms or relying on the paper record leaves infections undetected, and some are silent for years. At the bedside, order the infectious screen, treat the immunization record as unreliable, check serologies or restart vaccines as needed, and review development and growth.

Key principle: After international adoption, screen for hepatitis B and C, HIV, tuberculosis, and parasites, and treat vaccine records as unreliable.

Back to question Contents

BQ-039. Answer: E. Use a belt-positioning booster seat until the belt fits, usually near 4 feet 9 inches

Car seat stages follow size, not birthday. When a child outgrows a harness seat, a belt-positioning booster lifts him so the lap belt sits over the hips and the shoulder belt crosses the chest. A belt that crosses the neck and abdomen can injure the spine and abdominal organs in a crash. The belt generally fits at about 4 feet 9 inches, often at 8 to 12 years. At the bedside, recommend a booster, keep him in the back seat until 13, and recheck belt fit.

Key principle: Use a booster until the vehicle belt fits, and keep the child in the back seat until 13.

Back to question Contents

BQ-040. Answer: A. Stop swaddling now because she is starting to roll, since swaddling does not lower the risk

Swaddling does not reduce the risk of sudden unexpected infant death, and a swaddled infant who rolls onto the stomach cannot easily use her arms to turn back. Tight wrapping and positioners add risk without protection, and bed sharing is itself a risk. The AAP advises stopping the swaddle as soon as the baby shows signs of rolling. At the bedside, praise the flat crib in the parents' room, advise a wearable blanket or sleep sack, and keep the crib empty of soft items.

Key principle: Stop swaddling when the baby starts to roll, and keep the sleep space flat, firm, and empty.

Back to question Contents

BQ-041. Answer: E. Evaluate for a new treatable problem, such as dental or gastrointestinal disease

Children with special health care needs can have a second problem, and blaming a new symptom on the known condition is called diagnostic overshadowing. Pain on touching her cheek and crying with meals point to a dental or oral source, which is common and treatable in children with cerebral palsy. Assuming progression is the trap. A feeding tube would address calories and not pain. At the bedside, examine her mouth and teeth, look for reflux and constipation, and treat the cause found.

Key principle: Diagnostic overshadowing blames a new symptom on the known condition, so look for a second problem.

Back to question Contents

BQ-042. Answer: D. Typical, since median daytime dryness is reached at about 35 months in boys

Most children become ready to train between 18 and 30 months, and median daytime dryness arrives at about 32 months in girls and 35 in boys. A 3-year-old with accidents therefore sits within the ordinary range, and evaluation is reserved for a child who is not dry by day at 5 years. The trap is reading the calendar as the standard. At the bedside, reassure the parents, advise a child-oriented approach with rewards for success, and tell them never to punish accidents.

Key principle: Daytime accidents at age 3 are ordinary, and evaluation waits until a child is not dry by day at 5.

Back to question Contents

BQ-043. Answer: E. Withholding stool because of painful, firm stools

Stool refusal is common during toilet training. A firm stool hurts, the child learns to avoid it, and holding back makes the stool harder, which sustains the cycle. Large firm stools support this explanation and point to constipation as the driver. Organic disease is the trap, and nothing in this history suggests it. At the bedside, treat the constipation, pause training for a few weeks, avoid pressure or punishment, and resume when she shows readiness again.

Key principle: Stool refusal during toilet training is usually withholding from constipation, so treat the constipation and pause training.

Back to question Contents

BQ-044. Answer: B. Screen the mother with a validated depression tool at this visit and arrange follow-up

A struggling parent affects care long before the infant shows it, so the schedule calls for maternal depression screening with a validated tool at the 1-, 2-, 4-, and 6-month visits. Symptoms lasting beyond the first weeks are no longer the baby blues. The trap is treating the infant as the only patient, which leaves a treatable illness unaddressed. At the bedside, give the screening tool, respond kindly to the result, connect her to her own clinician or a mental health service, and plan to ask again at the next visit.

Key principle: Screen mothers for depression with a validated tool at the 1-, 2-, 4-, and 6-month infant visits.

Back to question Contents

BQ-045. Answer: A. Give hepatitis B vaccine together with hepatitis B immune globulin within 12 hours of birth

Perinatal hepatitis B infection becomes chronic in most infected newborns, and vaccine plus immune globulin prevents it when given early. For an infant of a surface antigen-positive mother, both products are given within 12 hours of birth at any weight. The 2,000 g rule applies to infants of negative mothers. Waiting for weight or for test results is the trap. At the bedside, give both products now in separate sites, record the doses, complete the series on schedule, and test for response later.

Key principle: An infant of a hepatitis B surface antigen-positive mother receives vaccine and immune globulin within 12 hours at any weight.

Back to question Contents

BQ-046. Answer: C. Invalid, because it was given before 24 weeks of age

The hepatitis B series has three limits: dose 2 at least 4 weeks after dose 1, dose 3 at least 8 weeks after dose 2 and 16 weeks after dose 1, and the final dose not before 24 weeks of age. Meeting the intervals is not enough, because the final dose given at 18 weeks falls short of the minimum age. A series is never restarted, so only the final dose is repeated. At the bedside, give a dose now or at 24 weeks as the catch-up rules direct, and document why.

Key principle: The final hepatitis B dose must be given at 24 weeks of age or older, and an invalid dose is repeated without restarting.

Back to question Contents

BQ-047. Answer: A. Withhold the rotavirus vaccine because it is contraindicated in this condition

Severe combined immunodeficiency is a contraindication to rotavirus vaccine because the weakened live virus can cause prolonged infection in a child who cannot clear it. Contraindications are conditions that raise the chance of a serious reaction, and the vaccine should not be given. The trap is withholding everything, which is not required for the inactivated vaccines, though immunology guidance should shape the plan. At the bedside, involve his immunologist, give no live vaccines, and record the contraindication in the chart.

Key principle: Severe combined immunodeficiency is a contraindication to rotavirus vaccine and to other live vaccines.

Back to question Contents

BQ-048. Answer: B. Give one dose today and no further doses

Catch-up schedules give fewer doses to older children because the disease risk falls with age. If the first Hib dose is given at 15 months or older, no further doses are needed. She is 16 months old, so a single dose completes her series. A series is never restarted, and no one is too old for a first dose at this age. The trap is a multi-dose plan that wastes visits and injections. At the bedside, give the dose, document it, and address her other missing vaccines on the same day.

Key principle: A first Hib dose at 15 months or older completes the series with no further doses.

Back to question Contents

BQ-049. Answer: C. The weakened vaccine viruses can replicate unchecked when immunity is badly impaired

Live attenuated vaccines work by replicating in the recipient, and a healthy immune system stops them after they teach it. In a child with severe immunodeficiency, the weakened virus can keep replicating and spread to organs and cause disease, which makes severe immunodeficiency a contraindication to these vaccines. Giving a partial dose or immune globulin does not remove the danger, and age is irrelevant. At the bedside, defer the live vaccines, ask the oncology team when it is safe, and make sure household contacts are fully vaccinated.

Key principle: Live vaccines are contraindicated in severe immunodeficiency because the weakened virus can replicate unchecked.

Back to question Contents

BQ-050. Answer: A. It is a precaution, since illness can blur a reaction's diagnosis and blunt the response

A precaution is a condition that might raise the risk of a serious reaction, blur the diagnosis, or blunt the response. Moderate or severe acute illness, with or without fever, fits that category, and a lethargic febrile child with pneumonia cannot be told apart from one who is reacting to a vaccine. Mild illness and antibiotics are not reasons to wait, and the response is blunted, not absent. At the bedside, treat the pneumonia, set the next visit date before the family leaves, and vaccinate when he has recovered.

Key principle: Moderate or severe acute illness is a precaution that defers vaccines until recovery.

Back to question Contents

BQ-051. Answer: B. Anaphylaxis to a vaccine

Anaphylaxis comes within minutes to an hour of a vaccine and involves skin, breathing, and circulation, so hives with wheeze and pallor fit. Mast cell mediators cause airway narrowing and vasodilation. Vasovagal syncope brings pallor and a slow pulse without hives or wheeze, and serum sickness appears days later. The trap is calling pallor a faint. At the bedside, give weight-based intramuscular epinephrine from the office supply at once, call emergency services, report to VAERS, and give no further doses of that vaccine.

Key principle: Hives with wheeze and pallor minutes after a vaccine is anaphylaxis, treated at once with intramuscular epinephrine.

Back to question Contents

BQ-052. Answer: E. It can signal events for further study but cannot give rates or prove causation

The Vaccine Adverse Event Reporting System is run jointly by CDC and FDA and accepts reports from anyone about any health problem after a vaccine. It is passive, so reports are incomplete, and without a count of doses given or of unvaccinated comparison children no rate can be calculated. Its value is detecting signals, which other systems, such as the Vaccine Safety Datalink, study. The trap is treating a report count as incidence. At the bedside, submit reports readily and explain that a report records a concern.

Key principle: A passive reporting system detects safety signals but cannot give rates or prove causation.

Back to question Contents

BQ-053. Answer: C. Document the refusal and the counseling, keep the relationship, and ask again at the next visit

Hesitant parents are not one group, and many move after a calm second conversation. Documented counseling, a continued relationship, and a repeated offer keep the door open, while dismissal is a last step, acceptable only after repeated documented counseling and compliance with state law on abandonment. A family that leaves rarely finds a clinician who discusses vaccines. The trap is giving up on the topic. At the bedside, record his reasons, ask what worries him, and invite him back with a firm recommendation.

Key principle: After a refusal, document counseling, keep the relationship, and ask again at the next visit.

Back to question Contents

BQ-054. Answer: D. Give measles, mumps, and rubella and varicella vaccines as separate injections

The combined vaccine is linked with a higher rate of fever and febrile seizure than separate injections in this age group, and a personal or family history of seizures is a precaution for it. For dose 1 at 12 to 47 months, separate injections are generally preferred. A family history of febrile seizures does not call for delay. The trap is skipping or postponing protection against measles and varicella. At the bedside, give both vaccines in separate sites, tell the father to expect low fever, and advise how to respond to a seizure.

Key principle: For dose 1 at 12 to 47 months, separate measles-mumps-rubella and varicella shots are preferred, especially with a family seizure history.

Back to question Contents

BQ-055. Answer: B. Stop human milk, start a galactose-free formula, and involve a metabolic specialist today

Human milk is the best food for most infants, but in classic galactosemia the infant cannot metabolize galactose, and the lactose in human milk and standard formula produces toxic metabolites that injure the liver, brain, and eyes. Human milk is unsafe for this infant. The trap is continuing breastfeeding in any form, or switching to a standard formula that also contains lactose. At the bedside, stop human milk, begin a galactose-free formula, and confirm the plan with a metabolic specialist.

Key principle: Classic galactosemia is a rare exception that makes human milk unsafe.

Back to question Contents

BQ-056. Answer: C. Give a daily vitamin D supplement, since she drinks less than about 27 ounces (800 mL) of formula

Infant formula is fortified so that an infant who drinks enough receives the daily vitamin D requirement. A formula-fed infant can skip the supplement only when she takes more than about 27 ounces (800 mL) of fortified formula a day. At 20 ounces she falls short. The trap is assuming fortification alone is enough regardless of volume. Missing vitamin D for months risks rickets. At the bedside, prescribe the standard daily supplement and recheck her intake at each visit, since the need ends once she drinks enough formula.

Key principle: Formula-fed infants need vitamin D until they drink more than about 27 ounces of fortified formula daily.

Back to question Contents

BQ-057. Answer: B. Vitamin D deficiency that impairs calcium and phosphate supply for mineralization of growing bone

Human milk carries little vitamin D, and darker skin or little sun lowers skin synthesis. Without vitamin D, the gut absorbs too little calcium and phosphate, so growing bone and cartilage fail to mineralize and soften, producing bowed legs, wide wrists, and a rosary at the costochondral junctions. Scurvy is the closest trap but brings bleeding gums and limb pain. At the bedside, confirm with alkaline phosphatase, vitamin D level, and wrist radiograph, treat by protocol, and then supplement.

Key principle: Rickets results from vitamin D deficiency, which impairs mineralization of growing bone.

Back to question Contents

BQ-058. Answer: A. Vitamin B12 deficiency

Vitamin B12 comes only from animal foods, so a mother who eats none passes little to her breastfed infant. B12 is needed for DNA synthesis in blood cells and for the nervous system, so deficiency causes macrocytic anemia, poor growth, and hypotonia. Iron deficiency is the closest trap, but it produces small red cells and does not explain the low tone. At the bedside, check a B12 level and metabolic markers, treat the infant by protocol, and counsel the mother on her diet and supplements.

Key principle: B12 deficiency in a breastfed infant of a mother who eats no animal products brings macrocytic anemia, poor growth, and hypotonia.

Back to question Contents

BQ-059. Answer: C. Vitamin C deficiency

Vitamin C is needed to make stable collagen, which supports blood vessels, gums, and the lining of bone. Without it, capillaries leak, gums bleed, and subperiosteal bleeding makes the limbs painful, so children refuse to walk. A very narrow diet with no fruit or vegetables sets the stage. Rickets and leukemia can cause bone pain, but the bleeding gums and diet history point to scurvy. At the bedside, confirm with a vitamin C level and radiographs if needed, give vitamin C, and involve a dietitian.

Key principle: Scurvy presents with limp, joint pain, and bleeding gums in a child with a very narrow diet.

Back to question Contents

BQ-060. Answer: D. Intestinal blood loss from cow's milk combined with displacement of iron-rich foods

Cow's milk before 12 months supplies more protein and minerals than an infant kidney handles easily, lacks adequate iron, and can cause small amounts of intestinal bleeding. At the same time, the large volume fills her stomach and displaces iron-rich foods. Over months, blood loss and low intake exhaust iron stores, and the marrow makes small red cells. The trap is lead, which can also cause small cells but does not explain occult blood. At the bedside, stop cow's milk, treat with iron by weight, and recheck.

Key principle: Cow's milk before 12 months causes anemia through intestinal blood loss and displacement of iron-rich foods.

Back to question Contents

BQ-061. Answer: B. Ingested spores germinate in the infant gut and release a toxin that blocks acetylcholine release

Honey can carry Clostridium botulinum spores. An infant's gut flora and acid are not yet able to prevent germination, so spores grow, make toxin, and block the release of acetylcholine at nerve endings, producing constipation followed by descending weakness. In older children and adults the illness more often follows preformed toxin, which is the trap here. At the bedside, hospitalize him with airway monitoring, give the specific treatment, and tell parents no honey before 12 months, even baked.

Key principle: Honey carries botulism spores, so none should be given before the first birthday.

Back to question Contents

BQ-062. Answer: E. Offer age-appropriate peanut, such as thinned smooth peanut butter, at home around 6 months without testing

Early peanut eating prevents peanut allergy, and the NIAID addendum sorts infants into three risk tiers. Severe eczema or egg allergy calls for testing first, while mild to moderate eczema allows introduction at home around 6 months with no test. She has mild to moderate disease, so extra testing delays a protective step. Whole peanuts are a choking hazard. At the bedside, show the parents the form to use, tell them how to recognize hives or vomiting, and advise continued regular exposure.

Key principle: An infant with mild to moderate eczema can start peanut at home around 6 months without testing.

Back to question Contents

BQ-063. Answer: B. 9 infants

The absolute risk reduction is the difference between the two rates: 13.7 minus 1.9 equals 11.8 percentage points, or 0.118. The number needed to treat is 1 divided by that value, about 8.5, which rounds to 9. Using the rate in the peanut-eating group alone gives 53, and using the avoidance group alone gives about 7, which are the common traps. At the bedside, share this effect with parents who hesitate, and remember that infants at highest risk gain even more in absolute terms.

Key principle: The number needed to treat is 1 divided by the absolute risk reduction.

Back to question Contents

BQ-064. Answer: C. Severe acute malnutrition by arm circumference

The World Health Organization defines severe acute malnutrition at 6 to 59 months by any one of three findings: a weight-for-height z-score under -3, a mid-upper arm circumference under 115 mm, or bilateral nutritional edema. Her z-score of -2.4 does not qualify alone, but her arm circumference of 112 mm does, because any single criterion is enough. The trap is classifying by weight-for-height only. At the bedside, assess appetite and infection to choose home or hospital care, and check electrolytes before refeeding.

Key principle: Any one criterion defines severe acute malnutrition: z-score under -3, arm circumference under 115 mm, or edema.

Back to question Contents

BQ-065. Answer: B. Insulin-driven movement of electrolytes into cells during rapid refeeding

In prolonged starvation, intracellular stores of phosphate, potassium, and magnesium are depleted while blood levels are held up. Sudden carbohydrate feeding raises insulin, which drives glucose and these electrolytes into cells and drops serum levels, causing weakness, edema, and arrhythmia. This is refeeding syndrome. The trap is attributing the change to fluid dilution alone. At the bedside, check electrolytes early, replace phosphate, potassium, and magnesium, and restart feeds at a measured pace with close monitoring.

Key principle: Restarting nutrition too fast in a starved child can drop phosphate, potassium, and magnesium.

Back to question Contents

BQ-066. Answer: C. Measure the fluoride content of the well water and assess his caries risk before any supplement

Fluoride protects the tooth surface, and its supplementation depends on how much the drinking water already provides and on the child's risk of decay. Private wells vary widely and are not regularly tested, so the amount is unknown. Supplements are not given before 6 months, and the later decision rests on water fluoride and caries risk. Starting drops by assumption is the trap. At the bedside, send a water sample, keep the smear of toothpaste and varnish by schedule, and decide on supplements when the result returns.

Key principle: Check the fluoride level of the main drinking water and assess caries risk before prescribing a supplement.

Back to question Contents

Answers: Domain 2. Fetal and Neonatal Care

Contents Questions for this subject

BQ-067. Answer: A. Placental transfusion of blood to the infant while the lungs aerate and pulmonary blood flow rises

At birth the placenta still holds a large share of the fetal blood volume, and flow through the cord continues for a short time. Leaving the cord intact for at least 60 seconds lets that blood move into the infant as the lungs fill with air and pulmonary flow rises, which supports the transition. Maternal surfactant does not cross to the fetus, and the umbilical oxygen content does not drive the first breath, so those options describe effects that do not occur. At the bedside, keep the baby skin to skin on the parent, dry and stimulate, and clamp after at least 60 seconds.

Key principle: Delayed cord clamping of at least 60 seconds allows placental transfusion in a baby who needs no resuscitation.

Back to question Contents

BQ-068. Answer: D. Send an umbilical arterial blood gas and a placental examination, and explain the scores

The Apgar score describes the infant at a moment in time and responds to resuscitation, so it is not a diagnosis of asphyxia and cannot predict one infant's outcome. A 5-minute score of 5 or lower calls for an umbilical arterial blood gas, which measures acid-base status directly, and a placental examination. Cooling needs evidence of encephalopathy and acidemia, which the score alone does not provide. At the bedside, send both samples, document the interventions, examine her tone and alertness over the next hours, and give the parents a plain explanation.

Key principle: A 5-minute Apgar score of 5 or lower prompts a cord arterial gas and placental examination, not a prediction.

Back to question Contents

BQ-069. Answer: C. Slow hepatic conjugation combined with low milk intake that reduces bilirubin excretion

Newborn red cells live a shorter time than adult cells, and an immature liver conjugates bilirubin slowly. A late-preterm infant with a weak suck and low intake adds poor excretion, because bilirubin is cleared through stool. The negative antiglobulin test and compatible blood groups argue against isoimmune hemolysis, which is the closest trap. Biliary obstruction produces conjugated bilirubin. At the bedside, plot the level on the hour-specific curve, support feeding every 2 to 3 hours, and recheck by the distance from the threshold.

Key principle: Late-preterm jaundice comes from slow conjugation and poor intake, so support feeding and plot the level.

Back to question Contents

BQ-070. Answer: B. Direct antiglobulin test on a sample of the infant's blood

Jaundice in the first 24 hours is hemolytic until proved otherwise. A mother with group O and an infant with group A allows maternal antibodies to coat the infant red cells, which shortens their survival and raises bilirubin quickly. The direct antiglobulin test detects that coating. Thyroid and metabolic tests explain prolonged jaundice, not an early rise, and a urine culture would be too late to guide the first decision. At the bedside, send the antiglobulin test, a serum bilirubin plotted on the hour-specific curve, and a blood count, and recheck at short intervals.

Key principle: Jaundice in the first 24 hours calls for a direct antiglobulin test and a plotted serum bilirubin.

Back to question Contents

BQ-071. Answer: C. A serum albumin of 2.8 g/dL

Bilirubin travels in the blood bound to albumin, and the free fraction crosses into the brain. A low albumin leaves more bilirubin unbound, so the neurotoxicity risk rises and the treatment threshold falls. The guideline lists a gestational age under 38 weeks, albumin under 3 g/dL, hemolytic disease, sepsis, and recent clinical instability as factors. Weight loss and mode of delivery are common distractors that do not change the curve. At the bedside, review the full risk list before every plot, because the wrong curve delays treatment.

Key principle: An albumin under 3 g/dL is a neurotoxicity risk factor that moves the infant to the lower threshold curve.

Back to question Contents

BQ-072. Answer: A. Repeat the bilirubin 6 to 12 hours after stopping and again the next day

Phototherapy converts bilirubin to forms excreted without conjugation, but red cell breakdown continues, so the level can climb again after treatment stops. Rebound is likelier when phototherapy began before 48 hours, when gestation is under 38 weeks, and in hemolytic disease, and this boy has all three. A repeat 6 to 12 hours after stopping and again the next day catches it. Early discharge with a late visit is the trap. At the bedside, keep him until the first repeat result and keep breastfeeding.

Key principle: After phototherapy in a baby with rebound risk, repeat the bilirubin 6 to 12 hours later and the next day.

Back to question Contents

BQ-073. Answer: C. Surfactant deficiency that raises alveolar surface tension and causes widespread atelectasis

Type II alveolar cells make surfactant late in gestation. Without it, surface tension rises, small alveoli collapse at the end of each breath, and the baby tires, which produces grunting, a rising oxygen need, and a granular radiograph. Retained lung fluid is the closest trap, because it also causes early fast breathing, but it belongs to term or late-preterm babies born without labor and clears in 1 to 2 days. At the bedside, give CPAP to a spontaneously breathing infant and arrange care in a unit that can ventilate.

Key principle: Respiratory distress syndrome in a preterm infant comes from surfactant deficiency and calls for CPAP and unit care.

Back to question Contents

BQ-074. Answer: B. Meconium aspiration syndrome

Fetal distress in a post-term baby can pass meconium into the amniotic fluid, and gasping draws it into the airways. The particles block small airways in a ball-valve pattern, which hyperinflates some segments and collapses others, and the chemical injury inactivates surfactant. Pneumonia is the closest trap because it can mimic any cause of distress, so infection stays on the list until excluded. Respiratory distress syndrome belongs to preterm babies. At the bedside, check saturations in the hand and foot for pulmonary hypertension, draw a culture, and arrange a unit that can ventilate.

Key principle: Stained fluid in a post-term baby with hyperinflated, patchy lungs points to meconium aspiration.

Back to question Contents

BQ-075. Answer: A. Sustained high pulmonary resistance with right-to-left shunting across the ductus

After birth the pulmonary arteries should relax. When resistance stays high, blood bypasses the lungs and flows right to left through the ductus, so postductal blood carries less oxygen than the right hand. The picture is cyanosis out of proportion to clear lung fields, with a gap between hand and foot readings. Critical heart disease also fails to improve with oxygen, which is the trap, so an echocardiogram settles the question. At the bedside, give oxygen, avoid handling that provokes crises, and move her to a unit that can ventilate.

Key principle: Cyanosis out of proportion to clear lungs, with a hand and foot saturation gap, signals pulmonary hypertension of the newborn.

Back to question Contents

BQ-076. Answer: D. Stop the antibiotics and continue routine newborn care

Early-onset sepsis is proved by culture, and a blood culture drawn before antibiotics usually turns positive within 36 to 48 hours when bacteremia is present. A sterile culture in a baby who stays well therefore allows the antibiotics to stop. Prolonged empiric treatment exposes the infant to harm without benefit. Waiting for a normal C-reactive protein is the trap, because that marker can neither diagnose nor exclude infection alone. At the bedside, stop the antibiotics, keep watching feeding and temperature, and teach the family the signs of illness.

Key principle: A sterile blood culture at 36 to 48 hours in a well infant means the antibiotics stop.

Back to question Contents

BQ-077. Answer: B. Feed her and repeat the glucose after the feed

Glucose falls in every newborn as the placental supply stops, and late-preterm and small-for-gestational-age infants have small glycogen stores, so both risk groups are screened. A value of 32 mg/dL at 3 hours lies in the actionable range of 25 to 40 mg/dL, but a vigorous baby without symptoms can be fed. Immediate intravenous dextrose is for symptoms or a value that stays low after feeding. At the bedside, give a feed, recheck after it, escalate if the value stays low, and keep her warm.

Key principle: A well, asymptomatic at-risk newborn with a low glucose gets a feed and a recheck.

Back to question Contents

BQ-078. Answer: A. Collect a critical sample for insulin and cortisol while the glucose is under 50 mg/dL

Hypoglycemia that persists beyond 48 hours is no longer a transitional state. Dependence on dextrose in a large infant suggests hyperinsulinism or an endocrine cause, and the hormones that explain the low glucose are present only while the glucose is low. A critical sample collected at that moment gives the diagnosis. Reassurance is the trap, since it delays treatment of a brain-threatening problem. At the bedside, draw the sample at the next value under 50 mg/dL, keep the infusion running, and involve pediatric endocrinology.

Key principle: Hypoglycemia lasting beyond 48 hours needs a critical sample drawn while the glucose is under 50 mg/dL.

Back to question Contents

BQ-079. Answer: D. Increased glucose consumption for heat production against small glycogen stores

A late-preterm baby has little fat, little brown fat, and a large skin surface, so heat is lost quickly. Producing heat burns glucose and oxygen, and the glycogen reserve in the liver is small, so the glucose falls within hours. Hyperinsulinism is the closest trap, but it would not explain the cold temperature, and it is sought only when low values persist past 48 hours. At the bedside, dry the baby, use skin-to-skin contact or a warmer, treat the glucose by protocol, and recheck after treatment.

Key principle: Cold stress feeds hypoglycemia because heat production burns glucose from small stores.

Back to question Contents

BQ-080. Answer: C. Admit her for a full sepsis evaluation and start empiric antibiotics after cultures are drawn

In a newborn the immune response is immature, and serious bacterial infection can produce few signs besides fever or poor feeding. A rectal temperature of 38.0 C (100.4 F) or more therefore calls for a sepsis evaluation, however well the baby looks. Good feeding and a normal examination are the trap, because they do not exclude bacteremia or meningitis. At the bedside, draw blood and urine cultures, consider spinal fluid, start ampicillin and gentamicin by protocol, and admit her.

Key principle: A rectal temperature of 38.0 C (100.4 F) or more in a newborn needs a sepsis evaluation, however well she looks.

Back to question Contents

BQ-081. Answer: E. Obtain a chest and abdominal radiograph with the tube in place and keep him unfed

In esophageal atresia the upper pouch ends blindly, so saliva pools and the first feed is regurgitated into the airway. A tube that will not pass is the clue. A radiograph with the tube in place shows the coiled tube in the pouch, and gas in the stomach points to a fistula. Another feed risks aspiration, and forceful passage risks perforation, which are the traps. At the bedside, keep the baby unfed, place a suction tube in the pouch, look for other anomalies, and call the surgeon.

Key principle: A feeding tube that will not pass in a drooling, choking newborn calls for a radiograph and a surgeon.

Back to question Contents

BQ-082. Answer: C. Cover the bowel with warm, moist sterile dressings and plastic wrap, place a gastric tube, and call the surgeon

In gastroschisis the abdominal wall fails to close beside the cord, so the bowel develops outside the abdomen without a sac. It loses heat and fluid quickly and can twist or lose its blood supply, and it is often thickened from exposure to amniotic fluid. Forced reduction can compress the vessels, which is the trap. At the bedside, wrap the bowel in warm, moist sterile dressings and plastic wrap, place a gastric tube to decompress the stomach, keep her warm, and arrange surgical care at once.

Key principle: Gastroschisis needs warm moist coverage, gastric decompression, and an immediate surgeon.

Back to question Contents

BQ-083. Answer: D. Bilateral choanal atresia

Newborns breathe mainly through the nose, so a bony or membranous block at the back of both nasal passages leaves the baby without an airway when the mouth is closed. Crying opens the mouth and the color returns, which is the signature of this lesion. Heart disease causes cyanosis that persists with crying. At the bedside, place an oral airway to keep the mouth open, confirm with a catheter that will not pass and with imaging, watch for other anomalies, and call the specialist team.

Key principle: Blue when quiet and pink when crying, with a catheter that will not pass the nose, is choanal atresia.

Back to question Contents

BQ-084. Answer: C. Explain that a few tests are needed, withhold a sex assignment, and call endocrinology and genetics

Ambiguous genitalia can be the first sign of congenital adrenal hyperplasia, in which an enzyme block lowers cortisol and aldosterone and causes salt wasting in the first weeks of life. Guessing the sex risks harm and a later reversal, and discharge without testing risks an adrenal crisis. At the bedside, tell the parents plainly that tests are needed, avoid gendered pronouns, consult endocrinology and genetics, and monitor electrolytes and glucose until the cause is known.

Key principle: Ambiguous genitalia is a medical urgency, so withhold a sex assignment and call endocrinology and genetics.

Back to question Contents

BQ-085. Answer: E. A girl born breech at 39 weeks to a mother with childhood hip dysplasia

Dysplasia of the hip is a developmental condition in which the femoral head sits poorly in a shallow socket. Breech position, female sex, and a family history each raise the risk, and the baby in this option has all three. Because the condition is found by examination and may be absent on the first day, a normal first examination does not close the question. At the bedside, repeat the Ortolani and Barlow maneuvers at every visit and consider imaging for a baby with these risks.

Key principle: Breech position, female sex, and a family history raise the risk of hip dysplasia and justify closer follow-up.

Back to question Contents

BQ-086. Answer: A. Arrange diagnostic audiology to confirm or exclude hearing loss by 3 months of age

Hearing loss in infancy delays speech and language unless it is found and treated early. The Joint Committee benchmark is screening by 1 month, confirmation by 3 months, and early intervention by 6 months. Waiting for a spontaneous cure is the trap, since each month lost is lost language exposure. At the bedside, send her to pediatric audiology now, ask about risk indicators including congenital cytomegalovirus, and keep her surveillance on schedule even after a pass.

Key principle: Follow the 1-3-6 benchmark: a failed hearing screen needs diagnostic audiology by 3 months of age.

Back to question Contents

BQ-087. Answer: C. Arrange a repeat dried blood spot, because the first was taken before 24 hours of feeding

The screen detects metabolic and endocrine disorders by measuring metabolites that accumulate only after feeding begins. Blood drawn too early may miss a disorder, so a specimen collected before 24 hours of milk feeding needs a repeat. Accepting the result is the trap, since a normal early screen is less reliable. At the bedside, contact the state program for its repeat window, collect the second spot promptly, and tell the parents why a repeat is routine in this situation.

Key principle: A newborn blood spot collected before 24 hours of milk feeding needs a repeat.

Back to question Contents

BQ-088. Answer: D. Coarctation of the aorta

Some obstructive lesions of the aorta depend on the ductus arteriosus to supply the lower body, and they give normal saturations until the ductus closes. Collapse then follows, with weak femoral pulses and a pressure gradient between arm and leg. The pulse oximetry screen has a pooled sensitivity of about 76 percent and most often misses coarctation, which is the trap in a passed screen. At the bedside, give prostaglandin on cardiology advice, correct shock, and arrange echocardiography and transfer.

Key principle: A passed oximetry screen never replaces the examination and the femoral pulses.

Back to question Contents

Answers: Domain 3. Adolescent Care

Contents Questions for this subject

BQ-089. Answer: A. State what stays private and the limits, such as serious risk of harm and abuse

Teens who expect a parent to be told skip visits, skip questions, and skip tests, so confidentiality is the condition that makes the rest of care possible. It has limits, though: serious risk of harm to self or others, abuse, and anything the law requires you to report. Stating those limits before any personal question lets him choose what to say and protects trust. A blanket promise is the trap, because it cannot be kept. At the bedside, start with what is going well and move toward personal topics.

Key principle: State the limits of confidentiality to the teen before the personal questions begin.

Back to question Contents

BQ-090. Answer: E. Authority of a minor to consent to contraception varies by state, so check state law

State law, not clinic habit, decides what a minor can consent to. The AAP counts 27 states that explicitly allow minors to consent to contraception and 19 that allow certain groups, so the rules are not uniform. Assuming one national rule is the trap in each distractor. All states, by contrast, allow minors to consent to testing and treatment for sexually transmitted infections. At the bedside, look up the current statute for your state, document the consent process, and tell her the limits of privacy before she discloses more.

Key principle: Consent for minors differs by state for contraception, so check state law before acting.

Back to question Contents

BQ-091. Answer: B. Ask which phone number or address is safe for her results and turn off portal release for them

A result posted to a parent's portal account ends confidentiality as surely as a phone call, and teens who fear disclosure avoid testing and stop returning. Because the law lets minors consent to testing for infection, you can offer a private route. Trusting a parent to ignore the portal is the trap, since the safeguard must be built into the record system. At the bedside, record a safe contact number, suppress portal and billing release for these results, and call her yourself when they arrive.

Key principle: Private results need a safe contact route and suppressed portal release, not trust in a parent.

Back to question Contents

BQ-092. Answer: D. Constitutional delay of growth and puberty

In constitutional delay, the hypothalamic-pituitary-gonadal axis activates late, so the growth spurt and puberty come late but complete normally. Clues are a family history of late maturation, a bone age that lags chronologic age, and steady growth. Pathologic causes need chronic symptoms, abnormal growth velocity, or specific features, and it is the most common cause and far more common in boys. At the bedside, measure LH, FSH, and bone age when evaluation is due, and reassure him that he will catch up.

Key principle: Constitutional delay is the commonest cause of late puberty, especially in boys with a family history and a delayed bone age.

Back to question Contents

BQ-093. Answer: E. Measure LH and FSH and obtain a bone age, and sort the cause as central or peripheral

The age for evaluation in a girl is no breast development by 13 years. Delay arises from central causes, from the hypothalamus and pituitary, or from peripheral causes, from the ovaries, and constitutional delay is common. LH and FSH separate those groups, and a bone age shows how much growth remains. Waiting until 15 years is the trap, since it delays the diagnosis of treatable causes. At the bedside, look for chronic disease, low energy intake, and a family history, and refer when the hormones point to a pathologic cause.

Key principle: No breast development by 13 years in a girl calls for evaluation with LH, FSH, and a bone age.

Back to question Contents

BQ-094. Answer: C. Early activation of the hypothalamic pulse generator that drives gonadotropin release

In central puberty the hypothalamic GnRH pulse generator switches on, raising LH and FSH, and the ovaries then make estrogen. A pubertal LH response in the stimulation test shows that the axis is active, which separates it from peripheral causes where gonadotropins stay low. Breast budding before about 8 years deserves evaluation, and the advanced bone age shows the growth cost. At the bedside, send her to pediatric endocrinology, arrange a bone age and, in a girl this young, consider brain imaging.

Key principle: Breast budding before about 8 years with a pubertal LH response means early central activation of the axis.

Back to question Contents

BQ-095. Answer: E. A transient imbalance between estrogen and androgen effect during mid puberty

In mid puberty the rising estrogen level can outpace the rising androgen effect, so glandular tissue grows under the areola for months. It is common, often tender, and usually fades within a year or two. The findings of normal testes, normal pubertal progress, and a firm disc under the areola argue against a tumor, Klinefelter syndrome, or a prolactin cause. At the bedside, measure the tissue, examine the testes, reassure him, and recheck in several months.

Key principle: Pubertal gynecomastia is common in mid puberty and usually fades, so measure it and reassure.

Back to question Contents

BQ-096. Answer: B. Urine human chorionic gonadotropin test

Pregnancy is the most common cause of secondary amenorrhea in a sexually capable adolescent, and it changes every later decision. The test is cheap and quick, and a girl may deny sexual activity from fear that a parent will hear. Hormone tests and imaging come afterward, when pregnancy is excluded. The hypothalamus, thyroid, and ovaries are the next causes. At the bedside, send the test privately, ask the HEEADSSS questions alone, and review her energy intake and training.

Key principle: Missed periods start with a pregnancy test, even when she denies sexual activity.

Back to question Contents

BQ-097. Answer: D. Suppressed GnRH pulsatility caused by low energy availability
When energy intake falls short of the energy used for training and basic function, the body conserves energy by switching off reproductive hormones. The hypothalamus reduces GnRH pulses, so LH and FSH fall, and the ovary makes little estrogen. The result is amenorrhea and risk to bone and heart. A prolactin tumor and ovarian failure are the traps, because they also stop periods but come with other features. At the bedside, restore energy, check vitals and an ECG, and tie clearance to train to recovery.
Key principle: Low energy availability suppresses GnRH pulses, so a missing period in an athlete is not a sign of fitness.
Back to question Contents

BQ-098. Answer: A. Physiologic anovulatory cycles in the early years after menarche
Early cycles are irregular because the hypothalamic-pituitary-ovarian axis is still maturing and many cycles are anovulatory. About 90 percent fall within 21 to 45 days, and most settle into 21 to 34 days by the third year. Polycystic ovary syndrome is the trap, and the label should be applied slowly in the early years after menarche because normal changes overlap with it. At the bedside, ask for a cycle diary, reassure the family, and reevaluate if the pattern leaves the expected range.
Key principle: Irregular cycles within 21 to 45 days in the first years after menarche are usually physiologic.
Back to question Contents

BQ-099. Answer: B. Admit her for monitored treatment of the bleeding and anemia, and send bleeding studies
Heavy bleeding with tachycardia, low blood pressure, and severe anemia means she is unstable and has lost a large blood volume. Anovulation is the commonest cause, and a bleeding disorder is the next. Admission lets you restore volume, treat the anemia, and start hormonal or antifibrinolytic therapy. Home treatment is the trap, and NSAIDs should be avoided when a bleeding disorder is possible. At the bedside, check orthostatic vitals, send a count, ferritin, and coagulation studies with hematology input, and exclude pregnancy.
Key principle: An adolescent who bleeds actively or is unstable with heavy menses is admitted.
Back to question Contents

BQ-100. Answer: C. Category 2, because advantages generally outweigh risks, so the pill is acceptable
The medical eligibility criteria rate each method against each condition: category 1 is no restriction, 2 is advantages generally outweigh risks, 3 is risks usually outweigh advantages, and 4 is an unacceptable risk. Under 35 years, combined pills are category 2 for a smoker, while migraine with aura is category 4. Choosing a higher category is the trap. At the bedside, counsel her to stop smoking, discuss the estrogen-free options, and add condoms for infection.
Key principle: Under 35 years, a smoker may use combined pills as category 2, with cessation counseling.
Back to question Contents

BQ-101. Answer: A. Collect chlamydia and gonorrhea tests at this visit and place the device without waiting

Most people need no extra screening for infection at placement, but a teen with risk factors who has not been tested should be tested then, and the device should not wait. The device carries no excess risk of pelvic infection in the patient who is tested and treated promptly if positive. Delaying placement is the trap, since each delay is a chance for pregnancy. At the bedside, collect a self-swab, place the device, call her with any positive result, and add condoms for infection protection.

Key principle: Test a teen with risk factors at IUD placement, and do not delay the device.

Back to question Contents

BQ-102. Answer: E. A copper-induced inflammatory reaction that is toxic to sperm and eggs

The copper ions release into the uterine cavity and produce a local inflammatory reaction that is toxic to sperm and eggs, which prevents fertilization. It is the most effective emergency method and gives long-term protection. Delay of ovulation is the mechanism of the emergency pill, which is the trap, and the pill works best the sooner it is taken. The device does not end an established pregnancy. At the bedside, confirm that she is not pregnant, place the device, and add condoms for infection.

Key principle: The copper IUD is the most effective emergency contraception because copper is toxic to sperm.

Back to question Contents

BQ-103. Answer: B. Ask how she feels and review all her options privately, without steering her

A positive test is a pivotal moment, and a teen who feels judged may delay care. Options counseling should be nondirective: parenting, adoption, and abortion are each presented, and the choice is hers. Steering is the trap, as is calling a parent against her wishes. Laws on minors' consent and parental involvement differ by state, so learn them. At the bedside, give the result privately, ask what she is feeling, encourage her to involve a trusted adult, and offer a follow-up in days.

Key principle: Give a positive pregnancy result privately and counsel on all options without steering.

Back to question Contents

BQ-104. Answer: D. A single intramuscular ceftriaxone dose, plus doxycycline unless chlamydia is excluded

Gonorrhea has become resistant to several drug classes, so a single injection of ceftriaxone is first-line and azithromycin cotreatment is no longer recommended. Because coinfection is common, treat for chlamydia too with doxycycline unless it is excluded. Azithromycin alone and oral cefixime are the traps. At the bedside, give the injection, take the dose from your local protocol, advise no sex until she and her partner are treated, offer expedited partner therapy where your state allows, and plan a retest.

Key principle: Treat gonorrhea with ceftriaxone alone and add doxycycline when chlamydia is not excluded.

Back to question Contents

BQ-105. Answer: E. A self-collected vaginal swab for nucleic acid amplification testing

Most chlamydia and gonorrhea in teens is silent, and adolescents carry the highest rates. A nucleic acid amplification test on a vaginal swab detects both organisms with high sensitivity, and the girl can collect it herself, which spares an examination she dislikes. Cytology is the trap, because it does not detect these organisms and cervical screening begins at 21 years. At the bedside, explain the swab, offer HIV screening, and arrange private results.

Key principle: A self-collected vaginal swab for nucleic acid amplification is the preferred test for most girls.

Back to question Contents

BQ-106. Answer: D. Pelvic inflammatory disease

Organisms from the cervix, usually chlamydia and gonorrhea, ascend into the uterus and tubes and cause inflammation, pain, and fever. Cervical motion tenderness with discharge in a sexually active teen favors this diagnosis, and a negative pregnancy test removes ectopic pregnancy. Appendicitis is the closest trap, so keep it in mind if the pain localizes. At the bedside, send testing for chlamydia and gonorrhea, start ceftriaxone, doxycycline, and metronidazole, the last two for 14 days, and admit her if she is ill or cannot take oral drugs.

Key principle: Lower abdominal pain, fever, and cervical motion tenderness in a sexually active teen mean pelvic inflammatory disease until proved otherwise.

Back to question Contents

BQ-107. Answer: E. Reinfection from an untreated or new partner

Chlamydia is common and silent in teens, and treatment does not protect against another exposure. Retesting at about 3 months is advised because reinfection is the usual problem, not treatment failure. A test of cure within weeks is not routine, since residual nucleic acid can give false-positive results, which is the trap. At the bedside, retreat her, ask about new partners and condom use, offer expedited partner therapy where legal, and retest again at 3 months.

Key principle: A positive retest at 3 months after chlamydia treatment usually means reinfection.

Back to question Contents

BQ-108. Answer: D. Interview her privately about coercion and safety, and follow the state reporting law

An infection in a young teen or a much older partner raises the question of sexual coercion or abuse, and silence can leave her in danger. A private interview lets her speak freely, and the law may require a report, which you must state to her honestly. Telling her mother or confronting the partner is the trap, because either may put her at risk. At the bedside, treat the infection, ask about safety at home and in the relationship, report as the law requires, and arrange follow-up.

Key principle: Infection in a young teen with a much older partner needs a private coercion screen and attention to reporting duties.

Back to question Contents

BQ-109. Answer: E. Give the first dose today and complete the series on the AAP schedule
HPV vaccination prevents infection and later cancer, and it works best before exposure. The schedule is in flux, with the AAP advising two doses starting at 9 to 12 years while the CDC revised to a single dose in 2026. Delay is the trap, since the first dose protects her whichever schedule applies. Cervical screening does not start until 21 years. At the bedside, give the first dose, check what your state and payers accept, and book the second dose on the AAP schedule.
Key principle: Do not let the schedule debate delay the first HPV dose at 11 to 12 years.
Back to question Contents

BQ-110. Answer: C. Offer brief counseling with motivational interviewing, and ask about impaired drivers
Screening places use on a spectrum from abstinence to disorder, and the response matches the level. Brief counseling and motivational interviewing suit a teen who uses without dependence, while severe use calls for referral. Ending the discussion misses an injury risk, and reporting punishes her for honesty. At the bedside, praise her candor, ask whether she rides with a driver who has used alcohol or cannabis, set a goal together, and screen again at the next visit.
Key principle: Match the response to the screening result: brief counseling for use, referral for severe use.
Back to question Contents

BQ-111. Answer: E. Counterfeit pills may contain fentanyl, so naloxone should be on hand
Counterfeit tablets made to look like prescription drugs may contain fentanyl, and a teen without opioid tolerance can stop breathing after a single pill. That risk, not slow harms, is the one that kills. The other effects are real but less urgent. At the bedside, state the risk plainly, offer naloxone and teach its use, give a nonjudgmental message about never taking pills from unknown sources, and arrange follow-up counseling.
Key principle: Teens who use pills should hear that counterfeit tablets can contain fentanyl and should have naloxone.
Back to question Contents

BQ-112. Answer: A. Restrictive eating disorder
Eating disorders occur in boys and athletes, and sports that reward low weight raise the risk. Restriction and compulsive exercise cause weight loss, and the body slows its heart rate and loses blood pressure on standing to conserve energy. Normal laboratory results do not rule out a disorder, which is the trap. Hyperthyroidism would raise the heart rate. At the bedside, plot the growth chart, check orthostatic vitals, temperature, and an ECG, and arrange team care.
Key principle: Weight loss with bradycardia and orthostasis in an athlete suggests a restrictive eating disorder, and normal laboratory results do not rule it out.
Back to question Contents

BQ-113. Answer: B. An insulin-driven shift of phosphate into cells

After prolonged starvation the body is depleted of phosphate, potassium, and magnesium. When carbohydrate is reintroduced, insulin rises and drives these ions into cells, and the serum level drops sharply while the demand for phosphate for energy production climbs. The result is weakness, edema, and risk of heart failure. Low phosphate, potassium, or sodium on admission warns of this, so feeding advances under monitoring. At the bedside, check electrolytes daily, replace phosphate, and slow the feeds.

Key principle: Refeeding drives phosphate into cells, so advance feeding under monitoring and replace electrolytes.

Back to question Contents

BQ-114. Answer: D. Family-based treatment in which the parents lead refeeding at home

Family-based treatment places the parents in charge of renourishing their child, and the illness is not their fault. It has the strongest evidence in adolescent anorexia nervosa when medical status allows outpatient care. A mental health clinician and a dietitian lead, while you follow weight, vital signs, and laboratory results. Excluding parents is the trap. At the bedside, set a target weight, review vitals at each visit, and admit her if she becomes unstable.

Key principle: Family-based treatment is first-line for adolescent anorexia nervosa when she is medically stable.

Back to question Contents

BQ-115. Answer: E. Relative energy deficiency in sport

Low energy availability suppresses the hypothalamic-pituitary-gonadal axis, lowering testosterone in a boy and impairing bone, immune function, and performance. The 2023 consensus describes a spectrum, affects male athletes, and overlaps with overtraining syndrome. Overtraining alone is the closest trap. At the bedside, calculate his energy intake against his training, check vitals and an ECG, involve a dietitian and a sports medicine team, and tie clearance to train to recovery.

Key principle: Male athletes with stress fractures and low testosterone may have relative energy deficiency in sport.

Back to question Contents

BQ-116. Answer: C. Explain that she is an adult, so results need her consent, and invite the mother to ask her

At 18 years, consent and privacy pass to the young adult, so a parent who attended every visit loses access unless the patient agrees. A structured transition builds readiness: teaching her to name her medicines, call for refills, and speak first at visits. Releasing results because of habit or payment is the trap. Refusing all future involvement ignores her wishes. At the bedside, ask whether she wants her mother involved and record her answer.

Key principle: At 18 years, privacy passes to the young adult, who decides what a parent may know.

Back to question Contents

Answers: Domain 4. Infectious Diseases

Contents Questions for this subject

BQ-117. Answer: E. Obtain a blood culture and spinal fluid, then admit him and start parenteral antibiotics

A newborn's immature immune defenses allow bacteremia and meningitis to progress before the examination changes, so appearance and early markers are unreliable at this age. In the 8 to 21 day band, the guideline calls for blood, urine, and spinal fluid in every well-appearing febrile infant, followed by admission and parenteral antibiotics, whatever the markers show. The closest wrong option is discharge on the strength of normal markers, which trusts tests that were never meant to clear a baby this young. At the bedside, tell the parents the workup is routine for his age and not a sign that he looks ill.

Key principle: In the 8 to 21 day band, a normal appearance and normal markers never replace spinal fluid and admission.

Back to question Contents

BQ-118. Answer: D. Hospitalize her, give parenteral antibiotics, and observe her until cultures are final

Abnormal inflammatory markers raise the chance of invasive bacterial infection, and without readable spinal fluid meningitis cannot be excluded. At 22 to 28 days, discharge after one empiric dose is allowed only when spinal fluid is obtained and reassuring. The closest wrong option sends her home after a single dose, which would partly treat meningitis and leave the team blind to it. At the bedside, admit her, start parenteral antibiotics, and repeat the puncture if the team judges it safe, while explaining to her parents why the failed sample changes the plan.

Key principle: A 22 to 28 day infant without interpretable spinal fluid is hospitalized, not sent home after one dose.

Back to question Contents

BQ-119. Answer: B. Treat with an oral antibiotic and recheck him in 12 to 24 hours

In older febrile infants the urinary tract is the usual hidden source, and normal inflammatory markers make bacteremia or meningitis unlikely. In the 29 to 60 day band, a positive urinalysis with normal markers is treated as a urinary infection with an oral antibiotic and a recheck in 12 to 24 hours. The closest wrong option sends him home untreated, which is the plan for a negative urinalysis and leaves a proven infection alone. At the bedside, prescribe by weight, write the weight on the order, and give clear return precautions.

Key principle: A 29 to 60 day infant with a positive urinalysis and normal markers gets oral antibiotics and a recheck in 12 to 24 hours.

Back to question Contents

BQ-120. Answer: D. Procalcitonin together with either absolute neutrophil count or C-reactive protein

Single markers each miss infections, but procalcitonin rises early and specifically in invasive bacterial disease, and neutrophil count or C-reactive protein adds sensitivity when paired with it. The guideline therefore prefers procalcitonin plus one of those two, with abnormal values of procalcitonin over 0.5 ng/mL, neutrophil count over 4,000 to 5,200 per mm3, or C-reactive protein over 20 mg/L. A plain white cell count is the closest wrong option because it is not recommended for risk stratification. At the bedside, fall back on neutrophil count, C-reactive protein, and temperature only when procalcitonin is unavailable.

Key principle: Stratify febrile infants with procalcitonin plus neutrophil count or C-reactive protein, never a plain white count.

Back to question Contents

BQ-121. Answer: A. Perform a full sepsis evaluation with blood, urine, and spinal fluid, then admit him and treat

The youngest newborns have the least ability to wall off infection, and group B Streptococcus, Escherichia coli, and herpes simplex virus can progress within hours. The guideline excludes infants 7 days old or younger, and most centers evaluate them fully with blood, urine, and spinal fluid before admission and treatment. The closest wrong option relies on a urinalysis and marker, which are not validated here. At the bedside, treat his appearance as irrelevant, begin parenteral antibiotics chosen by local protocol after cultures, and explain to the parents that this is routine for babies his age.

Key principle: A febrile infant 7 days old or younger gets a full sepsis workup and admission whatever he looks like.

Back to question Contents

BQ-122. Answer: C. Start acyclovir now and send cerebrospinal fluid PCR, surface swabs, and blood PCR

Herpes simplex virus spreads in newborns from skin to blood and brain and causes liver injury, so vesicles, maternal fever, white cells in spinal fluid without organisms, and a raised alanine aminotransferase all point to it. The guideline advises adding acyclovir without waiting for results when these features are present, alongside cerebrospinal fluid PCR, swabs of the mouth, eyes, and rectum, and blood PCR. The closest wrong option treats bacteria alone because the Gram stain is empty. At the bedside, give acyclovir by weight with the other antibiotics and admit him.

Key principle: A febrile newborn with herpes risk features gets acyclovir immediately, before any result returns.

Back to question Contents

BQ-123. Answer: D. Diagnose from the rash and start doxycycline without serologic testing
Early Lyme disease produces a skin lesion as spirochetes spread outward from the tick bite, and antibodies are often not yet detectable at that stage. Erythema migrans is therefore diagnosed by its appearance in an endemic area, and the guideline advises against testing for a typical lesion. Doxycycline, amoxicillin, or cefuroxime treats it. The closest wrong option sends serology and withholds treatment, which risks a false negative and delay. At the bedside, treat now, tell the family what spread to the heart or joints would look like, and reserve testing for atypical lesions.
Key principle: A typical erythema migrans is diagnosed by sight and treated without serologic testing.
Back to question Contents

BQ-124. Answer: C. Sepsis without septic shock
The Phoenix criteria, built from more than three million encounters, define pediatric sepsis as a score of 2 or higher in a child with suspected infection, and septic shock as sepsis with at least 1 cardiovascular point. This boy meets the first condition and not the second, so he has sepsis without shock. The closest wrong option calls any score of 2 shock and would send him down the one-hour pathway without a cardiovascular point to justify it. At the bedside, give antimicrobials as soon as possible after evaluation, within 3 hours, and reassess often.
Key principle: Phoenix sepsis is a score of 2 or higher, and shock requires at least 1 cardiovascular point.
Back to question Contents

BQ-125. Answer: D. As soon as possible after rapid evaluation, within 3 hours of recognition
Organ dysfunction without circulatory failure leaves a short window to make the evaluation count, so the pediatric Surviving Sepsis Campaign guidelines allow antimicrobials as soon as possible after proper evaluation, within 3 hours, while shock requires them within 1 hour of recognition. The closest wrong option applies the one-hour shock target to every febrile child, which ignores the stage-based approach. At the bedside, send cultures only if they cost no time, give a broad drug chosen by local protocol and his weight, and reassess for shock repeatedly.
Key principle: Without shock, antimicrobials follow proper evaluation within 3 hours, and with shock they follow within 1 hour.
Back to question Contents

BQ-126. Answer: B. Boluses of 10 to 20 mL/kg over the first hour, to a total of up to 40 mL/kg, reassessing after each
Hypotension means compensation has failed and circulating volume must be restored, but a setting without intensive care cannot manage the consequences of large volumes. For a hypotensive child there, the pediatric guidelines allow boluses of 10 to 20 mL/kg over the first hour, up to 40 mL/kg, with reassessment after each. The closest wrong option applies the 60 mL/kg ceiling that belongs to intensive care settings. At the bedside, give 200 to 400 mL per bolus for her weight, check lungs and liver after each, and request a vasoactive plan with the receiving team.
Key principle: In a setting without intensive care, a hypotensive child receives up to 40 mL/kg in 10 to 20 mL/kg boluses.
Back to question Contents

BQ-127. Answer: D. Volume given beyond her cardiac and pulmonary reserve, raising venous and capillary pressures

In septic shock the myocardium is depressed and the capillaries leak, so each bolus raises venous and capillary pressure more than it raises cardiac output. Once her reserve is exceeded, fluid backs up into the lungs and liver, which produces crackles, hepatomegaly, and hypoxemia. The closest wrong option blames new infection, which would not appear within minutes of the boluses. The pediatric guidelines advise looking again after every bolus. At the bedside, stop further boluses, support her breathing, and start epinephrine or norepinephrine for ongoing shock.

Key principle: New crackles and hepatomegaly after boluses signal fluid overload, which calls for stopping fluid and starting a vasoactive drug.

Back to question Contents

BQ-128. Answer: B. Tachycardia and vasoconstriction preserve pressure until compensation fails late in the course

Children defend blood pressure with a fast heart rate and tight peripheral vessels, so cool mottled skin and slow capillary refill appear well before the pressure drops. By the time a child is hypotensive, compensation has already failed. The closest wrong option treats a normal pressure as proof against shock, which delays care during the period when treatment works best. At the bedside, recognize shock from perfusion and behavior, give antimicrobials within 1 hour, and start measured 10 to 20 mL/kg boluses with reassessment.

Key principle: In children, hypotension is a late sign of shock, so diagnose it from perfusion and heart rate.

Back to question Contents

BQ-129. Answer: C. Obtain a catheterized specimen for urinalysis and culture before making the diagnosis

Skin and perineal bacteria colonize a bag specimen, so a positive bag culture often reflects contamination rather than bladder infection. The AAP guideline requires a suggestive urinalysis and at least 50,000 CFU/mL of a uropathogen from a catheter or suprapubic specimen. The closest wrong option accepts the bag result and commits a well infant to antibiotics and likely to unneeded imaging. At the bedside, catheterize, treat if the specimen confirms infection, and order renal and bladder ultrasonography afterward.

Key principle: Confirm a urinary infection only with a catheter or suprapubic specimen, never a bag.

Back to question Contents

BQ-130. Answer: A. Parvovirus B19 infection of erythroid precursors, halting red cell production

Parvovirus B19 infects red cell precursors in the marrow and stops their production for about a week. Healthy children barely notice because red cells live 120 days, but in sickle cell disease they live only 10 to 20, so hemoglobin falls fast. A reticulocyte count near zero with an unenlarged spleen separates this from sequestration and hemolysis. The closest wrong option, splenic sequestration, would show a large spleen and a high reticulocyte count. At the bedside, check the count, transfuse as needed, and isolate him from pregnant staff.

Key principle: In a child with sickle cell disease, parvovirus can stop red cell production and cause a profound anemia with no reticulocytes.

Back to question Contents

BQ-131. Answer: E. Voiding cystourethrography to look for reflux or obstruction of the right urinary tract

Obstruction or high-grade reflux causes urine to back up and dilate the collecting system, and these lesions predispose to scarring with repeat infection. The AAP guideline does not use voiding cystourethrography routinely after a first infection, but advises it when ultrasonography shows hydronephrosis, scarring, or signs of high-grade reflux or obstruction. The closest wrong option stops at the ultrasound. At the bedside, arrange the cystogram after the infection has been treated, involve pediatric urology if the study is abnormal, and explain the plan to the parents.

Key principle: Hydronephrosis on ultrasonography after a first febrile infection is a reason for voiding cystourethrography.

Back to question Contents

BQ-132. Answer: B. Oral cefpodoxime for the full course, with no routine blood tests or radiograph

Streptococcus pneumoniae is the usual cause of bacterial pneumonia in a healthy child, and beta-lactams that reach it are first-line. Amoxicillin at 90 mg/kg/day is preferred, but with a nonserious penicillin reaction the PIDS and IDSA guideline allows a cephalosporin such as cefuroxime, cefprozil, or cefpodoxime. Azithromycin is the closest wrong option because pneumococcal resistance reaches up to 40 percent in some places. At the bedside, dose by weight, write the weight on the prescription, and skip the chest radiograph, blood count, and cultures for a child who goes home.

Key principle: A child with mild pneumonia and a nonserious penicillin reaction can receive an oral cephalosporin instead of amoxicillin.

Back to question Contents

BQ-133. Answer: D. Intravenous ampicillin, with the dose chosen by weight and local protocol

Pneumococcus is the leading cause of lobar pneumonia, and penicillins act on it by blocking cell wall synthesis. For an immunized child sick enough for admission, with low local resistance and no empyema or other life-threatening complication, the guideline prefers intravenous ampicillin or penicillin G. Ceftriaxone or cefotaxime is for higher resistance or complications. The closest wrong option widens coverage with vancomycin, which adds toxicity without need. At the bedside, give ampicillin by weight, switch to oral amoxicillin when he is eating and improving, and narrow further if a culture returns.

Key principle: For an immunized child admitted with pneumonia and low local resistance, start intravenous ampicillin or penicillin G.

Back to question Contents

BQ-134. Answer: C. Explain that an antibiotic will not help and give a plan for symptoms and return precautions

A cold is caused by viruses, which do not respond to antibiotics, and unneeded drugs bring rashes, diarrhea, and resistance without benefit. Good stewardship means treating the child and the parent: say plainly that the illness is viral, explain what to expect, and name the signs that should bring him back. The closest wrong option prescribes amoxicillin to prevent a complication, a purpose it does not serve. Parents often hear no antibiotic as no care, so at the bedside give fever advice, fluids, and a clear threshold such as fever past day 5, trouble breathing, or poor drinking.

Key principle: Say no to antibiotics for a viral illness, and give the parent a plan along with the answer.

Back to question Contents

BQ-135. Answer: A. Mask her, move her to an airborne isolation room, send PCR and IgM, and call the health department

Measles virus spreads through the air and stays infectious in a room for hours, so a child with cough, coryza, conjunctivitis, Koplik spots, and a descending rash is dangerous to every unprotected person who shares space with her. She is infectious from 4 days before the rash to 4 days after. The closest wrong option uses droplet precautions and waits to report. At the bedside, mask her at once, use airborne isolation, send a throat or nasopharyngeal PCR and serum IgM, phone the health department, and list the exposed contacts.

Key principle: Suspected measles needs immediate masking, airborne isolation, and a phone call to the health department.

Back to question Contents

BQ-136. Answer: C. Immune globulin, which works up to 6 days after the exposure

Measles incubates for about two weeks, which leaves a window for passive antibody to blunt infection. Immune globulin works up to 6 days after exposure and is kept for contacts who cannot take the vaccine, such as infants under 6 months, pregnant women, and severely immunocompromised people. The closest wrong option gives MMR, which is outside its 72-hour window here, and he is too young for it besides. At the bedside, arrange immune globulin by weight through the health department, report the contact, and tell his mother which symptoms mean immediate return.

Key principle: Immune globulin protects exposed infants under 6 months for up to 6 days after measles contact.

Back to question Contents

BQ-137. Answer: E. Give a dose of MMR now; she still needs the routine doses at 12 to 15 months and 4 to 6 years

Maternal antibody wanes during the second half of the first year, and exposure risk abroad is high. The CDC recommends an early dose of MMR for infants 6 to 11 months old before international travel, and the health department may advise one during an outbreak. Because antibody from the mother can blunt the response at this age, the early dose does not count toward the routine series. The closest wrong option counts it and omits later doses. At the bedside, vaccinate now, schedule the 12 to 15 month and 4 to 6 year doses, and record the early dose.

Key principle: An infant 6 to 11 months old gets an early MMR dose before travel, and the routine doses still follow.

Back to question Contents

BQ-138. Answer: C. Persistent defective measles virus infection of neurons years after the original illness

Measles virus can persist in a mutated form in brain cells, then cause progressive inflammation and neuronal loss 7 to 10 years after the infection, a fatal disorder named subacute sclerosing panencephalitis. The slow decline, myoclonus, periodic complexes on the electroencephalogram, and the history of measles in infancy fit. The closest wrong option, acute disseminated encephalomyelitis, appears days to weeks after an infection and not years later. At the bedside, confirm with cerebrospinal fluid measles antibody, involve neurology, and recognize that MMR vaccination prevents the disease.

Key principle: Subacute sclerosing panencephalitis is a fatal late result of persistent measles virus, appearing 7 to 10 years after the infection.

Back to question Contents

BQ-139. Answer: D. Breakthrough varicella

Varicella-zoster virus produces new crops of lesions over several days, so macules, papules, vesicles, and crusts coexist, and prior vaccination changes severity more than appearance. Breakthrough disease in vaccinated children usually has fewer than 50 lesions and a mild course, but those with 50 or more lesions are as contagious as unvaccinated cases. The closest wrong option, hand, foot, and mouth disease, favors the palms, soles, and mouth. At the bedside, keep her home until every lesion has crusted, give acetaminophen and avoid aspirin, and ask about contacts who are pregnant or immunocompromised.

Key principle: Lesions in all stages together point to varicella, and vaccinated children often have a milder breakthrough illness.

Back to question Contents

BQ-140. Answer: B. Treat the illness as mumps, isolate him until 5 days after swelling began, and report

Mumps virus infects the parotid glands, and viral shedding peaks near the onset of swelling and falls quickly, so a swab taken after the peak or with poor technique can miss it. A negative RT-PCR does not exclude mumps in a patient with a compatible illness, so clinical judgment and public health guidance prevail. The closest wrong option relies on the negative result and sends him back among classmates. At the bedside, isolate him until 5 days after the swelling began, report to the health department, and confirm the vaccination record.

Key principle: A negative mumps PCR does not exclude mumps, and isolation lasts until 5 days after swelling begins.

Back to question Contents

BQ-141. Answer: A. Give MMR vaccine before discharge

Rubella is mild in children and adults, with up to half of infections causing no symptoms, but first-trimester infection can cause miscarriage or congenital rubella syndrome. A woman without immunity who becomes pregnant again is the route by which the harm occurs, so she is vaccinated right after delivery, and breastfeeding does not prevent it. The closest wrong option gives immune globulin, which does not produce lasting protection. At the bedside, vaccinate before discharge, and record the dose.

Key principle: A woman without rubella immunity receives MMR right after delivery, before she leaves the hospital.

Back to question Contents

BQ-142. Answer: B. Start oseltamivir now without waiting for the test result
Influenza replicates most in the first two days, so antivirals work best early, but severe disease continues beyond that. The CDC advises treating children who are hospitalized, severely ill, or at higher risk of complications as soon as possible, without waiting for a test. Asthma places this girl in the higher-risk group. The closest wrong option withholds the drug because she is past 48 hours, which confuses the point of greatest benefit with a cutoff. At the bedside, prescribe oseltamivir for 5 days by weight, give return precautions for fast breathing, and check on her again within a day.
Key principle: Treat a high-risk child with suspected influenza early, without waiting for a test or the 48-hour mark.
Back to question Contents

BQ-143. Answer: D. Placental antibody transfer needs time, and delivery within 14 days of vaccination is too soon
Maternal RSVpreF vaccine induces antibody that crosses the placenta, mostly in the weeks before birth, and levels in the infant need time to build. An infant born 14 days or fewer after the vaccine is treated as unprotected, and she should receive nirsevimab or clesrovimab. The closest wrong option claims the vaccine never reaches the fetus, which is the wrong explanation for a protection that works in other infants. At the bedside, give the antibody within the first week and ideally before discharge, because she is born between October and March.
Key principle: An infant born 14 days or fewer after maternal RSV vaccine has not yet received enough antibody and needs an RSV antibody.
Back to question Contents

BQ-144. Answer: B. Give nirsevimab or clesrovimab at this visit
RSV is the leading cause of hospitalization in the first year of life, and the season runs from fall into spring. The CDC recommends one long-acting antibody for infants under 8 months entering their first season when the mother was not vaccinated. An infant born from April through September who has not been dosed receives it at the first opportunity in October. The closest wrong option withholds it because his birth fell outside the season. At the bedside, give the dose today, record it, and teach hand hygiene and avoidance of sick contacts.
Key principle: An infant born April through September gets an RSV antibody in October if the mother was not vaccinated.
Back to question Contents

BQ-145. Answer: E. Multisystem inflammatory syndrome in children
A delayed hyperinflammatory response after SARS-CoV-2 infection injures the heart, gut, skin, and vessels together. The CDC definition requires fever, C-reactive protein over 3.0 mg/dL, hospitalization, no better explanation, new problems in at least two of five systems, and evidence of infection or exposure within 60 days. Shock, gastrointestinal symptoms, and raised troponin in an older child fit. Kawasaki disease is the closest wrong option because the two overlap, but it favors younger children without shock. At the bedside, draw serology before immune globulin, then obtain an electrocardiogram, echocardiogram, and troponin, and start treatment.
Key principle: Fever with shock, gut symptoms, a raised troponin, and recent SARS-CoV-2 exposure suggests MIS-C.
Back to question Contents

BQ-146. Answer: A. Oral valganciclovir, usually continued for 6 months, with audiology every 6 months

Cytomegalovirus damages the brain, retina, liver, and cochlea, and antiviral treatment started in the first month can limit hearing decline and improve development. Guidance advises treating symptomatic infants 30 days old or younger who have organ or central nervous system disease with valganciclovir, usually for 6 months. Hearing can fall later, so audiology continues every 6 months for 3 years, then yearly until age 5. The closest wrong option, acyclovir, is active against herpes and not CMV. At the bedside, dose by weight with blood count monitoring, and involve infectious diseases.

Key principle: Treat symptomatic congenital CMV in infants 30 days old or younger with valganciclovir, and follow hearing for years.

Back to question Contents

BQ-147. Answer: B. Give a second immune globulin infusion or pulse corticosteroids, with cardiology involvement

In Kawasaki disease, ongoing vasculitis can persist after the first infusion in 10 to 20 percent of patients, and every extra day of inflammation raises the chance of coronary damage. Persistent or recurrent fever after treatment calls for a second infusion of immune globulin, pulse corticosteroids, infliximab, or cyclosporine, with cardiology guidance. The closest wrong option waits, which risks aneurysm in a patient who still has active disease. At the bedside, repeat inflammatory markers and the echocardiogram, give the second agent by weight, and counsel the family that live vaccines are delayed after immune globulin.

Key principle: Persistent fever after the first immune globulin infusion needs a second agent without delay.

Back to question Contents

BQ-148. Answer: C. Congenital toxoplasmosis

Toxoplasma gondii crosses the placenta in roughly 30 percent of untreated maternal infections and damages the retina, brain, and ventricular drainage. The classic triad is chorioretinitis, hydrocephalus, and intracranial calcifications, and most affected newborns look well at birth. The closest wrong option, cytomegalovirus, causes calcification around the ventricles with a small head and hearing loss, and a normal screen argues against it. At the bedside, confirm with maternal and infant serology and PCR, obtain brain imaging and an eye examination, and start pyrimethamine, sulfadiazine, and folinic acid for 12 months.

Key principle: Chorioretinitis, hydrocephalus, and scattered calcifications form the classic triad of congenital toxoplasmosis.

Back to question Contents

BQ-149. Answer: B. Spinal fluid analysis, blood count, and long-bone films, with 10 days of intravenous penicillin G

Treponema pallidum crosses the placenta and can infect the brain, bones, and liver even when the newborn looks normal. A titer at least four times the mother's, or an untreated mother, places the infant in the confirmed or probable group, and the CDC requires cerebrospinal fluid analysis, a blood count, and long-bone radiographs, with 10 days of intravenous aqueous penicillin G. The closest wrong option waits for symptoms, which lets neurosyphilis go untreated. At the bedside, follow titers every 2 to 3 months until negative and test him for HIV.

Key principle: A newborn with a normal exam but an untreated mother or a fourfold titer needs full evaluation and penicillin G.

Back to question Contents

BQ-150. Answer: D. Maternal IgG that crossed the placenta, which does not establish infection in the infant

Maternal IgG antibody crosses the placenta late in pregnancy and persists in the infant for months, so an exposed newborn is nearly always antibody positive whether or not he is infected. For this reason, diagnosis in an exposed infant uses nucleic acid testing, at 2 to 3 weeks, 1 to 2 months, and 4 to 6 months, and also at birth in higher-risk infants. The closest wrong option reads the antibody as proof of infection. At the bedside, send nucleic acid testing, continue infant prophylaxis as risk requires, and reassure the family that the antibody result alone is not a diagnosis.

Key principle: Antibody tests cannot diagnose HIV in an exposed infant because maternal IgG persists, so use nucleic acid tests.

Back to question Contents

BQ-151. Answer: E. Offer formula, donor milk, or breastfeeding by shared decision-making, with infant prophylaxis

With fully suppressive therapy through pregnancy and lactation, transmission through milk falls under 1 percent but does not reach zero. Current DHHS guidance therefore moves from prohibition to shared decision-making and supports a parent who chooses to breastfeed. The closest wrong option calls transmission certain, which is false and undermines trust. At the bedside, continue her treatment, give the infant prophylaxis, monitor her viral load every 1 to 2 months, and use words such as: your treatment has made this baby's risk very low, so let us choose a feeding plan together.

Key principle: With suppression, breastfeeding with HIV carries under 1 percent risk, and the choice is made together.

Back to question Contents

BQ-152. Answer: D. Give a single dose of doxycycline now, within 72 hours of tick removal

Borrelia burgdorferi is passed to the host only after the tick has fed for a day or more, so a bite from a tick attached for under 36 hours carries little risk. The Ixodes tick here was attached about 48 hours in an endemic area, which meets the guideline's high-risk bite criteria. A single doxycycline dose within 72 hours of removal is advised. The closest wrong option uses serology, which would be negative this early. At the bedside, give the weight-based dose, teach the family to watch for a rash, and treat fully if one appears.

Key principle: Prophylaxis is one doxycycline dose for an Ixodes tick attached 36 hours or more, given within 72 hours of removal.

Back to question Contents

BQ-153. Answer: C. Lyme carditis

Spirochetes spread to the heart weeks after a bite and inflame the conduction system, most often producing varying degrees of atrioventricular block. A prior rash in an endemic state, fainting, bradycardia, and a prolonged PR interval fit. Acute rheumatic fever is the closest wrong option, but it follows streptococcal pharyngitis and brings joint pain and other criteria. At the bedside, admit him for monitoring because the PR interval is over 300 milliseconds, send serology, and give an antibiotic course of 14 to 21 days chosen with infectious diseases.

Key principle: Syncope or a prolonged PR interval weeks after a tick exposure in an endemic area suggests Lyme carditis.

Back to question Contents

BQ-154. Answer: A. Rickettsial infection of vascular endothelium causing widespread vasculitis and leakage

Rickettsia rickettsii infects the cells lining small blood vessels, producing vasculitis that explains the rash and petechiae, consumption of platelets, and leakage of fluid with hyponatremia. This is the tick-borne illness that kills a child within days. The closest wrong option points to immune complexes, which cause a different vasculitis and not an illness 7 days after a tick bite. At the bedside, do not wait for serology or for the rash to be classic, and start doxycycline at 2.2 mg/kg per dose twice daily, with a maximum of 100 mg per dose, since the drug is used at every age.

Key principle: Rocky Mountain spotted fever is a vasculitis of small vessels, and doxycycline is started on suspicion at any age.

Back to question Contents

BQ-155. Answer: C. Incomplete Kawasaki disease

Kawasaki disease is a vasculitis of medium-sized arteries, and infants often have too few of the five features to meet the complete definition. Fever for 5 days or more with a C-reactive protein of 3.0 mg/dL or higher or an ESR of 40 mm/hr or higher supports incomplete disease, and a positive echocardiogram settles treatment. The closest wrong option, adenovirus, can imitate Kawasaki disease, but a negative viral test and coronary dilation argue against it. At the bedside, give immune globulin and aspirin by local protocol, involve cardiology, and delay live vaccines afterward.

Key principle: Infants with prolonged fever, raised inflammatory markers, and coronary dilation have Kawasaki disease even with few features.

Back to question Contents

Answers: Domain 5. Mental and Behavioral Health

Contents Questions for this subject

BQ-173. Answer: A. Separation anxiety disorder
Anxiety in children often appears as bodily complaints and avoidance rather than a stated worry. Her pain tracks the school calendar and comes with clinging, fear for her parents, and sleeping near them, which defines separation anxiety. Generalized anxiety is the closest wrong option, but it spreads across many topics instead of centering on separation. Gastrointestinal causes would not switch off on weekends. At the bedside, ask what she worries will happen, check for school stressors, and refer for exposure-based cognitive behavioral therapy.
Key principle: Stomachaches that follow the school calendar with clinging and fear of separation point to separation anxiety.
Back to question Contents

BQ-174. Answer: B. Add a selective serotonin reuptake inhibitor to continued psychotherapy
Moderate to severe anxiety, or a partial response to therapy, calls for a medicine that adjusts serotonin signaling while the skills from therapy continue. A selective serotonin reuptake inhibitor is the first-line medicine class, and in the CAMS trial combined treatment outperformed either treatment alone. The closest wrong option is waiting another year, which leaves a functional impairment untreated. Benzodiazepines and antipsychotics are not first line, and a stimulant does not treat worry. At the bedside, start low, check on him within a week, ask about suicidal thinking at each visit, and keep therapy going.
Key principle: Partial response to therapy in moderate to severe anxiety is treated by adding an SSRI to continued cognitive behavioral therapy.
Back to question Contents

BQ-175. Answer: A. Anxiety screening with a validated questionnaire
Anxiety disorders are common in school-age children and respond to treatment, which is the logic for looking for them before the problem is reported. The Task Force recommended anxiety screening from age 8 through 18, so a 9-year-old qualifies. The closest wrong option is depression screening, which starts at 12. Evidence was insufficient for suicide risk screening in symptom-free youth. At the bedside, use a validated tool, act on a positive result, and keep asking about worries at later visits.
Key principle: Anxiety screening is recommended at ages 8 to 18, and depression screening starts at age 12.
Back to question Contents

BQ-176. Answer: B. A stimulant medication plus parent training in behavior management
Stimulants improve attention and impulse control, and parent training builds routines and skills that make the gains last. For children aged 6 to 11 the 2019 AAP guideline pairs a stimulant with parent training, and nonstimulants are second line. The closest wrong option is training alone, which is the preschool plan. Testing does not improve accuracy, and delaying treatment prolongs impairment. At the bedside, titrate the stimulant to effect and not to weight, check height, weight, blood pressure, and pulse at follow-up, and request school accommodations.
Key principle: At ages 6 to 11, ADHD is treated with a stimulant plus parent training.
Back to question Contents

BQ-177. Answer: A. Rebound sympathetic activity after abrupt withdrawal of an alpha-2 agonist
Alpha-2 agonists lower sympathetic outflow, and the body adapts to that suppression. When the drug stops abruptly, outflow surges and blood pressure rises over baseline. The timing after missed doses makes this the likeliest cause, while an adrenal tumor or renal artery disease would not appear on the third day of a missed medicine. At the bedside, restart the medicine, recheck her blood pressure, and counsel the family that this class is tapered and never stopped suddenly.
Key principle: Stopping an alpha-2 agonist abruptly causes rebound hypertension, so these drugs are tapered.
Back to question Contents

BQ-178. Answer: C. Autism spectrum disorder
Autism is diagnosed from behavior in two domains: persistent deficits in social communication and interaction, and restricted, repetitive patterns. This boy shows both, with poor reciprocity and nonverbal communication on one side and lining up, insistence on sameness, and sensory reactions on the other. A language disorder is the closest wrong option, but it would not explain the repetitive pattern or the social deficits. A normal hearing screen removes hearing loss. At the bedside, refer for diagnostic evaluation and early intervention together, and order a hearing evaluation.
Key principle: Social communication deficits plus restricted, repetitive behavior define autism spectrum disorder.
Back to question Contents

BQ-179. Answer: B. Administer the structured M-CHAT-R/F Follow-Up interview
The M-CHAT-R/F is built in two stages so that a checklist alone does not label a child. A total score of 3 to 7 is medium risk, and the Follow-Up interview clarifies the answers that were missed or misread. This score falls in that band. Rescreening later or reassuring the parents delays the process, and genetic testing comes after a positive result. At the bedside, complete the interview now, and if the Follow-Up score is 2 or higher, refer for diagnosis and early intervention at the same visit.
Key principle: A medium-risk M-CHAT-R/F score of 3 to 7 requires the Follow-Up interview.
Back to question Contents

BQ-180. Answer: C. Fluoxetine
Depression is linked to altered serotonin signaling, and selective serotonin reuptake inhibitors are the first-line medicine class in youth. Fluoxetine and escitalopram carry pediatric approval for depression. Paroxetine, the closest wrong option, is an SSRI but lacks that approval, and bupropion, amitriptyline, and venlafaxine are not recommended first line. At the bedside, start low, review her within about a week, ask about suicidal thinking and activation at every visit, and plan a trial that reaches the highest tolerated dose.
Key principle: Fluoxetine and escitalopram are the SSRIs with pediatric approval for depression.
Back to question Contents

BQ-181. Answer: E. Tourette syndrome

Tics are sudden, repeated movements or sounds, and they wax and wane and can be briefly suppressed. Tourette syndrome means several motor tics and at least one vocal tic lasting more than a year, which fits this boy. Persistent motor tic disorder is the closest wrong option, but it has no vocal tic. Provisional tics last under a year, and chorea and seizures do not match the suppressibility or the vocal component. At the bedside, screen for ADHD and obsessive-compulsive disorder, which often cause more trouble than the tics.

Key principle: Multiple motor tics plus at least one vocal tic for more than a year is Tourette syndrome.

Back to question Contents

BQ-182. Answer: D. About 4 percent of treated youth had suicidal thinking or behavior, twice placebo, and no suicides

The warning rests on an FDA analysis of 24 short-term placebo-controlled trials of nine drugs in 4,587 young patients. Suicidal thinking or behavior occurred in about 4 percent of treated patients, twice the placebo rate, and there were no suicides. Claiming deaths, as the closest wrong option does, overstates the finding. Untreated depression carries its own risk, and treatment fell after the warnings. At the bedside, explain this, then arrange a visit within a week of starting and ask about suicidal thinking and activation at every visit.

Key principle: The boxed warning describes a doubled rate of suicidal thinking without suicides, and it calls for close monitoring and not avoidance.

Back to question Contents

BQ-183. Answer: B. Arrange psychiatry consultation before any antidepressant is started

Antidepressants can trigger mania in a young person whose bipolar disorder has not been recognized, and a father with mania raises that possibility. Under GLAD-PC, a first-degree relative with bipolar disorder moves a depressed teen into the severe group for consultation. Starting fluoxetine, the closest wrong option, treats the surface problem and risks a switch. Waiting or reassurance leaves six weeks of impairment unaddressed. At the bedside, ask about past episodes of high energy and little sleep, arrange psychiatry input promptly, and set up close follow-up.

Key principle: Ask about bipolar disorder before any antidepressant, and consult psychiatry when a first-degree relative is affected.

Back to question Contents

BQ-184. Answer: D. Hypoglycemia

Agitation is a presentation, and some medical causes mimic psychiatric illness. In this boy, insulin without food during exercise lowers blood glucose, which causes confusion from low brain fuel and sweating and tremor from the counter-regulatory surge. Diabetic ketoacidosis, the closest wrong option, brings deep, fast breathing and high glucose. Psychosis and panic are diagnoses of exclusion in a diabetic child. At the bedside, check a glucose while de-escalating, give rapid glucose by the route he can safely take, and recheck.

Key principle: Check glucose in every agitated or confused child with diabetes before attributing the behavior to a psychiatric cause.

Back to question Contents

BQ-185. Answer: C. Cerebral hypoxia from impaired lung gas exchange
Brain function depends on a steady oxygen supply, and early hypoxia produces restlessness and combativeness before drowsiness. Pneumonia reduces gas exchange, which explains the low saturation and the work of breathing. The closest wrong option is a fever-related reaction, but high temperature alone does not produce this picture with a saturation of 82 percent. The normal glucose removes the metabolic cause. At the bedside, give oxygen, support breathing, and treat the pneumonia, and avoid sedating her before oxygenation is restored.
Key principle: Agitation in a child with respiratory distress is hypoxia until proven otherwise.
Back to question Contents

BQ-186. Answer: E. Keep her in sight, remove dangerous objects, and arrange an urgent mental health evaluation
An acute positive screen means a yes to any item and a yes to the acuity question, and it marks a child at imminent risk. The pathway calls for continuous observation, removal of dangerous objects from the room, and an urgent mental health evaluation. A brief assessment with home discharge, the closest wrong option, applies to a non-acute positive screen. A handout or a promise leaves risk unexamined. At the bedside, stay with her, involve her mother, and arrange emergency department evaluation unless a mental health clinician can be reached and an alternative plan is set.
Key principle: A yes on the acuity question makes the screen acute and calls for observation and urgent evaluation.
Back to question Contents

BQ-187. Answer: C. A positive screen that calls for the acuity question and further assessment
On this screening tool, a yes to any item or a refusal to answer counts as a positive screen, because a teen who declines to answer may be hiding thoughts. The next step is the acuity question, which sorts a non-acute positive from an acute positive. An immediate emergency transfer, the closest wrong option, skips that sorting. Calling the screen negative or waiting for a year leaves possible risk unexamined. At the bedside, thank him, ask whether he is having these thoughts now, and complete a brief suicide safety assessment before he leaves.
Key principle: A refusal to answer a suicide screening item is a positive screen.
Back to question Contents

BQ-188. Answer: E. A history of aggression toward other people
Prior aggression is the strongest predictor of an unsafe course in an agitated child. Adolescence, male sex, substance use, a neurodevelopmental condition, and trauma add risk but carry less weight. The closest wrong option is autism, which raises risk modestly and is also a reason to adapt the environment and not to escalate. Shouting describes the present, not the pattern. At the bedside, use a quiet room, one lead communicator, simple language, and choices, and keep medicine and restraint for imminent danger.
Key principle: Prior aggression is the strongest predictor of an unsafe course in an agitated child.
Back to question Contents

BQ-189. Answer: D. Locking up or removing the firearm and medicines at home, with her father's help

A safety plan lists warning signs, coping steps, supports, and professional contacts, but the item that carries the most weight is limiting access to lethal means. Many youth suicide attempts are impulsive, so distance from a firearm or medicines gives time for the urge to pass. A signed promise, the closest wrong option, is not a plan and does not change the risk in the home. At the bedside, ask the father directly, agree on where the items will go, give the 988 Lifeline number, and arrange follow-up before the family leaves.

Key principle: Restricting access to firearms and medicines is the most important part of a youth safety plan.

Back to question Contents

BQ-190. Answer: C. Selective mutism

Selective mutism is a form of anxiety in which a child who can speak is unable to speak in specific social settings. The pattern is fluent speech at home and silence at school, with freezing around unfamiliar adults. Normal hearing and language testing remove the language disorders. Oppositional defiance, the closest wrong option, would show active refusal and rule-breaking, while this boy is frightened. At the bedside, ask what he fears, coordinate with the school to lower speaking demands, and refer for exposure-based cognitive behavioral therapy.

Key principle: Fluent speech at home with silence in specific settings is selective mutism, an anxiety disorder.

Back to question Contents

BQ-191. Answer: E. Learned arousal or bladder inhibition in response to a full bladder

The alarm acts through conditioning. Each time urine triggers the sensor, the sound wakes him, and over weeks the sensation of a full bladder becomes linked to waking or to holding urine. Dryness follows in about two-thirds of children. Desmopressin, the closest wrong mechanism, works by reducing nighttime urine volume. At the bedside, explain that a parent must help him wake and change, continue for at least 2 to 3 months, and stop after 14 consecutive dry nights.

Key principle: A bed-wetting alarm works by conditioning, and a parent must help the child wake.

Back to question Contents

BQ-192. Answer: D. Nocturnal polyuria with normal bladder capacity

Monosymptomatic enuresis usually reflects excess nighttime urine production together with difficulty waking, and structural disease is uncommon. This boy has large overnight volumes and normal daytime volumes, so his bladder capacity is adequate and the problem is the volume. Detrusor overactivity, the closest wrong option, would cause daytime urgency and small voids. Obstruction and neurogenic bladder bring a weak stream, dribbling, or neurologic signs. At the bedside, discuss an alarm or desmopressin, which fits nighttime polyuria, and advise a void at bedtime.

Key principle: Bed-wetting with large overnight volumes and normal capacity reflects nocturnal polyuria and responds to desmopressin.

Back to question Contents

BQ-193. Answer: A. Treat the constipation first, then reassess the bed-wetting

A full rectum can press on the bladder and make bed-wetting worse, so constipation is a modifiable driver. The guidance for enuresis is to treat constipation first when it is present, and often the wetting improves. Starting desmopressin, the closest wrong option, treats the wrong problem first. A dipstick is the only required test, so imaging and urodynamics are not routine without red flags. At the bedside, start a stool regimen, advise a bedtime void and limited evening fluids, and reassess the nights in a few weeks.

Key principle: In enuresis with constipation, treat the constipation first.

Back to question Contents

BQ-194. Answer: E. Cannabis use affecting attention and memory

ADHD requires onset before age 12, so a new attention problem at 16 calls for a search for another cause. Substance use, including marijuana, can produce inattention, poor memory, and low motivation, and red eyes with increased appetite fit recent use. Depression, the closest wrong option, can look similar but does not explain red eyes and appetite. At the bedside, ask him privately about substance use, consider confidential toxicology, and screen for depression and anxiety before labeling ADHD.

Key principle: New inattention in an adolescent calls for a substance use history before an ADHD diagnosis.

Back to question Contents

BQ-195. Answer: A. Reduced central sympathetic outflow from alpha-2 receptor stimulation

Clonidine stimulates central alpha-2 receptors, which lowers sympathetic outflow, so heart rate and pressure fall and sleepiness follows. This is the expected pharmacology of the class and is the reason for the monitoring. Beta blockade, the closest wrong option, would slow the pulse without the central sedation. Stimulants and atomoxetine act on dopamine and norepinephrine reuptake and raise heart rate. At the bedside, check pulse and blood pressure, discuss a dose change with a slow titration, and taper to stop, because abrupt cessation causes rebound hypertension.

Key principle: Alpha-2 agonists lower sympathetic outflow, causing sleepiness, slow heart rate, and low blood pressure.

Back to question Contents

BQ-196. Answer: E. About 60 percent have minimal or mild tics within 6 years of evaluation

Tics usually begin in childhood, peak around ages 10 to 12, and improve in most adolescents as the brain matures. Follow-up studies show that about 60 percent have minimal or mild tics within 6 years of evaluation, and some have none. Claiming that they all vanish by 14, the closest wrong option, overstates the outlook. Medicine is not the main driver of improvement. At the bedside, offer watchful waiting while the tics cause no impairment, offer behavioral therapy if the family wants treatment, and screen for ADHD and anxiety.

Key principle: Tics peak around ages 10 to 12, and most adolescents improve.

Back to question Contents

BQ-197. Answer: D. A manic episode unmasked by the antidepressant

Antidepressants can precipitate mania in a young person who has an unrecognized bipolar disorder, and family history raises that risk. Her decreased need for sleep, pressured speech, flight of ideas, and reckless spending define a manic episode. Serotonin syndrome, the closest wrong option, would bring clonus, tremor, fever, and diarrhea. Normal toxicology removes intoxication. At the bedside, stop adding medicine, arrange urgent psychiatry consultation, ask about safety, and review whether the antidepressant should continue.

Key principle: Mania after starting an antidepressant suggests unrecognized bipolar disorder and needs psychiatry consultation.

Back to question Contents

BQ-198. Answer: A. Obtain a teacher rating scale and school observations

ADHD is diagnosed from symptoms that appear in more than one setting, with onset before age 12, using information from the parent and the teacher. Without a second informant, the history covers only the home. Neuropsychological testing, the closest wrong option, does not improve diagnostic accuracy and delays care. A medication response does not confirm the diagnosis. At the bedside, send rating scales to the teacher, review the school records, and screen for anxiety, depression, trauma, and learning problems.

Key principle: Diagnose ADHD only after confirming symptoms in more than one setting, with parent and teacher input.

Back to question Contents

BQ-199. Answer: E. Expected sympathomimetic effects of the stimulant

Stimulants increase dopamine and norepinephrine activity, which raises arousal and blunts appetite during the hours the drug is active. Reduced lunchtime appetite and delayed sleep are the commonest effects, and rebound hunger at dinner fits wearing off. An eating disorder, the closest wrong option, would show weight loss, body image concerns, and restriction beyond medication hours. At the bedside, plan an evening family meal and a bedtime routine, leave the dose alone, and plot height, weight, blood pressure, and pulse at each visit.

Key principle: Appetite loss and delayed sleep are expected stimulant effects, managed with routines and monitoring.

Back to question Contents

BQ-200. Answer: B. Reassess the diagnosis, adherence, comorbid conditions, and bipolar symptoms

A lack of response by 6 to 8 weeks at an adequate dose means the original assumptions need checking. GLAD-PC advises reassessing the diagnosis, adherence, coexisting conditions such as anxiety or substance use, and bipolar symptoms before changing the plan. Pushing past the highest tolerated dose, the closest wrong option, adds adverse effects without a new insight. A stimulant treats a different problem, and waiting prolongs illness. At the bedside, interview him and his mother again, screen for substance use, and consult mental health if the picture stays unclear.

Key principle: With no response by 6 to 8 weeks, reassess the diagnosis, adherence, comorbidity, and bipolar symptoms before changing treatment.

Back to question Contents

BQ-201. Answer: C. Major depressive disorder

Depression in youth often presents as irritability as well as low mood, with lost interest, sleep and appetite change, and falling grades. The duration, the loss of enjoyment, and the decline in function separate it from ordinary moodiness, which is the closest wrong option. A normal thyroid level removes a medical contributor. At the bedside, ask about suicidal thoughts directly, grade severity with a validated questionnaire, ask about family history of bipolar disorder and past high-energy episodes, and plan active monitoring or treatment by severity.

Key principle: Irritable mood, lost interest, and falling function for weeks suggest depression in a teen.

Back to question Contents

BQ-202. Answer: D. Sympathomimetic intoxication

Intoxication or withdrawal is a medical cause that looks psychiatric, and the clue is odd pupils with a fast pulse. Wide pupils, sweating, high heart rate and blood pressure, and mild fever form a sympathomimetic picture, as with stimulants. Anticholinergic toxicity, the closest wrong option, also dilates the pupils, but the skin is dry and flushed. A sedative would slow her. At the bedside, de-escalate verbally, keep her safe, use a benzodiazepine if needed, cool her, and guide toxicology by the history.

Key principle: Wide pupils with sweating, tachycardia, and agitation indicate sympathomimetic intoxication.

Back to question Contents

BQ-203. Answer: B. It prevents learning that the feared outcome does not occur

Avoidance brings relief, which rewards the behavior and teaches the brain that the situation was dangerous. Because she never meets the feared situation without the feared outcome, the fear is never corrected and spreads. Exposure-based cognitive behavioral therapy reverses this by letting her face feared situations in steps and learn what really happens. A permanent damage explanation, the closest wrong option, is not supported. At the bedside, build a graded list with her, practice small steps, and add an SSRI if anxiety is moderate to severe.

Key principle: Avoidance maintains anxiety by blocking corrective learning, and exposure-based cognitive behavioral therapy reverses it.

Back to question Contents

BQ-204. Answer: E. Refer the parents for parent training in behavior management

Defiance and tantrums are signals to look for ADHD, anxiety, trauma, language problems, or an unsafe home, and here those screens are negative. Parent training in behavior management is the main treatment for disruptive behavior in this age group. It teaches age-appropriate expectations, warm time together, praise for wanted behavior, and calm ignoring of minor misbehavior. An antipsychotic, the closest wrong option, carries metabolic and movement risks and is not first line. At the bedside, send the parents to a training program, set small goals, and review progress in a few weeks.

Key principle: Parent training in behavior management is the first treatment for disruptive behavior in young children.

Back to question Contents

BQ-205. Answer: D. Misuse and diversion of the prescription

In adolescents the stimulant has the strongest evidence of benefit, but misuse and diversion are real concerns because the drug is valued by peers. The guideline asks clinicians to watch for these and to screen for substance use. Height loss, the closest wrong option, is limited to about 2 cm at most, mostly in the first 2 years. Stimulants have not been shown to raise the risk of sudden death. At the bedside, limit the supply, count refills, lock the medicine at home, and ask about substance use at each visit.

Key principle: Misuse and diversion are the stimulant risks that most need attention in adolescents.

Back to question Contents

Answers: Domain 6. Psychosocial Issues

Contents Questions for this subject

BQ-156. Answer: D. Refer the family to social work and a medical-legal partnership for housing and legal help

Unstable housing exposes a child to mold, allergens, missed care, and stress, and these drive asthma exacerbations that no drug adjustment can fix. Asking about housing is part of the visit, and a positive answer deserves action the same day, such as social work and a medical-legal partnership that can address an eviction. The closest wrong option intensifies medication and ignores the cause, since adherence is already good. A family without money has a need, and a neglect report treats it as harm. At the bedside, name the referral, help with the forms, and write a letter on the housing hazards.

Key principle: A housing problem that drives illness is treated with referral to social work and legal aid in the same visit.

Back to question Contents

BQ-157. Answer: E. Food insecurity is unlikely, although a small share of food-insecure families are missed

A screen with 97 percent sensitivity detects nearly all food-insecure families, so a negative result makes food insecurity unlikely, but 3 percent are still missed and circumstances change. The AAP advises screening every family at health supervision visits and repeating it, because answers in one year may not hold in the next. The closest wrong option treats a negative screen as proof, which stops the questions that find a family before it runs short. At the bedside, ask the screen again at later visits, ask privately, and leave the door open by mentioning that many families run short at some point.

Key principle: A negative Hunger Vital Sign makes food insecurity unlikely but must be repeated at later visits.

Back to question Contents

BQ-158. Answer: B. Refer the mother for treatment and arrange close pediatric follow-up for the infant

Depression in a parent changes feeding, responsiveness, and the safety of the home, so the child's outcome depends on treating the parent. The AAP advises screening mothers for depression at visits in the first year and, when the screen is positive, referring the parent for treatment and following the infant closely. The closest wrong option normalizes her symptoms and delays care for six months. A depressed parent who denies harm does not meet the standard for a report. At the bedside, give her a named clinician or program, confirm she is safe, and see the infant again soon.

Key principle: A positive maternal depression screen leads to treatment referral for the parent and closer follow-up of the infant.

Back to question Contents

BQ-159. Answer: D. Exposure to ongoing parental conflict during an expected period of adjustment

What harms children in divorce is less the separation than conflict they watch or carry between homes. A few months of clinginess, sleep disruption, and school slippage is expected, and a child who is eating, playing, and attending school is adjusting. The closest wrong option, major depression, needs more pervasive and lasting symptoms than she shows. At the bedside, advise the parents to keep both households predictable, to stop using her as a messenger, and to shield her from arguments, and refer for counseling if the changes last beyond a few months or she stops eating or sleeping.

Key principle: After divorce, conflict the child sees drives most of the distress, and short-term regression is expected.

Back to question Contents

BQ-160. Answer: B. Complicated grief with risk of self-harm

Grief is expected after a parent dies, and it usually eases with support. It becomes complicated when it stays disabling or the child talks of joining the person who died, as the AAP clinical report on grief describes. Her withdrawal, falling grades, poor eating, and wish to be with her mother after eight months meet that description. The closest wrong option, normal grief, ignores the duration and the statement about joining her mother. At the bedside, ask about suicide directly, arrange an urgent mental health evaluation, and involve her aunt and school in a safety plan.

Key principle: Grief that stays disabling or includes wishing to join the person who died needs urgent mental health referral.

Back to question Contents

BQ-161. Answer: E. Repeated aggression, including rumors and exclusion, that involves a power imbalance

Bullying is aggression that is repeated and carries a power imbalance, and it can be direct, such as hitting and name-calling, or relational, such as rumors and exclusion. Here the captain's status gives the group power over him, and the behavior has lasted months. A single fight between equals is conflict, and the stomachaches are a typical early sign. The closest wrong option confuses online bullying with the whole definition. At the bedside, ask him directly about what happens and about low mood, tell his mother this is a health problem, and plan school contact.

Key principle: Bullying is repeated aggression with a power imbalance, direct or relational.

Back to question Contents

BQ-162. Answer: A. Within 30 days of placement, covering medical, dental, developmental, and mental health needs

Children in foster care arrive with more untreated disease, developmental delay, and mental health need than other patients, and records are often missing. The AAP statement asks for an initial assessment within 72 hours of placement and a full evaluation of medical, dental, developmental, mental health, and educational needs within 30 days, with a recheck at 60 to 90 days. The closest wrong option waits for the yearly visit and leaves needs unaddressed for months. At the bedside, treat the earache today, schedule the full evaluation within the next week, and keep him in one medical home if the placement changes.

Key principle: A child in foster care needs a full evaluation within 30 days of placement and a recheck at 60 to 90 days.

Back to question Contents

BQ-163. Answer: C. A significant mental health need is present in up to 80 percent of children in foster care

Children in foster care come from homes marked by neglect, abuse, and instability, so their burden of illness is higher than in any other group in a practice. The AAP statement reports that more than 70 percent have a history of abuse or neglect, up to 80 percent have a significant mental health need, and about one third have a chronic condition. The closest wrong option understates chronic illness. At the bedside, screen for acute mental health needs and suicide risk at entry, delay deeper trauma work until she has settled, and plan dental and developmental evaluations.

Key principle: Up to 80 percent of children in foster care have a significant mental health need, so screen at entry.

Back to question Contents

BQ-164. Answer: D. Treat the infection, give catch-up vaccines now, and help with Medicaid and a contact route

Homelessness fragments care, so each contact may be the only one, and acute visits are the best chance to close gaps. The AAP policy asks you to use acute visits to catch up on vaccines, to help enroll the family in Medicaid, and to ask about a working phone and a mailing address. The closest wrong option schedules a separate visit three months away, which a family without a phone is unlikely to keep. At the bedside, give a written summary of care, record the shelter's contact route, and give the vaccines with a clear plan.

Key principle: Use every visit for a homeless child to catch up vaccines, enroll in coverage, and secure a way to reach the family.

Back to question Contents

BQ-165. Answer: A. Use the trained video interpreter for the whole visit

Accurate communication underlies every screening question, and untrained interpreters omit, add, or change meaning, particularly when topics are sensitive. A teenager may also be unable to discuss a parent's stress, and her presence may inhibit candor. The AAP policy on immigrant families asks for trained interpreters, in person or by phone or video, and never relatives or the child. The closest wrong option mixes both, and still leaves routine items to an untrained helper. At the bedside, start the video interpreter at the beginning, speak to the mother, and ask about food, housing, and stress.

Key principle: Use a trained medical interpreter for every visit, never a relative or the child.

Back to question Contents

BQ-166. Answer: C. Peer victimization (bullying) presenting with somatic symptoms

Stomachaches, headaches, school avoidance, and falling grades are the usual first signs of bullying, and a child often denies it until asked directly in a safe way. Missing belongings and money, a changed route to school, and symptoms that cluster on school mornings point toward it, and the normal screening results argue against celiac disease or ulcer disease. The closest wrong option, functional constipation, would not explain the lost belongings. At the bedside, ask her directly after explaining the word, screen for depression, tell her parents it is a health problem, and plan contact with the school.

Key principle: Somatic complaints with school avoidance and falling grades should prompt a direct question about bullying.

Back to question Contents

BQ-167. Answer: D. He interprets the phrase literally and does not yet grasp that death is permanent

Preschool children think concretely and have not yet formed a concept of permanence, so the word sleep suggests a return and may make bedtime frightening. His repeated question and bedtime resistance follow logically from the explanation he was given. The closest wrong option labels the behavior as complicated grief, which needs disabling symptoms or talk of joining the person who died. At the bedside, advise using the word died, explaining that the body stopped working and cannot work again, answering the same question the same way each time, and linking the family to bereavement support.

Key principle: Young children take euphemisms literally, so use the word died and explain it in concrete terms.

Back to question Contents

BQ-168. Answer: A. Offer every parent private education and resource cards so that disclosure is not required

Partner violence affects about 1 in 4 children through what they see, hear, or live with afterward, and a parent who is hurt may tell the pediatrician first or never. The AAP prefers universal education over interrogation, with information and resource cards given to every parent, for example through the CUES approach, so that disclosure is not required. The closest wrong option screens only visible cases and misses most. At the bedside, speak with her alone, provide the National Domestic Violence Hotline number in a form she can carry safely, and avoid pressure to leave.

Key principle: Offer every parent private education and resources rather than demanding a disclosure.

Back to question Contents

BQ-169. Answer: E. Write brief factual entries in protected notes and confirm a safe way to reach her

A partner who is also a guardian may read the chart or the portal, and a visible record of a disclosure can put the mother at greater risk. The AAP advises brief, factual documentation, protected notes, and asking about the safest way to contact her. The closest wrong option records everything in the open chart. Declining to document anything leaves the child's care unsupported. Domestic violence alone does not trigger a child abuse report, although rules vary by state. At the bedside, give the hotline number privately, offer local resources, and arrange follow-up for the child's sleep.

Key principle: Document partner violence briefly, protect the record from the abuser, and confirm a safe way to contact the survivor.

Back to question Contents

BQ-170. Answer: C. The child welfare agency, usually through the foster parent or caseworker
A child in foster care is in the legal custody of the state, which delegates routine medical decisions to the agency and the foster parent or caseworker. The AAP statement states that consent for routine care usually comes through the child welfare agency. The closest wrong option makes care depend on a birth parent who cannot be reached, which delays protection for a child with an unknown vaccine history. At the bedside, confirm the consent document, give the overdue vaccines, and ask the agency for any records.
Key principle: Consent for routine care of a child in foster care usually comes through the child welfare agency.
Back to question Contents

BQ-171. Answer: E. Unmet social needs that call for support and resources rather than a neglect report
Poverty creates barriers such as lost transportation and unaffordable care, and these are needs and not abuse. Neglect requires harm, or risk of serious harm, to the child from a caregiver's failure to meet needs when help is available. This father is working, asking for help, and the child is growing well. The closest wrong option files a report for the missed visits and treats lack of money as neglect. At the bedside, screen for food, housing, and transportation needs, connect him to dental care and benefits, rebook visits at times he can attend, and reassess at follow-up.
Key principle: Poverty alone is a need to meet, not neglect, and reporting is reserved for actual harm.
Back to question Contents

BQ-172. Answer: D. Explain that care does not depend on immigration status, and ask about food, housing, and trauma
Fear of immigration consequences keeps families from care, and children who have migrated often carry trauma, interrupted schooling, and unmet food and housing needs. The AAP statement states that you do not need to ask about immigration status to give care and asks for trained interpreters, screening for social needs, assessment of development, and attention to trauma. The closest wrong option asks for documents and creates the fear that keeps families away. At the bedside, reassure her honestly, ask about the family's needs, check development and vaccines, and give a summary of resources.
Key principle: Provide care without asking immigration status, and screen for trauma and social needs with a trained interpreter.
Back to question Contents

Answers: Domain 7. Emergency and Critical Care

Contents Questions for this subject

BQ-206. Answer: E. Respiratory distress

The triangle sorts a child by appearance, work of breathing, and circulation to the skin. This girl interacts normally, so her appearance is normal, her work of breathing is abnormal, and her skin is normal, which is the pattern of respiratory distress. Respiratory failure, the closest wrong option, would add an abnormal appearance such as drowsiness or poor interaction. Shock would show abnormal skin. At the bedside, keep her in her father's lap, give oxygen in the form she tolerates, and avoid a struggle that could close the airway.

Key principle: Normal appearance with abnormal work of breathing and normal skin is respiratory distress.

Back to question Contents

BQ-207. Answer: B. Check a bedside glucose

An abnormal appearance with normal breathing and normal skin points to a primary brain or metabolic problem on the assessment triangle. Poor intake in a young child makes hypoglycemia a leading cause, and it is rapidly reversible and harmful if missed. Fluid, the closest wrong option, fits shock, but his skin is pink and warm. Ventilation and intubation address a breathing problem he does not have. At the bedside, check glucose in the primary survey, treat a low value at once, and look for other causes such as infection or ingestion.

Key principle: Check a glucose in every drowsy child, especially when only the appearance side of the triangle is abnormal.

Back to question Contents

BQ-208. Answer: A. Compensatory vasoconstriction and tachycardia that preserve pressure until late

In shock a child raises heart rate and constricts skin and gut vessels to protect the brain and heart, which keeps blood pressure normal while perfusion fails. Hypotension appears only after compensation is spent, so it is a late and dangerous finding. Her cool hands, slow capillary refill, weak pulses, and lethargy show the failure. Waiting for a low pressure, the closest wrong approach, delays rescue. At the bedside, treat perfusion and not the number, check glucose, give measured fluid boluses with reassessment, and start antimicrobials if infection is likely.

Key principle: Hypotension is a late sign of shock in children because compensation preserves pressure until it fails.

Back to question Contents

BQ-209. Answer: B. Give epinephrine 0.01 mg/kg intravenously as early as possible

Pulseless electrical activity is a nonshockable rhythm, and in children it usually follows hypoxia and acidosis that have starved the heart. Epinephrine, given as early as possible, improves coronary and cerebral perfusion while compressions and ventilation continue. Defibrillation, the closest wrong option, is reserved for ventricular fibrillation and pulseless ventricular tachycardia. Cardioversion needs a pulse, and pausing compressions wastes time. At the bedside, give 0.01 mg/kg, which is 0.26 mg for this boy, repeat every 3 to 5 minutes, and hunt for reversible causes.

Key principle: In pulseless electrical activity, give epinephrine early and keep compressing while looking for the cause.

Back to question Contents

BQ-210. Answer: A. Give oxygen and begin bag-mask ventilation

Bradycardia in an infant is usually caused by hypoxia, and the heart rate recovers when oxygen delivery is restored. Oxygen and ventilation come first. Compressions are added only if perfusion stays poor after effective ventilation, and epinephrine follows if the bradycardia persists. Starting epinephrine or compressions first, the closest wrong options, skip the treatment of the cause. Cardioversion is for tachyarrhythmia. At the bedside, clear secretions, give effective breaths, and reassess pulse and perfusion within seconds.

Key principle: Bradycardia with poor perfusion in an infant is treated first with oxygen and ventilation.

Back to question Contents

BQ-211. Answer: E. Closing ductus in a ductal-dependent heart lesion

In some heart defects the systemic circulation depends on the ductus arteriosus, and when it closes in the first weeks the baby collapses with poor pulses and heart failure. Normal sodium, potassium, and glucose argue against adrenal crisis, and the lack of fever makes sepsis less likely. Myocarditis, the closest wrong option, can look similar, but a collapse in the second week fits a duct. At the bedside, start prostaglandin on suspicion, give fluid cautiously, obtain access, and arrange urgent cardiology and transfer.

Key principle: A gray neonate with poor pulses in the second week of life has a ductal-dependent lesion until proven otherwise.

Back to question Contents

BQ-212. Answer: C. Obstructed venous return from rising pressure in the chest

A tension pneumothorax traps air that cannot escape, and the rising pressure compresses the great veins and the heart, so blood cannot return to fill the heart. This is obstructive shock, and the clues are distended neck veins, unequal breath sounds, and tracheal shift. Hemorrhage, the closest wrong option, would give flat neck veins. Neurogenic and septic shock do not fit the timing or the veins. At the bedside, relieve the obstruction at once with needle decompression, then a chest tube, without waiting for imaging.

Key principle: Obstructive shock is treated by relieving the obstruction, not by fluid.

Back to question Contents

BQ-213. Answer: A. Give intramuscular epinephrine immediately

Anaphylaxis is distributive shock and airway swelling driven by histamine and other mediators, and epinephrine reverses it by tightening vessels, relaxing the airway, and blunting mediator release. Intramuscular epinephrine is the first treatment, given immediately. Antihistamines and steroids, the closest wrong options, relieve itch but do not stop airway or circulatory collapse. Albuterol treats only wheeze. At the bedside, give epinephrine in the thigh, call emergency services, lay him flat, and repeat the dose if symptoms persist.

Key principle: Intramuscular epinephrine is the first treatment for anaphylaxis, and no other drug replaces it.

Back to question Contents

BQ-214. Answer: C. Cardiogenic shock

In cardiogenic shock the pump fails after a viral illness, so blood backs up into the lungs and liver while perfusion falls. Crackles, a gallop, and an enlarged liver are the clues, and the abdominal pain reflects liver congestion. Hypovolemic shock, the closest wrong option, would show dry mucosa and flat neck veins. Tamponade would give muffled heart sounds and clear lungs. At the bedside, give only small fluid volumes, obtain an echocardiogram, start an inotrope, and involve cardiology urgently.

Key principle: Crackles, a gallop, and an enlarged liver after a viral illness indicate cardiogenic shock.

Back to question Contents

BQ-215. Answer: D. Give broad-spectrum parenteral antimicrobials within one hour of recognition

Septic shock is a distributive failure in which infection drives vasodilation and capillary leak, and every hour of delay raises mortality. The 2020 Surviving Sepsis pediatric guidelines set antimicrobials within 1 hour of recognizing septic shock, with cultures drawn only if that costs no time. The three-hour window, the closest wrong option, applies to sepsis with organ dysfunction but no shock. At the bedside, give the antimicrobials now, reassess after each fluid bolus, and start epinephrine or norepinephrine if shock persists.

Key principle: In septic shock, give antimicrobials within 1 hour and do not wait for cultures or a lumbar puncture.

Back to question Contents

BQ-216. Answer: E. Airway resistance rises with the fourth power of the radius

Resistance to airflow rises with the fourth power of the radius, so halving the radius multiplies resistance sixteen times. An infant's airway is narrow to begin with, and a millimeter of swelling that is trivial in an adult can close it. The closest wrong option is a direct proportion, which would predict only a doubling. Infant chest walls are soft, not rigid, and the diaphragm tires quickly. At the bedside, suction the nose, give oxygen if needed, and watch for a slowing rate or drowsiness as a sign of failure.

Key principle: Airway resistance rises with the fourth power of the radius, so infants tolerate little swelling.

Back to question Contents

BQ-217. Answer: A. Alternate five back blows with five chest thrusts

A foreign body that blocks the airway needs a mechanical force to dislodge it. In an infant under 1 year, alternating five back blows with five chest thrusts creates that force while avoiding injury to the liver, which abdominal thrusts risk in a small child. A blind finger sweep, the closest wrong option, can push the object deeper. Water cannot clear a complete obstruction, and compressions begin only if he becomes unresponsive. At the bedside, repeat the cycle until the object comes out, and start CPR with a call for emergency help if he becomes limp.

Key principle: A choking infant under 1 year gets alternating back blows and chest thrusts, never abdominal thrusts.

Back to question Contents

BQ-218. Answer: E. Opioid

Opioids act on receptors that suppress the drive to breathe and constrict the pupils, so the combination of pinpoint pupils, slow shallow breathing, and sedation defines the opioid toxidrome. Sedative-hypnotics, the closest wrong option, also sedate and slow the rate, but the pupils stay normal. Anticholinergic and sympathomimetic poisoning widen the pupils, and cholinergic poisoning wets the airway. At the bedside, open the airway and ventilate first, give naloxone, call the poison center, and observe her for recurrence of sedation.

Key principle: Small pupils, slow breathing, and sedation indicate an opioid toxidrome, treated with ventilation and then naloxone.

Back to question Contents

BQ-219. Answer: C. Anticholinergic

Antihistamines in large amounts block muscarinic receptors, which stops sweating, dilates the pupils, slows the gut, and causes delirium and fever. Dry flushed skin separates this toxidrome from the sympathomimetic one, the closest wrong option, which has wide pupils too but sweats. Cholinergic poisoning produces wet secretions and small pupils. At the bedside, secure the airway and circulation, check glucose, obtain an ECG, call the poison center, and treat agitation with a benzodiazepine.

Key principle: Wide pupils, dry flushed skin, fever, and delirium indicate an anticholinergic toxidrome.

Back to question Contents

BQ-220. Answer: E. Esophageal button battery ingestion

A button battery lodged in the esophagus sets up a current in wet tissue, and burns can form within hours. A round object with a double contour, together with the access to a remote control, fits a battery. A coin, the closest wrong option, shows a single uniform outline. Magnets cause trouble farther down the bowel. At the bedside, treat this as an emergency, avoid feeding him, arrange immediate removal by a surgeon or endoscopist, and ask the poison center about interim measures.

Key principle: A button battery in the esophagus is an emergency that needs prompt radiography and removal.

Back to question Contents

BQ-221. Answer: B. A sustained-release calcium channel blocker

A few drugs can kill a toddler with one or two doses, including calcium channel blockers, sulfonylureas, tricyclic antidepressants, opioids, and clonidine. A sustained-release calcium channel blocker is absorbed over hours, so a well child can develop shock and bradycardia later. A second-generation antihistamine, the closest wrong option, causes mild effects. At the bedside, call the poison center, monitor the child in hospital for the expected period, check glucose and an ECG, and secure all medicines at home.

Key principle: One tablet of a calcium channel blocker can cause delayed collapse in a toddler who looks well.

Back to question Contents

BQ-222. Answer: A. Partial-thickness burn

Partial-thickness burns damage the epidermis and part of the dermis, so they blister, weep, and hurt, and the skin blanches. Superficial burns, the closest wrong option, are dry and red without blisters. Full-thickness burns are leathery and painless because the nerve endings are destroyed. Because her palm with fingers is about 1 percent of body surface, this burn is roughly 3 percent. At the bedside, cool the burn with running water and never ice, remove clothing and jewelry, cover it with a clean dry dressing, and ask whether the story fits the pattern.

Key principle: Blistering, weeping, painful skin that blanches is a partial-thickness burn.

Back to question Contents

BQ-223. Answer: D. A 7-year-old with a full-thickness burn of the palm

The American Burn Association advises immediate consultation for partial-thickness burns of 10 percent of body surface or more, any full-thickness burn, deep burns of the face, hands, feet, genitalia, perineum, or joints, inhalation injury, chemical burns, and high-voltage injury. A full-thickness burn of the palm meets two criteria. The partial-thickness burns, including the closest wrong option at 6 percent, fall short of the 10 percent threshold. At the bedside, cool and cover the burn, give analgesia, document the pattern, and arrange the consultation.

Key principle: Any full-thickness burn, or a deep burn of the hands, face, or feet, warrants immediate burn center consultation.

Back to question Contents

BQ-224. Answer: C. Observe her in the emergency department and reserve imaging for worsening

Under the PECARN rule for children 2 years and older, a single episode of vomiting without other high-risk findings places her in the intermediate tier, where about 0.9 percent have a clinically important brain injury. Observation allows detection of change while sparing radiation to a young brain. Immediate computed tomography, the closest wrong option, is for the high-risk tier or a child who worsens. Discharge without observation misses early deterioration. At the bedside, reassess serially, scan if symptoms progress, and give clear return instructions.

Key principle: Observe the intermediate-risk child in the emergency department instead of scanning by reflex.

Back to question Contents

BQ-225. Answer: D. A Glasgow Coma Scale score of 14

The PECARN head rule sorts risk into tiers. Altered mental status, a Glasgow Coma Scale score of 14 or less, or signs of a basilar skull fracture mark the high-risk tier, because they reflect brain dysfunction or a base-of-skull break. Loss of consciousness, vomiting, severe headache, and a severe mechanism, including the closest wrong option, define the intermediate tier at about 0.9 percent. At the bedside, order computed tomography for the high-risk tier, and observe the intermediate-risk child with serial examinations.

Key principle: A Glasgow Coma Scale score of 14 or less, altered mental status, or a basilar skull fracture is PECARN high risk.

Back to question Contents

BQ-226. Answer: E. Keep spinal immobilization, secure the airway, and image the cervical spine

A Glasgow Coma Scale score of 3 to 8 places a child in the PECARN cervical spine high-risk group, and altered mental status makes the neck examination unreliable. The rule had a sensitivity of 94.3 percent and a negative predictive value of 99.9 percent, though it is not a substitute for clinical judgment. Clearing the neck clinically, the closest wrong option, fails when the child cannot report pain. At the bedside, protect the airway, keep immobilization, involve the trauma team, and image the cervical spine.

Key principle: A child with a Glasgow Coma Scale score of 3 to 8 after trauma is high risk for cervical spine injury and stays immobilized until imaged.

Back to question Contents

BQ-227. Answer: D. Supplemental oxygen delays desaturation and hides hypoventilation

When a child breathes supplemental oxygen, the lungs hold a large oxygen reserve, so saturation stays normal for minutes after ventilation falls. Carbon dioxide rises meanwhile. Capnography detects a failing breath before saturation falls, which is why it is added to sedation monitoring. Trusting the oximeter, the closest wrong approach, gives false reassurance. At the bedside, open the airway, stimulate the child, give bag-mask breaths if effort stays poor, stop the drug, and call for help.

Key principle: Capnography detects hypoventilation before the oximeter does, especially when oxygen is being given.

Back to question Contents

Answers: Domain 8. Child Abuse and Neglect

Contents Questions for this subject

BQ-228. Answer: A. Make the report to child protective services now and inform the receiving hospital of the concern

A femur fracture in a baby who cannot walk is a high-risk injury, and a short fall rarely explains it. The reporting duty rests on reasonable suspicion, and it follows the physician when the family changes hospitals. The correct option meets that duty and protects the infant. The closest wrong option waits for another team to decide, which leaves the baby unprotected and the duty unmet. At the bedside, call the report in, record the intake worker and reference number, tell the receiving team, and arrange safe transfer or admission.

Key principle: Moving a child to another hospital does not remove the duty to report on reasonable suspicion.

Back to question Contents

BQ-229. Answer: E. Repeat the full skeletal survey in 2 to 3 weeks

Fresh fractures and early healing changes can be invisible on the first set of films, because callus and periosteal new bone take days to form. A second survey after an interval reveals injuries that the first one missed and helps date the existing ones. The closest wrong option, a repeat within a day, is too early for new healing changes. A single chest film misses limb injuries. At the bedside, order the second survey as a full series of views, and share the plan with the child protection team.

Key principle: A skeletal survey in a young infant is a full series that is repeated in 2 to 3 weeks.

Back to question Contents

BQ-230. Answer: C. Obtain a full skeletal survey

Abuse is often found in several children of the same household, and a young child cannot report injuries. Siblings under 2 years and twins have a real chance of hidden fractures even with a normal examination. The correct option finds them. The closest wrong option relies on symptoms, which are absent in many occult injuries. At the bedside, order a full series of views, involve the child abuse team, and make sure the child is in a protected setting.

Key principle: Twins and siblings under 2 years of an abused infant need a skeletal survey even when they look well.

Back to question Contents

BQ-231. Answer: D. Tearing of bridging veins by repeated acceleration and deceleration forces

In an infant the head is heavy and the neck is weak, so violent shaking or impact moves the brain relative to the skull and tears the bridging veins that cross the subdural space. Hemorrhages of different ages and retinal bleeding in several layers fit this pattern. The closest wrong option, a clotting deficiency, is a look-alike and needs testing but rarely explains this picture. At the bedside, stabilize, arrange brain imaging and a skeletal survey, and report.

Key principle: Subdural bleeding with multilayer retinal hemorrhages in an infant reflects inflicted acceleration and deceleration injury until proven otherwise.

Back to question Contents

BQ-232. Answer: B. Osteogenesis imperfecta

A collagen defect makes bone brittle, so fractures follow minimal force, and the same defect tints the sclerae, weakens the teeth, and in some families causes early hearing loss. The family history and the generalized bone thinning point to this disorder. The closest wrong option, inflicted trauma alone, ignores the extraskeletal signs. At the bedside, send bone health laboratory studies and genetic testing, keep the child safe during the work up, and involve the child protection team, since both problems can coexist.

Key principle: Blue sclerae, dental changes, and a family history make osteogenesis imperfecta the abuse look-alike to exclude.

Back to question Contents

BQ-233. Answer: D. Evaluate for abuse and test for a bleeding disorder at the same time

Bruising in a young infant is concerning whatever the family history, because infants who do not cruise rarely bruise. A bleeding disorder is a look-alike and can coexist with injury, so both pathways proceed together. The closest wrong option makes abuse depend on normal laboratory results, which delays protection and treats the disorder as the default. At the bedside, order a skeletal survey, head imaging, platelet count, and coagulation studies, report on reasonable suspicion, and speak plainly with the family.

Key principle: Evaluate abuse look-alikes in parallel with the abuse evaluation and not in sequence.

Back to question Contents

BQ-234. Answer: C. Congenital dermal melanocytosis

Pigment cells that stalled in the deep skin during migration produce flat blue-gray patches that are present from birth, stay unchanged, and cause no tenderness. A bruise changes and hurts, and these do not. The closest wrong option, inflicted bruising, is the concern the foster parent raises, but documentation from birth settles the question. At the bedside, document the patches with size and location in the record, tell the caregiver they are harmless, and reassess if the marks change.

Key principle: A blue-gray patch that is present from birth, unchanging, and painless is a birthmark and not a bruise.

Back to question Contents

BQ-235. Answer: E. Report to child protective services and arrange a scheduled specialist evaluation with trauma-focused mental health referral

A disclosure creates reasonable suspicion, so a report is required, but contact 5 days ago falls outside the 72-hour window for emergent evidence collection and prophylaxis. A scheduled visit with trained examiners and mental health referral serves her better. The closest wrong option applies an emergency pathway after its window has closed. Most examinations are normal, and a normal result neither confirms nor excludes abuse. At the bedside, record her words once, praise her for telling, and avoid repeating the interview.

Key principle: Contact beyond 72 hours moves care from emergent evidence collection to a report, scheduled evaluation, and mental health referral.

Back to question Contents

BQ-236. Answer: B. Sexual contact with an infected person

Trichomonas vaginalis is a sexually transmitted organism, and perinatal infection clears or presents in the first weeks, so a school-age child with active infection has most likely had sexual contact. Bacterial vaginosis and candidiasis, by contrast, are nonspecific. The closest wrong option, perinatal transmission, cannot explain a 5-year-old with new symptoms. At the bedside, treat the infection, report on reasonable suspicion, arrange evaluation by a child abuse specialist, and offer mental health referral.

Key principle: Trichomonas in a child past infancy raises concern for sexual abuse, whereas bacterial vaginosis and candidiasis do not.

Back to question Contents

BQ-237. Answer: D. Neglect

Neglect means a caregiver fails to meet basic needs for food, health care, supervision, or safety, and here every domain is affected. It is the most common form of confirmed maltreatment, with infants and toddlers at the highest rates. The closest wrong option, physical abuse, is the form most clinicians picture, but no injuries support it. At the bedside, treat the dental disease, catch up immunizations, address growth, ask about resources and child care, and report if the child is not improving despite help.

Key principle: Neglect is the most common form of confirmed maltreatment and shows as faltering growth, untreated disease, and inadequate supervision.

Back to question Contents

BQ-238. Answer: A. Obtain neuroimaging despite the normal neurologic examination

Among well-appearing infants evaluated for abuse, hidden head injury turned up in nearly 1 in 10 of those under 6 months, but in about 1 in 100 at 6 to 12 months. The risk is highest in the youngest, so the threshold for imaging is low. The closest wrong option waits for symptoms that may never come, and ultrasound misses small collections. At the bedside, obtain computed tomography or magnetic resonance imaging, complete the survey, and report.

Key principle: Image the brain of a young infant with suspected abuse even when the neurologic examination is normal.

Back to question Contents

BQ-239. Answer: C. Its location behind the ear

The TEN-4-FACESp rule flags bruises on the torso, ear, or neck, on the frenulum, angle of the jaw, fleshy cheek, eyelid, or conjunctiva, or with a patterned shape, and any bruise in an infant 4.99 months old or younger. Color cannot date a bruise, and side or size do not enter the rule. The closest wrong option, the color, tempts clinicians but has no validity. At the bedside, take a full history, examine the whole skin, consider a skeletal survey and further evaluation, and report if suspicion remains.

Key principle: Bruising behind the ear, on the torso or neck, or in the FACES regions of a child under 4 years deserves an abuse evaluation.

Back to question Contents

BQ-240. Answer: E. Inadequate nutrition provided at home
A child whose weight rises briskly when food is reliably offered, and stalls at home, has a problem of intake and not of absorption or metabolism. Medical causes rarely produce such a rapid, setting-dependent change. The closest wrong option, cow's milk protein allergy, would not improve on the same formula in hospital. At the bedside, watch a feed, document what was advised and understood, offer feeding support and social work help, and report if growth does not improve despite appropriate help.
Key principle: Brisk weight gain in hospital after stalled growth at home points to inadequate feeding and neglect.
Back to question Contents

BQ-241. Answer: C. Numerous hemorrhages in several retinal layers reaching the periphery
Violent acceleration and deceleration, with traction on the vitreous, produces bleeding that is widespread, multilayered, and extends to the ora serrata. A few posterior hemorrhages occur after birth or with other causes and carry little weight. The closest wrong option, a single hemorrhage, is a nonspecific finding. A fundus look without dilation is not an adequate screen. At the bedside, image the brain first, then request the dilated examination, and document the extent and layers.
Key principle: Retinal hemorrhages add weight when they are numerous, multilayered, and reach the periphery, and the eye examination never replaces brain imaging.
Back to question Contents

BQ-242. Answer: A. "I am required by law to report injuries like this for every baby, and a team will help work out what happened."
Families react best to honest, nonaccusatory language that presents the report as routine. The statement frames the duty as universal, avoids blame, and promises help, which keeps the family engaged. The closest wrong option sounds neutral, but it delays a report that reasonable suspicion already requires and leaves the infant unprotected. Accusing a person raises conflict and may be wrong. At the bedside, stay calm, document the exact conversation, call the agency, and ask a child abuse pediatrician to help when available.
Key principle: Tell the family plainly that the law requires a report for injuries like this, and avoid both accusation and delay.
Back to question Contents

BQ-243. Answer: B. Offer a resource card to every parent at the visit without requiring disclosure
Universal education gives every parent information and resources so that no one has to disclose to be helped, which is safe when an abuser is present. Direct questions need privacy, with no partner or verbal child aged 3 or older in the room. The closest wrong option exposes her to danger if the partner hears the questions. Violence between adults is not by itself a reason to report. At the bedside, use a discreet approach, offer the hotline number, and arrange a private moment at a later visit.
Key principle: When an abuser is present, offer universal resources without requiring disclosure, and ask direct questions only in private.
Back to question Contents

BQ-244. Answer: D. Her exact words in quotation marks with the question that prompted them, plus factual observations

The record is read later by investigators, lawyers, and courts, so exact words and the prompting question are the most valuable parts. They show that the question was open and not leading. The closest wrong option states a conclusion that a clinician cannot know, and it names a person without evidence. At the bedside, also record the time of the report call, the agency, the intake worker's name, and any reference number, and write that the findings are "concerning for" abuse.

Key principle: Document the child's exact words, the question asked, the observations, and the report details, and avoid conclusions about who did what.

Back to question Contents

Answers: Domain 9. Orthopedics and Sports Medicine

Contents Questions for this subject

BQ-245. Answer: C. Temporary loss of blood supply to the femoral head, with bone softening followed by healing

The femoral head loses its blood supply for a period, the bone softens and collapses, and it later heals with remodeling. This produces a limp with little pain, limited motion, and a small, dense head on films. The closest wrong option, slippage through the growth plate, describes the adolescent disorder in heavier children. At the bedside, obtain a radiograph of the pelvis, send to orthopedics, and do not label a limp lasting beyond a week or two as synovitis without an image.

Key principle: Perthes disease is a temporary loss of femoral head blood supply in a school-age child with a painless limp.

Back to question Contents

BQ-246. Answer: B. Transient synovitis of the hip

Self-limited hip inflammation follows viral illness in school-age children, and it produces a mild limp with a small effusion but no fever, a normal sedimentation rate, and a normal white count. None of the four Kocher predictors is present, so septic arthritis is very unlikely. The closest wrong option, a septic hip, would bring fever, refusal to bear weight, and abnormal markers. At the bedside, give an anti-inflammatory drug, arrange follow-up, and return the child the same day for fever or refusal to walk.

Key principle: A child with an irritable hip, no fever, normal markers, and weight bearing most likely has transient synovitis.

Back to question Contents

BQ-247. Answer: C. Obtain hip ultrasonography and involve orthopedics if an effusion is found

Two Kocher predictors are present, fever over 38.5 C (101.3 F) and refusal to bear weight, which places the chance of septic arthritis near 40 percent. At that level an ultrasound for effusion is the next step, followed by aspiration if fluid is found. The closest wrong option, ibuprofen with a delayed recheck, treats a possible septic joint as a viral limp. At the bedside, send blood cultures and inflammatory markers, call orthopedics early, and withhold antibiotics only until fluid is obtained when the child is stable.

Key principle: Two Kocher predictors call for hip ultrasonography and early orthopedic involvement.

Back to question Contents

BQ-248. Answer: A. About 93 percent

The four predictors are refusal to bear weight, fever over 38.5 C, a sedimentation rate over 40 mm/hour, and a white count over 12,000 per mm3. This girl has three of them, since her white count falls short of the threshold, and three predictors carry a probability of about 93 percent. The closest wrong option, 99.6 percent, applies to all four. The model came from one hospital, so treat it as a prompt and not a verdict. At the bedside, obtain ultrasound and call orthopedics for aspiration.

Key principle: Three Kocher predictors correspond to about a 93 percent chance of septic arthritis.

Back to question Contents

BQ-249. Answer: B. Kingella kingae

In children under 4 years, Kingella kingae is a leading cause of joint and bone infection and often produces a mild picture, with low-grade fever and only modest inflammatory markers. Culture in blood culture bottles improves recovery of this fastidious organism. The closest wrong option, Staphylococcus aureus, is the most common cause overall, but it is gram positive. At the bedside, aspirate the joint with orthopedics, send fluid and blood cultures, and choose treatment from the results.

Key principle: Kingella kingae causes a milder joint or bone infection than staphylococcus in children under 4 years.

Back to question Contents

BQ-250. Answer: D. Magnetic resonance imaging of the left leg

Bone infection spreads through the marrow before it changes the cortex, so plain films stay normal for 7 to 14 days. Magnetic resonance imaging shows marrow edema and abscesses early. The closest wrong option, repeating the films, delays a diagnosis in a febrile child who already has a probable one. At the bedside, give an antistaphylococcal antibiotic after cultures, aspirate or biopsy when it can be done safely, and involve orthopedics if a collection is found.

Key principle: Radiographs can stay normal for 7 to 14 days in osteomyelitis, so magnetic resonance imaging is the study of choice.

Back to question Contents

BQ-251. Answer: E. Switch to an oral antibiotic chosen by susceptibilities and arrange close follow-up

Children who improve quickly on intravenous therapy and have uncomplicated disease can move to oral treatment early, with outcomes similar to prolonged intravenous courses and fewer line complications. The closest wrong option, a long intravenous course, adds risk without benefit. Stopping all treatment ignores the need for a full course. At the bedside, pick an oral drug from the culture, give the dose by weight from the local protocol, and keep the follow-up visit, since monitoring matters more than the line.

Key principle: A child with uncomplicated osteomyelitis who is improving can switch early from intravenous to oral therapy.

Back to question Contents

BQ-252. Answer: D. Anteroposterior radiograph of the pelvis

Before about 4 months the femoral head is cartilage and ultrasound shows it well. After the ossific nucleus appears, ultrasound loses accuracy and a radiograph becomes the better test. Breech position and the exam findings suggest a dislocated left hip. The closest wrong option, ultrasound, is the right test only in younger infants. At the bedside, obtain the film, send to orthopedics, and explain that early treatment, usually a harness at this stage, offers the best result.

Key principle: After 4 months of age, a pelvic radiograph replaces ultrasonography for suspected hip dysplasia.

Back to question Contents

BQ-253. Answer: A. Stop weight bearing and arrange urgent orthopedic surgical fixation
The femoral head slips backward and downward off the neck through the growth plate, and body weight pushes it further. The usual repair is a single screw. The closest wrong option, therapy with weight bearing, risks a complete slip and avascular necrosis. The frog-leg view shows what the anteroposterior view hides. At the bedside, make her non-weight-bearing with crutches or a wheelchair, call orthopedics the same day, and tell the family that the other hip is at risk and follow-up matters.
Key principle: A confirmed slipped capital femoral epiphysis needs no weight bearing and urgent surgical fixation.
Back to question Contents

BQ-254. Answer: E. Type II
In a type II injury the line runs along the weak growth plate and then turns out through the metaphysis, leaving a triangular metaphyseal fragment. It is the most common type and usually heals well. The closest wrong option, type IV, crosses the metaphysis, plate, and epiphysis and enters the joint. Type I passes through the plate alone. At the bedside, reduce the fracture if it is displaced, immobilize with a cast, and warn the family that any plate injury can affect later growth.
Key principle: A Salter-Harris type II fracture runs through the growth plate and out through the metaphysis, sparing the epiphysis.
Back to question Contents

BQ-255. Answer: B. Salter-Harris type I fracture of the distal fibula
The growth plate is weaker than the ligaments around it, so a force that sprains an adult ankle can separate the plate of a child. A type I injury runs through the plate alone and often shows nothing on the first film. Localized tenderness over the plate with stable ligaments is the key. The closest wrong option, a ligament sprain, is the adult diagnosis. At the bedside, immobilize the ankle, restrict running, and recheck with repeat films in about a week.
Key principle: Tenderness over a growth plate with a normal film is a Salter-Harris type I fracture until shown otherwise.
Back to question Contents

BQ-256. Answer: D. Rising pressure within a closed fascial space that compromises tissue perfusion
Swelling inside a tight fascial compartment raises tissue pressure over capillary pressure, so muscle and nerve become ischemic while pulses stay normal. Pain out of proportion and pain with passive stretch are the early signs. The closest wrong option, arterial occlusion, would cause a pale cold hand and absent pulses. At the bedside, loosen or split the cast and padding to the skin, elevate to heart level, and call orthopedics at once for pressure measurement or fasciotomy.
Key principle: Pain out of proportion with passive stretch after casting is compartment syndrome, and pulses stay normal until late.
Back to question Contents

BQ-257. Answer: A. Toddler's fracture
A twisting force on a newly walking child produces a spiral fracture of the tibia without much swelling, and the child refuses to walk. The mechanism is typical of ambulatory toddlers. The closest wrong option, inflicted injury, is a real concern when a child cannot walk or the story does not fit, but here the age, mobility, and mechanism match. At the bedside, immobilize in a cast or splint, arrange orthopedic follow-up, and reconsider abuse if the history changes.
Key principle: A spiral tibial fracture after a twist in a child who has just learned to walk is the common innocent toddler's fracture.
Back to question Contents

BQ-258. Answer: C. Obtain magnetic resonance imaging of the spine to look for another cause
Idiopathic scoliosis usually involves a right thoracic curve in a girl, without pain. A boy, a left thoracic curve, and night pain suggest a secondary cause such as a spinal cord tumor or syrinx. The closest wrong option, bracing, treats the curve without finding its cause. At the bedside, order magnetic resonance imaging, examine the neurologic findings and skin of the back, and send to a spine specialist while the cause is sought.
Key principle: A painful curve, a left thoracic curve, a boy, or a neurologic sign needs magnetic resonance imaging for a secondary cause.
Back to question Contents

BQ-259. Answer: A. Teach a daily stretching and positioning program and examine the hips
A tight, fibrotic neck muscle tilts the head toward its side and turns the chin away, and a small lump may be felt in the muscle belly. Stretching and positioning, taught to the parents, correct most cases. The closest wrong option, imaging, is reserved for an acquired tilt, eye findings, or neurologic signs. Developmental hip dysplasia is more common with torticollis, and head flattening follows. At the bedside, teach the exercises, encourage tummy time, examine the hips, and reassess in a few weeks.
Key principle: Congenital muscular torticollis is treated with stretching and positioning, and the hips must be examined.
Back to question Contents

BQ-260. Answer: E. Begin serial casting with the Ponseti method in the first weeks of life
A fixed, rigid deformity is clubfoot, which differs from a flexible positional foot that corrects easily. Gentle serial manipulation and casting, started in the first weeks, correct most feet without major surgery. The closest wrong option, observation with stretching, treats it as positional and wastes the early window. At the bedside, send to an orthopedic surgeon or clinic that uses the Ponseti method, explain that casting is staged, and examine the hips and spine.
Key principle: A fixed clubfoot is corrected by Ponseti serial casting beginning in the first weeks of life.
Back to question Contents

BQ-261. Answer: C. Osgood-Schlatter disease
Repeated pull of the patellar tendon during a growth spurt inflames the tibial tubercle apophysis, producing pain with activity, local tenderness, and normal motion. Fragmentation is common and does not change management. The closest wrong option, Sinding-Larsen-Johansson disease, affects the lower pole of the patella instead. At the bedside, reassure the family, modify activity, add stretching and strengthening, and note that complete rest and surgery are rarely needed.
Key principle: Osgood-Schlatter disease is apophysitis of the tibial tubercle in a growing athlete and needs relative rest and no surgery.
Back to question Contents

BQ-262. Answer: B. Energy intake too low for training, lowering estrogen and thinning bone
When intake falls short of the energy that training uses, the body suppresses the hypothalamic signals for menses, estrogen falls, and bone thins, so stress fractures follow repeated load. Amenorrhea, restriction, and low weight together point to this. The closest wrong option, ovarian failure, would not explain the restriction or the training link. At the bedside, ask about eating, periods, and fatigue, plan nutritional rehabilitation, protect the bone, and delay a return to running until healing.
Key principle: Low energy availability in a female athlete causes amenorrhea, low bone density, and stress fractures.
Back to question Contents

BQ-263. Answer: E. Anterior cruciate ligament tear with hemarthrosis
A pivoting injury with a pop and rapid swelling suggests bleeding into the joint, which points to a cruciate tear or fracture. An increased Lachman translation with a soft end point confirms ligament failure. Adolescent girls tear this ligament more often than boys. The closest wrong option, patellofemoral pain, develops gradually and does not cause a tense effusion. At the bedside, protect the knee, obtain radiographs to exclude fracture, arrange magnetic resonance imaging, and send to orthopedics.
Key principle: A knee that swells within hours of a pivoting injury suggests hemarthrosis from a cruciate tear or a fracture.
Back to question Contents

BQ-264. Answer: C. Obtain echocardiography before clearing him for sports
A murmur that grows louder with standing, which reduces ventricular filling, suggests dynamic outflow obstruction as in hypertrophic cardiomyopathy, a leading cause of sudden death in athletes. An echocardiogram defines the anatomy. The closest wrong option, an electrocardiogram alone, cannot exclude structural disease and is not an adequate substitute. At the bedside, hold clearance until the study is complete, explain that it is a short pause, and recheck the murmur seated, standing, and with Valsalva.
Key principle: A systolic murmur that grows louder on standing requires echocardiography before an athlete is cleared.
Back to question Contents

BQ-265. Answer: D. Clear her for play without an electrocardiogram

The history and examination do most of the screening, and in an athlete with no red flags routine electrocardiography is not recommended, because false positives lead to unneeded testing and exclusion from sport. The correct option follows that guidance. The closest wrong option, a routine electrocardiogram, adds testing without proven benefit. At the bedside, finish the evaluation, document the plan, and update the history yearly, with a full examination every 2 to 3 years.

Key principle: Routine electrocardiography is not recommended for athletes who have no cardiac red flags on history and examination.

Back to question Contents

BQ-266. Answer: A. Allow light activity as tolerated and permit staged return to sport only after symptoms have cleared

A concussed athlete leaves play that day, and return to sport is staged and begins only when symptoms have cleared. Prolonged rest in a dark room is no longer advised. The closest wrong option applies strict rest, which can prolong recovery. Imaging is not needed without red flags such as worsening headache, vomiting, or neurologic signs. At the bedside, allow light activity, plan a gradual return, and recheck symptoms at each step, with school adjustments if needed.

Key principle: A concussed athlete leaves play that day and returns in stages after symptoms clear, without prolonged dark-room rest.

Back to question Contents

Answers: Domain 10. Eye, Ear, Nose, and Throat

Contents Questions for this subject

BQ-267. Answer: C. Give analgesia and watch for 48 to 72 hours, adding an antibiotic if she worsens or fails to improve

Acute otitis media is diagnosed by fluid plus bulging, and in a child of 2 years or older with mild illness in one ear, observation is an accepted option when follow-up is reliable. Pain control is needed either way. The closest wrong option, immediate amoxicillin-clavulanate, adds clavulanate with no recent amoxicillin use, and decongestants and antihistamines do not help. At the bedside, treat pain, explain the plan, and give the parent a way to return within 72 hours.

Key principle: Mild unilateral acute otitis media after age 2 years may be watched for 48 to 72 hours with pain control and reliable follow-up.

Back to question Contents

BQ-268. Answer: A. Amoxicillin-clavulanate at high dose for 10 days

Bulging drums with fever and bilateral disease at this age call for an antibiotic. Amoxicillin used in the past 30 days predicts resistant organisms that produce beta-lactamase, so clavulanate is added to the first-line drug. The closest wrong option, amoxicillin alone, repeats the drug that has just failed to prevent this infection. Topical drops do not reach the middle ear behind an intact drum. At the bedside, give the dose by weight from the local protocol, treat pain, and recheck if there is no improvement in 48 to 72 hours.

Key principle: Add clavulanate when amoxicillin was used within the past 30 days, or with purulent conjunctivitis or treatment failure.

Back to question Contents

BQ-269. Answer: C. Vasodilation from crying and vigorous handling

Crying, fever, and handling dilate the small vessels of the drum and redden it, but the middle ear is empty and the drum still moves. Acute otitis media needs fluid and bulging, and the drum is dull and immobile. The closest wrong option, early otitis media, would show a bulging or opaque drum with poor mobility. At the bedside, use pneumatic otoscopy to check movement, do not give an antibiotic for color alone, and reexamine when the child is calm if doubt remains.

Key principle: A red drum alone does not make acute otitis media, because crying and fever redden a normal drum.

Back to question Contents

BQ-270. Answer: E. Otitis media with effusion

Fluid persists behind the drum after a resolved infection, because the eustachian tube drains slowly. Without bulging, fever, or pain, the drum is neutral or retracted, amber or blue, and moves poorly. This is not infection. The closest wrong option, acute otitis media, requires a bulging drum with fluid. Chronic suppurative disease needs drainage through a perforation. At the bedside, watch for 3 months, do not give an antibiotic, antihistamine, decongestant, or steroid, and test hearing if fluid lasts 3 months.

Key principle: Fluid behind a retracted or neutral drum without bulging or pain is otitis media with effusion, and antibiotics do not treat it.

Back to question Contents

BQ-271. Answer: D. Arrange age-appropriate hearing testing
Persistent effusion can reduce hearing during the years when speech and learning develop. Guidelines call for a hearing test when fluid lasts 3 months, and its result guides referral for tubes. The closest wrong option, an antibiotic, does not clear an effusion that is not infected. Steroids, antihistamines, decongestants, and nasal steroid sprays do not help. At the bedside, send the child for audiology, reexamine every 3 to 6 months, and send to otolaryngology if hearing is reduced.
Key principle: Persistent effusion for 3 months calls for a hearing test, not for antibiotics or steroids.
Back to question Contents

BQ-272. Answer: B. Defer tubes because the middle ears are free of fluid today
Tubes help when fluid is present at the visit where candidacy is judged, because they ventilate and drain a middle ear that is full. Dry ears at that visit predict little benefit. The closest wrong option, placing tubes now, accepts the episode count and ignores the current ear findings. At the bedside, treat each new episode on its merits, reexamine if symptoms return, and reconsider tubes if effusion is documented.
Key principle: Offer tubes for recurrent acute otitis media only when fluid is present at the candidacy visit.
Back to question Contents

BQ-273. Answer: D. Topical antibiotic ear drops that are safe for use with tubes
Drainage through a tube is infected middle ear fluid draining out, and drops reach it directly through the open tube. A drop without ototoxic risk is chosen because the drum is open. The closest wrong option, oral amoxicillin, treats a local problem with systemic exposure. Systemic treatment is for fever, spread beyond the ear, or failure of drops. At the bedside, clean the canal, give the drops, recheck in 48 to 72 hours, and note that routine water precautions are not needed.
Key principle: Tube otorrhea in a well child is treated with non-ototoxic antibiotic drops alone.
Back to question Contents

BQ-274. Answer: B. Acute mastoiditis
Infection spreads from the middle ear into the mastoid air cells and forms a subperiosteal collection, which displaces the ear and swells the area behind it. The closest wrong option, otitis externa, hurts when the pinna is pulled but leaves the postauricular area normal. At the bedside, admit, obtain contrast computed tomography to look for abscess or intracranial spread, give intravenous antibiotics, and involve otolaryngology.
Key principle: A pinna pushed forward with postauricular swelling after otitis media is mastoiditis until proven otherwise.
Back to question Contents

BQ-275. Answer: D. Arrange diagnostic audiology now so that diagnosis is completed by 3 months of age

Early identification protects language, and national guidance sets screening by 1 month, diagnosis by 3 months, and enrollment in early intervention by 6 months. A startle can come from low-frequency or vibration cues and does not exclude loss. The closest wrong option, repeating the screen at 6 months, would miss the 3-month goal. At the bedside, send to an audiologist with pediatric experience, check for conditions that cause loss, and begin early intervention as soon as it is confirmed.

Key principle: A baby who fails the newborn hearing screen needs diagnostic audiology by 3 months, not reassurance.

Back to question Contents

BQ-276. Answer: A. Refer for formal audiologic testing

Hearing loss, including loss from persistent middle ear fluid, is a common and treatable cause of speech delay, and it must be excluded before other work up. Intervention is time sensitive. The closest wrong option, speech therapy alone, would proceed without knowing whether he hears. At the bedside, obtain a hearing test, review the ear examination, and refer for speech evaluation in parallel if the delay is significant.

Key principle: Test the hearing of any child with delayed speech before arranging other evaluations.

Back to question Contents

BQ-277. Answer: D. Perform a rapid antigen test for group A streptococcus

Group A streptococcus is rare under 3 years, and routine testing is not advised, but a household contact with proven infection changes his risk enough to test. The rapid test, with a backup culture if negative, identifies the cases that need treatment. The closest wrong option, treating without testing, exposes viral cases to antibiotics. A titer reflects past infection and does not diagnose a current one. At the bedside, swab the throat and confirm before prescribing.

Key principle: Under 3 years, test for streptococcal pharyngitis only when a household contact has proven infection.

Back to question Contents

BQ-278. Answer: A. Cephalexin for 10 days

Penicillin or amoxicillin is first line, treating to prevent rheumatic fever and abscess. A mild delayed rash is a nonsevere reaction, and a first-generation cephalosporin has a low chance of cross-reaction. The closest wrong option, azithromycin, is not preferred, because resistance is common. Clindamycin is kept for immediate-type reactions such as hives or anaphylaxis. At the bedside, give the course by weight from the local protocol, and tell the family she may return to school once afebrile and 12 hours into treatment.

Key principle: Use cephalexin for a nonsevere penicillin reaction and clindamycin for an immediate-type reaction.

Back to question Contents

BQ-279. Answer: D. Peritonsillar abscess

Infection breaks through the tonsil capsule into the space beside it and forms pus that pushes the tonsil and uvula to the opposite side, causing trismus and a muffled voice. Pain is one sided and severe. The closest wrong option, treatment failure of pharyngitis, misses the unilateral pressure signs. Oral antibiotics alone do not drain pus. At the bedside, arrange needle aspiration or drainage with otolaryngology, give antibiotics, and ensure hydration and an open airway.

Key principle: A unilateral tonsil with a displaced uvula, trismus, and a muffled voice is a peritonsillar abscess that needs drainage.

Back to question Contents

BQ-280. Answer: A. Watchful waiting with documentation of each future episode

Guidelines offer tonsillectomy for 7 documented episodes in 1 year, 5 per year for 2 years, or 3 per year for 3 years, and four episodes in one year falls short. Surgery carries risks and many children improve without it. The closest wrong option, tonsillectomy now, applies the benefit of surgery to a child who does not meet the counts. At the bedside, document each episode with temperature, nodes, exudate, or test, and revisit if the counts are reached or modifying factors appear.

Key principle: Tonsillectomy for recurrent infection requires documented episodes that meet the frequency counts, otherwise watch.

Back to question Contents

BQ-281. Answer: D. Order polysomnography before surgery

Sickle cell disease places a child in the higher-risk group, along with age under 2 years, obesity, Down syndrome, craniofacial and neuromuscular disease, and mucopolysaccharidoses. These children need objective testing, because surgery carries more risk and the diagnosis should be confirmed. The closest wrong option, surgery without testing, applies the shortcut meant for healthy children. At the bedside, order the sleep study, plan monitored postoperative care if surgery follows, and recheck symptoms 6 to 8 weeks afterward.

Key principle: Children at higher risk, including those with sickle cell disease, need polysomnography before adenotonsillectomy.

Back to question Contents

BQ-282. Answer: E. Use scheduled ibuprofen and acetaminophen

Pain after tonsillectomy is controlled with ibuprofen and acetaminophen. Codeine is avoided because ultra-rapid metabolism can cause severe or fatal respiratory depression, and the risk is higher in children with sleep apnea. The closest wrong option, codeine, offers the exact danger the guideline warns against. Aspirin raises bleeding and Reye risk. At the bedside, give a schedule by weight from local protocol, encourage fluids, and explain when to call for bleeding.

Key principle: After tonsillectomy, use ibuprofen and acetaminophen and avoid codeine.

Back to question Contents

BQ-283. Answer: C. Send him to the emergency department for evaluation by the operating surgeon

Bleeding can follow tonsillectomy in the first day and again when the scab separates around a week later. Bright red blood with pallor and tachycardia suggests significant loss, and a pause in bleeding may be temporary. The closest wrong option, reassurance, misses a potentially life-threatening bleed. Ibuprofen would add to bleeding risk. At the bedside, call ahead, send him for evaluation by the surgical team, and check hemoglobin and blood type.

Key principle: Any post-tonsillectomy bleeding with pallor or tachycardia needs emergency evaluation by the surgeon.

Back to question Contents

BQ-284. Answer: A. Acute bacterial sinusitis

A viral cold peaks around days 3 to 5 and fades over 10 days. Return of fever or worsening after initial improvement defines the worsening pattern of acute bacterial sinusitis, one of three recognized patterns along with persistent and severe illness. The closest wrong option, a plain viral infection, does not relapse after improving. At the bedside, treat with an antibiotic, avoid imaging and blood counts for uncomplicated disease, and reexamine the eyes for orbital spread.

Key principle: Fever or worsening after initial improvement defines the worsening pattern of acute bacterial sinusitis.

Back to question Contents

BQ-285. Answer: B. Amoxicillin-clavulanate for 10 days

Severe sinusitis is defined by a temperature of 39 C (102.2 F) or higher with purulent discharge for at least 3 days, and it is treated with an antibiotic. Child care, moderate to severe illness, and age under 2 years favor adding clavulanate. The closest wrong option, amoxicillin alone for a short course, undertreats a child with risk factors for resistant organisms. Azithromycin and trimethoprim-sulfamethoxazole fail too often. At the bedside, dose by weight from local protocol, treat for 10 days, and reassess at 72 hours.

Key principle: Severe sinusitis in a child in child care is treated with amoxicillin-clavulanate for 10 days.

Back to question Contents

BQ-286. Answer: D. Intracranial extension of the sinus infection

The frontal and ethmoid sinuses lie next to the dura and the orbit, and infection can spread through thin bone or through veins into the cranium, producing abscess, meningitis, or thrombosis. Severe headache, photophobia, seizure, or focal signs mark this. The closest wrong option, orbital cellulitis, would show lid swelling, a bulging eye, or painful movement, which are absent. At the bedside, obtain urgent contrast imaging of the head, start intravenous antibiotics, and involve neurosurgery and otolaryngology.

Key principle: Severe headache, photophobia, seizure, or focal deficits with sinusitis suggest intracranial extension.

Back to question Contents

BQ-287. Answer: B. IgE-mediated histamine release from mast cells after allergen exposure
In sensitized people, inhaled pollen cross-links IgE on mast cells, which release histamine and other mediators that cause itching, sneezing, and watery discharge. The seasonal pattern, itch, and pale turbinates fit. The closest wrong option, viral replication, produces a short illness with fever and discolored discharge. At the bedside, begin an intranasal steroid as first-line therapy, reduce exposure, and add an oral or topical antihistamine for itch if needed.
Key principle: Seasonal itching, sneezing, and clear discharge reflect IgE-mediated mast cell release, treated first with an intranasal steroid.
Back to question Contents

BQ-288. Answer: C. Nasal foreign body
A toddler explores with the nose, and an object left in place causes mucosal irritation and secondary infection, producing a one-sided, foul discharge. Allergic rhinitis and adenoid enlargement affect both sides and do not smell foul. The closest wrong option, sinusitis, would be bilateral with cough and fever in most cases. At the bedside, look in the nose with good light, remove the object with suction or forceps, and consider a battery or magnet an emergency.
Key principle: One-sided foul nasal discharge in a toddler is a foreign body until proven otherwise.
Back to question Contents

BQ-289. Answer: E. Evaluate for a bleeding disorder with laboratory testing
Most nosebleeds come from fragile vessels at the front of the septum and need only first aid. Frequent bleeds, bruising, a family bleeding history, or age under 2 years raise the chance of a bleeding disorder such as von Willebrand disease. The closest wrong option, reassurance about picking, ignores three risk features. At the bedside, order a platelet count and coagulation studies, ask about other bleeding, and send to hematology if abnormal.
Key principle: Frequent nosebleeds with bruising, a family bleeding history, or age under 2 years call for bleeding disorder testing.
Back to question Contents

BQ-290. Answer: C. Juvenile nasopharyngeal angiofibroma
A vascular tumor in the nasopharynx of an adolescent boy bleeds easily and obstructs one side of the nose. Common anterior bleeds from picking would be visible in the front of the septum. The closest wrong option, trauma to the septum, is the usual cause of nosebleeds but fails to explain the obstruction and the invisible source. At the bedside, avoid biopsy, obtain endoscopy and imaging, and send to otolaryngology.
Key principle: Recurrent one-sided nosebleeds with obstruction in an adolescent boy need a look at the nasopharynx.
Back to question Contents

BQ-291. Answer: E. Thyroglossal duct cyst

The thyroid descends from the base of the tongue along a duct that normally disappears, and a persisting remnant forms a cyst that moves with swallowing and tongue protrusion because it is attached to the hyoid. The closest wrong option, a branchial cleft cyst, lies laterally along the sternocleidomastoid. At the bedside, obtain ultrasound to confirm a normal thyroid gland in the usual position, and refer for excision.

Key principle: A midline neck mass that rises with swallowing and tongue protrusion is a thyroglossal duct cyst.

Back to question Contents

BQ-292. Answer: A. Bartonella henselae

A kitten scratch inoculates Bartonella henselae, and the regional nodes draining the site enlarge and become tender after a few weeks. The papule at the scratch site and the animal exposure complete the picture. The closest wrong option, Staphylococcus aureus, causes a warm, acute adenitis without a primary papule or cat exposure. At the bedside, confirm with serology, give supportive care, and refer if the node suppurates or the child has systemic illness.

Key principle: A tender regional node after kitten exposure with a scratch papule is cat-scratch disease from Bartonella henselae.

Back to question Contents

BQ-293. Answer: C. Nontuberculous mycobacterial lymphadenitis

Atypical mycobacteria infect nodes of young children in the head and neck, causing slow growth, little pain, no fever, and a violaceous skin change. The closest wrong option, staphylococcal adenitis, is tender and warm and responds to antibiotics within days. Failure of cephalexin supports a different organism. At the bedside, send to a surgeon for excision, avoid repeated antibiotic courses, and keep malignancy in mind if the node is hard, fixed, or enlarging.

Key principle: A subacute, nontender, violaceous node in a young child without response to antibiotics suggests nontuberculous mycobacteria.

Back to question Contents

BQ-294. Answer: E. Cortical suppression of the image from the misaligned eye

When the eyes point in different directions, the brain suppresses the image from the deviated eye to avoid double vision, and vision fails to develop in that eye although it is structurally normal. The correct option names that process. The closest wrong option, optic nerve atrophy, is a structural cause that would show a pale disc. Early treatment with glasses, patching, or both gives the best results. At the bedside, send to a pediatric ophthalmologist without delay.

Key principle: Amblyopia is reduced vision in a structurally normal eye because the brain suppresses its image.

Back to question Contents

BQ-295. Answer: C. Teach massage and plan probing if it persists at 12 to 18 months
A membrane at the lower end of the nasolacrimal duct has not yet opened, causing tearing with a white eye. Most obstructions open on their own during the first year. The closest wrong option, antibiotic drops, treats an infection that is not present and does not open the duct. Tearing with light sensitivity or a large, cloudy cornea would suggest glaucoma, which is absent. At the bedside, teach the massage, keep the area clean, and refer if it persists.
Key principle: Nasolacrimal duct obstruction is treated with massage, and probing is for persistence at 12 to 18 months.
Back to question Contents

BQ-296. Answer: A. Arrange ophthalmologist retinal examinations on the retinopathy of prematurity schedule
Preterm infants are at risk for abnormal retinal vessel growth that cannot be seen with a red reflex check, and treatment works best when found early. They follow a retinopathy of prematurity schedule set by gestational age and weight. The closest wrong option, routine reflex checks alone, relies on a screen that misses early disease. At the bedside, confirm the schedule with the neonatal team and follow results through discharge.
Key principle: Preterm infants need ophthalmologist retinal examinations on the retinopathy of prematurity schedule, not red reflex checks alone.
Back to question Contents

BQ-297. Answer: D. Attraction across separate bowel loops that traps the wall and causes necrosis
Magnets that sit in different loops pull toward each other through the bowel wall, and the trapped tissue loses its blood supply, causing pressure necrosis, fistula, or perforation. A single magnet usually passes. The closest wrong option, systemic metal toxicity, does not match how these objects injure. At the bedside, obtain urgent gastroenterology or surgical consultation, serial films, and a plan for endoscopic or operative removal before a perforation.
Key principle: Two or more magnets, or a magnet with metal, can perforate bowel by attracting across separate loops.
Back to question Contents

BQ-298. Answer: B. Inward collapse of floppy supraglottic tissue during inspiration
Immature, floppy supraglottic cartilage is drawn inward with each breath, causing inspiratory stridor that worsens when airflow increases or when lying supine and eases prone. This is the most common cause of chronic stridor under 2 years. The closest wrong option, a subglottic hemangioma, causes biphasic stridor that progresses. At the bedside, reassure the family, review feeding and growth, and refer for direct airway examination if stridor is persistent, biphasic, or accompanied by feeding trouble.
Key principle: Laryngomalacia produces inspiratory stridor that worsens supine and improves prone, and it usually resolves.
Back to question Contents

BQ-299. Answer: A. Call anesthesiology and otolaryngology to secure the airway before any throat examination

Swelling of the epiglottis can block the airway suddenly, and agitation or throat examination can trigger complete obstruction. It is now rare with Hib vaccine, but this child has no protection. The closest wrong option, a tongue depressor examination, is the very action that provokes closure. At the bedside, keep her calm and upright with a parent, call the airway team, give oxygen as tolerated, and secure the airway before cultures or imaging.

Key principle: In suspected epiglottitis, do not examine the throat, and secure the airway with the airway team.

Back to question Contents

Answers: Domain 11. Cardiology

Contents Questions for this subject

BQ-300. Answer: B. Venous hum, which is an innocent murmur

Blood returning through the jugular veins into the superior vena cava creates continuous turbulence when the child sits upright. Lying flat or compressing the neck veins changes that flow and abolishes the sound, which a structural continuous murmur would not do. The closest trap is patent ductus arteriosus, whose machinery murmur persists when supine and comes with bounding pulses. At the bedside, change his position, let the family hear the murmur vanish, and offer reassurance without ordering tests.

Key principle: A continuous murmur that vanishes when the child lies down or the neck veins are compressed is a venous hum.

Back to question Contents

BQ-301. Answer: E. Left-to-right shunting that keeps right ventricular volume high throughout the respiratory cycle

An atrial septal defect lets blood move from the left atrium to the right atrium, so the right ventricle carries extra volume and its ejection stays long. Inspiration normally adds venous return, but the two atria are connected, so respiration no longer changes the timing and the split is fixed. Right bundle conduction delay is the closest trap, because it widens the split but still varies with breathing. At the bedside, request an echocardiogram and refer the child to a pediatric cardiologist for planned closure.

Key principle: Fixed splitting of the second heart sound points to an atrial septal defect.

Back to question Contents

BQ-302. Answer: D. Falling pulmonary vascular resistance that lets more blood cross the defect from left to right

At birth, pulmonary vascular resistance is high and nearly equals systemic resistance, so little blood crosses a large ventricular septal defect and no murmur is heard. Over several weeks the resistance falls, the shunt grows, and the lungs flood, which produces fast breathing, sweating, and poor growth. Duct closure is the closest trap, but it causes shock in the first 2 weeks with absent pulses. At the bedside, request cardiology review and an echocardiogram, and begin medical therapy and calorie support.

Key principle: Heart failure from a large ventricular septal defect appears as pulmonary vascular resistance falls.

Back to question Contents

BQ-303. Answer: B. Coarctation of the aorta distal to the left subclavian artery

A narrowing of the descending aorta raises pressure in the arms and lowers it in the legs, and collateral vessels carry flow to the back, where they produce a murmur. Delayed, weak femoral pulses with strong radial pulses are the bedside sign that separates this from renal or primary disease. Renal artery stenosis is the closest trap because it causes secondary hypertension, but it does not change the femoral pulses. At the bedside, measure all four limbs, request an echocardiogram, and refer him to cardiology.

Key principle: Hypertension with weak femoral pulses is coarctation until proven otherwise.

Back to question Contents

BQ-304. Answer: D. Transposition of the great arteries
In transposition, the aorta arises from the right ventricle and the pulmonary artery from the left, so two parallel circuits exist and oxygenated blood recirculates in the lungs. Mixing happens only through the ductus and the atrial septum, so saturation barely changes with oxygen, and lung fields stay clear. Pulmonary hypertension is the closest trap, but it usually follows lung disease and brings distress. At the bedside, start prostaglandin E1, arrange an urgent echocardiogram, and call the cardiac center.
Key principle: Deep cyanosis with little distress and a poor response to oxygen suggests transposition of the great arteries.
Back to question Contents

BQ-305. Answer: C. Arrange an echocardiogram now, because the screen is positive
Screening exists to find duct-dependent lesions in babies who look well, and a lower-limb saturation that low suggests blood reaching the body through the ductus. A reading under 90 percent at either site fails the screen at once, and no retest applies. Repeating the test is the closest trap, because it delays imaging and the ductus may close. At the bedside, call neonatology, arrange an echocardiogram the same day, keep the infant in the hospital, and transfer if the center cannot image him.
Key principle: A newborn with a saturation under 90 percent on screening needs an echocardiogram without a retest.
Back to question Contents

BQ-306. Answer: A. Hold her in the knee-to-chest position, give oxygen, and calm her
Crying and agitation increase pulmonary resistance and lower systemic resistance, so more blood crosses the ventricular defect from right to left and less reaches the lungs. The knee-to-chest position raises systemic resistance and reduces that shunt. Prostaglandin E1 is the closest trap, since it helps newborns whose ducts are closing, not an older infant with open flow. At the bedside, keep the child with the parent, give oxygen, and call for help for morphine, fluids, or an agent that raises systemic resistance if the spell continues.
Key principle: A tet spell is treated by raising systemic resistance with knee-to-chest positioning, oxygen, and calm.
Back to question Contents

BQ-307. Answer: E. Diastolic runoff of blood from the aorta into the pulmonary circulation through the ductus
A persistent ductus in a preterm infant connects the aorta to the pulmonary artery. As pulmonary resistance falls, blood leaks out of the aorta throughout diastole, so diastolic pressure drops and the pulse pressure widens. The pulses become bounding and the lungs receive extra flow. Outflow obstruction is the closest trap, but it narrows the pulse pressure. At the bedside, request an echocardiogram, limit fluids, and ask neonatology and cardiology to plan treatment.
Key principle: A wide pulse pressure with bounding pulses in a preterm infant reflects aortic runoff through a patent ductus.
Back to question Contents

BQ-308. Answer: C. Atrioventricular septal defect

A defect of the endocardial cushions leaves a hole at the center of the heart, so blood moves left to right at atrial and ventricular levels, and the lungs are overloaded as resistance falls. Heart failure appears at weeks 2 to 8, and the superior axis is the clue that separates this lesion from a simple ventricular defect. Tetralogy is the closest trap, since it also occurs in Down syndrome, but it causes cyanosis. At the bedside, request an echocardiogram and refer her to pediatric cardiology.

Key principle: Heart failure in an infant with Down syndrome and a superior QRS axis suggests an atrioventricular septal defect.

Back to question Contents

BQ-309. Answer: E. An echocardiogram is advised soon, because many defects are silent in the first days

About half of infants with Down syndrome have a heart defect, most often an atrioventricular septal defect or a ventricular septal defect. In the first days, high pulmonary resistance hides shunts, so the examination and oximetry can be normal while a large lesion waits to declare itself. Relying on oximetry alone is the closest trap, because screening targets cyanotic lesions and misses shunts. At the bedside, order an echocardiogram before discharge or within the first weeks, and arrange cardiology follow-up.

Key principle: Every infant with Down syndrome needs an early echocardiogram, whatever the examination shows.

Back to question Contents

BQ-310. Answer: C. Closure of the ductus arteriosus that had been supplying the lower body

In a left-sided obstructive lesion, the right ventricle supplies the lower body through the ductus. When the ductus closes in the first week, perfusion collapses, the liver congests, and the lactate rises, even though the saturation was normal at screening. Sepsis is the closest trap, because it also makes a newborn gray, but it does not remove femoral pulses. At the bedside, check four-limb pressures, start prostaglandin E1 before the echocardiogram is complete, and call the cardiac center.

Key principle: A newborn can pass oximetry screening and still collapse when the ductus closes.

Back to question Contents

BQ-311. Answer: B. Wolff-Parkinson-White pattern with an accessory pathway

An accessory pathway bypasses the atrioventricular node, so part of the ventricle is activated early. This shortens the PR interval and produces the slurred delta wave at the start of the QRS. The same pathway can form a re-entry circuit, which explains sudden-onset, sudden-offset palpitations. Right bundle branch block is the closest trap, but its delay is at the end of the QRS. At the bedside, record a full 12-lead tracing and refer her to cardiology for risk assessment.

Key principle: A short PR interval with a delta wave in a child with abrupt palpitations indicates an accessory pathway.

Back to question Contents

BQ-312. Answer: A. Transient block of atrioventricular node conduction that breaks the re-entry circuit

Most childhood supraventricular tachycardia is a re-entry loop that uses the atrioventricular node as one limb. Adenosine briefly blocks the node, the circuit is interrupted, and the sinus node takes over. Because the drug lasts only seconds, a slow push fails, so it is given rapidly with an immediate saline flush. Sodium channel blockade is the closest trap, since it belongs to a different class of agents. At the bedside, give the dose fast through a large proximal vein with the rhythm strip running.

Key principle: Adenosine ends re-entrant supraventricular tachycardia by briefly blocking the atrioventricular node.

Back to question Contents

BQ-313. Answer: C. Synchronized cardioversion at 0.5 to 1 J/kg

In supraventricular tachycardia, a rate this fast shortens filling time, so cardiac output falls and perfusion fails. When the child is unstable, electrical termination is faster and more reliable than a drug. Waiting for access is the closest trap, because adenosine is the first choice only in a stable child. Verapamil in an infant can cause severe hypotension. At the bedside, sedate if there is time, deliver a synchronized shock at 0.5 to 1 J/kg, increase to 2 J/kg if needed, and place access during the preparation.

Key principle: An unstable child with supraventricular tachycardia is treated with synchronized cardioversion, not by waiting for access.

Back to question Contents

BQ-314. Answer: A. Obtain expert cardiology help before giving any antiarrhythmic drug

A wide QRS may be ventricular tachycardia or a supraventricular rhythm with aberrant conduction, and scar from a prior repair favors a ventricular origin. Drugs that suit one mechanism can cause collapse in the other, and this stable patient can afford the time to choose. Adenosine is the closest trap, because it is tried when a diagnosis of supraventricular tachycardia is certain. At the bedside, place pads, record a 12-lead tracing, monitor continuously, and call the pediatric cardiologist while preparing for synchronized cardioversion.

Key principle: In a stable child with a wide-complex tachycardia, ask for expert help before choosing a drug.

Back to question Contents

BQ-315. Answer: E. Long QT syndrome with torsades de pointes

Mutations in potassium or sodium channels delay ventricular repolarization, which allows torsades de pointes and sudden loss of consciousness. Swimming is a classic trigger for type 1, and an unexplained drowning in a relative suggests inheritance. A corrected QT of 490 ms scores highly. Vasovagal syncope is the closest trap, since a faint in water can look the same, but it gives warning and does not prolong the QT. At the bedside, keep her from sport, repeat the ECG, and refer her to cardiology for genetic testing.

Key principle: Exertional or swimming-related syncope with a long corrected QT interval and a family history is long QT syndrome.

Back to question Contents

BQ-316. Answer: B. A nonselective beta blocker such as nadolol

Sympathetic surges provoke torsades in long QT syndrome, and nonselective beta blockade blunts that response across both receptor types. Nadolol or propranolol is the mainstay, and these agents are favored over metoprolol, which has been linked to more breakthrough events. Amiodarone is the closest trap, since it is an antiarrhythmic, but it prolongs repolarization itself. At the bedside, begin the beta blocker with cardiology, give the family a list of drugs to avoid, and teach swimming supervision.

Key principle: A nonselective beta blocker is first-line treatment for long QT syndrome.

Back to question Contents

BQ-317. Answer: C. Delayed repolarization that lets early afterdepolarizations trigger ventricular tachycardia

Reduced potassium current prolongs the plateau of the action potential, and the membrane can reach threshold again before the cell has fully reset. These early afterdepolarizations trigger a burst of tachycardia, which twists around the baseline as torsades de pointes. A loud noise is a classic trigger in type 2. Scar-related re-entry is the closest trap, but this child has no surgical history. At the bedside, remove QT-prolonging drugs, check potassium and magnesium, and arrange cardiology review.

Key principle: Torsades de pointes arises from early afterdepolarizations after delayed repolarization.

Back to question Contents

BQ-318. Answer: B. Vasovagal syncope with a typical prodrome

Prolonged standing in heat pools blood in the legs, and a reflex then lowers the heart rate and the vessel tone, which reduces brain perfusion for a few seconds. The warning symptoms, the trigger, and the quick recovery all point to this benign mechanism, and a normal ECG supports it. Hypertrophic cardiomyopathy is the closest trap, but it causes faints during exertion without warning. At the bedside, explain the cause, advise fluids and salt, teach her to lie down at the first warning, and give return precautions.

Key principle: Syncope with a trigger, warning symptoms, and fast recovery is vasovagal when the ECG and history are normal.

Back to question Contents

BQ-319. Answer: D. Musculoskeletal chest wall pain

Chest pain in children comes from the chest wall, the lungs, the stomach, or anxiety far more often than from the heart, and cardiac causes are a small fraction in most community studies. Pain that can be reproduced by touch, changes with breathing, and appears with a normal examination fits the commonest cause. Myocarditis is the closest trap, but it brings fever, tiredness, and abnormal tests. At the bedside, reassure him, suggest simple analgesia, and ask him to return if exertional pain, fainting, or palpitations appear.

Key principle: Reproducible chest wall pain with a normal examination is most likely musculoskeletal and needs no testing.

Back to question Contents

BQ-320. Answer: B. Obtain an echocardiogram and hold her from vigorous exercise until it is reviewed

Exertional symptoms raise the possibility of a structural or coronary problem, such as a coronary anomaly or cardiomyopathy, which a resting ECG can miss. The risk is that the first event is the fatal one. A normal ECG is the closest trap, because it cannot exclude these lesions. At the bedside, arrange an echocardiogram and cardiology review, advise no vigorous exertion until they are complete, and explain that most results are reassuring.

Key principle: Exertional chest pain needs an echocardiogram and rest from exertion even when the resting ECG is normal.

Back to question Contents

BQ-321. Answer: D. Echocardiography

Viral injury to the heart muscle reduces contraction, so the ejection fraction is the key measure of severity. Echocardiography shows ventricular size, function, and any effusion at the bedside, without sedation or transport. Cardiac magnetic resonance can show inflammation and scar and is the closest trap, but it comes after the echocardiogram in a sick child. Stress testing is dangerous during acute inflammation. At the bedside, admit him to a monitored unit, obtain the echocardiogram first, and involve cardiology.

Key principle: Echocardiography is the first imaging test in suspected myocarditis.

Back to question Contents

BQ-322. Answer: A. Hypertrophic cardiomyopathy with outflow obstruction

A thickened septum narrows the outflow tract, and the gradient depends on ventricular volume. Standing lowers venous return and enlarges the obstruction, while squatting raises volume and reduces it. Together with the family death and ECG findings, this makes the diagnosis likely. Valvular aortic stenosis is the closest trap, since it also produces a harsh systolic murmur, but its murmur changes little with position and comes with an ejection click. At the bedside, stop sport, request an echocardiogram, and refer him to cardiology.

Key principle: A systolic murmur that grows louder on standing, with a family death, suggests hypertrophic cardiomyopathy.

Back to question Contents

BQ-323. Answer: C. Smaller ventricular volume that narrows the outflow tract and increases obstruction

Straining raises intrathoracic pressure and cuts venous return, so the ventricle fills less. A smaller cavity brings the thick septum and the mitral leaflet closer to each other, so the outflow tract narrows and the murmur increases. The opposite occurs with squatting or lying down, when volume rises. Higher afterload is the closest trap, since it would normally reduce the gradient. At the bedside, use position and strain to check the murmur, and avoid dehydration and vigorous exertion until cardiology advises.

Key principle: In hypertrophic cardiomyopathy, any maneuver that lowers ventricular volume makes the murmur louder.

Back to question Contents

BQ-324. Answer: E. Repeat the measurement with a larger cuff and confirm it by auscultation
A cuff bladder that is too narrow for the arm needs more pressure to compress the artery, so it overestimates the true pressure. The guideline asks for a bladder width of at least 40 percent of arm circumference, and for oscillometric readings that are high to be repeated by auscultation. Starting a drug is the closest trap, because it treats a number that may be false. At the bedside, fit the right cuff, seat him quietly for 3 to 5 minutes, and repeat the reading before classifying it.
Key principle: A cuff that is too small gives a falsely high reading, so fix the technique before classifying the pressure.
Back to question Contents

BQ-325. Answer: A. Measure it at every visit, because birth before 32 weeks is a risk condition
Preterm birth reduces nephron number, and that group has higher rates of later hypertension and kidney disease. The 2017 guideline sets routine measurement at every well visit from age 3 years, but in younger children it applies only when a risk condition exists, such as prematurity under 32 weeks or congenital heart disease. Waiting until age 3 is the closest trap. At the bedside, use an appropriate cuff, record the reading, and compare it with age, sex, and height reference values.
Key principle: Children under 3 years with a risk condition such as birth before 32 weeks need blood pressure checks at visits.
Back to question Contents

BQ-326. Answer: E. Send her to the emergency department for monitored blood pressure control
Severe elevation with headache, vomiting, drowsiness, and visual change signals hypertensive encephalopathy, where autoregulation of cerebral flow has failed. A symptomatic child, or an adolescent over 180/120 mmHg, needs emergency care with monitored and controlled reduction rather than rapid lowering. Outpatient follow-up is the closest trap, because it allows complications to progress. At the bedside, check the airway and glucose, call for emergency transfer, and avoid giving a rapid-acting drug in the office.
Key principle: A symptomatic child or a pressure over 180/120 mmHg in an adolescent needs emergency care.
Back to question Contents

BQ-327. Answer: C. Arrange cardiology reassessment with echocardiography, rhythm monitoring, and exercise testing before any return
Myocarditis can leave scar that causes ventricular arrhythmia during exertion, even when the child feels well. The 2025 statement describes the older fixed rest periods as arbitrary, and advises that return rest on symptoms, function, imaging, and exercise testing, under a cardiologist. A fixed clock is the closest trap, since it ignores the heart's actual state. At the bedside, schedule the cardiology visit, keep him out of sport until it is done, and explain that most recover and return.
Key principle: Return to sport after myocarditis is decided by testing in a cardiologist's hands, not by a fixed waiting period.
Back to question Contents

Answers: Domain 12. Pulmonology

Contents Questions for this subject

BQ-328. Answer: E. A short course of daily inhaled corticosteroid started at the first sign of a cold

Wheeze that appears only with viral infections in a young child is driven by airway inflammation during the infection, with normal airways between episodes. The 2020 expert panel conditionally recommends a short course of inhaled corticosteroid begun at the first sign of a cold, with the reliever as needed. Continuous daily treatment is the closest trap, since it exposes a well child to long-term drug for no added benefit. At the bedside, check the holding chamber technique, write a plan, and review the pattern at each visit.

Key principle: For wheeze with colds only in a child under 5, a short inhaled corticosteroid course at illness onset is an option.

Back to question Contents

BQ-329. Answer: B. Her exacerbations place her at high risk, so a daily controller is warranted despite few symptoms

Asthma assessment has two parts: impairment, meaning current symptoms, and risk, meaning future attacks. Two oral steroid courses and an urgent visit in a year define a high-risk pattern, even when the child feels well between episodes. Calling the disease mild because of low impairment is the closest trap. A viral trigger does not lower the risk, since colds are the most common cause of attacks. At the bedside, start a daily low-dose inhaled corticosteroid, check technique, and write an action plan.

Key principle: Asthma severity is judged by risk as well as symptoms, and two oral steroid courses in a year signal high risk.

Back to question Contents

BQ-330. Answer: C. Give an oral corticosteroid now and repeat albuterol by spacer

An attack combines bronchospasm, which a bronchodilator relieves in minutes, with airway inflammation, which a systemic steroid reduces over hours. Because the benefit is delayed, the steroid must start early rather than after the bronchodilator fails. Delaying it is the closest trap. Magnesium is reserved for a child who does not respond to the first hour of treatment. At the bedside, give oxygen, repeat the bronchodilator, add ipratropium if the attack is severe, and reassess after each dose.

Key principle: Give the systemic corticosteroid early in an asthma attack, because it takes hours to work.

Back to question Contents

BQ-331. Answer: A. Give intravenous magnesium sulfate and arrange intensive care review
Severe airflow limitation persisting after the first hour of bronchodilator, anticholinergic, and steroid treatment signals a child at risk of respiratory failure. Magnesium relaxes airway smooth muscle by a different mechanism, and intensive care input prepares for escalation. Repeating the same therapy is the closest trap, since it delays the next step while she tires. At the bedside, continue oxygen and bronchodilators, set up intravenous access, call senior and intensive care staff, and watch for drowsiness or a quiet chest.
Key principle: Consider intravenous magnesium and intensive care when an asthma attack persists after the first hour of treatment.
Back to question Contents

BQ-332. Answer: E. Airflow so reduced that little air moves, signaling impending respiratory failure
In severe obstruction, the airways narrow so much that the flow is too small to produce a wheeze, and exhaustion further reduces the effort. The quiet chest therefore means worse disease, and drowsiness and falling saturation confirm that he is tiring. Improvement is the closest trap, since a louder wheeze after treatment usually reflects better air movement. At the bedside, give oxygen, repeat the bronchodilator, add ipratropium and a steroid, call intensive care, and obtain a blood gas.
Key principle: A silent chest in an exhausted asthmatic child signals impending respiratory failure.
Back to question Contents

BQ-333. Answer: A. Spirometry with a bronchodilator response
Asthma is chronic airway inflammation with variable obstruction, so the airflow improves after a bronchodilator and the lung function can be normal between attacks. From about 5 years, spirometry before and after a bronchodilator shows that reversibility. A chest radiograph is the closest trap, since it is often normal in asthma and does not show variable obstruction. At the bedside, record the history of triggers and atopy, perform spirometry, and start a trial of treatment with a planned review.
Key principle: From age 5 years, reversible airflow limitation on spirometry supports the diagnosis of asthma.
Back to question Contents

BQ-334. Answer: C. Tracheomalacia
In tracheomalacia, the cartilage of the trachea is soft, so the airway collapses during expiration when pressure in the chest rises. Crying, feeding, and colds increase the flow and the collapse, which makes a single-pitched expiratory sound that does not respond to a bronchodilator. Asthma is the closest trap, since both cause wheeze, but asthma wheeze is polyphonic and responds to treatment. At the bedside, reassure the parents about growth, check for feeding problems, and refer for airway endoscopy if the sound worsens.
Key principle: A monophonic expiratory wheeze that does not respond to a bronchodilator in a thriving infant suggests tracheomalacia.
Back to question Contents

BQ-335. Answer: B. Provide nasal suction and give fluids by nasogastric tube or admit for hydration

Bronchiolitis narrows and blocks the small airways with swollen cells and mucus, and the main danger is dehydration or tiring rather than low oxygen. Supportive care is the treatment, and fluids are given by tube or vein when the infant cannot drink enough. Albuterol is the closest trap, since the wheeze suggests a bronchodilator, but it does not change the course. At the bedside, clear the nose, set up feeding support, watch her work of breathing, and tell parents the cough may last two weeks or more.

Key principle: Bronchiolitis with poor intake is treated with suction and fluids, not with bronchodilators, steroids, or antibiotics.

Back to question Contents

BQ-336. Answer: D. Give a long-acting RSV monoclonal antibody before or early in the season

Respiratory syncytial virus is the leading cause of bronchiolitis admission in infancy, and the youngest infants are at highest risk because their airways are small. A monoclonal antibody gives passive protection for the season, and maternal vaccination in pregnancy gives a similar effect. Ribavirin is the closest trap, since it is an antiviral, but it is not used for prevention. At the bedside, check which products the national schedule offers and give the dose before he is exposed.

Key principle: Passive protection with a long-acting monoclonal antibody, or maternal vaccination, prevents severe RSV disease in infants.

Back to question Contents

BQ-337. Answer: B. Give a single dose of dexamethasone and nebulized epinephrine

Viral inflammation narrows the subglottic airway, and the swelling produces the bark and the stridor. Dexamethasone reduces inflammation over hours in every grade of croup, while nebulized epinephrine constricts mucosal vessels and relieves stridor at rest within minutes. Dexamethasone alone is the closest trap, because it leaves the stridor at rest untreated for hours. At the bedside, keep her calm on a parent's lap, give both drugs, and observe for a period because the effect of epinephrine fades.

Key principle: Croup with stridor at rest is treated with dexamethasone plus nebulized epinephrine, followed by observation.

Back to question Contents

BQ-338. Answer: D. Bacterial tracheitis

A bacterial infection, often with Staphylococcus aureus, complicates the viral illness and produces thick sloughing exudate that narrows the trachea. Toxic appearance, high fever, and little response to croup therapy separate it from viral croup. Epiglottitis is the closest trap, but it is uncommon in immunized children and causes drooling and a muffled voice rather than a barking cough. At the bedside, keep him calm, call anesthesia and otolaryngology to secure the airway in a controlled setting, and start intravenous antibiotics.

Key principle: A toxic child with croup who fails treatment has bacterial tracheitis until proven otherwise.

Back to question Contents

BQ-339. Answer: E. Treat at home with oral amoxicillin and give return precautions
Pneumonia is infection of the lung tissue, and in a preschool child with focal findings, Streptococcus pneumoniae is the leading bacterial cause. Amoxicillin covers it, and mild disease in an immunized child can be managed at home. Azithromycin is the closest trap, since it targets atypical organisms and is not first choice for presumed pneumococcus. A chest film is not needed for a child who goes home. At the bedside, explain how to give the drug, advise fluids, and review her if fever lasts beyond 48 hours.
Key principle: Mild community-acquired pneumonia in an immunized child is treated at home with amoxicillin, without routine imaging.
Back to question Contents

BQ-340. Answer: B. Mycoplasma pneumoniae
Atypical organisms infect the airway lining and interstitium rather than filling the alveoli, so the onset is gradual, the cough is dry, and the child looks better than the film suggests. School-age children are the usual hosts. Streptococcus pneumoniae is the closest trap, since it causes most bacterial pneumonia overall, but it usually brings abrupt fever, a lobar pattern, and a sicker child. At the bedside, treat with a macrolide, advise fluids, and arrange review if fever persists after 48 to 72 hours.
Key principle: A school-age child with gradual onset, headache, and dry cough most likely has Mycoplasma pneumonia.
Back to question Contents

BQ-341. Answer: C. Obtain chest imaging to look for a parapneumonic effusion and arrange admission
Bacteria can spread from the infected lung to the pleural space, forming an effusion that causes pain, dullness, and persistent fever. Failure to improve after 48 to 72 hours of treatment, or worsening effort, calls for imaging and admission. Changing or doubling the oral drug is the closest trap, since it treats the lung and not the collection of fluid. At the bedside, give oxygen, obtain a chest film and ultrasound, start intravenous antibiotics, and ask a pediatric surgeon about drainage.
Key principle: Persistent fever with rising effort after 48 to 72 hours of treatment needs imaging for effusion and admission.
Back to question Contents

BQ-342. Answer: E. Sweat chloride measurement
A defective chloride channel stops salt and water moving across sweat gland ducts, so sweat is rich in chloride. Pancreatic ducts block, which causes fat malabsorption, greasy stools, and poor growth despite a good appetite. The sweat test measures that salt loss and is the diagnostic standard. Celiac serology is the closest trap, because it explains malabsorption in older children, not a 6-week-old. At the bedside, arrange the test at an accredited laboratory and send her to a cystic fibrosis center if it is raised.
Key principle: Poor growth with greasy stools and salty skin needs a sweat chloride test, whatever the newborn screen showed.
Back to question Contents

BQ-343. Answer: C. Defective chloride transport across epithelium that reduces water in the airway surface liquid

The CFTR protein moves chloride and water onto the airway surface. When it fails, the surface liquid is dehydrated, mucus becomes thick, and the cilia cannot clear it, so bacteria settle and inflammation follows. Ciliary dyskinesia is the closest trap, since it also causes a wet cough with sinus disease, but the cilia themselves are normal here. At the bedside, start airway clearance, refer the family to a cystic fibrosis center, and plan nutrition and enzyme care.

Key principle: Cystic fibrosis mucus is thick because chloride and water movement across the epithelium fails.

Back to question Contents

BQ-344. Answer: D. Cystic fibrosis

Thick secretions in the sinuses cause polyps, in the lungs cause bronchiectasis and clubbing, and in the pancreas cause fat malabsorption and poor growth. This multisystem pattern is typical, and newborn screening is not perfect, so a normal result does not exclude disease. Primary ciliary dyskinesia is the closest trap, since it shares the wet cough and sinus disease, but it does not cause pancreatic malabsorption. At the bedside, order a sweat chloride test, measure growth, and send the family to a cystic fibrosis center if it is high.

Key principle: Nasal polyps, clubbing, and poor growth in a child call for a sweat test even after a normal newborn screen.

Back to question Contents

BQ-345. Answer: B. Send her to a cystic fibrosis center to begin multidisciplinary care

A sweat chloride of 60 mmol/L or higher with two causative variants confirms the diagnosis, and early care slows lung damage and protects growth even before symptoms appear. Care involves airway clearance, nutrition, and pancreatic assessment by a specialist team. Waiting for symptoms is the closest trap, since the damage begins silently. At the bedside, explain the diagnosis, arrange the center visit this week, review stool and growth, and discuss modulator treatment with the specialist.

Key principle: A confirmed diagnosis of cystic fibrosis should lead to specialist center care before symptoms develop.

Back to question Contents

BQ-346. Answer: E. Give the recommended immunizations and RSV prevention, and keep the home free of smoke

Damaged and immature lungs have less reserve, so a viral infection that causes a cold in a healthy infant can cause respiratory failure here. Prevention works at the source: immunizations, passive RSV protection, and avoiding tobacco smoke reduce infections and their severity. Home monitoring is the closest trap, since it records events but prevents none. At the bedside, review the immunization record, arrange RSV protection, confirm a smoke-free home, and teach the family early signs of tiring.

Key principle: Immunization, RSV prevention, and a smoke-free home are the main protections for an infant with bronchopulmonary dysplasia.

Back to question Contents

BQ-347. Answer: D. A videofluoroscopic swallow study

Weak, uncoordinated swallowing lets liquid pass into the airway, and the gravity-dependent segments, often the right lower lobe, become infected repeatedly. The history of coughing with feeds in a neurologically impaired child points to aspiration. Sweat testing is the closest trap, since recurrent pneumonia can reflect cystic fibrosis, but the story and normal growth point elsewhere. At the bedside, arrange the swallow study with a speech therapist, change feeding texture, and consider nutrition support while awaiting results.

Key principle: Recurrent pneumonia in a child who chokes with feeds suggests aspiration, and a swallow study is the next test.

Back to question Contents

BQ-348. Answer: C. Arrange further evaluation, including consideration of bronchoscopy with lavage

A wet cough that lasts over 4 weeks without pointers is often protracted bacterial bronchitis, which clears within 2 weeks of the right antibiotic. When the cough survives 4 weeks of appropriate treatment, the diagnosis is in doubt, and a structural problem, aspiration, or an underlying lung disease must be sought. Repeating the same antibiotic is the closest trap. At the bedside, request pediatric pulmonology review, check for pointers again, and consider bronchoscopy to examine and sample the airways.

Key principle: A wet cough that persists after 4 weeks of appropriate antibiotic treatment needs further evaluation.

Back to question Contents

BQ-349. Answer: D. Give five back blows alternating with five chest thrusts

A foreign body that completely blocks the airway stops all air movement, so the infant cannot cough or cry and quickly becomes hypoxic. Back blows and chest thrusts raise intrathoracic pressure and can expel the object, whereas abdominal thrusts can injure an infant's liver. A blind finger sweep is the closest trap, since it can push the object deeper. At the bedside, call for help, deliver the cycles, look in the mouth only for a visible object, and start resuscitation if he becomes unresponsive.

Key principle: Use back blows and chest thrusts for a choking infant, and abdominal thrusts only after age 1 year.

Back to question Contents

BQ-350. Answer: E. A ball-valve obstruction that lets air in on inspiration but traps it on expiration

The bronchus widens on inspiration and narrows on expiration. A partly obstructing object lets air pass in but not out, so the distal lung inflates and pushes the mediastinum away on the film. Complete obstruction is the closest trap, since it causes collapse, not hyperinflation. A normal film does not exclude an aspirated object, because many foods are not visible. At the bedside, keep him fasting, avoid agitation, and arrange rigid bronchoscopy with an experienced team.

Key principle: A one-sided hyperinflated lung after choking points to a ball-valve foreign body.

Back to question Contents

BQ-351. Answer: A. Laryngomalacia

In laryngomalacia, the soft supraglottic tissue falls into the airway during inspiration, which produces a fluttering sound that worsens with crying and feeding and when supine. It is the most common cause of infant stridor and tends to improve as the child grows. Vocal cord paralysis is the closest trap, but it gives a weak cry. At the bedside, reassure the parents, keep her upright after feeds, track growth, and refer for endoscopy if feeding trouble, poor growth, cyanosis, or retractions appear.

Key principle: Fluttering inspiratory stridor that is worse supine in a thriving infant is laryngomalacia.

Back to question Contents

BQ-352. Answer: E. Subglottic hemangioma

Infantile hemangiomas grow rapidly in the first months of life, and a lesion in the subglottis narrows the airway in both phases of breathing. A skin hemangioma in the lower face and neck raises the likelihood that an airway lesion is also present. Recurrent croup is the closest trap, but true croup is rare under 6 months and does not progress over weeks. At the bedside, arrange laryngoscopy and bronchoscopy, avoid agitation, and plan treatment with otolaryngology and a hemangioma specialist.

Key principle: Biphasic stridor with a hemangioma of the lower face suggests a subglottic hemangioma.

Back to question Contents

BQ-353. Answer: A. Higher risk, because he is younger than 60 days

The lower-risk category for a brief resolved unexplained event requires every criterion to be met, including an age over 60 days, a gestation of at least 32 weeks, one event, a duration under 1 minute, and no concerning findings. This infant fails the age criterion, so the other normal findings cannot place him in the lower-risk group. Focusing on the short duration is the closest trap. At the bedside, perform a careful history and examination, consider further evaluation guided by findings, and discuss monitored observation.

Key principle: Any infant who fails a single criterion, such as age under 60 days, is outside the lower-risk BRUE group.

Back to question Contents

BQ-354. Answer: E. Carboxyhemoglobin absorbs light like oxyhemoglobin, so the oximeter overestimates saturation

Carbon monoxide binds hemoglobin with high affinity and blocks oxygen delivery, but a two-wavelength oximeter cannot tell carboxyhemoglobin from oxyhemoglobin, so the reading looks normal while tissues are starved. Cold hands usually lower the reading, so that is the closest trap. The setting and the sibling's symptoms make the diagnosis. At the bedside, give high-flow oxygen, remove the family from the source, measure carboxyhemoglobin with a co-oximeter on a blood sample, and call poison control.

Key principle: A normal pulse oximeter reading does not exclude carbon monoxide poisoning.

Back to question Contents

BQ-355. Answer: D. Hypercapnic failure from weakness of the respiratory pump

Respiratory failure can arise from the lung, which causes low oxygen, or from the pump, which causes carbon dioxide retention. Weak respiratory muscles cannot move enough air, so carbon dioxide rises while the lungs themselves are clear. Oxygen can raise the saturation and hide the problem, which makes oxygen alone the closest trap. At the bedside, check a gas rather than relying on the oximeter, start noninvasive ventilation, call for senior and intensive care help, and avoid sedatives.

Key principle: Hypercapnia with clear lungs in a child with weak muscles is pump failure, and oximetry can hide it.

Back to question Contents

Answers: Domain 13. Gastroenterology

Contents Questions for this subject

BQ-356. Answer: C. Obtain an urgent upper gastrointestinal contrast study with surgical consultation

Green vomit means bile is passing a point of obstruction. In malrotation, the midgut twists on its narrow mesenteric stalk and strangles its own blood supply, and bowel can die within hours. A contrast study that shows an abnormal duodenojejunal position confirms the problem, and the surgeon must be involved at once. The closest wrong option is pyloric ultrasonography, which fits nonbilious projectile vomiting at a later age. Acid suppression and formula changes treat reflux and allergy. At the bedside, keep the baby fasting, resuscitate, and call surgery while imaging is arranged.

Key principle: Bilious vomiting in an infant is malrotation with volvulus until imaging proves otherwise.

Back to question Contents

BQ-357. Answer: A. Abdominal ultrasonography

In intussusception, a segment of bowel telescopes into the segment beyond it, trapping the mesentery and causing venous congestion, colicky pain, and later bloody mucus. Lethargy between spells is the clue at this age. Ultrasonography shows the target sign with high accuracy and spares radiation. The closest wrong option is a barium study first, which is a treatment-oriented contrast test and not the preferred initial diagnostic step. Currant-jelly stool is a late sign. At the bedside, start intravenous access, image with ultrasound, and arrange an air or contrast enema reduction with surgical backup.

Key principle: Ultrasonography is the first study for suspected intussusception, and enema reduction is the usual treatment.

Back to question Contents

BQ-358. Answer: E. Hirschsprung disease

In Hirschsprung disease, neural crest cells fail to reach the distal bowel, so the aganglionic segment cannot relax and a proximal obstruction develops. Delayed passage of meconium beyond 48 hours, a tight empty rectum, and explosive stool on withdrawal of the finger form the classic triad. Functional constipation is the closest wrong option, but it begins later and has a rectum filled with stool. Congenital hypothyroidism causes sluggish bowels without the gush. At the bedside, obtain a contrast enema and confirm with rectal suction biopsy showing absent ganglion cells.

Key principle: A delayed first stool with a tight empty rectum in a newborn is Hirschsprung disease until biopsy shows ganglion cells.

Back to question Contents

BQ-359. Answer: C. Transient relaxation of an immature lower esophageal sphincter with large liquid feeds

In young infants, the lower esophageal sphincter relaxes often, the esophagus is short, and the diet is liquid and large relative to stomach size, so milk flows back easily. This physiologic reflux is effortless and does not harm a thriving baby. The closest wrong option is pyloric hypertrophy, which causes forceful, projectile vomiting, hunger, and weight loss at 2 to 6 weeks. At the bedside, reassure the family, offer smaller and more frequent feeds, avoid acid suppression, and explain that it usually fades by about 12 months.

Key principle: Effortless regurgitation in a thriving infant is physiologic reflux from an immature sphincter, not disease.

Back to question Contents

BQ-360. Answer: A. Cow's milk protein allergy

Cow's milk proteins can trigger an immune response in the infant gut, causing inflammation, irritability, vomiting, mucus in the stool, and poor weight gain. The timing after the formula started is the clue. The closest wrong option is pathologic acid reflux, which shares the symptoms, but the 2018 guideline advises first trying a hydrolysate or amino acid formula for 2 to 4 weeks before any acid suppressant. Colic does not impair growth. At the bedside, change to an extensively hydrolyzed formula, reassess, and plan a later supervised reintroduction.

Key principle: Reflux-like symptoms with poor weight gain after starting cow's milk formula suggest cow's milk protein allergy.

Back to question Contents

BQ-361. Answer: B. Oral erythromycin given for pertussis prophylaxis

Macrolide antibiotics act as motilin agonists and may drive gastric hypertrophy of the pylorus when given in the first weeks of life, which raises the risk several-fold. Male sex and being firstborn add risk. Vitamin K, phototherapy, breastfeeding, and vaccines have no such link. The closest wrong option, phototherapy, treats a common newborn problem and does not affect the gastric outlet. At the bedside, avoid erythromycin in neonates when an alternative exists, and teach families that nonbilious projectile vomiting at 2 to 6 weeks needs same-day assessment.

Key principle: Macrolide exposure in the first 2 weeks of life, male sex, and being firstborn raise the risk of pyloric stenosis.

Back to question Contents

BQ-362. Answer: A. Oral rehydration solution in small, frequent amounts, with replacement of ongoing losses

Gut sodium-glucose cotransport stays intact in most diarrheal illness, so a glucose-electrolyte solution is absorbed even when stool losses continue. Oral rehydration is as effective as intravenous fluid in moderate dehydration and less invasive. The closest wrong option is the intravenous bolus, which is kept for shock, altered consciousness, or failure of oral and nasogastric therapy. Juice has too little sodium and too much sugar, which worsens osmotic losses. At the bedside, give a spoonful or syringe every few minutes, add each loose stool volume, restart food early, and reassess in a few hours.

Key principle: Oral rehydration solution in small, frequent amounts is first-line for mild to moderate dehydration.

Back to question Contents

BQ-363. Answer: C. Rapid isotonic crystalloid bolus with reassessment after each bolus
Severe volume loss lowers cardiac preload, so the body clamps down the skin and kidneys, and acidosis causes deep breathing. Perfusion fails before blood pressure does, so hypotension means compensation has been lost. An isotonic bolus restores circulating volume fastest. The closest wrong option is nasogastric rehydration, which is correct for moderate dehydration but too slow in shock. Hypotonic fluid risks hyponatremia. At the bedside, give the bolus by weight from the local protocol, reassess perfusion after each, check electrolytes, and repeat if the child remains in shock.
Key principle: Shock from dehydration needs an isotonic bolus, with reassessment after every bolus.
Back to question Contents

BQ-364. Answer: B. Supportive care with fluids, withholding antibiotics and antimotility drugs
Shiga toxin-producing Escherichia coli releases toxin that injures endothelium in the colon and kidney. Antibiotics can provoke more toxin release, and antimotility drugs prolong exposure, so both raise the risk of hemolytic uremic syndrome. The closest wrong option is azithromycin, which feels reasonable for bloody diarrhea but has no proven benefit when the cause is unknown. At the bedside, keep her hydrated, avoid nonsteroidal drugs, check blood count and creatinine over the next several days, and teach the family to return for pallor, edema, or low urine output.
Key principle: Withhold antibiotics and antimotility drugs in bloody diarrhea while Shiga toxin-producing infection is possible.
Back to question Contents

BQ-365. Answer: E. Toxin-mediated endothelial injury causing microvascular thrombosis
Shiga toxin binds endothelial cells in the glomerulus and small vessels, causing swelling and fibrin-platelet thrombi. Red cells shear through these narrowed channels, which creates schistocytes and hemolysis, and platelets are used up. Kidney injury follows from glomerular thrombosis. The closest wrong option is immune complex deposition, which causes nephritis with hypertension and cola-colored urine, but not anemia with fragments and thrombocytopenia. At the bedside, admit him, give careful fluids, monitor electrolytes, blood pressure, and creatinine, and transfuse only when needed.
Key principle: Hemolytic uremic syndrome follows Shiga toxin endothelial injury and presents as anemia, thrombocytopenia, and renal failure.
Back to question Contents

BQ-366. Answer: A. Liquid stool leaking around an impacted rectal mass
Withholding stool lets the rectum absorb water, so the stool hardens, and the rectum then stretches and loses its sensation. Softer stool from higher in the colon bypasses the mass and leaks out without warning, which is overflow encopresis. Hirschsprung disease is the closest wrong option, but it begins in the newborn period, has a tight empty rectum, and rarely soils. At the bedside, explain to the family that the child cannot feel the leakage, clear the impaction with polyethylene glycol, and keep maintenance treatment going for months.
Key principle: Soiling in a constipated child is overflow around impacted stool, not diarrhea or misbehavior.
Back to question Contents

BQ-367. Answer: D. Meckel diverticulum

A Meckel diverticulum is a remnant of the omphalomesenteric duct, and about half contain acid-secreting gastric mucosa that ulcerates the adjacent ileum. This causes large, painless bleeding that looks maroon or brick-red in a toddler. The closest wrong option is a juvenile polyp, which usually bleeds in small amounts of bright red blood. Intussusception causes colicky pain and lethargy. At the bedside, stabilize with intravenous access and fluids, check hemoglobin, and arrange a technetium pertechnetate scan to find the ectopic gastric mucosa.

Key principle: Large painless red-brown rectal bleeding in a toddler suggests a Meckel diverticulum.

Back to question Contents

BQ-368. Answer: B. Juvenile polyp

Juvenile polyps are hamartomatous growths in the rectosigmoid colon, with a rich blood supply that bleeds when stool passes. The bleeding is small, painless, bright red, and recurrent in children aged 2 to 6 years. A fissure is the closest wrong option, but it causes pain with passing hard stool and a visible tear. Meckel diverticulum bleeds more heavily and is darker. At the bedside, confirm that growth and hemoglobin are normal, then refer for colonoscopy to remove the polyp and to check for others.

Key principle: Painless small-volume bright red rectal bleeding in a preschooler suggests a juvenile polyp.

Back to question Contents

BQ-369. Answer: E. Food protein-induced proctocolitis

Food protein-induced proctocolitis is an immune reaction in the rectum and colon to cow's milk proteins that pass into breast milk. It produces blood-streaked stool in a well-looking, thriving infant. The closest wrong option is an anal fissure, but the perianal skin is intact and the stool is loose, not hard. Intussusception brings colicky pain and lethargy. At the bedside, remove cow's milk protein from the mother's diet for 2 to 4 weeks, counsel about calcium intake, and plan a later reintroduction.

Key principle: Blood-streaked stool in a well, breastfed infant is usually food protein-induced proctocolitis.

Back to question Contents

BQ-370. Answer: A. Daily polyethylene glycol to soften stool, with a barrier ointment

Hard stool tears the anal mucosa, the tear hurts, and the toddler withholds, which makes the next stool harder. This cycle keeps the fissure from healing. The goal is soft stool every day, so polyethylene glycol is the maintenance drug supported by guideline. The closest wrong option is colonoscopy, which is for chronic diarrhea or weight loss. A midline posterior fissure with normal growth fits functional constipation. At the bedside, treat the constipation for months, offer warm baths, and explain that the tear usually heals in a few weeks.

Key principle: Anal fissure in a toddler is treated by breaking the cycle of hard stool and withholding.

Back to question Contents

BQ-371. Answer: C. Measure an IgG-based celiac antibody, such as tissue transglutaminase IgG
Tissue transglutaminase IgA depends on the child making IgA, so IgA deficiency produces false negatives. Celiac disease is more common in IgA deficiency than in the general population. An IgG-based antibody, or deamidated gliadin IgG, bypasses the deficiency. The closest wrong option is accepting the negative result, which would miss the disease in a child with strong symptoms and a first-degree relative. At the bedside, keep gluten in the diet, send the IgG test, and plan endoscopy with biopsies if it is positive.
Key principle: A negative IgA-based celiac test is unreliable when total IgA is absent, so use an IgG-based test.
Back to question Contents

BQ-372. Answer: B. Normal selective eating of toddlerhood
Toddlers slow their growth, become wary of new foods, and eat less, so selective eating is common and normal. Growth on the curve separates it from a feeding disorder. The closest wrong option is avoidant restrictive food intake disorder, which involves weight loss, nutritional deficiency, or marked interference with life. Repeated calm exposure, up to ten or more times, helps acceptance, and forcing or bribing backfires. At the bedside, set regular meals with no grazing, offer one accepted food at each meal, and recheck growth.
Key principle: A toddler who grows well and eats a limited variety has normal selective eating, and growth separates it from a disorder.
Back to question Contents

BQ-373. Answer: C. Upper endoscopy with multiple duodenal and bulb biopsies
Gluten triggers an immune attack on the small bowel mucosa, flattening the villi and causing malabsorption of iron. When the antibody level is lower than 10 times normal, serology alone is not specific enough, and histology is required. The closest wrong option is the gluten-free diet, which would heal the mucosa and make the biopsy uninformative. Colonoscopy does not sample the duodenum. At the bedside, keep her on gluten, arrange the endoscopy with multiple sampling sites, and treat the iron deficiency.
Key principle: A TGA-IgA under 10 times normal needs duodenal biopsies while the child still eats gluten.
Back to question Contents

BQ-374. Answer: B. Crohn disease
Crohn disease is a transmural, patchy inflammation that can involve any part of the gut, often the terminal ileum, and it creates fistulas and perianal disease. Chronic inflammation and poor intake slow growth, which may precede bowel symptoms. The closest wrong option is ulcerative colitis, which begins at the rectum, affects only mucosa, and produces bloody diarrhea with urgency without perianal fistulas. At the bedside, exclude infection, then arrange upper endoscopy, ileocolonoscopy with biopsies, and small-bowel imaging.
Key principle: Perianal fistulas with weight loss and growth failure point to Crohn disease.
Back to question Contents

BQ-375. Answer: C. Ulcerative colitis

Inflammation that starts in the rectum and spreads up the colon causes bloody stools, urgency, and nocturnal symptoms in ulcerative colitis. A high calprotectin level confirms mucosal inflammation, and negative stool studies reduce the chance of infection. The closest wrong option is Crohn disease, which more often causes weight loss, perianal disease, and pain without gross blood. At the bedside, check albumin and inflammatory markers, assess severity, avoid antimotility drugs, and arrange colonoscopy with biopsies through gastroenterology.

Key principle: Persistent bloody diarrhea with urgency and negative stool studies suggests ulcerative colitis.

Back to question Contents

BQ-376. Answer: A. Hepatitis B vaccine and hepatitis B immune globulin

Hepatitis B virus passes from mother to infant at birth through blood and secretions, and an infected newborn has a high chance of chronic infection. Immune globulin gives immediate passive antibodies, and the vaccine begins active protection. The two together prevent most transmission. The closest wrong option is vaccine alone, which would leave a gap before the infant's own antibodies develop. At the bedside, give both products at different sites within 12 hours, complete the series, and test for antibody and antigen after the series.

Key principle: Infants of surface antigen positive mothers need hepatitis B vaccine and immune globulin within 12 hours of birth.

Back to question Contents

BQ-377. Answer: D. Fecal-oral spread through contaminated hands, food, or objects

Hepatitis A virus is shed in stool, survives on surfaces, and enters the mouth through unwashed hands. Young children are often asymptomatic or mildly ill, so they spread it quietly in child care and diaper areas. The closest wrong option is blood exposure, which applies to hepatitis B and C, not to a child care outbreak. At the bedside, reinforce handwashing and diaper hygiene, report to the health department, and offer vaccine to unvaccinated contacts within the recommended window.

Key principle: Hepatitis A spreads by the fecal-oral route, and vaccination prevents it.

Back to question Contents

BQ-378. Answer: E. Breast milk jaundice

Breast milk can increase the intestinal reabsorption of bilirubin, and the immature liver conjugates it slowly, so unconjugated bilirubin stays elevated for weeks. A direct level under 1.0 mg/dL, normal stool color, pale urine, and good growth fit this benign pattern. The closest wrong option is biliary atresia, which causes conjugated jaundice, pale stool, and dark urine. At the bedside, fractionate the bilirubin as done, reassure the family, continue breastfeeding, and recheck if the jaundice persists beyond the third week or stools lose color.

Key principle: Prolonged jaundice in a thriving breastfed infant with a direct bilirubin under 1.0 mg/dL is breast milk jaundice.

Back to question Contents

BQ-379. Answer: A. Progressive duct obliteration scars the liver, and success falls after about 60 days

In biliary atresia, an inflammatory process destroys the bile ducts, so bile backs up into the liver, causing cholestasis, fibrosis, and eventually cirrhosis. The Kasai hepatoportoenterostomy drains bile from the porta hepatis only while enough liver remains healthy, and success is best within the first 60 days of life. The closest wrong option is kernicterus, which is a risk with unconjugated hyperbilirubinemia, not conjugated. At the bedside, expedite surgical transfer, give fat-soluble vitamins, and keep monitoring for cholangitis.

Key principle: Biliary atresia scars the liver progressively, so the Kasai operation works best within 60 days of life.

Back to question Contents

BQ-380. Answer: E. Intravenous fluids and antibiotics with urgent surgical consultation

Young children cannot describe their pain, so the appendix often perforates before the diagnosis is made. Perforation spills bacteria into the peritoneum, causing peritonitis, sepsis, and ileus. The ultrasound already shows the appendix and free fluid. The closest wrong option is oral antibiotics alone, which are inadequate for a toxic child with perforation. Barium delays care. At the bedside, give a bolus and broad-spectrum intravenous antibiotics, give analgesia, and arrange surgery after resuscitation.

Key principle: A toxic child with suspected perforated appendicitis needs resuscitation, antibiotics, and surgery.

Back to question Contents

BQ-381. Answer: B. Urine pregnancy test

In an adolescent girl with lower abdominal pain, the differential includes appendicitis, ovarian torsion, pelvic inflammatory disease, and ectopic pregnancy. A ruptured ectopic pregnancy can cause shock, and the first quick test separates this branch from the rest. The closest wrong option is CT first, which exposes a possibly pregnant patient to radiation before the test that would change the choice of imaging. At the bedside, obtain the pregnancy test, then ultrasonography, and involve surgery or gynecology based on the result.

Key principle: Test for pregnancy first in an adolescent girl with acute lower abdominal pain.

Back to question Contents

BQ-382. Answer: D. Reassurance with a pain diary and no further testing

Functional abdominal pain arises from visceral hypersensitivity and altered gut-brain signaling, so the pain is real but no structural cause is found. Features that argue for functional pain are periumbilical location, a pattern tied to stress, normal growth, and no alarm signs. The closest wrong option is the laboratory panel, which has low yield and adds worry when no red flags exist. At the bedside, validate the pain, explain the mechanism, encourage school attendance, and return for weight loss, nocturnal pain, bleeding, or fever.

Key principle: Functional abdominal pain without red flags needs reassurance and a clear return plan, not extensive testing.

Back to question Contents

Answers: Domain 14. Neurology

Contents Questions for this subject

BQ-383. Answer: E. About one in three children has another febrile seizure

A developing brain is more prone to seizure when temperature rises rapidly, and a family history adds risk. Roughly one third of children have a second febrile seizure, and recurrence is more likely with onset before 18 months, a lower temperature at onset, and a first-degree relative with febrile seizures. The closest wrong option is antipyretic prevention, because antipyretics bring comfort but do not prevent recurrence. At the bedside, explain that the tendency fades by school age, review seizure first aid, and discourage routine daily antiseizure drugs.

Key principle: About one third of children have a recurrence after a first febrile seizure, and antipyretics do not prevent it.

Back to question Contents

BQ-384. Answer: A. Complex febrile seizure

Febrile seizures arise when a rapid rise in temperature meets an immature brain. The seizure is complex when it is focal, lasts 15 minutes or longer, or recurs within 24 hours, and this one is both focal and prolonged. The closest wrong option is a simple febrile seizure, which is generalized, brief, and followed by quick recovery. Complex features raise the risk of later epilepsy and the need to search for a cause. At the bedside, find the fever source, consider whether meningitis needs to be excluded, and arrange follow-up.

Key principle: A febrile seizure that is focal or lasts 15 minutes or longer is complex.

Back to question Contents

BQ-385. Answer: C. Perform a lumbar puncture

Seizure with fever in an infant who has not received Hib or pneumococcal vaccines may signal bacterial meningitis, and the infant has no fever source. Meningitis is the one cause that changes immediate treatment. The 2011 guideline lists lumbar puncture as an option in this situation, and the missing vaccines make it the appropriate choice. The closest wrong option is the electroencephalogram, which does not predict recurrence or distinguish infection. At the bedside, obtain blood cultures, perform the puncture if safe, start antibiotics if the result is abnormal, and admit.

Key principle: Consider lumbar puncture after a febrile seizure in an infant with missing Hib or pneumococcal vaccines.

Back to question Contents

BQ-386. Answer: E. Give intramuscular midazolam

A seizure lasting 5 minutes or longer rarely stops on its own, and each minute raises the risk of neuronal injury and hypoxia. Benzodiazepines act quickly on gamma-aminobutyric acid receptors to stop the seizure. The route should be whatever is fastest, and intramuscular midazolam is effective without a vein. The closest wrong option is phenytoin first, which is a second-line agent and acts too slowly as initial therapy. At the bedside, give the benzodiazepine by local protocol dose, support the airway, and prepare second-line medicine if the seizure continues.

Key principle: Status epilepticus needs a benzodiazepine by the fastest available route.

Back to question Contents

BQ-387. Answer: A. Infantile epileptic spasms

Infantile spasms arise from a disorganized cortical network in the first year of life, and the EEG shows hypsarrhythmia. Clusters on waking and loss of developmental skills are the hallmark. The closest wrong option is Sandifer posturing, which follows feeds and has a normal EEG. Benign sleep myoclonus occurs only during sleep. Delay in treatment worsens developmental outcome. At the bedside, arrange an urgent video EEG and MRI, begin hormonal therapy or vigabatrin quickly, and evaluate for tuberous sclerosis and metabolic causes.

Key principle: Clusters of spasms on waking with developmental regression at 3 to 12 months are infantile spasms until EEG proves otherwise.

Back to question Contents

BQ-388. Answer: B. Vigabatrin

Tuberous sclerosis causes overactivity of the mTOR pathway, producing cortical tubers, ash leaf macules, cardiac rhabdomyomas, and early epileptic spasms. Vigabatrin blocks the breakdown of gamma-aminobutyric acid and is preferred in this setting. The closest wrong option is valproate, which is a broad-spectrum drug but is not preferred for infantile spasms. Carbamazepine can worsen generalized epilepsies. At the bedside, start vigabatrin by protocol dose, monitor for visual field changes, and involve neurology, genetics, and cardiology.

Key principle: Vigabatrin is the preferred first treatment for infantile spasms caused by tuberous sclerosis.

Back to question Contents

BQ-389. Answer: D. Self-limited epilepsy with centrotemporal spikes

This epilepsy of school-age children is an age-related, self-limited condition with focal seizures arising near the Rolandic region, usually from sleep. Facial twitching, drooling, and speech arrest with preserved awareness, together with centrotemporal spikes, define it. The closest wrong option is night terrors, which cause screaming, no memory, and a normal EEG. Most children have few seizures and outgrow the condition by adolescence. At the bedside, reassure the family, discuss whether medication is needed, and give safety advice.

Key principle: Nocturnal facial twitching with drooling and centrotemporal spikes defines self-limited epilepsy that resolves by adolescence.

Back to question Contents

BQ-390. Answer: B. Valproate carries a major risk of birth defects, so alternatives are preferred

Valproate interferes with folate metabolism and fetal development, raising the risk of neural tube defects, other malformations, and lower cognitive outcomes in exposed fetuses. It works well in juvenile myoclonic epilepsy, which is why clinicians are tempted to use it. The closest wrong option is the claim that it is safe in pregnancy, which is false. At the bedside, offer another suitable drug, discuss reliable contraception and folic acid, and counsel about triggers such as sleep loss.

Key principle: Counsel every adolescent girl about valproate teratogenicity before starting it.

Back to question Contents

BQ-391. Answer: E. The drug aggravates generalized spike-wave discharges in this syndrome
Absence seizures come from thalamocortical circuits oscillating at 3 Hz. Carbamazepine acts on sodium channels and can enhance this oscillation, so it can increase or trigger absence and myoclonic seizures. The closest wrong option is a low dose, which would not worsen the EEG pattern in this way. Ethosuximide controls absence as well as valproate, with fewer attention problems. At the bedside, stop carbamazepine under neurology guidance, start ethosuximide, and review the diagnosis in the classification of seizure type.
Key principle: Carbamazepine can worsen absence epilepsy, so choose a drug by seizure type.
Back to question Contents

BQ-392. Answer: A. Electroencephalogram
A first unprovoked seizure may be an isolated event or the start of epilepsy. An EEG helps classify the seizure and estimate recurrence risk, because epileptiform discharges raise the chance of another. The closest wrong option is immediate daily medication, which does not change long-term outcome after a first seizure with a normal examination. Lumbar puncture is for suspected infection. At the bedside, arrange the EEG, consider MRI if focal features or abnormal examination appear, and teach seizure first aid and safety.
Key principle: After a first unprovoked seizure, an EEG helps classify it and estimate recurrence.
Back to question Contents

BQ-393. Answer: D. Ibuprofen taken at the start of the headache
Migraine involves activation of trigeminovascular pathways with release of neuropeptides that cause pain and sensitivity. Early treatment works best, before central sensitization is established. The 2019 guideline supports ibuprofen as the first-line acute therapy in children and notes that triptans help adolescents. The closest wrong option is daily amitriptyline, which is not needed with only two attacks a month and has shown little advantage over placebo. At the bedside, give an early-dose plan, limit analgesic use to avoid overuse, and keep a headache diary.
Key principle: Treat a migraine attack early with ibuprofen as first-line therapy.
Back to question Contents

BQ-394. Answer: A. Blood cultures and prompt intravenous antibiotics, with lumbar puncture when safe
Meningeal inflammation from bacteria in the subarachnoid space causes fever, headache, neck stiffness, and vomiting, and the outcome depends on how fast antibiotics begin. Delay raises the risk of death and hearing loss. The closest wrong option is imaging first, which is useful only when raised intracranial pressure is suspected and should not delay antibiotics. At the bedside, obtain cultures, give antibiotics, perform the puncture when it is safe, and admit to a monitored bed.
Key principle: Fever with headache and neck stiffness needs prompt antibiotics and a lumbar puncture when safe.
Back to question Contents

BQ-395. Answer: B. Medication-overuse headache

Frequent analgesic use sensitizes pain pathways, and the headache returns as each dose wears off, creating a daily cycle. The pattern of near-daily analgesic use for months fits this diagnosis. The closest wrong option is a raised-pressure cause, but there is no morning vomiting, papilledema, or focal sign. At the bedside, confirm a normal examination, limit analgesics to a few days a week, start preventive measures and a headache diary, and arrange follow-up.

Key principle: Daily headache with analgesic use most days of the week suggests medication-overuse headache.

Back to question Contents

BQ-396. Answer: C. A temporary disturbance of brain function without visible structural injury

A blow to the head or body causes an acute neurometabolic cascade, with ion shifts and altered energy use in neurons, which disturbs function without damaging structure. Standard imaging is therefore normal. The closest wrong option is a tiny bleed, which would usually be visible and would not fit the normal scan. At the bedside, advise brief relative rest for the first day or two, then symptom-limited return to school and graded activity, with return to contact sport after being symptom-free.

Key principle: Concussion is a functional injury, so a normal scan does not rule it out.

Back to question Contents

BQ-397. Answer: B. Admit for monitoring of respiratory and autonomic function and start immunotherapy

After an infection, antibodies can attack peripheral nerve myelin or axons, causing ascending weakness and loss of reflexes in Guillain-Barre syndrome. The danger lies in weakness reaching the respiratory muscles and in autonomic instability. The closest wrong option is waiting for loss of walking, which risks missing a treatable respiratory failure. At the bedside, check forced vital capacity, monitor heart rate and blood pressure, perform lumbar puncture or nerve studies if needed, and give intravenous immunoglobulin or plasma exchange.

Key principle: Ascending weakness with absent reflexes after infection needs admission and monitoring of breathing.

Back to question Contents

BQ-398. Answer: D. Leakage of enzyme through fragile muscle membranes that lack dystrophin

Dystrophin links the muscle cell cytoskeleton to the membrane, so its absence makes fibers fragile during contraction. They leak creatine kinase and are replaced by fat and fibrous tissue, which gives calf enlargement and proximal weakness with a Gowers sign. The closest wrong option is anterior horn cell degeneration, which produces weakness with absent reflexes and a normal creatine kinase. At the bedside, confirm with genetic testing, arrange care with a neuromuscular team, and discuss corticosteroids and cardiac and respiratory surveillance.

Key principle: Duchenne dystrophy markedly elevates creatine kinase through leaky dystrophin-deficient membranes.

Back to question Contents

BQ-399. Answer: C. A toxin that blocks acetylcholine release at the neuromuscular junction
Clostridium botulinum spores can colonize the infant gut and release a toxin that cleaves proteins needed for vesicle release, so nerve endings cannot release acetylcholine. This produces descending flaccid weakness, constipation, ptosis, and poor suck. The closest wrong option is receptor autoantibodies, which cause myasthenia in older patients and not an acute descending paralysis. At the bedside, admit for respiratory monitoring, send stool for toxin, give botulism immune globulin, and notify the health department.
Key principle: Infant botulism causes descending weakness by blocking acetylcholine release.
Back to question Contents

BQ-400. Answer: B. Refer now for early intervention and therapy while confirming the diagnosis
Cerebral palsy results from a fixed injury to the developing brain, and the early signs can be detected well before walking. Combining history, MRI, and the General Movements Assessment allows accurate diagnosis before 6 months in high-risk infants. The closest wrong option is waiting until 12 months, which loses the period when early therapy has the most effect. At the bedside, arrange early intervention and rehabilitation, screen for hip displacement, feeding, and vision problems, and support the family.
Key principle: Diagnose high-risk infants for cerebral palsy before 6 months and refer for therapy without waiting.
Back to question Contents

BQ-401. Answer: D. Use latex-free gloves and equipment
Children with open neural tube defects undergo repeated exposure to latex through surgery and catheterization, and they develop high rates of latex sensitization and anaphylaxis. Latex-free practice from birth lowers that risk. The closest wrong option is supine positioning, which presses on the sac and risks rupture. At the bedside, keep him prone or side-lying, protect the sac with a moist sterile dressing, arrange early neurosurgical repair, and watch for hydrocephalus.
Key principle: Handle infants with myelomeningocele using latex-free supplies from birth.
Back to question Contents

BQ-402. Answer: C. Administer a validated autism screening tool today
Autism spectrum disorder affects social communication, and early signs appear in the second year. The American Academy of Pediatrics recommends screening with a validated tool at 18 and 24 months, and concerns from parents should prompt it at once. The closest wrong option is waiting until age 4, which would delay early intervention that improves outcomes. At the bedside, screen, check hearing and refer for audiology, and start early intervention referral without waiting for a final diagnosis.
Key principle: Screen for autism at 18 and 24 months with a validated tool, and act on parental concern at once.
Back to question Contents

BQ-403. Answer: E. Formal audiologic evaluation

Hearing underlies speech development, and chronic middle ear fluid or sensorineural loss can limit exposure to sounds. Because the child shows good social skills, hearing loss is more likely than autism. The closest wrong option is waiting, which would let a treatable cause persist. At the bedside, order audiology, enroll in early intervention for speech and language, and consider genetic or developmental evaluation if there are other concerns.

Key principle: Test hearing first in any child with a language delay.

Back to question Contents

BQ-404. Answer: B. Shunt malfunction

When the shunt fails to drain cerebrospinal fluid, ventricular pressure rises, causing headache, vomiting, and drowsiness, and in advanced cases downward gaze. The absence of fever and the normal wound argue against infection. The closest wrong option is shunt infection, which more often presents with fever and in the first months after surgery. At the bedside, obtain urgent neuroimaging compared with a prior baseline and a shunt series, call neurosurgery, and keep the child fasting.

Key principle: Headache, vomiting, and sleepiness in a child with a shunt is malfunction until proven otherwise.

Back to question Contents

Answers: Domain 15. Skin/Dermatology

Contents Questions for this subject

BQ-405. Answer: E. Apply a mid-potency steroid to red patches until smooth and moisturize daily
Atopic dermatitis involves a leaky skin barrier and immune activation, so itch and scratching damage the skin and keep the cycle going. Early, adequate anti-inflammatory treatment of flares restores the barrier, while emollients maintain it. Undertreatment from steroid fear is the commonest cause of failure. The closest wrong option is the high-potency regimen, which is excessive for body skin of an infant. At the bedside, match potency to site, explain that eczema is controlled and not cured, and arrange follow-up.
Key principle: Treat eczema flares early with a steroid matched to the site and moisturize every day.
Back to question Contents

BQ-406. Answer: A. Start antiviral therapy for herpes simplex today and send a swab for PCR
Herpes simplex virus spreads rapidly across skin where the barrier is damaged and local immunity is impaired, causing eczema herpeticum with uniform vesicles, fever, and ill appearance. Delay risks keratitis, bacteremia, and dissemination. The closest wrong option is cephalexin alone, because secondary staphylococcal infection can coexist, but antibiotics do not treat the virus. At the bedside, stop the steroid on affected skin, swab a fresh vesicle, give acyclovir by weight without waiting for the result, and ask ophthalmology to review if the eye is near.
Key principle: A fever with uniform punched-out vesicles in an eczema flare is eczema herpeticum and needs same-day antiviral therapy.
Back to question Contents

BQ-407. Answer: B. Erythema toxicum neonatorum
Erythema toxicum is a common, benign newborn eruption in which eosinophil-rich inflammation develops around hair follicles in the first days of life. It spares the palms and soles and clears within about a week without treatment. The closest wrong option is transient neonatal pustular melanosis, which is present at birth and leaves brown spots. Neonatal herpes would bring grouped vesicles, fever, or lethargy. At the bedside, reassure the parents, examine for systemic signs, and obtain a smear only if the diagnosis is uncertain.
Key principle: A well newborn with blotches and pustules on day 3 that spare palms and soles has erythema toxicum.
Back to question Contents

BQ-408. Answer: E. Admit for sepsis evaluation, send herpes PCR, and begin intravenous acyclovir
Neonatal herpes simplex infection spreads from the skin to the brain and organs, and early mortality and neurologic injury are high if treatment is delayed. Vesicles on a red base with fever and lethargy in a newborn are herpes until proven otherwise. The closest wrong option is topical therapy, which cannot treat disseminated disease. At the bedside, send skin, mouth, eye, and blood PCR and a cerebrospinal fluid study, start intravenous acyclovir immediately, and give antibacterial coverage until cultures are negative.
Key principle: Newborn vesicles with illness need sepsis evaluation and intravenous acyclovir without waiting for results.
Back to question Contents

BQ-409. Answer: C. Seborrheic dermatitis
Seborrheic dermatitis in infants is linked to oil gland activity and Malassezia yeast, and it appears at 2 to 8 weeks as greasy scale. It does not itch. The closest wrong option is atopic dermatitis, which causes itch, dryness, and typical weeping patches on cheeks and extensor surfaces. Tinea capitis is rare at this age. Most cases fade by 6 to 12 months. At the bedside, soften scale with oil, wash gently, and use a low-potency steroid or antifungal cream only for stubborn areas.
Key principle: Greasy scale on the scalp and folds in a young infant without itch is seborrheic dermatitis.
Back to question Contents

BQ-410. Answer: D. An exfoliative toxin from Staphylococcus aureus
Staphylococcus aureus can release a toxin that cleaves desmoglein 1 in the superficial epidermis, causing flaccid blisters that rupture easily in bullous impetigo. The disease spreads through small breaks in skin. The closest wrong option is herpes simplex virus, which produces grouped, painful vesicles. A drug reaction would have systemic illness and mucosal erosions. At the bedside, treat with an oral antistaphylococcal antibiotic when lesions are multiple, cover lesions, wash hands, and remind the family to return if fever or widespread peeling occurs.
Key principle: Bullous impetigo is caused by a staphylococcal toxin that splits the superficial epidermis.
Back to question Contents

BQ-411. Answer: C. A delayed allergic reaction to mite antigens that lingers after treatment
Scabies itch is a hypersensitivity response to mites, eggs, and feces in the skin, and the immune reaction continues after the mites are dead. Itch can last 2 to 4 weeks despite effective treatment. The closest wrong option is treatment failure, but the lack of new burrows and no spread to the household favors a post-treatment reaction. At the bedside, reassure the family, offer a topical steroid or antihistamine for comfort, and re-treat only if new burrows or contacts appear.
Key principle: Itch after scabies treatment can persist for weeks because it is an allergic response.
Back to question Contents

BQ-412. Answer: D. Observation, because the lesions resolve on their own, with hygiene advice
Molluscum contagiosum is a poxvirus infection of the epidermis, and the immune system clears it within months to a couple of years. Treatment can hurt and scar, and no treatment is clearly better than waiting. The closest wrong option is oral antivirals, which do not work against this virus. At the bedside, advise against scratching and sharing towels, allow school attendance with covered lesions when practical, and reserve treatment for lesions that cause symptoms or distress.
Key principle: Molluscum usually resolves on its own, so observation is an acceptable plan.
Back to question Contents

BQ-413. Answer: B. Kerion from tinea capitis
A kerion is an intense inflammatory reaction to a dermatophyte infection of the hair follicle, caused by the host immune response. It forms a tender, boggy mass that looks like an abscess. The closest wrong option is a bacterial abscess, but failure to respond to antibiotics and the surrounding hair loss with broken hairs favor a fungal cause. At the bedside, send a hair culture, treat with an oral antifungal agent, avoid incision, and consider a short steroid course for severe inflammation.
Key principle: A boggy scalp mass with hair loss that fails antibiotics is a kerion from tinea capitis.
Back to question Contents

BQ-414. Answer: E. Evaluation for androgen excess, including bone age and hormone testing
Acne in a child between ages 1 and 7 years is unusual because oil gland activity is normally low before adrenarche. Early androgen exposure from premature adrenarche, a tumor, or an enzyme deficiency can produce acne, body odor, pubic hair, and rapid growth. The closest wrong option is reassurance, which would miss a treatable endocrine cause. At the bedside, measure growth and pubertal stage, order bone age, and involve pediatric endocrinology for hormone evaluation.
Key principle: Acne with pubic hair and rapid growth before age 7 years requires evaluation for androgen excess.
Back to question Contents

BQ-415. Answer: D. Oral isotretinoin through a registered risk-management program
Severe nodulocystic acne comes from plugged follicles, Cutibacterium acnes, and deep inflammation, and it leaves permanent scars. Isotretinoin shrinks oil glands and treats all these factors, so it is indicated for severe or scarring acne and for disease that fails other therapy. The closest wrong option is more antibiotics, which add resistance without altering the course. At the bedside, enroll in the program, check baseline lipids and liver tests, counsel about dryness and mood, and screen for depression.
Key principle: Isotretinoin is the treatment for severe nodular or scarring acne and for acne that fails other treatment.
Back to question Contents

BQ-416. Answer: A. Rapid growth until about 5 months of age, then slow shrinking over years
Infantile hemangioma is a benign tumor of endothelial cells that proliferates quickly between about 1 and 3 months of age, slows by 5 months, and then involutes over years. The closest wrong option is steady growth, which would suggest a different vascular lesion such as a malformation. Most lesions need no treatment. At the bedside, measure and photograph the lesion, assess for high-risk sites, and plan follow-up to check that growth is slowing.
Key principle: Infantile hemangiomas proliferate during the first months, plateau by about 5 months, and then involute.
Back to question Contents

BQ-417. Answer: E. Glaucoma and seizures
A port-wine stain is a capillary malformation, and when it involves the forehead or upper eyelid it can signal Sturge-Weber syndrome, in which similar vascular abnormalities affect the eye and brain surface. This predisposes to glaucoma and seizures. The closest wrong option is airway obstruction and heart failure, which belong to extensive infantile hemangioma. At the bedside, arrange an ophthalmology examination for intraocular pressure, neurology review with MRI as indicated, and discuss laser therapy for the skin.
Key principle: A port-wine stain on the forehead or upper eyelid raises the risk of glaucoma and seizures in Sturge-Weber syndrome.
Back to question Contents

BQ-418. Answer: D. Neurofibromatosis type 1
Neurofibromatosis type 1 results from loss of neurofibromin, which normally restrains cell growth, and it produces cafe-au-lait macules, axillary freckling, neurofibromas, and optic glioma. Six or more macules larger than 5 mm in a prepubertal child support the diagnosis. The closest wrong option is McCune-Albright syndrome, which has larger, irregular macules and endocrine abnormalities. At the bedside, check the eyes, blood pressure, and spine, arrange genetics and ophthalmology review, and follow yearly.
Key principle: Six or more cafe-au-lait macules with axillary freckling indicate neurofibromatosis type 1.
Back to question Contents

BQ-419. Answer: E. Dermal melanocytes that stopped migrating on their way to the epidermis
During fetal development, melanocytes migrate from the neural crest to the epidermis, and some remain in the dermis, which makes the skin look blue-gray because of how light scatters. This congenital dermal melanocytosis is common in dark-skinned infants and usually fades in childhood. The closest wrong option is bleeding from birth trauma, which would change color over days. At the bedside, document the location and size at the first visit, which helps avoid confusion with abuse later, and reassure the parents.
Key principle: Congenital dermal melanocytosis looks like a bruise but is a stable, benign pigment finding that should be documented.
Back to question Contents

BQ-420. Answer: B. Abdominal ultrasonography to look for hepatic hemangiomas
Multiple cutaneous infantile hemangiomas increase the chance of vascular tumors in the liver, and a large hepatic lesion can shunt blood, cause heart failure, and produce hypothyroidism by inactivating thyroid hormone. Five or more skin lesions should prompt screening. The closest wrong option is observation alone, which misses a lesion that may be silent. At the bedside, order ultrasonography of the liver, check thyroid function if a significant liver lesion is found, and arrange specialist follow-up.
Key principle: Five or more cutaneous hemangiomas should lead to screening for liver hemangiomas.
Back to question Contents

BQ-421. Answer: C. Urgent referral to ophthalmology and a hemangioma specialist
A hemangioma over the eyelid can block vision during the critical period of visual development, producing amblyopia, and it can distort the eye by pressure. Lesions near the eye are high risk because the growth phase occurs when the visual system is most sensitive. The 2019 guideline names oral propranolol as the first-line systemic medicine, started by a specialist. The closest wrong option is waiting for natural involution, which would risk permanent visual loss. At the bedside, arrange the same-week assessment and document the pupil clearance.
Key principle: A periocular hemangioma that threatens vision needs urgent specialist treatment, usually with propranolol.
Back to question Contents

BQ-422. Answer: D. Mast cell release of histamine with transient leakage from dermal vessels
Urticaria arises when mast cells release histamine and other mediators, which dilates vessels and lets fluid leak into the dermis, forming wheals. Each wheal fades within 24 hours without bruising or scale. Viral infections are the commonest trigger. The closest wrong option is immune complex vasculitis, whose lesions persist longer than 24 hours and leave purpura. At the bedside, reassure the family, treat with a daily second-generation antihistamine, advise no laboratory testing, and warn about signs of anaphylaxis.
Key principle: Urticaria is mast cell mediated, and each wheal lasts less than 24 hours.
Back to question Contents

BQ-423. Answer: A. Intramuscular epinephrine
Peanut exposure triggers IgE-mediated mast cell degranulation, and the mediators cause hives, airway swelling, bronchospasm, vomiting, and hypotension. Epinephrine reverses the vasodilation, edema, and bronchospasm within minutes. The closest wrong option is an antihistamine alone, which relieves itch but does not reverse airway or circulatory compromise. At the bedside, give epinephrine in the thigh, lay the child flat, call emergency services, add oxygen and fluids, and observe for biphasic reactions.
Key principle: Hives with airway, gastrointestinal, or circulatory involvement is anaphylaxis and needs intramuscular epinephrine at once.
Back to question Contents

BQ-424. Answer: D. A non-allergic rash that often follows aminopenicillins in mononucleosis
In infectious mononucleosis, the altered immune response to the virus makes a rash common after ampicillin or amoxicillin, and the eruption is not usually mediated by IgE. It generally appears days after starting the drug, and mucous membranes are spared. The closest wrong option is true penicillin allergy, which usually presents with hives or anaphylaxis soon after dosing. At the bedside, stop the antibiotic, describe the rash accurately in the chart, and refer for allergy assessment before labeling him penicillin-allergic.
Key principle: A morbilliform rash after amoxicillin in mononucleosis does not prove penicillin allergy.
Back to question Contents

BQ-425. Answer: E. Serum sickness-like reaction
A serum sickness-like reaction is a drug hypersensitivity that follows exposure to certain antibiotics, especially cefaclor, by one to three weeks. It causes fever, arthralgia, and urticarial plaques that often have a purple center. The closest wrong option is Stevens-Johnson syndrome, which would show painful dusky lesions with mucosal erosions. At the bedside, stop the drug, treat with antihistamines and short courses of steroids if severe, and record the drug in the allergy list.
Key principle: Fever, joint pain, and urticarial plaques 1 to 3 weeks after cefaclor suggest a serum sickness-like reaction.
Back to question Contents

BQ-426. Answer: C. Alopecia areata

Alopecia areata is an autoimmune attack on hair follicles in the growth phase, causing a smooth patch with exclamation point hairs and no scale. The closest wrong option is tinea capitis, which causes scale, broken hairs, and nodes. Trichotillomania produces irregular patches with hairs of varied length. Because the condition is autoimmune, check for nail pitting and consider thyroid disease if other signs exist. At the bedside, confirm the absence of scale, avoid steroid cream on any scaly bald patch without fungal testing, and discuss regrowth and referral.

Key principle: A smooth round bald patch with no scale and exclamation point hairs is alopecia areata.

Back to question Contents

Answers: Domain 16. Hematology-Oncology

Contents Questions for this subject

BQ-427. Answer: A. Give oxygen, a cephalosporin plus a macrolide, and incentive spirometry, with early hematology input

Sickled cells obstruct pulmonary microvessels, and infection, fat embolism, and shallow breathing from pain all injure the lung, so oxygenation falls quickly. A new infiltrate with fever and hypoxemia defines acute chest syndrome, which calls for oxygen, a cephalosporin plus a macrolide to cover typical and atypical pathogens, and incentive spirometry. Treating this as simple pneumonia with oral amoxicillin leaves atypical organisms uncovered and sends home a child who can worsen within hours. Admit her, give analgesia carefully to avoid hypoventilation, give fluids by need rather than excess, and involve hematology early because transfusion may be needed.

Key principle: A new infiltrate with fever or hypoxemia in sickle cell disease is acute chest syndrome and needs oxygen, dual antibiotics, and admission.

Back to question Contents

BQ-428. Answer: B. Acute splenic sequestration crisis

Sickled cells trapped in the splenic red pulp enlarge the organ and pool a large part of the red cell mass, so hemoglobin falls while the marrow answers with a high reticulocyte count. Sudden pallor, shock, and a new left upper quadrant mass in a young child with sickle cell anemia fit this pattern. Aplastic crisis also gives profound anemia, but the reticulocyte count drops to almost nothing and the spleen does not enlarge. Give isotonic fluid at once, send blood for crossmatch, and transfuse red cells by simple transfusion while monitoring perfusion.

Key principle: Sudden pallor with a big spleen and a high reticulocyte count in a young child with sickle cell anemia is sequestration.

Back to question Contents

BQ-429. Answer: D. Increased fetal hemoglobin that inhibits hemoglobin S polymerization

Hydroxyurea raises fetal hemoglobin, and the higher mean corpuscular volume on the blood count is the visible sign that it is working. Fetal hemoglobin interferes with polymerization of hemoglobin S when red cells give up oxygen, so fewer cells sickle, stiffen, and stick in small vessels. It does not change the genotype, so the idea of replacing hemoglobin S with hemoglobin A describes transfusion or transplant instead. Offer it to every child with sickle cell anemia from 9 months, monitor blood counts for low neutrophils and platelets, and continue penicillin prophylaxis and vaccines.

Key principle: Hydroxyurea works mainly by raising fetal hemoglobin, which blocks hemoglobin S polymerization.

Back to question Contents

BQ-430. Answer: E. Send him to hematology to consider chronic transfusion therapy for stroke prevention

Narrowed intracranial arteries in sickle cell anemia force blood through at higher speed, so an elevated Doppler velocity marks a child at high risk of overt stroke before any symptom appears. A velocity of 200 cm per second or more is elevated, and it should send the child to a hematologist to consider chronic transfusion. Screening with CT or MRI is not the tool for a child without symptoms, and aspirin or yearly repeat screening ignores the risk. Arrange the specialist visit promptly, continue hydroxyurea and penicillin, and keep the annual Doppler schedule going between ages 2 and 16.

Key principle: A Doppler velocity of 200 cm per second or more in a child with sickle cell anemia means prompt hematology review for chronic transfusion.

Back to question Contents

BQ-431. Answer: A. Glucose-6-phosphate dehydrogenase deficiency

Without glucose-6-phosphate dehydrogenase, red cells cannot neutralize oxidant stress, so hemoglobin denatures into Heinz bodies and the spleen bites out damaged cells. An X-linked pattern, hemolysis after fava beans or a new drug, a negative antiglobulin test, and bite cells point to this enzyme defect. Warm autoimmune hemolysis looks similar clinically but gives a positive direct antiglobulin test and spherocytes. Stop the oxidant exposure, watch hemoglobin and urine output, transfuse if he is unstable, and give the family a written list of drugs and foods to avoid.

Key principle: Dark urine and hemolysis after fava beans or a new drug with bite cells and Heinz bodies means G6PD deficiency.

Back to question Contents

BQ-432. Answer: D. Viral infection of red cell precursors that halts production

Parvovirus B19 infects red cell precursors and shuts down the marrow for about a week. A person with normal red cell survival barely notices, but a child with a short red cell life span falls into aplastic crisis because replacement stops. The hallmark is severe anemia with a reticulocyte count near zero and a spleen of unchanged size. Sequestration also drops hemoglobin, but the spleen enlarges and reticulocytes rise. Support him, send parvovirus testing, transfuse if he is symptomatic or unstable, and expect recovery as the infection clears.

Key principle: Severe anemia with almost no reticulocytes in a chronic hemolytic state signals parvovirus-induced aplastic crisis.

Back to question Contents

BQ-433. Answer: B. 25 percent

Beta thalassemia trait is autosomal recessive in its severe form, and the child with major disease inherits a mutated beta globin gene from each parent. Two carriers give a 1 in 4 chance of an affected child, a 1 in 2 chance of another carrier, and a 1 in 4 chance of an unaffected noncarrier in each pregnancy. The trap is 50 percent, which is the chance of a carrier child and not of major disease. Offer genetic counseling, explain that the chance is the same for every pregnancy, and discuss prenatal and preconception options before the next pregnancy.

Key principle: Two carriers of beta thalassemia have a 25 percent chance of a child with thalassemia major in each pregnancy.

Back to question Contents

BQ-434. Answer: A. Factor VIII and factor IX activity assays

A prolonged activated partial thromboplastin time with normal prothrombin time and platelets points to a deficiency in the intrinsic pathway, and correction on mixing excludes an inhibitor. In a boy who has just begun cruising, with hemarthrosis and bleeding after circumcision, X-linked hemophilia A or B is the leading cause, and factor assays make the diagnosis and set severity. Aspirating the knee risks more bleeding and adds nothing, and a single ristocetin test would miss hemophilia. Treat early with factor replacement, protect the joint, and send him to a hemophilia center.

Key principle: A prolonged aPTT that corrects on mixing in a boy with a joint bleed needs factor VIII and IX assays.

Back to question Contents

BQ-435. Answer: D. Send von Willebrand factor antigen, activity, and a factor VIII level

Von Willebrand factor carries factor VIII and tethers platelets to damaged vessels, so a shortage produces mucosal bleeding, nosebleeds, and heavy menses even when routine screening tests are normal. Her history and her mother's history fit the most common inherited bleeding disorder. Platelet transfusion treats a platelet number problem she does not have, and a factor VIII assay alone would miss most cases. Send the full von Willebrand panel, repeat it when she is well if borderline because stress raises levels, treat the anemia, and involve hematology.

Key principle: Heavy menses and mucosal bleeding with normal screening tests call for von Willebrand factor antigen, activity, and factor VIII.

Back to question Contents

BQ-436. Answer: C. Apply local measures, give corticosteroids, and arrange hematology follow-up

Immune thrombocytopenia follows a viral illness when antibodies coat platelets and the spleen removes them. Most children need only observation, but persistent mucosal bleeding is an indication to treat, and corticosteroids, immune globulin, or anti-D are the choices. Observation alone is right for minor skin bleeding, and treating the number without bleeding is the common error. Here the nosebleed is the reason to act. Use pressure and local measures first, give corticosteroids after confirming a normal smear and blood count, and make sure hematology follows her closely.

Key principle: Treat immune thrombocytopenia when bleeding is mucosal or significant, and not by the platelet count alone.

Back to question Contents

BQ-437. Answer: A. Deficiency of vitamin K-dependent clotting factors

Newborns have low vitamin K stores, and breast milk contains little, so factors II, VII, IX, and X remain low in a baby who got no prophylaxis. The prothrombin time rises first because factor VII has the shortest half-life, then the activated partial thromboplastin time lengthens, while platelets and fibrinogen stay normal. Disseminated intravascular coagulation also prolongs both tests but lowers platelets and fibrinogen. Give intravenous vitamin K at once, support the baby with blood products and neurosurgical review for the hemorrhage, and counsel families that the newborn injection prevents this.

Key principle: Intracranial bleeding in a breastfed infant with no vitamin K injection and a prolonged prothrombin time is vitamin K deficiency bleeding.

Back to question Contents

BQ-438. Answer: C. Benign ethnic neutropenia with normal marrow reserve

Some people of African, Middle Eastern, and other ancestries have a lower normal neutrophil count with a normal marrow response to infection. This boy's absolute neutrophil count is about 990 per mm3, which is mild neutropenia, and he has no infections, ulcers, or other cell line changes, and a father with the same finding. Cyclic and severe congenital neutropenia cause repeated serious infections and mouth ulcers. Reassure the family, record the finding in the chart so no one repeats an extensive workup, and send him to hematology if infections or ulcers appear.

Key principle: A well child of African ancestry with mild isolated neutropenia and no infections most likely has benign ethnic neutropenia.

Back to question Contents

BQ-439. Answer: A. Acute hemolytic transfusion reaction

Recipient antibodies against donor red cell antigens activate complement and destroy the transfused cells inside the vessels, releasing hemoglobin and triggering shock. Fever, chills, back pain, dark urine, and hypotension early in a unit fit this reaction. Febrile nonhemolytic reactions give fever and chills but not back pain, dark urine, or hypotension. Stop the transfusion immediately, keep the intravenous line open with saline, send the unit and a new patient sample to the blood bank, support blood pressure and urine output, and watch for kidney injury and clotting problems.

Key principle: Fever, back pain, dark urine, and hypotension during a transfusion mean acute hemolysis until the blood bank proves otherwise.

Back to question Contents

BQ-440. Answer: E. Neuroblastoma

Neuroblastoma arises from sympathetic nervous tissue and secretes catecholamines, which appear as raised urine metabolites. A young child with a firm abdominal mass across the midline, bruising around the eyes from orbital metastases, and opsoclonus-myoclonus fits this tumor. Wilms tumor gives a smooth flank mass in a well child, often with hypertension, and does not cause the eye movements. Arrange urgent imaging and tissue diagnosis, send him to an oncology center the same day, and avoid steroids or delay while waiting for a clinic slot.

Key principle: An abdominal mass with periorbital bruising, opsoclonus-myoclonus, and raised urine catecholamines in a toddler is neuroblastoma.

Back to question Contents

BQ-441. Answer: A. Start vigorous intravenous hydration without potassium or phosphate, check G6PD status, and give rasburicase

Burkitt lymphoma cells turn over so quickly that they release potassium, phosphate, and nucleic acids before chemotherapy even begins, causing high uric acid, hyperkalemia, hyperphosphatemia, low calcium, and kidney injury. The first step is aggressive hydration, with no potassium or phosphate in the fluid, and rasburicase for the high uric acid. Rasburicase is dangerous in G6PD deficiency, so check status first. Starting chemotherapy or restricting fluids worsens the lysis. Monitor electrolytes and urine output closely, and involve oncology and nephrology.

Key principle: Tumor lysis is managed with vigorous hydration, rasburicase after a G6PD check, and no added potassium or phosphate.

Back to question Contents

BQ-442. Answer: B. Retinoblastoma

A tumor in the retina reflects light as a white pupil, and a growing mass can also pull the eye into strabismus. A white reflex in a flash photograph with an absent red reflex is retinoblastoma until proven otherwise. Congenital cataract and persistent fetal vasculature can also cause leukocoria, but delay costs both sight and life when the tumor is the cause. Retinopathy of prematurity does not fit a term infant. Send her the same day to an ophthalmologist experienced in pediatric eye tumors, and do not wait for the next routine visit.

Key principle: A white pupillary reflex in a child is retinoblastoma until an ophthalmologist excludes it.

Back to question Contents

Answers: Domain 17. Allergy and Immunology

Contents Questions for this subject

BQ-443. Answer: C. Give intramuscular epinephrine into the anterolateral thigh and call emergency services

Mast cell and basophil mediators released after an IgE-mediated exposure act on skin and gut together, and involvement of two organ systems after a known allergen defines anaphylaxis even when breathing and blood pressure are normal. Epinephrine is the first treatment, given at the first sign, in the anterolateral thigh. An antihistamine eases itch but does not stop progression, and giving it first costs minutes. Use the autoinjector, call emergency services, lay her down, repeat the dose in 5 to 15 minutes if needed, and observe her afterward.

Key principle: Hives with vomiting after a known allergen is anaphylaxis, and epinephrine is given first.

Back to question Contents

BQ-444. Answer: B. 0.22 mg

Epinephrine reverses vasodilation, bronchospasm, and mucosal swelling, and the weight-based dose for anaphylaxis is 0.01 mg/kg intramuscularly, up to a maximum of 0.3 mg. For 22 kg that is 0.22 mg, or 0.22 mL of a 1 mg/mL solution. The trap is 0.15 mg, which is the strength of a pediatric autoinjector and falls short for this girl. A tenfold arithmetic slip gives 0.022 mg or 2.2 mg. Inject into the anterolateral thigh, call emergency services, and repeat in 5 to 15 minutes if symptoms persist.

Key principle: Intramuscular epinephrine for anaphylaxis is 0.01 mg/kg, to a maximum of 0.3 mg in children.

Back to question Contents

BQ-445. Answer: E. Introduce peanut-containing foods at home with other solids, without testing

Early oral exposure to peanut in the gut builds tolerance, and delay raises the risk of allergy. The 2017 NIAID addendum sorts infants by eczema, and one with mild to moderate eczema and no egg allergy can start peanut at about 6 months at home, without prior testing. Testing first is for severe eczema or egg allergy, and delaying until age 1 is outdated advice. A food panel gives false positives and needless avoidance. Mix smooth peanut butter or peanut puff into other foods, and continue regular exposure afterward.

Key principle: An infant with mild to moderate eczema can start peanut at about 6 months at home without testing.

Back to question Contents

BQ-446. Answer: C. Sensitization without clinical allergy, so she can keep eating egg
A blood IgE result shows that the immune system has made antibody to egg, not that egg causes disease. A child who eats egg regularly without symptoms is tolerant, whatever the number. Calling this allergy and stopping egg risks turning a tolerant child into an avoider, and repeating panels yearly adds nothing. A supervised challenge is reserved for doubtful reactions. Tell the parents to continue egg in the diet, avoid screening panels in well children, and seek care if hives or vomiting ever follow egg.
Key principle: A positive IgE test in a child who eats the food without symptoms is sensitization, not allergy.
Back to question Contents

BQ-447. Answer: A. Needing a second dose of epinephrine to control symptoms
A biphasic reaction is a recurrence of symptoms between 1 and 48 hours after resolution, without new exposure. Risk rises with severe reactions and when more than one dose of epinephrine was needed, which suggests a stronger initial mediator burst. Sex, age, and a family history of hay fever do not predict it. One symptom-free hour has a high negative predictive value but does not remove risk, and this boy has the strongest risk feature. Observe him longer, prescribe two autoinjectors, teach the family an action plan, and send him to an allergist.
Key principle: Severe anaphylaxis and the need for more than one epinephrine dose raise the risk of a biphasic reaction.
Back to question Contents

BQ-448. Answer: B. Food protein-induced enterocolitis syndrome
Food protein-induced enterocolitis syndrome is a non-IgE reaction in which a food such as rice, cow's milk, or soy provokes repeated vomiting 1 to 4 hours after eating, with pallor and lethargy, and sometimes shock from fluid shifts. The absence of hives, wheeze, and lip swelling argues against IgE-mediated anaphylaxis. Pyloric stenosis appears with projectile vomiting at 3 to 6 weeks and not after a single food. Treat dehydration with intravenous fluid, not epinephrine, avoid the trigger food, and arrange allergy follow-up for a feeding plan.
Key principle: Repeated vomiting 1 to 4 hours after a food, with pallor and no hives, is food protein-induced enterocolitis.
Back to question Contents

BQ-449. Answer: E. Start a daily intranasal corticosteroid and teach aim away from the septum
Allergen exposure drives local inflammation in the nasal mucosa, which causes congestion that antihistamines treat poorly. For persistent or moderate to severe rhinitis, an intranasal corticosteroid is preferred. First-generation antihistamines such as diphenhydramine cause sedation and poor school performance and are advised against, and montelukast is not a first choice. A second-generation antihistamine alone is less effective for congestion. Aim the tip toward the outer wall of each nostril, tell the family benefit builds over days, and reserve immunotherapy for disease uncontrolled on drugs.
Key principle: Persistent allergic rhinitis is best treated with a daily intranasal corticosteroid, not a sedating antihistamine.
Back to question Contents

BQ-450. Answer: C. Retained nasal foreign body

An object lodged in one nostril obstructs drainage and provokes inflammation and secondary infection, so discharge is one-sided, foul, and often bloody. Allergic rhinitis and viral infections affect both sides and bring itch or sneezing, or fever and thick discharge. Sinusitis is usually bilateral and follows a persistent cold. Examine the nostril with a light, remove the object with suitable instruments or positive-pressure technique, and consider a button battery or magnet as an emergency requiring urgent removal. Check the other nostril and the ears for additional objects.

Key principle: One-sided foul nasal discharge in a toddler is a foreign body until proven otherwise.

Back to question Contents

BQ-451. Answer: A. Increase the second-generation antihistamine up to four times the standard dose

Chronic spontaneous urticaria results from mast cell activation without an outside trigger, and a cause is rarely found. The 2021 international guideline sets stepwise treatment: a second-generation antihistamine, then up to four times the standard dose, then omalizumab for those still uncontrolled. Jumping to omalizumab skips a step, and prolonged oral steroids carry harm without benefit. A food panel in a child without a trigger yields false positives. Raise the dose, reassure the family, review in a few weeks, and send her to an allergist if uncontrolled.

Key principle: Chronic spontaneous urticaria is treated by raising the second-generation antihistamine dose before adding omalizumab.

Back to question Contents

BQ-452. Answer: E. Serum C4 level

Hereditary C1 inhibitor deficiency results from deficiency or dysfunction of C1 inhibitor, which allows bradykinin to build up and fluid to leak into deep tissues. Recurrent swelling without hives or itch, abdominal pain from bowel wall edema, no response to antihistamine or epinephrine, and a family history fit it. Tryptase and IgE tests look for mast cell or allergic causes that do not apply here. A low C4 is the screen, and it is followed by C1 inhibitor level and function. Send C4, send him to an immunologist, and avoid treating each episode with antihistamines alone.

Key principle: Recurrent swelling without hives and without response to epinephrine calls for a C4 level to screen for hereditary C1 inhibitor deficiency.

Back to question Contents

BQ-453. Answer: C. Bradykinin accumulating from reduced breakdown

Angiotensin-converting enzyme also breaks down bradykinin, so an inhibitor lets it accumulate and increases vascular permeability, producing swelling of the lips and tongue without hives or itch. This pattern differs from allergic swelling, which comes with welts, and histamine is not the mediator. Antihistamines and steroids therefore do little. Stop the drug, protect the airway, watch closely because the tongue can swell enough to obstruct, and choose a different class of blood pressure and kidney medication with his nephrologist.

Key principle: Lip and tongue swelling without hives on an ACE inhibitor is bradykinin-mediated, so the drug is stopped.

Back to question Contents

BQ-454. Answer: A. Complete blood count with differential

Severe combined immunodeficiency leaves an infant without functional T cells, so thrush, chronic diarrhea, poor growth, and pneumonia appear in the first months. Many infants are found by newborn screening, but a normal screen should not stop action in a sick baby. The first test is a count with a differential, because a low lymphocyte count in an infant is abnormal and not reassuring. Other tests are not the first step here. Hold live vaccines, including rotavirus, send him urgently to immunology, and use irradiated, leukoreduced blood products if a transfusion is needed.

Key principle: In a sick infant with thrush and poor growth, check the lymphocyte count first, whatever the newborn screen showed.

Back to question Contents

BQ-455. Answer: E. Killing of ingested microbes by phagocytes

Deep abscesses with Staphylococcus aureus and invasive Aspergillus infection, with normal immunoglobulins and lymphocytes, point to a phagocyte defect such as chronic granulomatous disease, in which cells engulf microbes but cannot generate the oxidative burst to kill them. An antibody defect brings sinopulmonary infections with encapsulated bacteria, and a T cell defect brings severe viral and opportunistic infections. Send an oxidative burst test, start antimicrobial prophylaxis, and send him to immunology promptly, since early diagnosis changes outcomes.

Key principle: Recurrent deep abscesses with staphylococci and molds suggest a phagocyte killing defect.

Back to question Contents

BQ-456. Answer: C. Arrange evaluation by an immunologist because two warning signs apply

Most children with frequent infections have normal immunity, but some have a primary immunodeficiency, and warning-sign lists help find them. Four or more ear infections in a year and two or more pneumonias in a year are two signs, and seeing an immunologist is advised when two or more apply. Reassurance would miss a possible antibody defect, and prophylactic antibiotics or surgery treat the effects without finding a cause. Treat the list as a prompt and not a diagnosis. Send the infection history with the referral.

Key principle: Two or more warning signs of primary immunodeficiency warrant an immunology evaluation.

Back to question Contents

BQ-457. Answer: A. Stop the drug permanently and arrange urgent hospital care

Fever, painful blistering skin, and mucosal erosions on day 9 of a sulfonamide are the signs of a severe delayed drug reaction, in which T cell mediated injury attacks skin and mucous membranes. These reactions are rare but dangerous. Antihistamines and topical steroids do not address the process, and continuing or rechallenging the drug can be fatal. Stop the drug, document the reaction as severe, give supportive care in a unit able to manage skin and fluid losses, and never rechallenge.

Key principle: Blistering rash with mucosal lesions and fever on a new drug means stopping it for good and urgent care.

Back to question Contents

BQ-458. Answer: C. Start empiric elimination of common triggers such as milk and repeat endoscopy

Eosinophilic esophagitis is driven by food antigens through a mechanism that IgE and skin prick tests do not capture, so a positive test does not identify the culprit. The guideline therefore recommends against allergy-test-guided elimination, and instead uses empiric removal of common foods, from milk alone up to six foods, with repeat endoscopy to judge response. Food IgG panels measure exposure, not allergy, and symptoms alone track inflammation poorly. Remove the foods in stages, repeat the biopsy after each change, and involve a dietitian to protect growth.

Key principle: Dietary treatment of eosinophilic esophagitis uses empiric elimination judged by repeat endoscopy, not allergy tests.

Back to question Contents

Answers: Domain 18. Endocrinology

Contents Questions for this subject

BQ-459. Answer: B. Start an intravenous insulin infusion at 2 units per hour with no bolus
In ketoacidosis, insulin lack lets fat break down into ketone acids, and a rapid fall in glucose and osmolality too early raises the risk of cerebral injury. Insulin follows about an hour after fluids begin, as an infusion of 0.05 to 0.1 units/kg/h with no bolus, and a pH over 7.15 gets the lower rate. For 40 kg, 0.05 units/kg/h is 2 units per hour. A bolus or 8 units per hour is too aggressive. Monitor glucose hourly, add dextrose when glucose approaches 250 to 300 mg/dL, and keep the infusion running until the acid clears.
Key principle: Start insulin as an infusion, without a bolus, about an hour after fluids, using the lower rate when pH exceeds 7.15.
Back to question Contents

BQ-460. Answer: D. Begin potassium replacement and hold insulin until the level rises
Total body potassium is depleted in ketoacidosis even when serum values look normal, and insulin drives potassium into cells, so serum levels fall further once it starts. A level of 3.0 mmol/L with ECG changes risks arrhythmia if insulin begins first. The rule is to treat low potassium before insulin. Bicarbonate is not routine and lowers potassium more, and hypotonic fluid is not appropriate. Give potassium with cardiac monitoring, recheck the level, and start the infusion once it is safely higher.
Key principle: When potassium is low in ketoacidosis, replace it before starting insulin.
Back to question Contents

BQ-461. Answer: E. Give mannitol or 3% saline now and obtain imaging afterward
Cerebral injury is the most dangerous complication of ketoacidosis treatment, and it tends to appear several hours into treatment, just when laboratory values look better. New headache, vomiting, irritability, drowsiness, a falling heart rate, and falling saturation suggest raised intracranial pressure. Give hyperosmolar therapy immediately and image afterward, because delay costs brain. A scan first wastes time, more insulin accelerates osmolar shifts, and a fluid bolus may add to the swelling. Reduce fluid rate, elevate the head, call intensive care, and secure the airway.
Key principle: Suspected cerebral injury in ketoacidosis is treated with mannitol or 3% saline first and imaging afterward.
Back to question Contents

BQ-462. Answer: D. Insulin secretion that blocks glucose release, lipolysis, and ketogenesis
Insulin stops the liver from releasing glucose and blocks fat breakdown, so the body has neither glucose nor ketones during a fast. A low glucose with detectable insulin and low ketones and fatty acids is the signature of hyperinsulinism, which threatens the brain because the usual alternate fuel is missing. In a fatty acid oxidation defect fatty acids would be high, and ketotic states show high ketones with low insulin. Treat with a glucose bolus and a high glucose infusion rate, keep the critical sample, and arrange urgent endocrine care.
Key principle: Hypoglycemia with low ketones and detectable insulin means hyperinsulinism.
Back to question Contents

BQ-463. Answer: B. Basal insulin, metformin, and a lifestyle program together
Insulin resistance in puberty has exhausted this boy's beta cell reserve, so glucose is high enough to cause symptoms. The American Academy of Pediatrics guideline starts insulin when A1C is 9 percent or higher or random glucose is 250 mg/dL or more, and adds metformin and a lifestyle program. Metformin and lifestyle alone suit milder cases with A1C under 9 percent. Lifestyle alone or a sulfonylurea leaves him under-treated. Start basal insulin, teach the family, check A1C every 3 months, and screen for blood pressure, lipids, and fatty liver.
Key principle: Type 2 diabetes in youth with A1C of 9 percent or higher starts with insulin plus metformin and lifestyle change.
Back to question Contents

BQ-464. Answer: D. Tissue transglutaminase IgA with total IgA, and thyroid antibodies with TSH
Type 1 diabetes shares immune susceptibility with celiac disease and autoimmune thyroid disease, so these conditions cluster, and a mother with thyroiditis adds to his risk. Screening at diagnosis uses tissue transglutaminase IgA with a total IgA and thyroid antibodies with TSH. Lipid profile and albumin screening belong later as complication surveillance, and antinuclear antibody or adrenal tests are not routine. Send the tests before discharge, repeat them periodically, and reassure the family that finding these conditions early is the aim.
Key principle: Screen children with new type 1 diabetes for celiac disease and thyroid autoimmunity.
Back to question Contents

BQ-465. Answer: A. Congenital hypothyroidism
Thyroid hormone drives metabolism and brain growth, so a deficit slows gut movement, prolongs jaundice, thickens the tongue, and weakens the cry. Babies are rarely symptomatic at birth, which is why newborn screening exists. Prolonged jaundice with constipation, a large tongue, a hoarse cry, and an umbilical hernia is the classic pattern. Biliary atresia gives pale stools and conjugated jaundice, and Hirschsprung disease lacks the other features. Send TSH and free T4 and start levothyroxine without delay, because lost developmental potential cannot be recovered.
Key principle: Prolonged jaundice, constipation, a large tongue, and a hoarse cry in a young infant suggest congenital hypothyroidism.
Back to question Contents

BQ-466. Answer: C. Hashimoto thyroiditis
Autoimmune destruction of thyroid tissue lowers hormone output, and the pituitary responds with a very high TSH. Slowed growth despite weight gain, fatigue, cold intolerance, constipation, and a firm painless goiter are typical. Graves disease causes the opposite picture with a suppressed TSH. Central hypothyroidism has a low or normal TSH, so a TSH of 38 mIU/L excludes it. Confirm with thyroid peroxidase antibodies, treat with levothyroxine, and expect catch-up growth over a year or more.
Key principle: A high TSH with a low free T4 and a firm goiter in a school-age child is Hashimoto thyroiditis.
Back to question Contents

BQ-467. Answer: E. Autoantibodies that stimulate the thyrotropin receptor

In Graves disease, antibodies bind the thyrotropin receptor and drive thyroid hormone production without regard to the pituitary, so TSH is suppressed while free T4 is high. A diffuse goiter, proptosis, and typical symptoms in a teenage girl fit. Thyroiditis releases stored hormone through damage and is often transient and painful, and pituitary adenomas give a high TSH. Confirm with thyroid-stimulating antibodies, start an antithyroid drug, add a beta blocker for symptoms if needed, and send her to an endocrinologist.

Key principle: Graves disease results from antibodies that stimulate the thyrotropin receptor, suppressing TSH.

Back to question Contents

BQ-468. Answer: D. Send a karyotype and 17-hydroxyprogesterone, monitor electrolytes, and involve a specialist team

Ambiguous genitalia can come from androgen excess, as in congenital adrenal hyperplasia, or from other differences of sex development, and some forms carry a risk of salt loss in the second week. The first steps are to avoid guessing, send a karyotype and 17-hydroxyprogesterone, order pelvic imaging, follow sodium and potassium, and involve a multidisciplinary team. Assigning sex, early surgery, or discharge with a late review ignores the danger of adrenal crisis. Tell the parents the baby is healthy and that you are working out the pattern.

Key principle: A newborn with ambiguous genitalia needs a karyotype, 17-hydroxyprogesterone, electrolyte monitoring, and a team, not a guess.

Back to question Contents

BQ-469. Answer: C. Familial short stature

Children who inherit short parents' genes track a low percentile at a normal growth rate, with a bone age that matches chronologic age. This girl grows 6 cm per year, over the 4 cm threshold that would prompt a search, and her parents are short with normal puberty timing. Constitutional delay shows a delayed bone age, and growth hormone deficiency or hypothyroidism show slow growth. Turner syndrome would normally show a slow growth rate. Plot her on the growth chart, compare with parental heights, and reassure the family while continuing routine monitoring.

Key principle: A child tracking a low percentile at a normal speed with a matching bone age has familial short stature.

Back to question Contents

BQ-470. Answer: A. Reexamine her every 4 to 6 months to watch for progression

Early breast budding in girls aged 7 to 8 is most often central and slowly progressive, and many girls do not need treatment. Current guidance favors watching a girl of this age with stage 2 breasts, no rapid growth, and no neurologic signs, with examination every 4 to 6 months. Starting a GnRH agonist is for rapid progression, advanced stage, or very early onset. Brain MRI is not routine at this age without neurologic findings. Measure height and stage each visit, and act if puberty advances quickly.

Key principle: A girl aged 7 to 8 with slow, early breast development is watched, not treated, unless progression is fast.

Back to question Contents

BQ-471. Answer: E. Constitutional delay of growth and puberty

In constitutional delay, the pubertal clock runs late, so the child is short for age, the bone age lags, and the growth rate is low but normal for bone age. The family pattern of a father who grew until 19 supports it, and it is the commonest cause of delayed puberty in boys. Klinefelter syndrome and hypothyroidism are less frequent and give other signs, and isolated GnRH deficiency usually lacks a lagging family pattern. Reassure the family, track growth and testicular size every 6 months, and check TSH if growth lags further.

Key principle: Constitutional delay is the most common cause of delayed puberty in a short boy with a delayed bone age.

Back to question Contents

BQ-472. Answer: B. Renal wasting of phosphate

When the kidney wastes phosphate, bone cannot mineralize even though vitamin D and calcium are normal, so rickets develops with low phosphate and a normal or mildly raised parathyroid hormone. The X-linked pattern with a short, bowed father and a failure to respond to vitamin D alone fits X-linked hypophosphatemia. Dietary vitamin D deficiency would show a low 25-hydroxyvitamin D and high parathyroid hormone. Check urine phosphate handling and fibroblast growth factor 23, and send the family to a metabolic bone clinic, since treatment uses phosphate and active vitamin D.

Key principle: Rickets with a normal vitamin D level and low phosphate suggests renal phosphate wasting.

Back to question Contents

BQ-473. Answer: C. Hypoparathyroidism as part of DiGeorge syndrome

Without parathyroid hormone, bone and kidney do not hold calcium, so calcium falls while phosphate rises. A young infant with seizure, Chvostek sign, long QT, and heart and jaw anomalies fits DiGeorge syndrome, in which the parathyroid glands are underdeveloped. Vitamin D deficiency and dietary calcium lack raise parathyroid hormone, and phosphate wasting lowers phosphate. Give intravenous calcium with cardiac monitoring for the seizure, check ionized calcium, magnesium, and phosphate, and arrange cardiac evaluation and chromosome 22q11.2 testing.

Key principle: Low calcium, high phosphate, and low parathyroid hormone in an infant with heart and jaw anomalies suggest DiGeorge syndrome.

Back to question Contents

BQ-474. Answer: A. Ketotic hypoglycemia of early childhood

A thin toddler has limited glycogen and muscle mass, so a long fast or vomiting illness exhausts glucose, and fat breakdown makes abundant ketones while insulin is appropriately low. The critical sample shows high ketones, low insulin, and a normal cortisol response, which is the pattern of ketotic hypoglycemia. Hyperinsulinism would show low ketones with detectable insulin, and adrenal insufficiency would show a low cortisol. Treat with glucose first, then prevent recurrence with frequent meals and sick-day carbohydrate, and expect improvement by school age.

Key principle: Hypoglycemia with high ketones, low insulin, and normal cortisol after a fast is ketotic hypoglycemia.

Back to question Contents

Answers: Domain 19. Nephrology, Fluids, and Electrolytes

Contents Questions for this subject

BQ-475. Answer: D. Movement of water into brain cells that adapted to high osmolality

During sustained hypernatremia, brain cells defend their volume by accumulating osmotically active solutes, so intracellular osmolality rises along with plasma. A rapid fall in plasma sodium leaves those solutes behind, and water follows the gradient into the cells, producing cerebral edema and seizures. Cell shrinkage is the closest trap, because it describes the hypernatremic state itself rather than the hazard of correction. At the bedside, give isotonic fluid for perfusion, replace the deficit evenly over about two days, and recheck the sodium every few hours.

Key principle: A fast fall in sodium after chronic hypernatremia moves water into adapted brain cells, so correct over about two days.

Back to question Contents

BQ-476. Answer: E. Intravenous calcium to stabilize the cardiac membrane

Tumor lysis releases intracellular potassium faster than the kidney can clear it, and a high extracellular level depolarizes the cardiac membrane, producing peaked T waves and a wide QRS. Calcium raises the threshold potential and protects the myocardium within minutes without changing the potassium level. Insulin with glucose is the closest trap because it lowers the level, but it leaves the heart exposed while it works. At the bedside, give calcium under monitoring, then shift and remove potassium, treat the tumor lysis, and involve nephrology early.

Key principle: When hyperkalemia changes the ECG, give calcium first to protect the heart, then shift and remove potassium.

Back to question Contents

BQ-477. Answer: C. A small intravenous bolus of 3 percent saline

Hypotonic fluid plus the antidiuretic hormone drive of pain, nausea, and surgery dilutes the plasma sodium, and the fall in osmolality moves water into brain cells and causes seizures. A small bolus of hypertonic saline raises the sodium just enough to reverse the swelling. Fluid restriction alone is the closest trap, because it is correct for mild asymptomatic cases but acts too slowly for a seizure. At the bedside, give the bolus, stop the hypotonic fluid, switch to an isotonic solution, and recheck the sodium frequently so the rise stays controlled.

Key principle: A seizure from hyponatremia is treated with a small bolus of 3 percent saline, not fluid restriction.

Back to question Contents

BQ-478. Answer: D. Failure of the distal tubule to excrete acid into the urine

A normal anion gap with a high chloride means the acidosis comes from bicarbonate loss or failed acid excretion rather than an added acid. A urine pH over 5.5 during systemic acidosis shows that the distal tubule is not secreting hydrogen ion, which is the signature of renal tubular acidosis, and potassium wastes alongside. Stool bicarbonate loss is the closest trap because it also leaves a normal gap, but the urine would be maximally acidic and diarrhea is absent. At the bedside, confirm with a urine anion gap, check calcium and renal ultrasound, and begin alkali replacement.

Key principle: A normal-gap acidosis with an inappropriately high urine pH points to a tubular acid excretion defect.

Back to question Contents

BQ-479. Answer: A. Small frequent amounts of oral rehydration solution by spoon or syringe
Gastroenteritis in a child with moderate dehydration still leaves the gut able to absorb sodium and glucose together, so a balanced oral solution replaces losses as effectively as the vein. Giving a spoonful every minute or two overcomes vomiting, and an antiemetic can help if it persists. Intravenous fluid is the closest trap, but it is reserved for shock or failed oral therapy. High-sugar drinks worsen osmotic diarrhea. At the bedside, teach the parents the spoon method, reassess in a few hours, and return to normal feeding as soon as he rehydrates.
Key principle: Oral rehydration solution given in small frequent amounts is first-line for mild and moderate dehydration.
Back to question Contents

BQ-480. Answer: D. Post-infectious glomerulonephritis
Streptococcal antigen-antibody complexes deposit in the glomeruli 1 to 3 weeks after pharyngitis and activate complement, causing nephritis with sodium retention, edema, and hypertension. Red cell casts and a low C3 confirm a glomerular source. IgA nephropathy is the closest trap, but it bleeds within a day or two of the infection and keeps a normal C3. At the bedside, check a streptococcal serology, restrict salt, treat the hypertension, and repeat C3 in 8 to 12 weeks, because a level that stays low beyond 12 weeks calls for another diagnosis.
Key principle: Cola urine, hypertension, and low C3 two weeks after pharyngitis make post-infectious glomerulonephritis.
Back to question Contents

BQ-481. Answer: A. IgA nephropathy
Mucosal infection triggers production of galactose-deficient IgA, which deposits in the mesangium and causes bleeding within one or two days of the illness, which is why it is called synpharyngitic. Dysmorphic cells and protein place the bleeding in the glomerulus, and a normal C3 separates it from post-infectious disease. Post-infectious glomerulonephritis is the closest trap, but it follows an infection by one to three weeks with hypertension and low C3. At the bedside, check blood pressure and creatinine, monitor protein over time, and involve nephrology if proteinuria persists.
Key principle: Visible hematuria one to two days after a cold with a normal C3 suggests IgA nephropathy.
Back to question Contents

BQ-482. Answer: E. Abnormal type IV collagen in the glomerular and cochlear basement membranes
Alport syndrome results from a variant in a type IV collagen gene, most often on the X chromosome, so the glomerular basement membrane thins, splits, and leaks red cells, and the same collagen supports the cochlea. That explains hematuria, hearing loss, and a family history of renal failure in male relatives. IgA deposition is the closest trap because it also causes hematuria, but it does not affect hearing or run in families this way. At the bedside, obtain an audiogram, urine protein, and genetic testing, and begin early nephrology care.
Key principle: Hematuria with hearing loss and affected male relatives reflects a type IV collagen defect.
Back to question Contents

BQ-483. Answer: B. Orthostatic proteinuria

In orthostatic proteinuria, upright posture alters glomerular hemodynamics, so protein spills during the day and disappears when the child lies flat overnight. A first-morning ratio under 0.2 mg/mg in a child over 2 years is normal, which proves the variant is benign. Focal segmental glomerulosclerosis is the closest trap, but it keeps protein in every sample. At the bedside, reassure the family, avoid biopsy, and repeat a first-morning sample in several months to confirm the pattern.

Key principle: A normal first-morning urine protein-to-creatinine ratio with daytime proteinuria shows benign orthostatic proteinuria.

Back to question Contents

BQ-484. Answer: C. Urinary loss of immunoglobulins and complement, with edema fluid as a growth medium

Heavy proteinuria drains immunoglobulins and complement factors, which opsonize encapsulated bacteria, and the ascitic fluid supplies a protein-poor medium where bacteria multiply. That is why pneumococcal peritonitis is the classic infection of nephrotic children. Steroid immunosuppression contributes but is the closest trap, because infection occurs in untreated children too. At the bedside, send blood and fluid cultures, begin antibiotics with pneumococcal coverage, admit the child, and confirm that pneumococcal vaccines are current.

Key principle: Nephrotic children lose immunoglobulins and complement in urine, so pneumococcal peritonitis is a leading risk.

Back to question Contents

BQ-485. Answer: E. Reduced glomerular perfusion after NSAID blockade of prostaglandin-mediated dilation

In a dry child the kidney protects filtration by dilating the afferent arteriole through prostaglandins. An NSAID removes that dilation, so glomerular pressure and filtration fall in a prerenal pattern, and the urinalysis stays bland. Microangiopathic injury is the closest trap, but it would bring anemia, low platelets, and blood in the urine. At the bedside, restore volume with isotonic fluid, stop the NSAID and any other nephrotoxin, follow urine output and daily weights, and recheck creatinine and potassium within a day.

Key principle: An NSAID in a volume-depleted child blocks prostaglandin-mediated afferent dilation and causes prerenal injury.

Back to question Contents

BQ-486. Answer: D. Acute kidney injury by the urine output criterion, with creatinine lagging

Creatinine rises hours to days after filtration falls, so urine output is the first signal when it can be measured. The KDIGO definition counts output under 0.5 mL/kg/hour for 6 hours as injury, and this boy has met that threshold for 8 hours. Trusting the unchanged creatinine is the closest trap. At the bedside, check volume status, bladder scan for obstruction, stop nephrotoxins, review his medications, and follow output, weight, potassium, and creatinine closely over the next day.

Key principle: Urine output under 0.5 mL/kg/hour for 6 hours defines kidney injury even when creatinine has not yet risen.

Back to question Contents

BQ-487. Answer: B. Renal and bladder ultrasound with urinalysis and creatinine
In a young child, hypertension is secondary until proven otherwise, and kidney disease, including reflux scarring and congenital anomalies, is a leading source. The 2017 AAP guideline recommends renal ultrasound for any child under 6 with hypertension. Calling it primary hypertension is the closest trap, because that diagnosis fits the overweight teenager. At the bedside, send urinalysis, electrolytes, and creatinine with the imaging, consider ambulatory monitoring, and involve nephrology or cardiology for the secondary evaluation.
Key principle: Hypertension in a child under 6 is secondary until proven otherwise, and renal ultrasound is the first test.
Back to question Contents

BQ-488. Answer: C. Stage 1 hypertension
For adolescents 13 years and older, the 2017 AAP guideline uses fixed thresholds: elevated is 120 to 129 systolic with diastolic under 80, stage 1 is 130/80 to 139/89, and stage 2 is 140/90 or higher. Her repeated readings fall in the stage 1 band. Elevated blood pressure is the closest trap, because it applies only to systolic values under 130. White coat hypertension needs ambulatory monitoring to establish. At the bedside, arrange ambulatory monitoring, counsel on weight and diet, and screen for target-organ effects and secondary causes.
Key principle: In adolescents 13 and older, 130/80 to 139/89 mm Hg on repeated visits is stage 1 hypertension.
Back to question Contents

BQ-489. Answer: D. Reduced intestinal calcium absorption from vitamin D deficiency
Without enough vitamin D, the gut absorbs calcium poorly, serum calcium falls, and neuromuscular excitability rises until tetany or seizure occurs. The metaphyseal changes show rickets, which supports a nutritional cause. Hypoparathyroidism is the closest trap because it also lowers calcium, but bone changes of rickets and the breastfeeding history point to deficiency. At the bedside, give intravenous calcium with ECG monitoring, then start vitamin D replacement, check magnesium, phosphate, and parathyroid hormone, and counsel the family on supplementation.
Key principle: Seizure with low ionized calcium and rickets in a breastfed infant reflects vitamin D deficiency.
Back to question Contents

BQ-490. Answer: A. Aldosterone-driven renal potassium loss in a volume-depleted infant
Vomiting removes gastric acid and chloride, which leaves a hypochloremic alkalosis and contracts the extracellular volume. Volume depletion raises aldosterone, and the kidney exchanges sodium for potassium and hydrogen, so most of the potassium is lost in urine rather than vomitus. Direct gastric loss is the closest trap, because gastric fluid contains little potassium. At the bedside, correct volume with isotonic saline plus dextrose, add potassium once he voids, recheck electrolytes, and delay pyloromyotomy until the alkalosis has resolved.
Key principle: Hypokalemia in pyloric stenosis comes from renal potassium wasting driven by volume depletion, not from the vomitus.
Back to question Contents

Answers: Domain 20. Genitourinary System

Contents Questions for this subject

BQ-491. Answer: C. Place a bladder catheter, then arrange a voiding cystourethrogram with urology

Posterior urethral valves obstruct the male urethra, so the bladder wall thickens and back pressure dilates both ureters and kidneys, risking kidney failure. A catheter relieves the obstruction at once, and a voiding study shows the valve and any reflux. Watching for transient dilation is the closest trap, because that pattern is unilateral and mild. At the bedside, drain the bladder, check electrolytes and creatinine, involve urology early, and monitor for postobstructive diuresis and infection.

Key principle: A boy with bilateral hydronephrosis, a thick bladder, and a weak stream needs bladder drainage and urgent urology.

Back to question Contents

BQ-492. Answer: B. Continuous antibiotic prophylaxis with referral to urology for follow-up

Reflux sends infected urine up to the kidney, and each pyelonephritis can scar parenchyma. The RIVUR trial found that prophylaxis lowered recurrent infection in children with reflux, and it fits a child with frequent recurrences or high-grade reflux. Reassurance alone is the closest trap, since many children outgrow low-grade reflux but her infections are recurring. At the bedside, prescribe the prophylactic agent by weight, treat each fever with prompt culture, address bowel and bladder habits, and arrange urology review.

Key principle: Prophylaxis suits the child with frequent infections or high-grade reflux, though it does not reduce scarring on average.

Back to question Contents

BQ-493. Answer: E. Referral to pediatric urology for orchiopexy

Spontaneous descent rarely occurs after about 6 months of age, and a testis that stays outside the scrotum is exposed to a warmer environment that impairs germ cell development and raises long-term cancer risk. The 2014 AUA guideline sends the boy to surgery referral by 6 months of corrected age, with orchiopexy within the next year, and advises against ultrasound and hormonal therapy. Imaging is the closest trap, because it cannot show whether a testis is viable. At the bedside, refer now and counsel the family about fertility and cancer surveillance.

Key principle: An undescended testis at 6 months of corrected age goes to urology for surgery without imaging or hormones.

Back to question Contents

BQ-494. Answer: B. Retractile testis
The cremasteric reflex is brisk in young boys and pulls a normally descended testis up toward the canal in the cold or on stimulation. A testis that can be guided into the scrotum and stays there without tension is retractile, not undescended. An undescended testis is the closest trap, because it also sits in the canal, but it cannot be brought down. At the bedside, examine in a warm room with the child relaxed, reassure the family, and recheck yearly because some retractile testes ascend and then need surgery.
Key principle: A testis that can be brought down without tension and stays there is retractile and needs only annual examination.
Back to question Contents

BQ-495. Answer: E. Incarcerated inguinal hernia
A persistent processus vaginalis lets bowel enter the inguinal canal, and a loop that becomes trapped swells, loses venous drainage, and can infarct, especially in premature infants. Irritability, vomiting, and a firm tender groin mass that reaches the scrotum fit incarceration. Torsion of the cord is the closest trap, but the testes are palpable in a normal position and the reflex is present. At the bedside, give analgesia, attempt gentle steady reduction, keep him fasting, and call surgery urgently, since an irreducible hernia needs prompt repair.
Key principle: A firm, tender groin mass with irritability and vomiting in an infant is an incarcerated hernia until proven otherwise.
Back to question Contents

BQ-496. Answer: D. Delay circumcision and arrange pediatric urology repair in early childhood
Hypospadias is incomplete closure of the urethral folds, and the foreskin tissue is often needed to reconstruct the urethra. Circumcision removes that tissue and complicates repair. A karyotype is the closest trap, because it is indicated when hypospadias comes with an undescended testis, which he does not have. At the bedside, examine the genitalia fully, hold circumcision, and involve urology so that repair is planned in early childhood.
Key principle: In hypospadias, do not circumcise, because the foreskin may be needed for repair.
Back to question Contents

BQ-497. Answer: B. Torsion of the appendix testis
The appendix testis is a small embryologic remnant on the upper pole that can twist and infarct in boys aged 7 to 14, producing a localized tender nodule that looks blue through the skin. The gradual onset, preserved cremasteric reflex, and absence of nausea separate it from torsion of the cord. Testicular torsion is the closest trap, and the two can be hard to separate when the scrotum is very swollen. At the bedside, give rest and analgesia, follow the course, and ask for imaging or urology if the picture is uncertain.
Key principle: A blue dot at the upper pole with a present cremasteric reflex indicates torsion of the appendix testis.
Back to question Contents

BQ-498. Answer: D. Treat as epididymitis and image the urinary tract for an anomaly
Epididymitis in a prepubertal boy is uncommon and often reflects reflux of infected urine through an ectopic ureter or other urinary tract anomaly, unlike teenagers, in whom sexually transmitted organisms are typical. Gradual onset, a preserved cremasteric reflex, and pyuria support it over torsion. Treating with antibiotics alone is the closest trap, because it leaves a structural cause unfound. At the bedside, send a urine culture, start appropriate therapy, obtain renal and bladder ultrasound, and ask urology to review for an anomaly.
Key principle: Epididymitis in a young boy calls for treatment and a search for a urinary tract anomaly.
Back to question Contents

BQ-499. Answer: A. Patent processus vaginalis allowing peritoneal fluid to enter the scrotum
During fetal life the processus vaginalis carries the testis into the scrotum and then closes. If a narrow channel stays open, peritoneal fluid flows in when the child is upright and drains back when he lies down, so the swelling fluctuates in size and transilluminates. Torsion is the closest trap, but it is painful and does not change with position. At the bedside, reassure the parents, watch for a bulge that suggests a hernia, and plan surgical review if it persists beyond about 12 to 18 months.
Key principle: A fluctuating transilluminating scrotal swelling reflects a patent processus vaginalis.
Back to question Contents

BQ-500. Answer: D. Obtain urine by catheterization for urinalysis and culture
Urine from a bag collects skin flora, so a positive culture is unreliable and a negative one is useful only for ruling out. Catheter or suprapubic specimens come directly from the bladder and give a trustworthy culture. The 2011 AAP guideline expects a urinalysis plus culture before treating a febrile infant or toddler. The bagged specimen is the closest trap because it is convenient. At the bedside, catheterize with sterile technique, send analysis and culture, and start treatment when the analysis is abnormal.
Key principle: A febrile toddler who cannot void on request needs a catheter specimen for urinalysis and culture.
Back to question Contents

BQ-501. Answer: C. Repeat the ultrasound at 3 to 7 days of age
In the first day of life a newborn is relatively dehydrated and urine flow is low, so the pelvis can look smaller than it really is and a true dilation may be missed. An ultrasound after the first 48 hours, ideally at days 3 to 7, gives a more reliable picture. Accepting the normal first scan is the closest trap. At the bedside, schedule the repeat before discharge or soon after, and move to urology and a voiding study if the dilation is significant or bilateral.
Key principle: Image the kidneys again at days 3 to 7, because an early newborn scan can understate dilation.
Back to question Contents

BQ-502. Answer: A. Evaluate for a disorder of sex development with karyotype and hormone tests
Hypospadias combined with an undescended testis raises the chance of a disorder of sex development, because both depend on normal androgen production and action during fetal life. A karyotype, hormone levels such as 17-hydroxyprogesterone, and pelvic ultrasound identify conditions such as congenital adrenal hyperplasia that can cause adrenal crisis. Reassurance is the closest trap, since either finding alone is often isolated. At the bedside, hold sex assignment decisions, check electrolytes and glucose, and involve endocrinology, genetics, and urology promptly.
Key principle: Hypospadias with an undescended testis should prompt evaluation for a disorder of sex development.
Back to question Contents

BQ-503. Answer: E. A nuclear renal scan to measure split function, with urology review
Ureteropelvic junction obstruction narrows the outflow of the renal pelvis, so urine backs up into the pelvis and calyces while the ureter stays thin. The normal voiding study excludes reflux. A diuretic nuclear scan shows how much each kidney contributes to function and whether drainage is impaired, which guides the choice between observation and pyeloplasty. Repeating the voiding study is the closest trap. At the bedside, arrange the scan, teach the family to report fever, and have urology monitor with serial ultrasounds.
Key principle: Moderate unilateral hydronephrosis with a normal ureter fits ureteropelvic junction obstruction and needs a renal scan.
Back to question Contents

BQ-504. Answer: D. A spot urine calcium-to-creatinine ratio
Excess calcium in the urine can irritate the urinary tract and form microcrystals, and it is a common treatable cause of isolated hematuria in children. Round cells without casts or protein point away from the glomerulus, and a normal family evaluation lowers the chance of a hereditary nephropathy. Biopsy is the closest trap, because it is invasive and reserved for glomerular disease. At the bedside, send the ratio, encourage hydration, restrict excess salt, and recheck the urinalysis periodically for new protein or hypertension.
Key principle: A urine calcium-to-creatinine ratio finds hypercalciuria, a common cause of isolated hematuria in children.
Back to question Contents

BQ-505. Answer: E. Voiding cystourethrography
Reflux can coexist with a normal ultrasound, and repeated infection of a refluxing kidney leads to scarring. The 2011 AAP guideline advises a voiding study when a child has a recurrent febrile infection, even after a normal first ultrasound. Relying on the normal scan is the closest trap. A scan during active infection would show inflammation rather than scarring. At the bedside, arrange the voiding study once the infection has settled and involve urology if reflux is high grade.
Key principle: A second febrile urinary tract infection is an indication for voiding cystourethrography.
Back to question Contents

BQ-506. Answer: B. Infertility and testicular cancer

A testis held in the groin or abdomen is warmer than the scrotum, and the heat damages germ cells and predisposes the gonad to malignant change. Early orchiopexy moves the testis to a cooler site and lowers, but does not remove, those risks. Pubertal delay is the closest trap because endocrine problems are more familiar, but a single undescended testis rarely disturbs overall hormone output. At the bedside, refer for surgery by the recommended age and teach self-examination of both testes in adolescence.

Key principle: Untreated cryptorchidism raises the risks of infertility and testicular cancer.

Back to question Contents

Answers: Domain 21. Genetics, Dysmorphology, and Metabolic Disorders

Contents Questions for this subject

BQ-507. Answer: D. Coarctation of the aorta
Lymphedema of the hands and feet with a webbed neck suggests Turner syndrome (45,X), in which left-sided obstructive lesions are common. A narrowing of the distal arch reduces flow to the lower body, which explains weak femoral pulses and an arm-to-leg pressure gradient. Atrioventricular septal defect is the closest trap, because it is the typical lesion of trisomy 21. At the bedside, obtain an echocardiogram, four-limb blood pressures, a karyotype, and a renal ultrasound, and plan endocrine follow-up for growth and ovarian function.
Key principle: Turner syndrome brings coarctation and bicuspid aortic valve, so check pulses and four-limb pressures.
Back to question Contents

BQ-508. Answer: E. Hypoparathyroidism from small or absent parathyroid tissue
Deletion of 22q11.2 disrupts development of the third and fourth pharyngeal pouches, which form the parathyroid glands, the thymus, and the outflow tract of the heart. Small parathyroid glands cannot maintain calcium once the maternal supply ends, so hypocalcemic seizures appear in the first days. Maternal vitamin D deficiency is the closest trap, but it would not explain the heart and palate. At the bedside, give calcium, check parathyroid hormone and magnesium, order a microarray and lymphocyte counts, and use irradiated blood products until the thymus is assessed.
Key principle: Cleft palate, a conotruncal heart defect, and neonatal hypocalcemia point to 22q11.2 deletion and hypoparathyroidism.
Back to question Contents

BQ-509. Answer: D. Loss of function of the paternally inherited genes at 15q11-q13
Genes in this region are imprinted, so only the paternal copy is active in the relevant brain tissue. Loss of the paternal contribution by deletion or maternal uniparental disomy removes any working copy, producing neonatal hypotonia and later hyperphagia and obesity. Loss of the maternal copy is the closest trap, because it affects the same region but causes Angelman syndrome, with seizures and frequent laughter. At the bedside, confirm with methylation testing, control food access, and involve endocrinology, nutrition, and a genetics team.
Key principle: Prader-Willi syndrome arises from loss of paternal 15q11-q13 function, whereas loss of the maternal copy causes Angelman syndrome.
Back to question Contents

BQ-510. Answer: A. Williams syndrome
A deletion at 7q11.23 includes the elastin gene, which explains the supravalvular narrowing of the aorta, and it is also associated with high calcium in infancy and an outgoing personality. Hypercalcemia explains the irritability and constipation. Angelman syndrome is the closest trap because it also has a happy demeanor, but it comes with seizures, ataxia, and absent speech, not a high calcium or aortic lesion. At the bedside, obtain an echocardiogram, monitor calcium, order a microarray or FISH, and involve cardiology and genetics.
Key principle: An elfin face, friendly manner, supravalvular aortic stenosis, and high calcium describe Williams syndrome.
Back to question Contents

BQ-511. Answer: E. Fragile X syndrome
An expanded CGG repeat in the FMR1 gene silences the gene, which deprives neurons of a protein needed for synaptic function. This makes fragile X the leading inherited cause of intellectual disability, with X-linked transmission through carrier mothers and facial features that emerge after puberty. Klinefelter syndrome is the closest trap since it also causes learning problems in boys, but it brings tall stature and small testes and is not inherited through the maternal line. At the bedside, order FMR1 testing, offer genetic counseling to the mother, and refer for behavioral and educational support.
Key principle: Intellectual disability with a long face, large ears, and maternal male relatives points to fragile X syndrome.
Back to question Contents

BQ-512. Answer: B. Chromosomal microarray
Many causes of unexplained delay are small deletions or duplications that a karyotype cannot resolve, and microarray detects them in a meaningful fraction of children. For this reason it is recommended as the first-tier test for delay, autism, or multiple congenital anomalies. Karyotype is the closest trap, but it is best kept for a recognizable aneuploidy such as Down syndrome. At the bedside, order the microarray, prepare the family for the possibility of a result of uncertain significance, and proceed to exome sequencing only if the array is normal and suspicion remains.
Key principle: Chromosomal microarray is the first-tier test for unexplained delay, autism, or multiple anomalies.
Back to question Contents

BQ-513. Answer: C. Stop lactose feeds and start antibiotics covering Gram-negative bacteria
Galactosemia blocks conversion of galactose from lactose, so galactose-1-phosphate accumulates and injures the liver, kidney, and brain, and infants are prone to Escherichia coli sepsis. Reducing substances in urine with a negative glucose test suggest a sugar other than glucose. Continuing breast milk is the closest trap because it supplies lactose. At the bedside, give a lactose-free soy or elemental formula, treat sepsis promptly, correct coagulopathy, confirm with an enzyme assay, and consult a metabolic center.
Key principle: Jaundice, vomiting, and reducing substances in a milk-fed newborn call for stopping lactose and treating Gram-negative sepsis.
Back to question Contents

BQ-514. Answer: B. Organic acidemia

A block in amino acid breakdown lets organic acids accumulate, which widens the anion gap and consumes bicarbonate, and the metabolic strain can raise ammonia as well. Newborns seldom make ketones, so heavy ketonuria with a wide gap is a strong clue. A urea cycle defect is the closest trap, but it produces high ammonia with respiratory alkalosis and a normal gap. At the bedside, stop protein, give glucose to reverse catabolism, send urine organic acids and acylcarnitines, and transfer to a metabolic center.

Key principle: A wide-gap acidosis with large ketones in a newborn suggests an organic acidemia.

Back to question Contents

BQ-515. Answer: C. Impaired fatty acid oxidation limiting ketone production during fasting

During fasting the body normally burns fatty acids in the liver and makes ketones as a fuel for the brain. A defect in that pathway forces reliance on glucose, which runs out within hours, leaving hypoglycemia without ketones and a fatty liver. Hyperinsulinism is the closest trap, since it also gives low ketones, but it does not explain the liver or the sibling's death. At the bedside, give intravenous glucose at once, send acylcarnitines and urine organic acids before and after treatment, and avoid fasting thereafter.

Key principle: Hypoglycemia without ketones during fasting suggests a fatty acid oxidation defect.

Back to question Contents

BQ-516. Answer: B. It may be falsely reassuring because of early collection, so diagnostic testing is needed

A sample drawn before about 24 hours of age, after transfusion, or during parenteral nutrition can miss a disorder, because feeding and metabolite accumulation have not yet occurred. A screening test is also built to flag, not to exclude, disease. Accepting the result is the closest trap. At the bedside, send ammonia, gas, glucose, lactate, plasma amino acids, urine organic acids, and acylcarnitines with the cultures, stop protein, give glucose, and call a metabolic center without waiting for a repeat card.

Key principle: A normal newborn screen does not exclude disease in a sick infant, especially if collected before 24 hours.

Back to question Contents

BQ-517. Answer: E. Send confirmatory plasma amino acids and contact the metabolic specialist

Phenylalanine hydroxylase deficiency lets phenylalanine accumulate, which damages the developing brain when untreated, yet the infant looks well in the first weeks. A screen is a call to action, not a diagnosis, so quantitative plasma amino acids confirm it and a specialist guides dietary therapy. Reassurance is the closest trap, because the healthy appearance is exactly what the condition produces. At the bedside, contact the family within a day, arrange the blood test and specialist visit, and keep breastfeeding or standard formula until directed.

Key principle: A positive screen for phenylketonuria needs prompt confirmatory plasma amino acids and a metabolic referral.

Back to question Contents

Answers: Domain 22. Rheumatology

Contents Questions for this subject

BQ-518. Answer: A. Systemic juvenile idiopathic arthritis

Systemic arthritis is driven by innate immune activation, with interleukin-1 and interleukin-6 producing quotidian fever, an evanescent rash that appears with the fever, and high inflammatory markers; arthritis may follow weeks later. Leukemia is the closest trap since it can also cause fever and bone pain, but the rash timing and the fever pattern fit systemic disease, and a smear still needs review. At the bedside, exclude infection and malignancy with a smear and cultures, watch for macrophage activation syndrome, and ask rheumatology to consider an interleukin-1 or interleukin-6 inhibitor.

Key principle: Quotidian fever with an evanescent salmon rash and high ferritin suggests systemic juvenile idiopathic arthritis.

Back to question Contents

BQ-519. Answer: B. Uncontrolled macrophage and T cell activation with massive cytokine release

Macrophage activation syndrome is a hyperinflammatory state in which activated macrophages and T cells release large amounts of cytokines and consume blood cells, so counts and fibrinogen fall and ferritin soars. The falling ESR reflects the drop in fibrinogen and is a clue that the disease is changing rather than improving. Drug toxicity is the closest trap, but it would not raise ferritin so sharply. At the bedside, check triglycerides, liver enzymes, and fibrinogen, rule out infection, and call rheumatology urgently, since high-dose steroids and anakinra are used.

Key principle: Persistent fever with falling counts, a falling ESR, and soaring ferritin in systemic arthritis signals macrophage activation syndrome.

Back to question Contents

BQ-520. Answer: D. Obtain an echocardiogram and treat with intravenous immunoglobulin

Kawasaki disease is a vasculitis that can inflame the coronary arteries, and infants often present with prolonged fever and few features. Raised inflammatory markers, anemia, low albumin, and sterile pyuria meet the supportive laboratory criteria of the 2017 AHA algorithm. Waiting for classic criteria is the closest trap, because treatment within 10 days of onset lowers the risk of coronary aneurysm. At the bedside, obtain the echocardiogram, give 2 g/kg of immunoglobulin as a single infusion, start aspirin per protocol, and repeat imaging in follow-up.

Key principle: Infants with fever for 7 days and supportive laboratory findings should be treated for incomplete Kawasaki disease.

Back to question Contents

BQ-521. Answer: A. Medium aneurysm

Coronary artery size is adjusted for body surface area using a Z-score, because arteries of a small child cannot be compared with fixed values. A Z-score of 5 to under 10 marks a medium aneurysm, and the absolute diameter is under 8 mm, so the giant category is excluded. Small aneurysm, which covers 2.5 to under 5, is the closest trap. At the bedside, give immunoglobulin, discuss antiplatelet and possible anticoagulant therapy with cardiology, and arrange serial imaging, because the category sets the follow-up intensity.

Key principle: A coronary Z-score of 5 to under 10 is a medium aneurysm, and 10 or more or 8 mm or more is giant.

Back to question Contents

BQ-522. Answer: B. Urinalysis and blood pressure now, then repeat urine checks for months

In IgA vasculitis, IgA-containing immune complexes deposit in small vessels of the skin, gut, joints, and kidney. Purpura with a normal platelet count is not a bleeding disorder, so transfusion has no role. Starting steroids to prevent nephritis is the closest trap, because it does not change the renal course. Nephritis can appear weeks to months later, so at the bedside check urine and blood pressure at the first visit, repeat them over several months, and watch for bloody stool or severe pain that suggests intussusception.

Key principle: In IgA vasculitis, monitor urine and blood pressure for months because nephritis can appear late.

Back to question Contents

BQ-523. Answer: D. PFAPA syndrome

PFAPA is a periodic autoinflammatory syndrome of early childhood with regular fevers, aphthae, pharyngitis, and adenitis, and attacks end spontaneously with normal growth between them. A dramatic response to one dose of steroid is a typical clue. Cyclic neutropenia is the closest trap since it also produces periodic fever and mouth ulcers, but the counts here were normal during attacks. At the bedside, document attacks with cultures and counts, exclude other causes, and discuss a single steroid dose at onset or, when attacks are disabling, tonsillectomy.

Key principle: Regular fevers every few weeks with aphthae, pharyngitis, adenitis, and normal growth indicate PFAPA.

Back to question Contents

BQ-524. Answer: C. Familial Mediterranean fever

Variants in the MEFV gene make the inflammasome overactive, so short, self-limited bursts of neutrophil-driven inflammation of the peritoneum, pleura, or joints recur, and the child is well between attacks. Ancestry from the Mediterranean, Middle East, or Armenia and an affected sibling support the diagnosis. PFAPA is the closest trap, because it also recurs, but it brings aphthae, pharyngitis, and adenitis, and attacks last longer. At the bedside, confirm with MEFV testing, start daily colchicine to prevent attacks and amyloidosis, and monitor adherence and urine protein.

Key principle: Short recurrent attacks of fever and serositis in a child of Mediterranean, Middle Eastern, or Armenian ancestry suggest familial Mediterranean fever.

Back to question Contents

BQ-525. Answer: B. Juvenile dermatomyositis

Juvenile dermatomyositis is a vasculopathy that injures skin and muscle capillaries, causing heliotrope eyelids, Gottron papules, and proximal weakness with raised muscle enzymes. Lupus is the closest trap because it can also cause a facial rash and weakness, but it does not produce Gottron papules or a marked creatine kinase rise. At the bedside, check aldolase and other muscle enzymes, obtain MRI or biopsy if needed, start corticosteroids and methotrexate promptly with rheumatology, and watch for calcinosis and swallowing problems.

Key principle: Heliotrope eyelids, Gottron papules, and proximal weakness indicate juvenile dermatomyositis.

Back to question Contents

BQ-526. Answer: D. Consumption of complement by immune complexes activating the classical pathway

In lupus, autoantibodies bind nuclear antigens and form immune complexes that deposit in tissues and activate the classical complement pathway, so C3 and C4 are used up faster than the liver can replace them. Low levels therefore signal active disease. Urinary loss is the closest trap, since she has proteinuria, but the amounts lost are small compared with consumption. At the bedside, check urine protein-to-creatinine ratio and creatinine, arrange renal biopsy with nephrology, start hydroxychloroquine, and monitor complement as a marker of activity.

Key principle: Low C3 and C4 in active lupus reflect complement consumption by immune complexes.

Back to question Contents

BQ-527. Answer: E. Intra-articular triamcinolone hexacetonide

Oligoarthritis involves inflamed synovium confined to a few joints, so local therapy can control disease while avoiding systemic drug exposure. The 2021 ACR guideline strongly recommends intra-articular glucocorticoid injection, with triamcinolone hexacetonide preferred, as part of initial therapy. Oral prednisone is the closest trap, because it works but exposes the child to systemic effects when a single joint is involved. At the bedside, arrange the injection with the rheumatologist, continue the eye screening, and escalate to methotrexate if the response is inadequate.

Key principle: Intra-articular triamcinolone hexacetonide is strongly recommended as initial therapy for oligoarticular arthritis.

Back to question Contents

BQ-528. Answer: B. Growing pains

Growing pains are a benign pattern of muscle aching in both legs of young children, thought to follow heavy daytime activity, and they resolve by morning with a normal examination. The absence of limp, joint swelling, constitutional symptoms, and a single bone focus separates them from dangerous causes. Leukemia is the closest trap, because bone pain can be its first sign, but it brings pallor, weight loss, and persistent daytime pain. At the bedside, reassure the family, offer massage and analgesia, and arrange a blood count if red flags develop.

Key principle: Bilateral evening leg pain that is gone by morning in a well, active child is growing pains.

Back to question Contents

Answers: Domain 23. Ethics

Contents Questions for this subject

BQ-529. Answer: C. Use a qualified medical interpreter for the discussion

Informed permission requires that the parents understand the risks, benefits, and alternatives, and medical terms are difficult to translate accurately. A child interpreter may omit or soften frightening information and should not carry the burden of a parent's decision. A phone app is the closest trap, because it is quick but unreliable for consent-level detail. At the bedside, call an in-person, video, or telephone interpreter, document the interpreter's name or identification number, and use teach-back to confirm that the parents understand.

Key principle: Use a qualified medical interpreter, never a sibling, for a consent discussion.

Back to question Contents

BQ-530. Answer: A. Treat under implied consent while continuing to reach the parents

The emergency exception allows treatment without permission when delay would threaten life or cause serious harm and no one with authority can be reached in time. Bleeding from splenic rupture is such a threat. Waiting for a parent is the closest trap, since it follows the usual rule but risks death. At the bedside, treat what is necessary to stabilize the child, document the reasons and the attempts to contact the family, and explain events to the parents as soon as they are reached.

Key principle: Treat without permission only when delay would risk life and no one with authority can be reached in time.

Back to question Contents

BQ-531. Answer: B. Keep talking, involve the ethics service, and seek legal help given the imminent risk

Parents may decline many choices for their child, but their authority has limits when a proven treatment would prevent death or serious harm. Without insulin, ketoacidosis is fatal, so the child's interests come first. Honoring the refusal is the closest trap, because it follows the usual deference to parents. At the bedside, listen for the fears driving the refusal, offer interpreters or trusted advisers, call the ethics service and hospital leadership at once, and request legal or child protection authority to treat if the risk is imminent.

Key principle: When parents refuse a proven life-saving treatment, the child's interests come first, and ethics and legal help are sought.

Back to question Contents

BQ-532. Answer: A. Appreciating how the information applies to himself

Capacity is decision-specific and has four parts: understanding, appreciation, reasoning, and expressing a choice. This boy can recite the information and state a stable choice, but he denies that the risks apply to him, which is a failure of appreciation. Understanding is the closest trap, because it is the part most often tested, yet he demonstrates it. At the bedside, explore his beliefs with open questions, use his own symptoms and goals to personalize the risks, reassess capacity over time, and involve his parents in the discussion.

Key principle: Capacity includes appreciating that the information applies to oneself, not only repeating it.

Back to question Contents

BQ-533. Answer: D. Tell her you must involve her parents and arrange urgent mental health evaluation

Confidentiality is the rule in adolescent care, but it yields when a teen is at serious risk of harming herself or others. When it must be broken, honesty preserves trust: tell her first, explain what will be shared and with whom, and limit the disclosure to what safety requires. Secrecy with later follow-up is the closest trap, because it honors her request but leaves her in danger tonight. At the bedside, stay with her, secure the means, involve her parents, and arrange same-day mental health assessment.

Key principle: Break confidentiality for serious risk of self-harm, tell the teen first, and share only what safety requires.

Back to question Contents

BQ-534. Answer: A. Review what the proxy portal shows and ask how she wants results and bills handled

Adolescents avoid care when they expect disclosure, and the electronic portal and insurance statements are common routes of leakage. The federal privacy rule allows a minor's confidentiality to be protected when she lawfully consents to a service, so settings can be adjusted. Telling her mother is the closest trap, because it solves the portal problem by removing the minor's control. At the bedside, check what the proxy sees, set the confidential note and result release options, ask the billing office about suppressing mailings, and record a private contact number.

Key principle: Check portal and billing settings before the visit ends so lawful confidential care actually stays confidential.

Back to question Contents

BQ-535. Answer: D. Explain that palliative care adds symptom control and family support alongside treatment

Palliative care is a layer of support for the child and family that begins at diagnosis of a serious illness and runs together with disease-directed care. Parents who hear it as abandonment often decline, so naming what it does helps. Offering hospice is the closest trap, because hospice is for a child expected to die and is a different model. At the bedside, describe pain and symptom control, help with hard choices, and sibling support, state that treatment continues, and offer to arrange a first visit with no commitment.

Key principle: Describe palliative care as added support alongside treatment, from diagnosis of a serious illness.

Back to question Contents

BQ-536. Answer: C. A time-limited trial is acceptable, as withdrawing is ethically equivalent to not starting

Decisions about life-sustaining treatment rest on the child's best interests and shared decision-making. Professional bodies, including the AAP, regard withholding and withdrawing as ethically and morally equivalent, so a trial can be started and stopped when it no longer helps. The belief that starting commits everyone is the closest trap, and it makes families fear any trial. At the bedside, set the goals and the time limit of the trial in advance, explain that comfort care continues in every plan, and meet again to review progress.

Key principle: A time-limited trial of life support is acceptable because withholding and withdrawing are ethically equivalent.

Back to question Contents

BQ-537. Answer: E. Request an ethics service consultation

When parents and clinicians disagree about a major decision and the discussion stalls, a structured ethics consultation can clarify values, facts, and options, and it supports both parties. Skipping to a court order is the closest trap, since it may eventually be needed, but it is adversarial and should follow efforts to reach agreement. At the bedside, request the consult early, offer a palliative care team to help with goals, keep communicating with the family, and document the discussions and the child's current care plan.

Key principle: Call the ethics service early when parents and the team disagree and talks stall.

Back to question Contents

BQ-538. Answer: A. It covers resuscitation only; other care continues and the plan can be revisited

A do-not-resuscitate order is a plan about one intervention, cardiopulmonary resuscitation, and says nothing about other care. Chemotherapy, antibiotics, nutrition, and symptom relief continue unless the family and team choose otherwise, and the order can be reviewed as the situation changes. The belief that it ends all treatment is the closest trap and causes many families to refuse. At the bedside, repeat the explanation in plain words, ask what the boy wishes to know, document the order, and review it at each major change.

Key principle: A do-not-resuscitate order covers one intervention, not withdrawal of care, and it can be revisited.

Back to question Contents

BQ-539. Answer: D. Honor his refusal and do not enroll him

Research that offers no direct benefit places a child at some risk for the benefit of others, so his dissent carries far more weight than in treatment of a sick child. Parental permission is necessary but not sufficient when a capable child refuses. Enrolling on parental permission alone is the closest trap, because it applies the rule for necessary treatment to optional research. At the bedside, thank him, explain that saying no will not change his care, and document that he was not enrolled.

Key principle: A child's refusal to join research without direct benefit should be honored.

Back to question Contents

Answers: Domain 24. Patient Safety, Quality Improvement, and Research Methods

Contents Questions for this subject

BQ-540. Answer: B. Near miss

A near miss is an error that is caught before it reaches the patient. Here the pharmacist's independent check served as a defensive layer, and the harm did not occur. A preventable adverse event is the closest trap, since the same error would have caused one had the barrier failed, but no harm reached the child. At the bedside, thank the pharmacist, report the event through the safety system, and review the order process, for example standard concentrations and dose-range checking, so that the same hole does not line up again.

Key principle: An error intercepted before it reaches the patient is a near miss, and it should be reported.

Back to question Contents

BQ-541. Answer: A. Perform a root cause analysis of system and process failures

Most harm in hospitals arises from system design, in which weaknesses in several layers of defense line up, as the Swiss cheese model describes. A structured root cause analysis identifies those weaknesses, such as marking, time-out, and communication failures, and leads to redesign. Blaming the individual is the closest trap, because it feels decisive but leaves the hazards in place. At the bedside, disclose to the family, support the staff involved under a just culture approach, and track the corrective actions.

Key principle: Use root cause analysis to find system failures behind a sentinel event instead of blaming individuals.

Back to question Contents

BQ-542. Answer: E. A standardized handoff format combined with team training

Handoffs fail when information is unstructured, hurried, or interrupted. The I-PASS bundle combined a standard format with team training and, in a study across nine pediatric residency programs, reduced medical errors by 23 percent and preventable adverse events by 30 percent. A reminder to be careful is the closest trap, because it is inexpensive, yet exhortation without system change rarely works. At the bedside, adopt a structured handoff with illness severity, action list, situation awareness, and synthesis by the receiver, and train and audit the team.

Key principle: A structured handoff bundle with team training reduces errors more than reminders to be careful.

Back to question Contents

BQ-543. Answer: C. Ask the institutional review board to determine whether review is required
Quality improvement answers a local question about whether a change improves care, while research seeks generalizable knowledge, and the difference affects oversight. Intent to share results does not by itself turn a project into research, which is why the first trap, assuming research, is wrong. The decision belongs to the review board, not the investigators. At the bedside, submit a brief description, follow the board's determination, and use SQUIRE 2.0 guidelines to report the work if it is published.
Key principle: When it is unclear whether a project is quality improvement or research, ask the review board first.
Back to question Contents

BQ-544. Answer: A. The proportion given antibiotics who prove to have no bacterial infection
A balancing measure shows whether a change has caused trouble elsewhere in the system. A drive to treat quickly can lead to many children without infection receiving antibiotics, with cost and resistance, and this measure would reveal it. Mortality is the closest trap, because it is an outcome measure that tells whether patients are better off, not what the change has disrupted. At the bedside, track the measures on run charts, plot each change on the chart, and review the data at team meetings to adjust the process.
Key principle: A balancing measure detects unintended harm elsewhere, unlike process and outcome measures.
Back to question Contents

BQ-545. Answer: D. A sustained shift that is unlikely to be random variation
On a run chart, random variation scatters points on both sides of the median. A sustained run on one side, or a steady trend, signals a nonrandom change, which here follows the intervention and so supports a real improvement. Calling it noise is the closest trap, since a run chart needs no statistical test to show a shift. At the bedside, annotate the date of each change, continue plotting, check that the gain holds, and consider whether another factor, such as a season, might also explain the change.
Key principle: A sustained run of points on one side of the median signals real change, not random variation.
Back to question Contents

BQ-546. Answer: C. Case-control study
A case-control study starts with children who have the outcome and compares their exposure history with a matched group without it, so a rare condition can be studied with a modest number of participants. A cohort study is the closest trap, because it is stronger for incidence and risk, but it would need to follow tens of thousands of exposed pregnancies to find a few cases. In practice, select controls carefully and reduce recall bias by using records rather than memory when possible.
Key principle: Case-control designs suit rare outcomes, whereas cohort designs suit common outcomes and incidence.
Back to question Contents

BQ-547. Answer: E. Recall bias

Mothers of affected infants search their memory harder for possible causes, so they report exposures more completely than mothers of healthy infants, producing an apparent association. Equal prescribing in the records shows that the actual exposure did not differ. Selection bias is the closest trap, but it concerns how participants are chosen, not how they remember. In practice, use objective records, blinded interviewers, and structured questions to reduce this error, and interpret interview-based associations with caution.

Key principle: Case mothers recalling exposures better than control mothers is recall bias.

Back to question Contents

BQ-548. Answer: B. 87 percent

Specificity is the proportion of unaffected children who test negative. Among the 150 children without the condition, 20 test positive, so 130 test negative, and 130 divided by 150 is about 87 percent. The value of 90 percent is the sensitivity, 45 of 50, which is the closest trap; 69 percent is the positive predictive value and 96 percent is the negative predictive value. In practice, a highly specific test is better for ruling in disease, and predictive values still depend on how common the condition is in the population tested.

Key principle: Specificity equals true negatives divided by all unaffected children.

Back to question Contents

BQ-549. Answer: E. 25

The absolute risk reduction is the control event rate minus the treated rate, 12 percent minus 8 percent, or 4 percentage points. The number needed to treat is 1 divided by the absolute risk reduction, so 1 divided by 0.04 is 25. The value 33 is the closest trap because it is the relative risk reduction expressed as a percentage, 4 divided by 12, and it is commonly mistaken for a number needed to treat. In practice, quote absolute numbers, such as about 4 fewer hospitalizations per 100 children, when advising families.

Key principle: The number needed to treat is 1 divided by the absolute risk reduction.

Back to question Contents

BQ-550. Answer: B. The data show no clear difference, because the interval includes 1

A confidence interval shows the range of effects compatible with the data. For a ratio, the value 1 means no difference, so an interval that includes 1 cannot exclude no effect, even though the point estimate favors treatment. Reading the 20 percent reduction as significant is the closest trap, because it uses the point estimate alone. A type I error is a false-positive finding, which cannot be shown here. In practice, ask whether the interval also includes clinically important benefit and whether a larger study is needed.

Key principle: A ratio confidence interval that includes 1 does not show a clear difference.

Back to question Contents